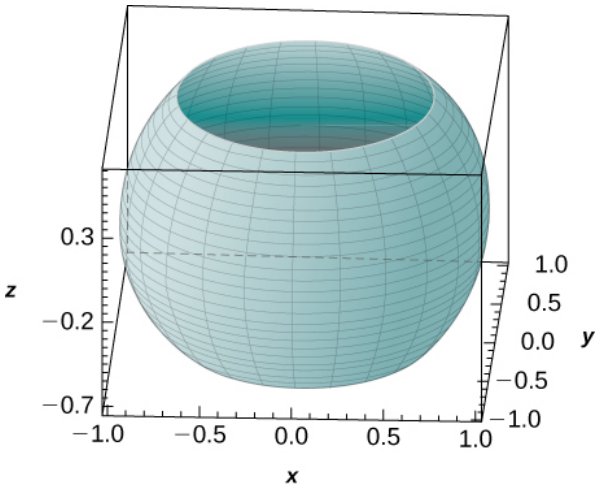
46. Convert the integral $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} dz\,dy\,dx$ into an integral in spherical coordinates and evaluate it.

47. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in spherical coordinates $\int_{\pi/2}^{\pi}\int_{5\pi}^{\pi/6}\int_{0}^{2}\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$. Find the volume V of the solid. Round your answer to three decimal places.

Answer

$V=\frac{4\pi\sqrt{3}}{3}\approx 7.255$



48. [T] Use a CAS to graph the solid whose volume is given by the iterated integral in spherical coordinates as $\int_{0}^{2\pi}\int_{3\pi/4}^{\pi/4}\int_{0}^{1}\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$. Find the volume V of the solid. Round your answer to three decimal places.

49. [T] Use a CAS to evaluate the integral $\iiint_E (x^2+y^2)\,dV$ where E lies above the paraboloid $z=x^2+y^2$ and below the plane $z=3y$.

Answer

$\frac{343\pi}{32}$

50. [T]

a. Evaluate the integral $\iiint_E e^{\sqrt{x^2+y^2+z^2}}\,dV$, where E is bounded by spheres $4x^2+4y^2+4z^2=1$ and $x^2+y^2+z^2=1$.

b. Use a CAS to find an approximation of the previous integral. Round your answer to two decimal places.

51. Express the volume of the solid inside the sphere $x^2+y^2+z^2=16$ and outside the cylinder $x^2+y^2=4$ as triple integrals in cylindrical coordinates and spherical coordinates, respectively.

Answer

$\int_{0}^{2\pi}\int_{2}^{4}\int_{-\sqrt{16-r^2}}^{\sqrt{16-r^2}} r\,dz\,dr\,d\theta$ and $\int_{\pi/6}^{5\pi/6}\int_{0}^{2\pi}\int_{2\csc\phi}^{4}\rho^2\sin\rho\,d\rho\,d\theta\,d\phi$

52. Express the volume of the solid inside the sphere $x^2+y^2+z^2=16$ and outside the cylinder $x^2+y^2=4$ that is located in the first octant as triple integrals in cylindrical coordinates and spherical coordinates, respectively.

53. The power emitted by an antenna has a power density per unit volume given in spherical coordinates by $p(\rho,\theta,\varphi)=\frac{P_0}{\rho^2}\cos^2\theta\,\sin^4\varphi$, where P_0 is a constant with units in watts. The total power within a sphere B of radius r meters is defined as $P=\iiint_B p(\rho,\theta,\varphi)\,dV$. Find the total power P.

Answer

$P=\frac{32P_0\pi}{3}$ watts

54. Use the preceding exercise to find the total power within a sphere B of radius 5 meters when the power density per unit volume is given by $p(\rho,\theta,\varphi)=\frac{30}{\rho^2}\cos^2\theta\sin^4\varphi$.

55. A charge cloud contained in a sphere B of radius r centimeters centered at the origin has its charge density given by $q(x,y,z) = k\sqrt{x^2+y^2+z^2}\,\frac{\mu C}{cm^3}$, where $k>0$. The total charge contained in B is given by $Q = \iiint_B q(x,y,z)\,dV$. Find the total charge Q.

Answer

$Q = kr^4\pi\mu C$

56. Use the preceding exercise to find the total charge cloud contained in the unit sphere if the charge density is $q(x,y,z) = 20\sqrt{x^2+y^2+z^2}\,\frac{\mu C}{cm^3}$.

CONTRIBUTORS

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

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15.6: Calculating Centers of Mass and Moments of Inertia

Learning Objectives

- Use double integrals to locate the center of mass of a two-dimensional object.
- Use double integrals to find the moment of inertia of a two-dimensional object.
- Use triple integrals to locate the center of mass of a three-dimensional object.

We have already discussed a few applications of multiple integrals, such as finding areas, volumes, and the average value of a function over a bounded region. In this section we develop computational techniques for finding the center of mass and moments of inertia of several types of physical objects, using double integrals for a lamina (flat plate) and triple integrals for a three-dimensional object with variable density. The density is usually considered to be a constant number when the lamina or the object is homogeneous; that is, the object has uniform density.

Center of Mass in Two Dimensions

The center of mass is also known as the center of gravity if the object is in a uniform gravitational field. If the object has uniform density, the center of mass is the geometric center of the object, which is called the centroid. Figure 15.6.1 shows a point P as the center of mass of a lamina. The lamina is perfectly balanced about its center of mass.

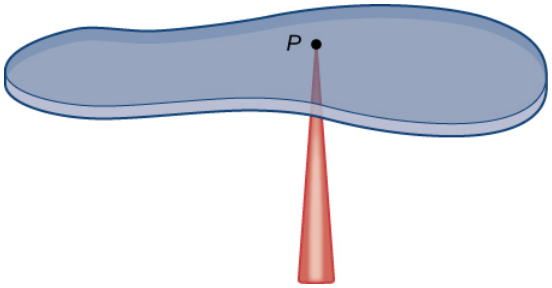


Figure 15.6.1: A lamina is perfectly balanced on a spindle if the lamina's center of mass sits on the spindle.

To find the coordinates of the center of mass $P(\bar{x}, \bar{y})$ of a lamina, we need to find the moment M_x of the lamina about the x-axis and the moment M_y about the y-axis. We also need to find the mass m of the lamina. Then

$$\bar{x} = \frac{M_y}{m}$$

and

$$\bar{y} = \frac{M_x}{m}.$$

Refer to Moments and Centers of Mass for the definitions and the methods of single integration to find the center of mass of a one-dimensional object (for example, a thin rod). We are going to use a similar idea here except that the object is a two-dimensional lamina and we use a double integral.

If we allow a constant density function, then $\bar{x} = \frac{M_y}{m}$ and $\bar{y} = \frac{M_x}{m}$ give the *centroid* of the lamina.

Suppose that the lamina occupies a region R in the xy-plane and let $\rho(x, y)$ be its density (in units of mass per unit area) at any point (x, y). Hence,

$$\rho(x, y) = \lim_{\Delta A \to 0} \frac{\Delta m}{\Delta A}$$

where Δm and ΔA are the mass and area of a small rectangle containing the point (x, y) and the limit is taken as the dimensions of the rectangle go to 0 (see the following figure).

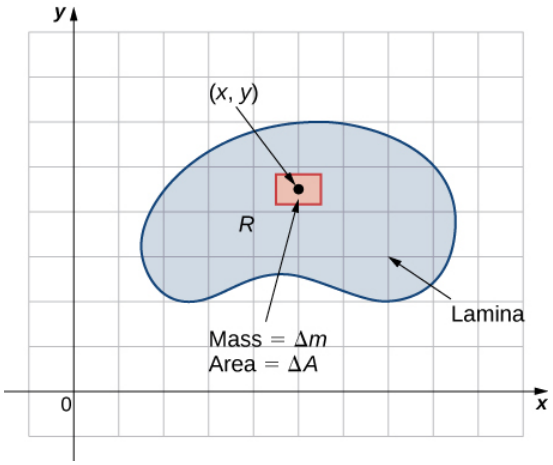


Figure 15.6.2: The density of a lamina at a point is the limit of its mass per area in a small rectangle about the point as the area goes to zero.

Just as before, we divide the region R into tiny rectangles R_{ij} with area ΔA and choose (x_{ij}^*, y_{ij}^*) as sample points. Then the mass m_{ij} of each R_{ij} is equal to $\rho(x_{ij}^*, y_{ij}^*)\Delta A$ (Figure 15.6.2). Let k and l be the number of subintervals in x and y respectively. Also, note that the shape might not always be rectangular but the limit works anyway, as seen in previous sections.

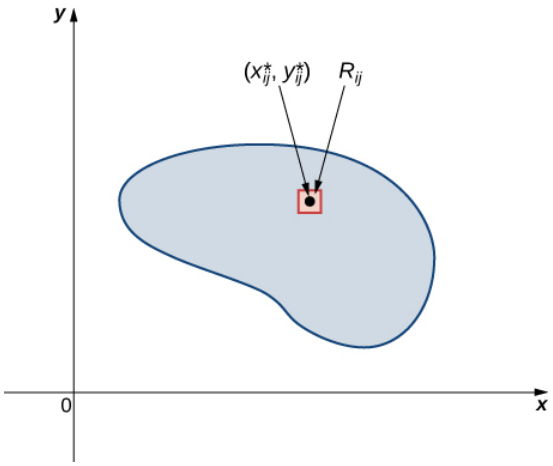


Figure 15.6.3: Subdividing the lamina into tiny rectangles R_{ij} each containing a sample point (x_{ij}^*, y_{ij}^*).

Hence, the mass of the lamina is

$$m = \lim_{k,l\to\infty} \sum_{i=1}^{k} \sum_{j=1}^{l} m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k} \sum_{j=1}^{l} \rho(x_{ij}^*, y_{ij}^*)\Delta A = \iint_R \rho(x,y)dA.$$

Let's see an example now of finding the total mass of a triangular lamina.

Example 15.6.1: Finding the Total Mass of a Lamina

Consider a triangular lamina R with vertices $(0,0)$, $(0,3)$, $(3,0)$ and with density $\rho(x,y) = xy\, kg/m^2$. Find the total mass.

Solution

A sketch of the region R is always helpful, as shown in the following figure.

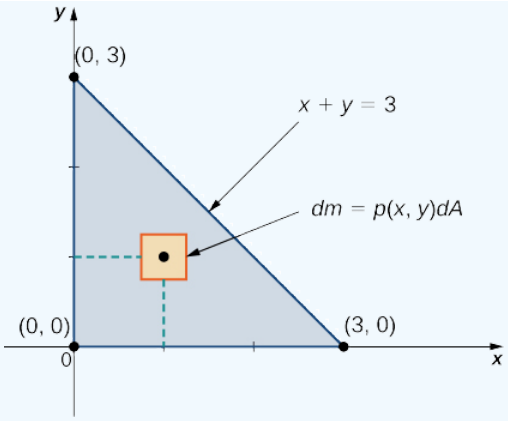


Figure 15.6.4: A lamina in the xy-plane with density $\rho(x,y) = xy$.

Using the expression developed for mass, we see that

$$m = \iint_R dm = \iint_R \rho(x,y)dA = \int_{x=0}^{x=3}\int_{y=0}^{y=3-x} xy\,dy\,dx = \int_{x=0}^{x=3}\left[x\frac{y^2}{2}\Big|_{y=0}^{y=3}\right]dx$$
$$= \int_{x=0}^{x=3}\frac{1}{2}x(3-x)^2dx = \left[\frac{9x^2}{4} - x^3 + \frac{x^4}{8}\right]\Big|_{x=0}^{x=3} = \frac{27}{8}.$$

The computation is straightforward, giving the answer $m = \frac{27}{8}\,kg$.

? Exercise 15.6.1

Consider the same region R as in the previous example, and use the density function $\rho(x,y) = \sqrt{xy}$. Find the total mass.

Answer

$\frac{9\pi}{8}\,kg$

Now that we have established the expression for mass, we have the tools we need for calculating moments and centers of mass. The moment M_z about the x-axis for R is the limit of the sums of moments of the regions R_{ij} about the x-axis. Hence

$$M_x = \lim_{k,l\to\infty}\sum_{i=1}^{k}\sum_{j=1}^{l}(y_{ij}^*)m_{ij} = \lim_{k,l\to\infty}\sum_{i=1}^{k}\sum_{j=1}^{l}(y_{ij}^*)\rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R y\rho(x,y)\,dA$$

Similarly, the moment M_y about the y-axis for R is the limit of the sums of moments of the regions R_{ij} about the y-axis. Hence

$$M_y = \lim_{k,l\to\infty}\sum_{i=1}^{k}\sum_{j=1}^{l}(x_{ij}^*)m_{ij} = \lim_{k,l\to\infty}\sum_{i=1}^{k}\sum_{j=1}^{l}(x_{ij}^*)\rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R x\rho(x,y)\,dA$$

✓ Example 15.6.2: Finding Moments

Consider the same triangular lamina R with vertices $(0,0),\ (0,3),\ (3,0)$ and with density $\rho(x,y) = xy$. Find the moments M_x and M_y.

Solution

Use double integrals for each moment and compute their values:

$$M_x = \iint_R y\rho(x,y)\,dA = \int_{x=0}^{x=3}\int_{y=0}^{y=3-x} xy^2\,dy\,dx = \frac{81}{20},$$

$$M_y = \iint_R x\rho(x,y)\,dA = \int_{x=0}^{x=3}\int_{y=0}^{y=3-x} x^2y\,dy\,dx = \frac{81}{20},$$

The computation is quite straightforward.

Exercise 15.6.2

Consider the same lamina R as above and use the density function $\rho(x,y)=\sqrt{xy}$. Find the moments M_x and M_y.

Answer

$M_x=\dfrac{81\pi}{64}$ and $M_y=\dfrac{81\pi}{64}$

Finally we are ready to restate the expressions for the center of mass in terms of integrals. We denote the *x*-coordinate of the center of mass by $\bar{x}$ and the *y*-coordinate by $\bar{y}$. Specifically,

$$\bar{x}=\frac{M_y}{m}=\frac{\iint_R x\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA}$$

and

$$\bar{y}=\frac{M_x}{m}=\frac{\iint_R y\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA}$$

Example 15.6.3: center of mass

Again consider the same triangular region R with vertices $(0,0),\ (0,3),\ (3,0)$ and with density function $\rho(x,y)=xy$. Find the center of mass.

Solution

Using the formulas we developed, we have

$$\bar{x}=\frac{M_y}{m}=\frac{\iint_R x\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA}=\frac{81/20}{27/8}=\frac{6}{5},$$

$$\bar{y}=\frac{M_x}{m}=\frac{\iint_R y\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA}=\frac{81/20}{27/8}=\frac{6}{5}.$$

Therefore, the center of mass is the point $\left(\frac{6}{5},\frac{6}{5}\right)$.

Analysis

If we choose the density $\rho(x,y)$ instead to be uniform throughout the region (i.e., constant), such as the value 1 (any constant will do), then we can compute the centroid,

$$x_c=\frac{M_y}{m}=\frac{\iint_R x\,dA}{\iint_R dA}=\frac{9/2}{9/2}=1,$$

$$y_c=\frac{M_x}{m}=\frac{\iint_R y\,dA}{\iint_R dA}=\frac{9/2}{9/2}=1.$$

Notice that the center of mass $\left(\frac{6}{5},\frac{6}{5}\right)$ is not exactly the same as the centroid $(1,1)$ of the triangular region. This is due to the variable density of R If the density is constant, then we just use $\rho(x,y)=c$ (constant). This value cancels out from the formulas, so for a constant density, the center of mass coincides with the centroid of the lamina.

Exercise 15.6.3

Again use the same region R as above and use the density function $\rho(x,y)=\sqrt{xy}$. Find the center of mass.

Answer

$$\bar{x} = \frac{M_y}{m} = \frac{81\pi/64}{9\pi/8} = \frac{9}{8} \text{ and } \bar{y} = \frac{M_x}{m} = \frac{81\pi}{9\pi/8} = \frac{0}{8}.$$

Once again, based on the comments at the end of Example 15.6.3, we have expressions for the centroid of a region on the plane:

$$x_c = \frac{M_y}{m} = \frac{\iint_R x\,dA}{\iint_R dA} \text{ and } y_c = \frac{M_x}{m} = \frac{\iint_R y\,dA}{\iint_R dA}.$$

We should use these formulas and verify the centroid of the triangular region referred to in the last three examples.

Example 15.6.4: Finding Mass, Moments, and Center of Mass

Find the mass, moments, and the center of mass of the lamina of density $\rho(x, y) = x + y$ occupying the region R under the curve $y = x^2$ in the interval $0 \le x \le 2$ (see the following figure).

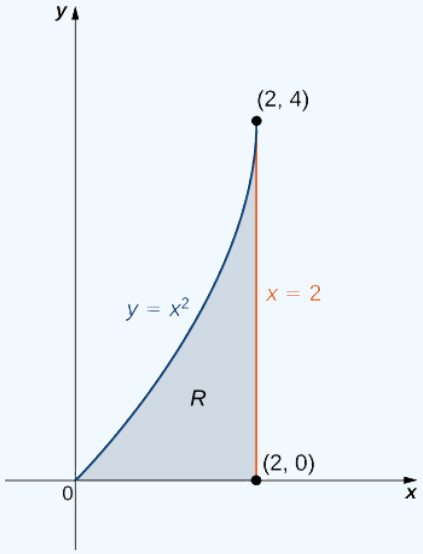


Figure 15.6.5: Locating the center of mass of a lamina R with density $\rho(x, y) = x + y$.

Solution

First we compute the mass m. We need to describe the region between the graph of $y = x^2$ and the vertical lines $x = 0$ and $x = 2$:

$$m = \iint_R dm = \iint_R \rho(x, y)\,dA = \int_{x=0} x = 2 \int_{y=0}^{y=x^2} (x + y)dy\,dx = \int_{x=0}^{x=2} \left[xy + \frac{y^2}{2} \Big|_{y=0}^{y=x^2} \right] dx$$

$$= \int_{x=0}^{x=2} \left[x^3 + \frac{x^4}{2} \right] dx = \left[\frac{x^4}{4} + \frac{x^5}{10} \right] \Big|_{x=0}^{x=2} = \frac{36}{5}.$$

Now compute the moments M_x and M_y:

$$M_x = \iint_R y\rho(x, y)\,dA = \int_{x=0}^{x=2} \int_{y=0}^{y=x^2} y(x + y)\,dy\,dx = \frac{80}{7},$$

$$M_y = \iint_R x\rho(x, y)\,dA = \int_{x=0}^{x=2} \int_{y=0}^{y=x^2} x(x + y)\,dy\,dx = \frac{176}{15}.$$

Finally, evaluate the center of mass,

$$\bar{x} = \frac{M_y}{m} = \frac{\iint_R x\rho(x, y)\,dA}{\iint_R \rho(x, y)\,dA} = \frac{176/15}{36/5} = \frac{44}{27},$$

$$\bar{y} = \frac{M_x}{m} = \frac{\iint_R y\rho(x, y)\,dA}{\iint_R \rho(x, y)\,dA} = \frac{80/7}{36/5} = \frac{100}{63}.$$

Hence the center of mass is $(\bar{x}, \bar{y}) = \left(\frac{44}{27}, \frac{100}{63}\right)$.

Exercise 15.6.4

Calculate the mass, moments, and the center of mass of the region between the curves $y = x$ and $y = x^2$ with the density function $\rho(x, y) = x$ in the interval $0 \le x \le 1$.

Answer

$\bar{x} = \frac{M_y}{m} = \frac{1/20}{1/12} = \frac{3}{5}$ and $\bar{y} = \frac{M_x}{m} = \frac{1/24}{1/12} = \frac{1}{2}$

Example 15.6.5: Finding a Centroid

Find the centroid of the region under the curve $y = e^x$ over the interval $1 \le x \le 3$ (Figure 15.6.6).

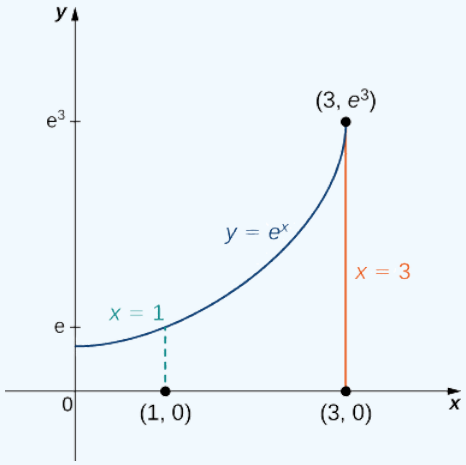


Figure 15.6.6: Finding a centroid of a region below the curve $y = e^x$.

Solution

To compute the centroid, we assume that the density function is constant and hence it cancels out:

$$x_c = \frac{M_y}{m} = \frac{\iint_R x\,dA}{\iint_R dA} \text{ and } y_c = \frac{M_x}{m} = \frac{\iint_R y\,dA}{\iint_R dA},$$

$$x_c = \frac{M_y}{m} = \frac{\iint_R x\,dA}{\iint_R dA} = \frac{\int_{x=1}^{x=3}\int_{y=0}^{y=e^x} x\,dy\,dx}{\int_{x=1}^{x=3}\int_{y=0}^{y=e^x} dy\,dx} = \frac{\int_{x=1}^{x=3} xe^x dx}{\int_{x=1}^{x=3} e^x dx} = \frac{2e^3}{e^3 - e} = \frac{2e^2}{e^2 - 1},$$

$$y_c = \frac{M_x}{m} = \frac{\iint_R y\,dA}{\iint_R dA} = \frac{\int_{x=1}^{x=3}\int_{y=0}^{y=e^x} y\,dy\,dx}{\int_{x=1}^{x=3}\int_{y=0}^{y=e^x} dy\,dx} = \frac{\int_{x=1}^{x=3} \frac{e^{2x}}{2} dx}{\int_{x=1}^{x=3} e^x dx} = \frac{\frac{1}{4}e^2(e^4 - 1)}{e(e^2 - 1)} = \frac{1}{4}e(e^2 + 1).$$

Thus the centroid of the region is

$$(x_c, y_c) = \left(\frac{2e^2}{e^2 - 1}, \frac{1}{4}e(e^2 + 1)\right).$$

Exercise 15.6.5

Calculate the centroid of the region between the curves $y = x$ and $y = \sqrt{x}$ with uniform density in the interval $0 \le x \le 1$.

Answer

$$x_c = \frac{M_y}{m} = \frac{1/15}{1/6} = \frac{2}{5} \text{ and } y_c = \frac{M_x}{m} = \frac{1/12}{1/6} = \frac{1}{2}$$

Moments of Inertia

For a clear understanding of how to calculate moments of inertia using double integrals, we need to go back to the general definition in Section 6.6. The moment of inertia of a particle of mass m about an axis is mr^2 where r is the distance of the particle from the axis. We can see from Figure 15.6.3 that the moment of inertia of the subrectangle R_{ij} about the x-axis is $(y_{ij}^*)^2\rho(x_{ij}^*, y_{ij}^*)\Delta A$. Similarly, the moment of inertia of the subrectangle R_{ij} about the y-axis is $(x_{ij}^*)^2\rho(x_{ij}^*, y_{ij}^*)\Delta A$. The moment of inertia is related to the rotation of the mass; specifically, it measures the tendency of the mass to resist a change in rotational motion about an axis.

The moment of inertia I_x about the x-axis for the region R is the limit of the sum of moments of inertia of the regions R_{ij} about the x-axis. Hence

$$I_x = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (y_{ij}^*)^2 m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (y_{ij}^*)^2 \rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R y^2\rho(x,y)\,dA.$$

Similarly, the moment of inertia I_y about the y-axis for R is the limit of the sum of moments of inertia of the regions R_{ij} about the y -axis. Hence

$$I_y = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (x_{ij}^*)^2 m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (x_{ij}^*)^2 \rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R x^2\rho(x,y)\,dA.$$

Sometimes, we need to find the moment of inertia of an object about the origin, which is known as the polar moment of inertia. We denote this by I_0 and obtain it by adding the moments of inertia I_x and I_y. Hence

$$I_0 = I_x + I_y = \iint_R (x^2+y^2)\rho(x,y)\,dA.$$

All these expressions can be written in polar coordinates by substituting $x = r\cos\theta$, $y = r\sin\theta$, and $dA = r\,dr\,d\theta$. For example, $I_0 = \iint_R r^2\rho(r\cos\theta, r\sin\theta)\,dA$.

Example 15.6.6: Finding Moments of Inertia for a Triangular Lamina

Use the triangular region R with vertices $(0,0)$, $(2,2)$, and $(2,0)$ and with density $\rho(x,y) = xy$ as in previous examples. Find the moments of inertia.

Solution

Using the expressions established above for the moments of inertia, we have

$$I_x = \iint_R y^2\rho(x,y)\,dA = \int_{x=0}^{x=2}\int_{y=0}^{y=x} xy^3\,dy\,dx = \frac{8}{3},$$

$$I_y = \iint_R x^2\rho(x,y)\,dA = \int_{x=0}^{x=2}\int_{y=0}^{y=x} x^3y\,dy\,dx = \frac{16}{3},$$

$$I_0 = \iint_R (x^2+y^2)\rho(x,y)\,dA = \int_0^2\int_0^x (x^2+y^2)xy\,dy\,dx = I_x + I_y = 8$$

Exercise 15.6.6

Again use the same region R as above and the density function $\rho(x,y) = \sqrt{xy}$. Find the moments of inertia.

Answer

$$I_x = \int_{x=0}^{x=2}\int_{y=0}^{y=x} y^2\sqrt{xy}\,dy\,dx = \frac{64}{35}$$

and

$$I_y = \int_{x=0}^{x=2} \int_{y=0}^{y=x} x^2 \sqrt{xy}\, dy\, dx = \frac{64}{35}.$$

Also,

$$I_0 = \int_{x=0}^{x=2} \int_{y=0}^{y=x} (x^2 + y^2) \sqrt{xy}\, dy\, dx = \frac{128}{21}$$

As mentioned earlier, the moment of inertia of a particle of mass m about an axis is mr^2 where r is the distance of the particle from the axis, also known as the **radius of gyration**.

Hence the radii of gyration with respect to the x-axis, the y-axis and the origin are

$$R_x = \sqrt{\frac{I_x}{m}},\ R_y = \sqrt{\frac{I_y}{m}},\ and\ R_0 = \sqrt{\frac{I_0}{m}},$$

respectively. In each case, the radius of gyration tells us how far (perpendicular distance) from the axis of rotation the entire mass of an object might be concentrated. The moments of an object are useful for finding information on the balance and torque of the object about an axis, but radii of gyration are used to describe the distribution of mass around its centroidal axis. There are many applications in engineering and physics. Sometimes it is necessary to find the radius of gyration, as in the next example.

Example 15.6.7: Finding the Radius of Gyration for a Triangular Lamina

Consider the same triangular lamina R with vertices $(0,0)$, $(2,2)$, and $(2,0)$ and with density $\rho(x,y) = xy$ as in previous examples. Find the radii of gyration with respect to the x-axis the y-axis and the origin.

Solution

If we compute the mass of this region we find that $m = 2$. We found the moments of inertia of this lamina in Example 15.6.4. From these data, the radii of gyration with respect to the x-axis, y-axis and the origin are, respectively,

$$R_x = \sqrt{\frac{I_x}{m}} = \sqrt{\frac{8/3}{2}} = \sqrt{\frac{8}{6}} = \frac{2\sqrt{3}}{3}, \tag{15.6.1}$$

$$R_y = \sqrt{\frac{I_y}{m}} = \sqrt{\frac{16/3}{2}} = \sqrt{\frac{8}{3}} = \frac{2\sqrt{6}}{3}, \tag{15.6.2}$$

$$R_0 = \sqrt{\frac{I_0}{m}} = \sqrt{\frac{8}{2}} = \sqrt{4} = 2. \tag{15.6.3}$$

Exercise 15.6.7

Use the same region R from Example 15.6.7 and the density function $\rho(x,y) = \sqrt{xy}$. Find the radii of gyration with respect to the x-axis, the y-axis, and the origin.

Hint

Follow the steps shown in the previous example.

Answer

$R_x = \frac{6\sqrt{35}}{35}$, $R_y = \frac{6\sqrt{15}}{15}$, and $R_0 = \frac{4\sqrt{42}}{7}$.

Center of Mass and Moments of Inertia in Three Dimensions

All the expressions of double integrals discussed so far can be modified to become triple integrals.

Definition

If we have a solid object Q with a density function $\rho(x, y, z)$ at any point (x, y, z) in space, then its mass is

$$m = \iiint_Q \rho(x, y, z)\, dV.$$

Its moments about the xy-plane the xz-plane and the yz-plane are

$$M_{xy} = \iiint_Q z\rho(x, y, z)\, dV,\ M_{xz} = \iiint_Q y\rho(x, y, z)\, dV,\ M_{yz} = \iiint_Q x\rho(x, y, z)\, dV.$$

If the center of mass of the object is the point $(\bar{x}, \bar{y}, \bar{z})$, then

$$\bar{x} = \frac{M_{yz}}{m},\ \bar{y} = \frac{M_{xz}}{m},\ \bar{z} = \frac{M_{xy}}{m}.$$

Also, if the solid object is homogeneous (with constant density), then the center of mass becomes the centroid of the solid. Finally, the moments of inertia about the yz-plane, xz-plane, and the xy-plane are

$$I_x = \iiint_Q (y^2 + z^2)\, \rho(x, y, z)\, dV,$$

$$I_y = \iiint_Q (x^2 + z^2)\, \rho(x, y, z)\, dV,$$

$$I_z = \iiint_Q (x^2 + y^2)\, \rho(x, y, z)\, dV.$$

Example 15.6.8: Finding the Mass of a Solid

Suppose that Q is a solid region bounded by $x + 2y + 3z = 6$ and the coordinate planes and has density $\rho(x, y, z) = x^2yz$. Find the total mass.

Solution

The region Q is a tetrahedron (Figure 15.6.7) meeting the axes at the points $(6, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$. To find the limits of integration, let $z = 0$ in the slanted plane $z = \frac{1}{3}(6 - x - 2y)$. Then for x and y find the projection of Q onto the xy-plane, which is bounded by the axes and the line $x + 2y = 6$. Hence the mass is

$$m = \iiint_Q \rho(x, y, z)\, dV = \int_{x=0}^{x=6} \int_{y=0}^{y=1/2(6-x)} \int_{z=0}^{z=1/3(6-x-2y)} x^2yz\, dz\, dy\, dx = \frac{108}{35}$$

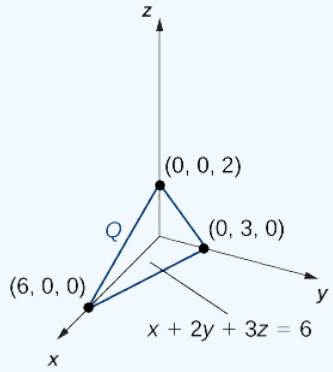


Figure 15.6.7: Finding the mass of a three-dimensional solid Q.

Exercise 15.6.8

Consider the same region Q (Figure 15.6.7), and use the density function $\rho(x, y, z) = xy^2z$. Find the mass.

Hint

Follow the steps in the previous example.

Answer

$\frac{54}{35} = 1.543$

Example 15.6.9: Finding the Center of Mass of a Solid

Suppose Q is a solid region bounded by the plane $x + 2y + 3z = 6$ and the coordinate planes with density $\rho(x, y, z) = x^2yz$ (see Figure 15.6.7). Find the center of mass using decimal approximation.

Solution

We have used this tetrahedron before and know the limits of integration, so we can proceed to the computations right away. First, we need to find the moments about the xy-plane, the xz-plane, and the yz-plane:

$$M_{xy} = \iiint_Q z\rho(x, y, z)\, dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} x^2yz^2\, dz\, dy\, dx = \frac{54}{35} \approx 1.543,$$

$$M_{xz} = \iiint_Q y\rho(x, y, z)\, dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} x^2y^2z\, dz\, dy\, dx = \frac{81}{35} \approx 2.314,$$

$$M_{yz} = \iiint_Q x\rho(x, y, z)\, dV = \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} x^3yz\, dz\, dy\, dx = \frac{243}{35} \approx 6.943.$$

Hence the center of mass is

$$\bar{x} = \frac{M_{yz}}{m},\ \bar{y} = \frac{M_{xz}}{m},\ \bar{z} = \frac{M_{xy}}{m},$$

$$\bar{x} = \frac{M_{yz}}{m} = \frac{243/35}{108/35} = \frac{243}{108} = 2.25,$$

$$\bar{y} = \frac{M_{xz}}{m} = \frac{81/35}{108/35} = \frac{81}{108} = 0.75,$$

$$\bar{z} = \frac{M_{xy}}{m} = \frac{54/35}{108/35} = \frac{54}{108} = 0.5$$

The center of mass for the tetrahedron Q is the point $(2.25, 0.75, 0.5)$.

Exercise 15.6.9

Consider the same region Q (Figure 15.6.7) and use the density function $\rho(x, y, z) = xy^2z$. Find the center of mass.

Hint

Check that $M_{xy} = \frac{27}{35}$, $M_{xz} = \frac{243}{140}$, and $M_{yz} = \frac{81}{35}$. Then use m from a previous checkpoint question.

Answer

$\left(\frac{3}{2}, \frac{9}{8}, \frac{1}{2}\right)$

We conclude this section with an example of finding moments of inertia I_x, I_y, and I_z.

Example 15.6.10: Finding the Moments of Inertia of a Solid

Suppose that Q is a solid region and is bounded by $x + 2y + 3z = 6$ and the coordinate planes with density $\rho(x, y, z) = x^2yz$ (see Figure 15.6.7). Find the moments of inertia of the tetrahedron Q about the yz-plane, the xz-plane, and the xy-plane.

Solution

Once again, we can almost immediately write the limits of integration and hence we can quickly proceed to evaluating the moments of inertia. Using the formula stated before, the moments of inertia of the tetrahedron Q about the yz-plane, the xz-plane, and the xy-plane are

$$I_x = \iiint_Q (y^2 + z^2)\rho(x, y, z)\, dV,$$

$$I_y = \iiint_Q (x^2 + z^2)\rho(x, y, z)\, dV,$$

and

$$I_z = \iiint_Q (x^2 + y^2)\rho(x, y, z)\, dV \; with \; \rho(x, y, z) = x^2yz.$$

Proceeding with the computations, we have

$$I_x = \iiint_Q (y^2 + z^2)x^2\rho(x, y, z)\, dV$$

$$= \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} (y^2 + z^2)x^2yz\, dz\, dy\, dx = \frac{117}{35} \approx 3.343,$$

$$I_y = \iiint_Q (x^2 + z^2)x^2\rho(x, y, z)\, dV$$

$$= \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} (x^2 + z^2)x^2yz\, dz\, dy\, dx = \frac{684}{35} \approx 19.543,$$

$$I_z = \iiint_Q (x^2 + y^2)x^2\rho(x, y, z)\, dV$$

$$= \int_{x=0}^{x=6} \int_{y=0}^{y=\frac{1}{2}(6-x)} \int_{z=0}^{z=\frac{1}{3}(6-x-2y)} (x^2 + y^2)x^2yz\, dz\, dy\, dx = \frac{729}{35} \approx 20.829.$$

Thus, the moments of inertia of the tetrahedron Q about the yz-plane, the xz-plane, and the xy-plane are $117/35$, $684/35$, and $729/35$, respectively.

Exercise 15.6.10

Consider the same region Q (Figure 15.6.7), and use the density function $\rho(x, y, z) = xy^2z$. Find the moments of inertia about the three coordinate planes.

Answer

The moments of inertia of the tetrahedron Q about the yz-plane, the xz-plane, and the xy-plane are $99/35$, $36/7$ and $243/35$, respectively.

Key Concepts

Finding the mass, center of mass, moments, and moments of inertia in double integrals:

- For a lamina R with a density function $\rho(x, y)$ at any point (x, y) in the plane, the mass is

$$m = \iint_R \rho(x, y)\, dA.$$

- The moments about the x-axis and y-axis are

$$M_x = \iint_R y\rho(x,y)\,dA \,and\, M_y = \iint_R x\rho(x,y)\,dA.$$

- The center of mass is given by $\bar{x} = \dfrac{M_y}{m}$, $\bar{y} = \dfrac{M_x}{m}$.
- The center of mass becomes the centroid of the plane when the density is constant.
- The moments of inertia about the x-axis, y-axis, and the origin are

$$I_x = \iint_R y^2\rho(x,y)\,dA,\ I_y = \iint_R x^2\rho(x,y)\,dA,\ and\, I_0 = I_x + I_y = \iint_R (x^2+y^2)\rho(x,y)\,dA.$$

Finding the mass, center of mass, moments, and moments of inertia in triple integrals:

- For a solid object Q with a density function $\rho(x,y,z)$ at any point (x,y,z) in space, the mass is

$$m = \iiint_Q \rho(x,y,z)\,dV.$$

- The moments about the xy-plane, the xz-plane, and the yz-plane are

$$M_{xy} = \iiint_Q z\rho(x,y,z)\,dV,\ M_{xz} = \iiint_Q y\rho(x,y,z)\,dV,\ M_{yz} = \iiint_Q x\rho(x,y,z)\,dV$$

- The center of mass is given by $\bar{x} = \dfrac{M_{yz}}{m}$, $\bar{y} = \dfrac{M_{xz}}{m}$, $\bar{z} = \dfrac{M_{xy}}{m}$.
- The center of mass becomes the centroid of the solid when the density is constant.
- The moments of inertia about the yz-plane, the xz-plane, and the xy-plane are

$$I_x = \iiint_Q (y^2+z^2)\,\rho(x,y,z)\,dV,\ I_y = \iiint_Q (x^2+z^2)\,\rho(x,y,z)\,dV,$$
$$I_z = \iiint_Q (x^2+y^2)\,\rho(x,y,z)\,dV.$$

Key Equations

- **Mass of a lamina**

$$m = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} \rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R \rho(x,y)\,dA$$

- **Moment about the *x*-axis**

$$M_x = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (y_{ij}^*) m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (y_{ij}^*)\rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R y\rho(x,y)\,dA$$

- **Moment about the *y*-axis**

$$M_y = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (x_{ij}^*) m_{ij} = \lim_{k,l\to\infty} \sum_{i=1}^{k}\sum_{j=1}^{l} (x_{ij}^*)\rho(x_{ij}^*, y_{ij}^*)\,\Delta A = \iint_R x\rho(x,y)\,dA$$

- **Center of mass of a lamina**

$$\bar{x} = \frac{M_y}{m} = \frac{\iint_R x\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA} \,and\, \bar{y} = \frac{M_x}{m} = \frac{\iint_R y\rho(x,y)\,dA}{\iint_R \rho(x,y)\,dA}$$

Glossary

radius of gyration

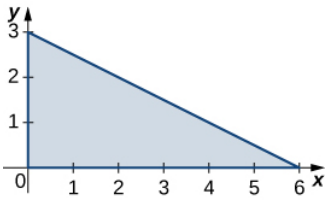
the distance from an object's center of mass to its axis of rotation

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15.6E: Exercises for Section 15.6

In exercises 1 - 12, the region R occupied by a lamina is shown in a graph. Find the mass of R with the density function ρ.

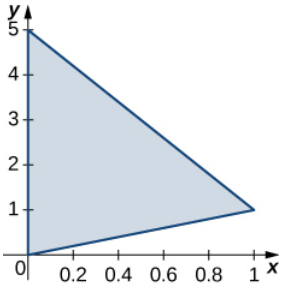
1. R is the triangular region with vertices $(0,0),\ (0,3)$, and $(6,0);\ \rho(x,y)=xy$.



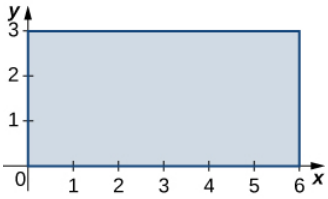
Answer

$\frac{27}{2}$

2. R is the triangular region with vertices $(0,0),\ (1,1)$, and $(0,5);\ \rho(x,y)=x+y$.



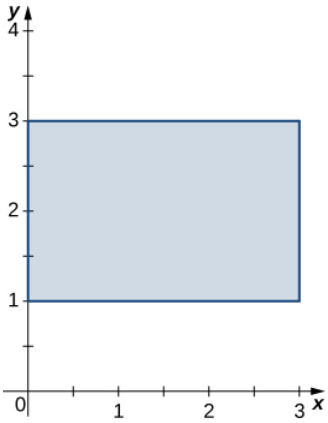
3. R is the rectangular region with vertices $(0,0),\ (0,3),\ (6,3)$ and $(6,0);\ \rho(x,y)=\sqrt{xy}$.



Answer

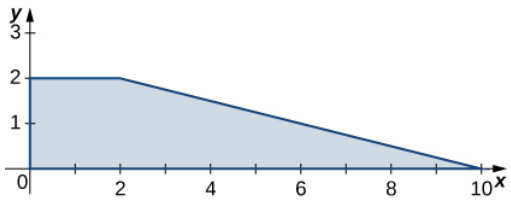
$24\sqrt{2}$

4. R is the rectangular region with vertices $(0,1),\ (0,3),\ (3,3)$ and $(3,1);\ \rho(x,y)=x^2y$.



5. R is the trapezoidal region determined by the lines $y=-\frac{1}{4}x+\frac{5}{2},\ y=0,\ y=2$, and $x=0;\ \rho(x,y)=3xy$.

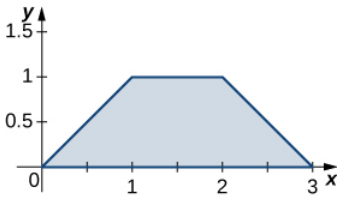
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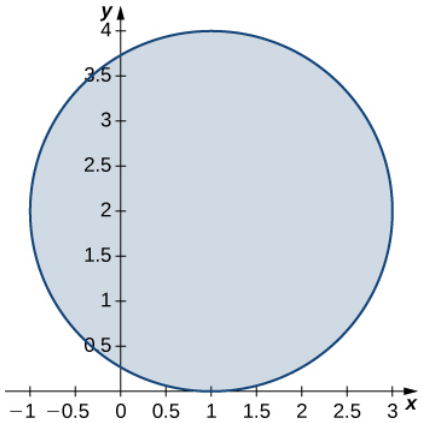
Answer

76

6. R is the trapezoidal region determined by the lines $y = 0,\ y = 1,\ y = x$ and $y = -x + 3;\ \rho(x, y) = 2x + y$.



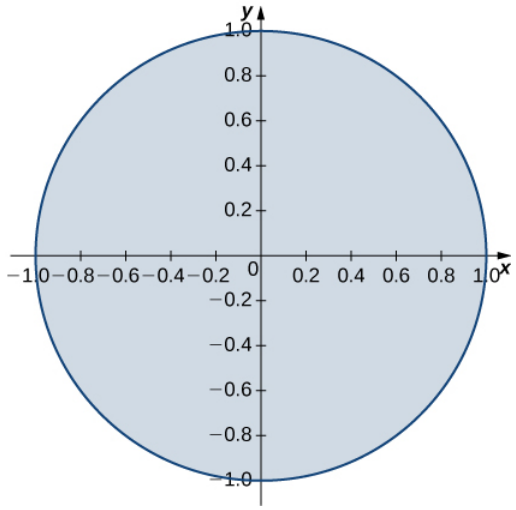
7. R is the disk of radius 2 centered at $(1, 2);\ \rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.



Answer

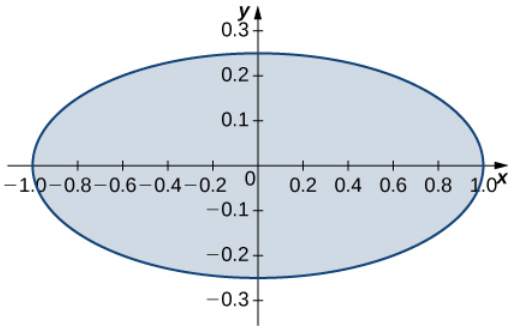
8π

8. R is the unit disk; $\rho(x, y) = 3x^4 + 6x^2y^2 + 3y^4$.



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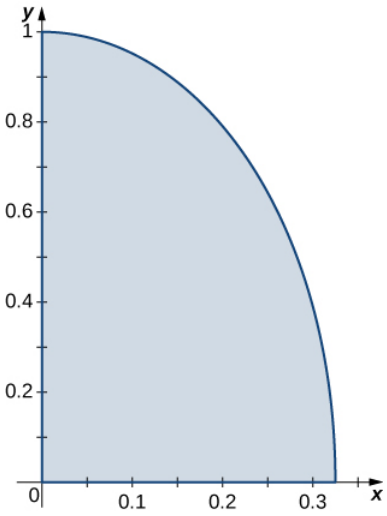
9. R is the region enclosed by the ellipse $x^2 + 4y^2 = 1;\ \rho(x, y) = 1$.



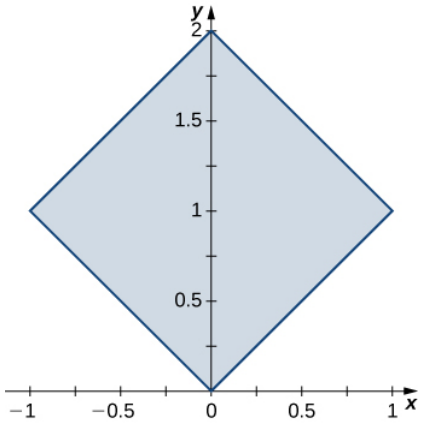
Answer

$\frac{\pi}{2}$

10. $R = \{(x, y) \mid 9x^2 + y^2 \leq 1,\ x \geq 0,\ y \geq 0\};\ \rho(x, y) = \sqrt{9x^2 + y^2}$.



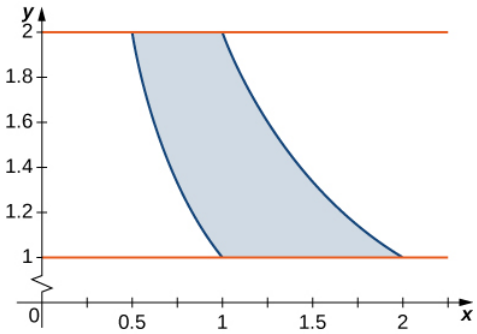
11. R is the region bounded by $y = x,\ y = -x,\ y = x + 2,\ y = -x + 2;\ \rho(x, y) = 1$.



Answer

2

12. R is the region bounded by $y = \frac{1}{x},\ y = \frac{2}{x},\ y = 1$, and $y = 2;\ \rho(x, y) = 4(x + y)$.



In exercises 13 - 24, consider a lamina occupying the region R and having the density function ρ given in the preceding group of exercises. Use a computer algebra system (CAS) to answer the following questions.

a. Find the moments M_x and M_y about the x-axis and y-axis, respectively.

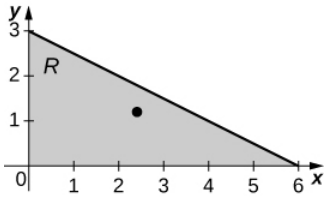
b. Calculate and plot the center of mass of the lamina.

c. [T] Use a CAS to locate the center of mass on the graph of R.

13. [T] R is the triangular region with vertices $(0,0),\ (0,3)$, and $(6,0);\ \rho(x,y)=xy$.

Answer

a. $M_x=\frac{81}{5},\ M_y=\frac{162}{5}$;
b. $\bar{x}=\frac{12}{5},\ \bar{y}=\frac{6}{5}$;
c.

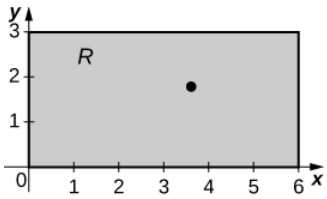


14. [T] R is the triangular region with vertices $(0,0),\ (1,1)$, and $(0,5);\ \rho(x,y)=x+y$.

15. [T] R is the rectangular region with vertices $(0,0),\ (0,3),\ (6,3)$, and $(6,0);\ \rho(x,y)=\sqrt{xy}$.

Answer

a. $M_x=\frac{216\sqrt{2}}{5},\ M_y=\frac{432\sqrt{2}}{5}$;
b. $\bar{x}=\frac{18}{5},\ \bar{y}=\frac{9}{5}$;
c.



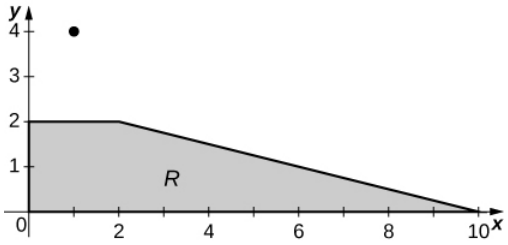
16. [T] R is the rectangular region with vertices $(0,1),\ (0,3),\ (3,3)$, and $(3,1);\ \rho(x,y)=x^2y$.

17. [T] R is the trapezoidal region determined by the lines $y=-\frac{1}{4}x+\frac{5}{2},\ y=0,\ y=2$, and $x=0;\ \rho(x,y)=3xy$.

Answer

a. $M_x=\frac{368}{5},\ M_y=\frac{1552}{5}$;
b. $\bar{x}=\frac{92}{95},\ \bar{y}=\frac{388}{95}$;

c.



18. [T] R is the trapezoidal region determined by the lines $y = 0,\ y = 1,\ y = x$, and $y = -x + 3;\ \rho(x, y) = 2x + y$.

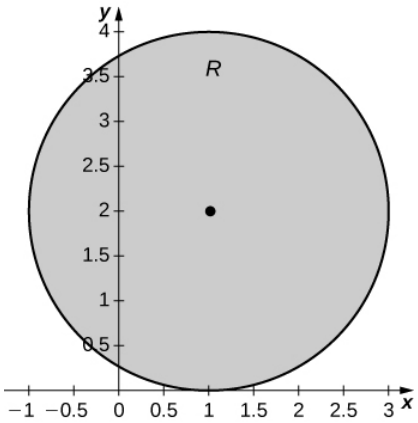
19. [T] R is the disk of radius 2 centered at $(1, 2);\ \rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.

Answer

a. $M_x = 16\pi,\ M_y = 8\pi$;

b. $\bar{x} = 1,\ \bar{y} = 2$;

c.



20. [T] R is the unit disk; $\rho(x, y) = 3x^4 + 6x^2y^2 + 3y^4$.

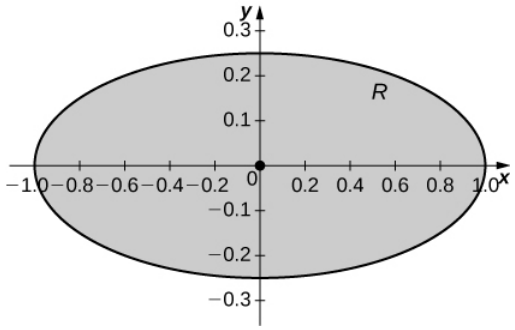
21. [T] R is the region enclosed by the ellipse $x^2 + 4y^2 = 1;\ \rho(x, y) = 1$.

Answer

a. $M_x = 0,\ M_y = 0)$;

b. $\bar{x} = 0,\ \bar{y} = 0$;

c.

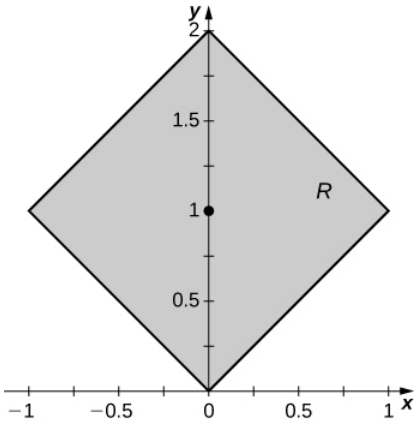


22. [T] $R = \{(x, y) \mid 9x^2 + y^2 \le 1,\ x \ge 0,\ y \ge 0\};\ \rho(x, y) = \sqrt{9x^2 + y^2}$.

23. [T] R is the region bounded by $y = x,\ y = -x,\ y = x + 2$, and $y = -x + 2;\ \rho(x, y) = 1$.

Answer

a. $M_x = 2,\ M_y = 0$);
b. $\bar{x} = 0,\ \bar{y} = 1$;
c.



24. [T] R is the region bounded by $y = \frac{1}{x},\ y = \frac{2}{x},\ y = 1$, and $y = 2;\ \rho(x, y) = 4(x + y)$.

In exercises 25 - 36, consider a lamina occupying the region R and having the density function ρ given in the first two groups of Exercises.

a. Find the moments of inertia I_x, I_y and I_0 about the x-axis, y-axis, and origin, respectively.

b. Find the radii of gyration with respect to the x-axis, y-axis, and origin, respectively.

25. R is the triangular region with vertices $(0, 0),\ (0, 3)$, and $(6, 0);\ \rho(x, y) = xy$.

Answer

a. $I_x = \frac{243}{10},\ I_y = \frac{486}{5}$, and $I_0 = \frac{243}{2}$;
b. $R_x = \frac{3\sqrt{5}}{5},\ R_y = \frac{6\sqrt{5}}{5}$, and $R_0 = 3$

26. R is the triangular region with vertices $(0, 0),\ (1, 1)$, and $(0, 5);\ \rho(x, y) = x + y$.

27. R is the rectangular region with vertices $(0, 0),\ (0, 3),\ (6, 3)$, and $(6, 0);\ \rho(x, y) = \sqrt{xy}$.

Answer

a. $I_x = \frac{2592\sqrt{2}}{7},\ I_y = \frac{648\sqrt{2}}{7}$, and $I_0 = \frac{3240\sqrt{2}}{7}$;
b. $R_x = \frac{6\sqrt{21}}{7},\ R_y = \frac{3\sqrt{21}}{7}$, and $R_0 = \frac{3\sqrt{106}}{7}$

28. R is the rectangular region with vertices $(0, 1),\ (0, 3),\ (3, 3)$, and $(3, 1);\ \rho(x, y) = x^2y$.

29. R is the trapezoidal region determined by the lines $y = -\frac{1}{4}x + \frac{5}{2},\ y = 0,\ y = 2$, and x = 0; \space \rho (x,y) = 3xy\).

Answer

a. $I_x = 88,\ I_y = 1560$, and $I_0 = 1648$;
b. $R_x = \frac{\sqrt{418}}{19},\ R_y = \frac{\sqrt{7410}}{10}$, and $R_0 = \frac{2\sqrt{1957}}{19}$

30. R is the trapezoidal region determined by the lines $y = 0,\ y = 1,\ y = x$, and y = -x + 3; \space \rho (x,y) = 2x + y\).

31. R is the disk of radius 2 centered at $(1, 2);\ \rho(x, y) = x^2 + y^2 - 2x - 4y + 5$.

Answer

a. $I_x = \frac{128\pi}{3},\ I_y = \frac{56\pi}{3}$, and $I_0 = \frac{184\pi}{3}$;
b. $R_x = \frac{4\sqrt{3}}{3},\ R_y = \frac{\sqrt{21}}{2}$, and $R_0 = \frac{\sqrt{69}}{3}$

32. R is the unit disk; $\rho(x,y) = 3x^4 + 6x^2y^2 + 3y^4$.

33. R is the region enclosed by the ellipse $x^2 + 4y^2 = 1;\ \rho(x,y) = 1$.

Answer

a. $I_x = \frac{\pi}{32},\ I_y = \frac{\pi}{8}$, and $I_0 = \frac{5\pi}{32}$;
b. $R_x = \frac{1}{4},\ R_y = \frac{1}{2}$, and $R_0 = \frac{\sqrt{5}}{4}$

34. $R = \{(x,y) \mid 9x^2 + y^2 \le 1,\ x \ge 0,\ y \ge 0\};\ \rho(x,y) = \sqrt{9x^2 + y^2}$.

35. R is the region bounded by $y = x,\ y = -x,\ y = x + 2$, and $y = -x + 2;\ \rho(x,y) = 1$.

Answer

a. $I_x = \frac{7}{3},\ I_y = \frac{1}{3}$, and $I_0 = \frac{8}{3}$;
b. $R_x = \frac{\sqrt{42}}{6},\ R_y = \frac{\sqrt{6}}{6}$, and $R_0 = \frac{2\sqrt{3}}{3}$

36. R is the region bounded by $y = \frac{1}{x},\ y = \frac{2}{x},\ y = 1$, and $y = 2;\ \rho(x,y) = 4(x + y)$.

37. Let Q be the solid unit cube. Find the mass of the solid if its density ρ is equal to the square of the distance of an arbitrary point of Q to the xy-plane.

Answer

$m = \frac{1}{3}$

38. Let Q be the solid unit hemisphere. Find the mass of the solid if its density ρ is proportional to the distance of an arbitrary point of Q to the origin.

39. The solid Q of constant density 1 is situated inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the sphere $x^2 + y^2 + z^2 = 1$. Show that the center of mass of the solid is not located within the solid.

40. Find the mass of the solid $Q = \{(x,y,z) \mid 1 \le x^2 + z^2 \le 25,\ y \le 1 - x^2 - z^2\}$ whose density is $\rho(x,y,z) = k$, where $k > 0$.

41. [T] The solid $Q = \{(x,y,z) \mid x^2 + y^2 \le 9,\ 0 \le z \le 1,\ x \ge 0,\ y \ge 0\}$ has density equal to the distance to the xy-plane. Use a CAS to answer the following questions.

a. Find the mass of Q.

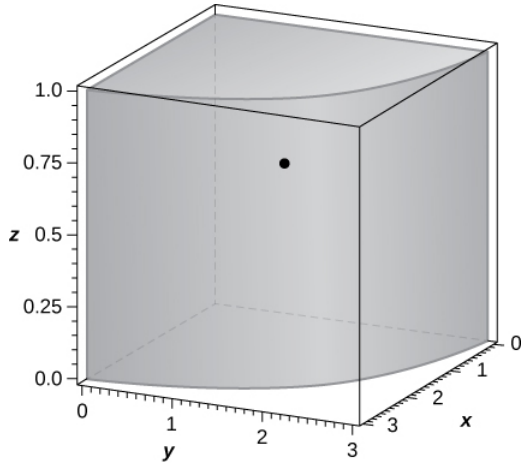
b. Find the moments $M_{xy},\ M_{xz}$ and M_{yz} about the xy-plane, xz-plane, and yz-plane, respectively.

c. Find the center of mass of Q.

d. Graph Q and locate its center of mass.

Answer

a. $m = \frac{9\pi}{4}$;
b. $M_{xy} = \frac{3\pi}{2},\ M_{xz} = \frac{81}{8},\ M_{yz} = \frac{81}{8}$;
c. $\bar{x} = \frac{9}{2\pi},\ \bar{y} = \frac{9}{2\pi},\ \bar{z} = \frac{2}{3}$;
d.



42. Consider the solid $Q = \{(x, y, z) \,|\, 0 \le x \le 1,\ 0 \le y \le 2,\ 0 \le z \le 3\}$ with the density function $\rho(x, y, z) = x + y + 1$.

a. Find the mass of Q.

b. Find the moments $M_{xy},\ M_{xz}$ and M_{yz} about the xy-plane, xz-plane, and yz-plane, respectively.

c. Find the center of mass of Q.

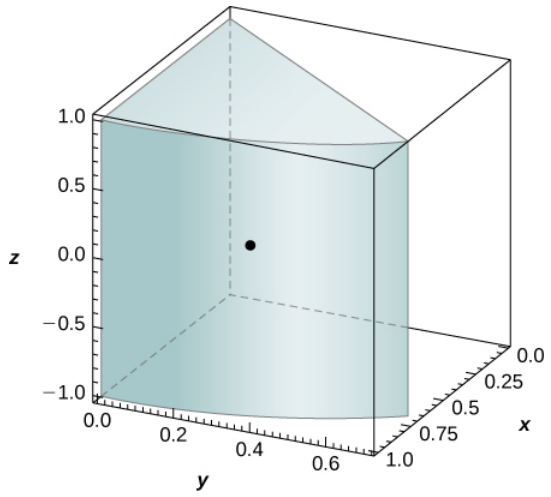
43. [T] The solid Q has the mass given by the triple integral $\int_{-1}^{1} \int_{0}^{\pi/4} \int_{0}^{1} r^2 \, dr \, d\theta \, dz.$

Use a CAS to answer the following questions.

- Show that the center of mass of Q is located in the xy-plane.
- Graph Q and locate its center of mass.

Answer

$\bar{x} = \frac{3\sqrt{2}}{2\pi}$, $\bar{y} = \frac{3(2-\sqrt{2})}{2\pi}$, $\bar{z} = 0$; 2. the solid Q and its center of mass are shown in the following figure.



44. The solid Q is bounded by the planes $x + 4y + z = 8,\ x = 0,\ y = 0$, and $z = 0$. Its density at any point is equal to the distance to the xz-plane. Find the moments of inertia of the solid about the xz-plane.

45. The solid Q is bounded by the planes $x + y + z = 3,\ x = 0,\ y = 0$, and $z = 0$. Its density is $\rho(x, y, z) = x + ay$, where $a > 0$. Show that the center of mass of the solid is located in the plane $z = \frac{3}{5}$ for any value of a.

46. Let Q be the solid situated outside the sphere $x^2 + y^2 + z^2 = z$ and inside the upper hemisphere $x^2 + y^2 + z^2 = R^2$, where $R > 1$. If the density of the solid is $\rho(x, y, z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$, find R such that the mass of the solid is $\frac{7\pi}{2}$.

47. The mass of a solid Q is given by $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} (x^2+y^2+z^2)^n \, dz \, dy \, dx$, where n is an integer. Determine n such the mass of the solid is $(2-\sqrt{2})\pi$.

Answer

$n = -2$

48. Let Q be the solid bounded above the cone $x^2+y^2=z^2$ and below the sphere $x^2+y^2+z^2-4z=0$. Its density is a constant $k > 0$. Find k such that the center of mass of the solid is situated 7 units from the origin.

49. The solid $Q = \{(x,y,z) \mid 0 \le x^2+y^2 \le 16, \ x \ge 0, \ y \ge 0, \ 0 \le z \le x\}$ has the density $\rho(x,y,z) = k$. Show that the moment M_{xy} about the xy-plane is half of the moment M_{yz} about the yz-plane.

50. The solid Q is bounded by the cylinder $x^2+y^2=a^2$, the paraboloid $b^2 - z = x^2+y^2$, and the xy-plane, where $0 < a < b$. Find the mass of the solid if its density is given by $\rho(x,y,z) = \sqrt{x^2+y^2}$.

51. Let Q be a solid of constant density k, where $k > 0$, that is located in the first octant, inside the circular cone $x^2+y^2 = 9(z-1)^2$, and above the plane $z = 0$. Show that the moment M_{xy} about the xy-plane is the same as the moment M_{yz} about the xz-plane.

52. The solid Q has the mass given by the triple integral $\int_0^1 \int_0^{\pi/2} \int_0^{r^3} (r^4 + r) \, dz \, d\theta \, dr$.

a. Find the density of the solid in rectangular coordinates.

b. Find the moment M_{xy} about the xy-plane.

53. The solid Q has the moment of inertia I_x about the yz-plane given by the triple integral $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_{\frac{1}{2}(x^2+y^2)}^{\sqrt{x^2+y^2}} (y^2+z^2)(x^2+y^2) \, dz \, dx \, dy$.

a. Find the density of Q.

b. Find the moment of inertia I_z about the xy-plane.

Answer

a. $\rho(x,y,z) = x^2+y^2$;
b. $\frac{16\pi}{7}$

54. The solid Q has the mass given by the triple integral $\int_0^{\pi/4} \int_0^{2 \sec\theta} \int_0^1 (r^3 \cos\theta \sin\theta + 2r) \, dz \, dr \, d\theta$.

a. Find the density of the solid in rectangular coordinates.

b. Find the moment M_{xz} about the xz-plane.

55. Let Q be the solid bounded by the xy-plane, the cylinder $x^2+y^2=a^2$, and the plane $z = 1$, where $a > 1$ is a real number. Find the moment M_{xy} of the solid about the xy-plane if its density given in cylindrical coordinates is $\rho(x,y,z) = \frac{d^2 f}{dr^2}(r)$, where f is a differentiable function with the first and second derivatives continuous and differentiable on $(0,a)$.

Answer

$M_{xy} = \pi(f(0) - f(a) + af'(a))$

56. A solid Q has a volume given by $\iint_D \int_a^b dA \, dz$, where D is the projection of the solid onto the xy-plane and $a < b$ are real numbers, and its density does not depend on the variable z. Show that its center of mass lies in the plane $z = \frac{a+b}{2}$.

57. Consider the solid enclosed by the cylinder $x^2+z^2=a^2$ and the planes $y = b$ and $y = c$, where $a > 0$ and $b < c$ are real numbers. The density of Q is given by $\rho(x,y,z) = f'(y)$, where f is a differential function whose derivative is continuous on (b,c). Show that if $f(b) = f(c)$, then the moment of inertia about the xz-plane of Q is null.

58. [T] The average density of a solid Q is defined as $\rho_{ave} = \frac{1}{V(Q)} \iiint_Q \rho(x,y,z)\, dV = \frac{m}{V(Q)}$, where $V(Q)$ and m are the volume and the mass of Q, respectively. If the density of the unit ball centered at the origin is $\rho(x,y,z) = e^{-x^2-y^2-z^2}$, use a CAS to find its average density. Round your answer to three decimal places.

59. Show that the moments of inertia $I_x,\ I_y$, and I_z about the yz-plane, xz-plane, and xy-plane, respectively, of the unit ball centered at the origin whose density is $\rho(x,y,z) = e^{-x^2-y^2-z^2}$ are the same. Round your answer to two decimal places.

Answer

$I_x = I_y = I_z \approx 0.84$

Contributors

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15.7: Change of Variables in Multiple Integrals

Learning Objectives

- Determine the image of a region under a given transformation of variables.
- Compute the Jacobian of a given transformation.
- Evaluate a double integral using a change of variables.
- Evaluate a triple integral using a change of variables.

Recall from Substitution Rule the method of integration by substitution. When evaluating an integral such as

$$\int_2^3 x(x^2-4)^5 dx,$$

we substitute $u = g(x) = x^2 - 4$. Then $du = 2x\,dx$ or $x\,dx = \frac{1}{2}du$ and the limits change to $u = g(2) = 2^2 - 4 = 0$ and $u = g(3) = 9 - 4 = 5$. Thus the integral becomes

$$\int_0^5 \frac{1}{2}u^5 du$$

and this integral is much simpler to evaluate. In other words, when solving integration problems, we make appropriate substitutions to obtain an integral that becomes much simpler than the original integral.

We also used this idea when we transformed double integrals in rectangular coordinates to polar coordinates and transformed triple integrals in rectangular coordinates to cylindrical or spherical coordinates to make the computations simpler. More generally,

$$\int_a^b f(x)dx = \int_c^d f(g(u))g'(u)du,$$

Where $x = g(u)$, $dx = g'(u)du$, and $u = c$ and $u = d$ satisfy $c = g(a)$ and $d = g(b)$.

A similar result occurs in double integrals when we substitute

- $x = f(r,\theta) = r\,\cos\,\theta$
- $y = g(r,\theta) = r\,\sin\,\theta$, and
- $dA = dx\,dy = r\,dr\,d\theta$.

Then we get

$$\iint_R f(x,y)dA = \iint_S (r\,\cos\,\theta,\, r\,\sin\,\theta)r\,dr\,d\theta$$

where the domain R is replaced by the domain S in polar coordinates. Generally, the function that we use to change the variables to make the integration simpler is called a transformation or mapping.

Planar Transformations

A planar transformation T is a function that transforms a region G in one plane into a region R in another plane by a change of variables. Both G and R are subsets of R^2. For example, Figure 15.7.1 shows a region G in the uv-plane transformed into a region R in the xy-plane by the change of variables $x = g(u,v)$ and $y = h(u,v)$, or sometimes we write $x = x(u,v)$ and $y = y(u,v)$. We shall typically assume that each of these functions has continuous first partial derivatives, which means g_u, g_v, h_u, and h_v exist and are also continuous. The need for this requirement will become clear soon.

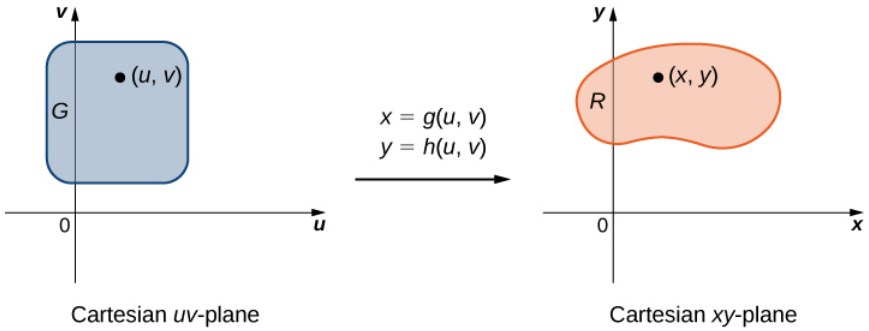


Figure 15.7.1: The transformation of a region G in the uv-plane into a region R in the xy-plane.

Definition: one-to-one transformation

A transformation $T: G \to R$, defined as $T(u, v) = (x, y)$, is said to be a one-to-one transformation if no two points map to the same image point.

To show that T s a one-to-one transformation, we assume $T(u_1, v_1) = T(u_2, v_2)$ and show that as a consequence we obtain$(u_1, v_1) = (u_2, v_2)$. If the transformation T is one-to-one in the domain G, then the inverse T^{-1} exists with the domain R such that $T^{-1} \circ T$ and $T \circ T^{-1}$ are identity functions.

Figure 15.7.2 shows the mapping $T(u, v) = (x, y)$ where x and y are related to u and v by the equations $x = g(u, v)$ and $y = h(u, v)$. The region G is the domain of T and the region R is the range of T, also known as the *image* of G under the transformation T.

Example 15.7.1A: Determining How the Transformation Works

Suppose a transformation T is defined as $T(r, \theta) = (x, y)$ where $x = r \cos \theta$, $y = r \sin \theta$. Find the image of the polar rectangle $G = \{(r, \theta) | 0 \le r \le 1,\ 0 \le \theta \le \pi/2\}$ in the $r\theta$-plane to a region R in the xy-plane. Show that T is a one-to-one transformation in G and find $T^{-1}(x, y)$.

Solution

Since r varies from 0 to 1 in the $r\theta$-plane, we have a circular disc of radius 0 to 1 in the xy-plane. Because θ varies from 0 to $\pi/2$ in the $r\theta$-plane, we end up getting a quarter circle of radius 1 in the first quadrant of the xy-plane (Figure 15.7.2). Hence R is a quarter circle bounded by $x^2 + y^2 = 1$ in the first quadrant.

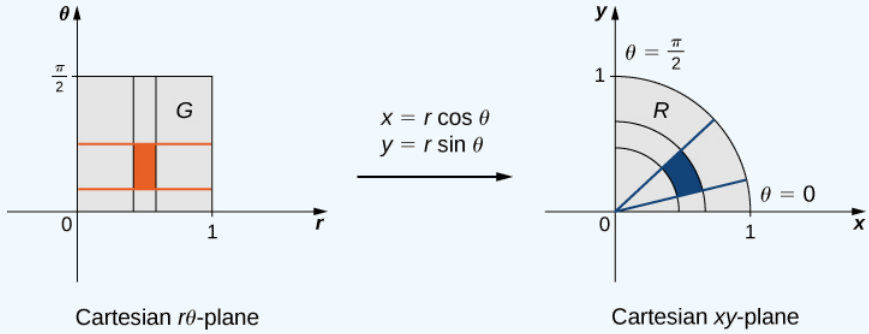


Figure 15.7.2: A rectangle in the $r\theta$-plane is mapped into a quarter circle in the xy-plane.

In order to show that T is a one-to-one transformation, assume $T(r_1, \theta_1) = T(r_2, \theta_2)$ and show as a consequence that $(r_1, \theta_1) = (r_2, \theta_2)$. In this case, we have

$$T(r_1, \theta_1) = T(r_2, \theta_2),$$

$$(x_1, y_1) = (x_1, y_1),$$

$$(r_1 \cos \theta_1, r_1 \sin \theta_1) = (r_2 \cos \theta_2, r_2 \sin \theta_2),$$

$$r_1 \cos \theta_1 = r_2 \cos \theta_2,\ r_1 \sin \theta_1 = r_2 \sin \theta_2.$$

Dividing, we obtain

$$\frac{r_1 \cos\theta_1}{r_1 \sin\theta_1} = \frac{r_2 \cos\theta_2}{r_2 \sin\theta_2}$$

$$\frac{\cos\theta_1}{\sin\theta_1} = \frac{\cos\theta_2}{\sin\theta_2}$$

$$\tan\theta_1 = \tan\theta_2$$

$$\theta_1 = \theta_2$$

since the tangent function is one-one function in the interval $0 \le \theta \le \pi/2$. Also, since $0 \le r \le 1$, we have $r_1 = r_2$, $\theta_1 = \theta_2$. Therefore, $(r_1, \theta_1) = (r_2, \theta_2)$ and T is a one-to-one transformation from G to R.

To find $T^{-1}(x, y)$ solve for r, θ in terms of x, y. We already know that $r^2 = x^2 + y^2$ and $\tan\theta = \frac{y}{x}$. Thus $T^{-1}(x, y) = (r, \theta)$ is defined as $r = \sqrt{x^2 + y^2}$ and $\tan^{-1}(\frac{y}{x})$.

Example 15.7.1B: Finding the Image under T

Let the transformation T be defined by $T(u, v) = (x, y)$ where $x = u^2 - v^2$ and $y = uv$. Find the image of the triangle in the uv-plane with vertices $(0, 0)$, $(0, 1)$, and $(1, 1)$.

Solution

The triangle and its image are shown in Figure 15.7.3. To understand how the sides of the triangle transform, call the side that joins $(0, 0)$ and $(0, 1)$ side A, the side that joins $(0, 0)$ and $(1, 1)$ side B, and the side that joins $(1, 1)$ and $(0, 1)$ side C.

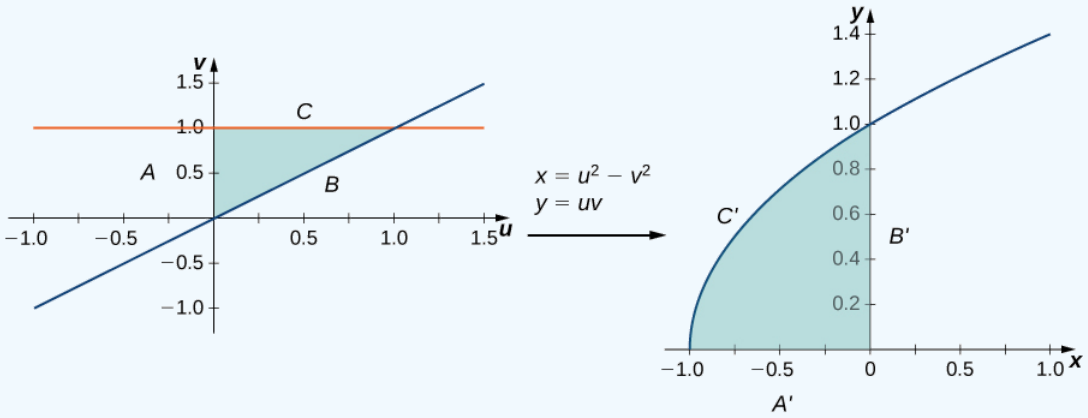


Figure 15.7.3: A triangular region in the uv-plane is transformed into an image in the xy-plane.

- For the side $A: u = 0,\ 0 \le v \le 1$ transforms to $x = -v^2,\ y = 0$ so this is the side A' that joins $(-1, 0)$ and $(0, 0)$.
- For the side $B: u = v,\ 0 \le u \le 1$ transforms to $x = 0,\ y = u^2$ so this is the side B' that joins $(0, 0)$ and $(0, 1)$.
- For the side $C: 0 \le u \le 1,\ v = 1$ transforms to $x = u^2 - 1,\ y = u$ (hence $x = y^2 - 1$ so this is the side C' that makes the upper half of the parabolic arc joining $(-1, 0)$ and $(0, 1)$.

All the points in the entire region of the triangle in the uv-plane are mapped inside the parabolic region in the xy-plane.

Exercise 15.7.1

Let a transformation T be defined as $T(u, v) = (x, y)$ where $x = u + v,\ y = 3v$. Find the image of the rectangle $G = \{(u, v): 0 \le u \le 1,\ 0 \le v \le 2\}$ from the uv-plane after the transformation into a region R in the xy-plane. Show that T is a one-to-one transformation and find $T^{-1}(x, y)$.

Hint

Follow the steps of Example 15.7.1B.

Answer

$T^{-1}(x, y) = (u, v)$ where $u = \frac{3x-y}{3}$ and $v = \frac{y}{3}$

Jacobians

Recall that we mentioned near the beginning of this section that each of the component functions must have continuous first partial derivatives, which means that g_u, g_v, h_u and h_v exist and are also continuous. A transformation that has this property is called a C^1 transformation (here C denotes continuous). Let $T(u, v) = (g(u, v),\ h(u, v))$, where $x = g(u, v)$ and $y = h(u, v)$ be a one-to-one C^1 transformation. We want to see how it transforms a small rectangular region S, Δu units by Δv units, in the uv-plane (Figure 15.7.4).

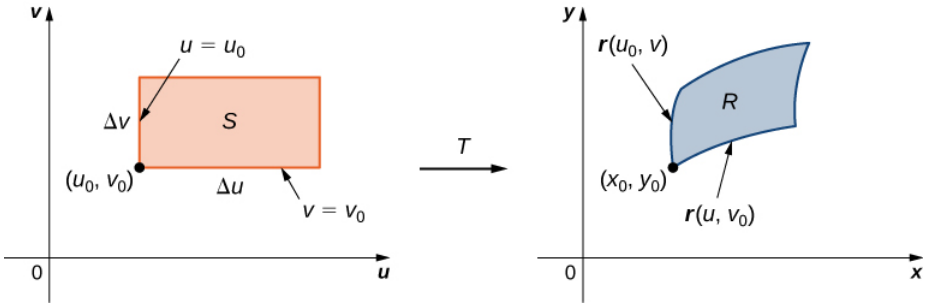


Figure 15.7.4: A small rectangle S in the uv-plane is transformed into a region R in the xy-plane.

Since $x = g(u, v)$ and $y = h(u, v)$, we have the position vector $r(u, v) = g(u, v)i + h(u, v)j$ of the image of the point (u, v). Suppose that $(u_0, v_0)$is the coordinate of the point at the lower left corner that mapped to $(x_0, y_0) = T(u_0, v_0)$ The line $v = v_0$ maps to the image curve with vector function $r(u, v_0)$, and the tangent vector at (x_0, y_0) to the image curve is

$$r_u = g_u(u_0, v_0)i + h_v(u_0, v_0)j = \frac{\partial x}{\partial u}i + \frac{\partial y}{\partial u}j.$$

Similarly, the line $u = u_0$ maps to the image curve with vector function $r(u_0, v)$, and the tangent vector at (x_0, y_0) to the image curve is

$$r_v = g_v(u_0, v_0)i + h_u(u_0, v_0)j = \frac{\partial x}{\partial v}i + \frac{\partial y}{\partial v}j.$$

Now, note that

$$r_u = \lim_{\Delta u \to 0} \frac{r(u_0 + \Delta u, v_0) - r(u_0, v_0)}{\Delta u} \; so \; r(u_0 + \Delta u, v_0) - r(u_0, v_0) \approx \Delta u r_u.$$

Similarly,

$$r_v = \lim_{\Delta v \to 0} \frac{r(u_0, v_0 + \Delta v) - r(u_0, v_0)}{\Delta v} \; so \; r(u_0, v_0 + \Delta v) - r(u_0, v_0) \approx \Delta v r_v.$$

This allows us to estimate the area ΔA of the image R by finding the area of the parallelogram formed by the sides $\Delta v r_v$ and $\Delta u r_u$. By using the cross product of these two vectors by adding the **k**th component as 0, the area ΔA of the image R (refer to The Cross Product) is approximately $|\Delta u r_u \times \Delta v r_v| = |r_u \times r_v| \Delta u \Delta v$. In determinant form, the cross product is

$$r_u \times r_v = \begin{vmatrix} i & j & k \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} k = \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \right) k$$

Since $|k| = 1$, we have

$\Delta A \approx |r_u \times r_v| \Delta u \Delta v = \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \right) \Delta u \Delta v.$

Definition: Jacobian

The *Jacobian* of the C^1 transformation $T(u, v) = (g(u, v),\ h(u, v))$ is denoted by $J(u, v)$ and is defined by the 2×2 determinant

$$J(u,v)=\left|\frac{\partial(x,y)}{\partial(u,v)}\right|=\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial y}{\partial u}\\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v}\end{vmatrix}=\left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right).$$

Using the definition, we have

$$\Delta A\approx J(u,v)\Delta u\Delta v=\left|\frac{\partial(x,y)}{\partial(u,v)}\right|\Delta u\Delta v.$$

Note that the Jacobian is frequently denoted simply by

$$J(u,v)=\frac{\partial(x,y)}{\partial(u,v)}.$$

Note also that

$$\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial y}{\partial u}\\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v}\end{vmatrix}=\left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v}-\frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right)=\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}.$$

Hence the notation $J(u,v)=\frac{\partial(x,y)}{\partial(u,v)}$ suggests that we can write the Jacobian determinant with partials of x in the first row and partials of y in the second row.

Example 15.7.2A: Finding the Jacobian

Find the Jacobian of the transformation given in Example 15.7.1A.

Solution

The transformation in the example is $T(r,\theta)=(r\cos\theta,\ r\sin\theta)$ where $x=r\cos\theta$ and $y=r\sin\theta$. Thus the Jacobian is

$$J(r,\theta)=\frac{\partial(x,y)}{\partial(r,\theta)}=\begin{vmatrix}\frac{\partial x}{\partial r} & \frac{\partial x}{\partial\theta}\\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial\theta}\end{vmatrix}=\begin{vmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{vmatrix}=r\cos^2\theta+r\sin^2\theta=r(\cos^2\theta+\sin^2\theta)=r.$$

Example 15.7.2B: Finding the Jacobian

Find the Jacobian of the transformation given in Example 15.7.1B.

Solution

The transformation in the example is $T(u,v)=(u^2-v^2,uv)$ where $x=u^2-v^2$ and $y=uv$. Thus the Jacobian is

$$J(u,v)=\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}=\begin{vmatrix}2u & -2v\\ v & u\end{vmatrix}=2u^2+2v^2.$$

Exercise 15.7.2

Find the Jacobian of the transformation given in the previous checkpoint: $T(u,v)=(u+v,2v)$.

Hint

Follow the steps in the previous two examples.

Answer

https://math.libretexts.org/@go/page/2615

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = 2$$

Change of Variables for Double Integrals

We have already seen that, under the change of variables $T(u,v) = (x,y)$ where $x = g(u,v)$ and $y = h(u,v)$, a small region ΔA in the xy-plane is related to the area formed by the product $\Delta u \Delta v$ in the uv-plane by the approximation

$$\Delta A \approx J(u,v)\Delta u,\ \Delta v.$$

Now let's go back to the definition of double integral for a minute:

$$\iint_R f(x,y) fA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}, y_{ij}) \Delta A.$$

Referring to Figure 15.7.5, observe that we divided the region S in the uv-plane into small subrectangles S_{ij} and we let the subrectangles R_{ij} in the xy-plane be the images of S_{ij} under the transformation $T(u,v) = (x,y)$.

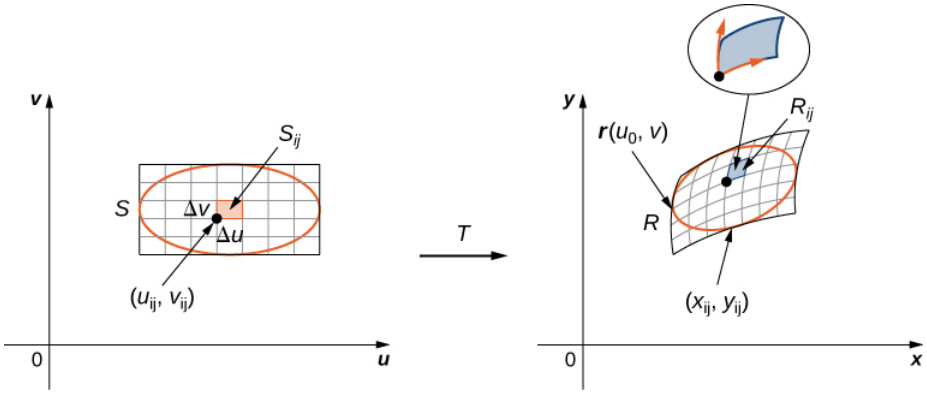


Figure 15.7.5: The subrectangles S_{ij} in the uv-plane transform into subrectangles R_{ij} in the xy-plane.

Then the double integral becomes

$$\iint_R = f(x,y)dA = \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}, y_{ij}) \Delta A$$

$$= \lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(g(u_{ij}, v_{ij}),\ h(u_{ij}, v_{ij}))|J(u_{ij}, v_{ij})|\Delta u \Delta v.$$

Notice this is exactly the double Riemann sum for the integral

$$\iint_S f(g(u,v),\ h(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv.$$

Change of Variables for Double Integrals

Let $T(u,v) = (x,y)$ where $x = g(u,v)$ and $y = h(u,v)$ be a one-to-one C^1 transformation, with a nonzero Jacobian on the interior of the region S in the uv-plane it maps S into the region R in the xy-plane. If f is continuous on R, then

$$\iint_R f(x,y)dA = \iint_S f(g(u,v),\ h(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv.$$

With this theorem for double integrals, we can change the variables from (x,y) to (u,v) in a double integral simply by replacing

$$dA = dx\, dy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv$$

when we use the substitutions $x = g(u, v)$ and $y = h(u, v)$ and then change the limits of integration accordingly. This change of variables often makes any computations much simpler.

Example 15.7.3: Changing Variables from Rectangular to Polar Coordinates

Consider the integral

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2+y^2}\, dy\, dx.$$

Use the change of variables $x = r\cos\theta$ and $y = r\sin\theta$, and find the resulting integral.

Solution

First we need to find the region of integration. This region is bounded below by $y = 0$ and above by $y = \sqrt{2x - x^2}$ (Figure 15.7.6).

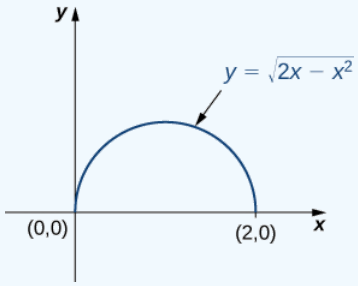


Figure 15.7.6: Changing a region from rectangular to polar coordinates.

Squaring and collecting terms, we find that the region is the upper half of the circle $x^2 + y^2 - 2x = 0$, that is $y^2 + (x-1)^2 = 1$. In polar coordinates, the circle is $r = 2\cos\theta$ so the region of integration in polar coordinates is bounded by $0 \le r \le \cos\theta$ and $0 \le \theta \le \frac{\pi}{2}$.

The Jacobian is $J(r, \theta) = r$, as shown in Example 15.7.2A. Since $r \ge 0$, we have $|J(r, \theta)| = r$.

The integrand $\sqrt{x^2 + y^2}$ changes to r in polar coordinates, so the double iterated integral is

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2+y^2}\, dy\, dx = \int_0^{\pi/2} \int_0^{2\cos\theta} r|j(r,\theta)|\, dr\, d\theta = \int_0^{\pi/2} \int_0^{2\cos\theta} r^2\, dr\, d\theta.$$

Exercise 15.7.3

Considering the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)\, dy\, dx$, use the change of variables $x = r\cos\theta$ and $y = r\sin\theta$ and find the resulting integral.

Hint

Follow the steps in the previous example.

Answer

$$\int_0^{\pi/2} \int_0^1 r^3\, dr\, d\theta$$

Notice in the next example that the region over which we are to integrate may suggest a suitable transformation for the integration. This is a common and important situation.

Example 15.7.4: Changing Variables

Consider the integral

$$\iint_R (x-y)dy\,dx,$$

where R is the parallelogram joining the points $(1,2)$, $(3,4)$, $(4,3)$, and $(6,5)$ (Figure 15.7.7). Make appropriate changes of variables, and write the resulting integral.

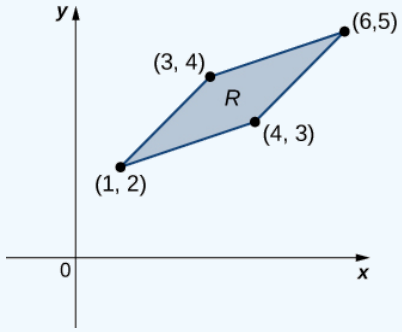


Figure 15.7.7: The region of integration for the given integral.

Solution

First, we need to understand the region over which we are to integrate. The sides of the parallelogram are $x-y+1$, $x-y-1=0$, $x-3y+5=0$ and $x-3y+9=0$ (Figure 15.7.8). Another way to look at them is $x-y=-1$, $x-y=1$, $x-3y=-5$, and $x-3y=9$.

Clearly the parallelogram is bounded by the lines $y=x+1$, $y=x-1$, $y=\frac{1}{3}(x+5)$, and $y=\frac{1}{3}(x+9)$.

Notice that if we were to make $u=x-y$ and $v=x-3y$, then the limits on the integral would be $-1\le u\le 1$ and $-9\le v\le -5$.

To solve for x and y, we multiply the first equation by 3 and subtract the second equation, $3u-v=(3x-3y)-(x-3y)=2x$. Then we have $x=\frac{3u-v}{2}$. Moreover, if we simply subtract the second equation from the first, we get $u-v=(x-y)-(x-3y)=2y$ and $y=\frac{u-v}{2}$.

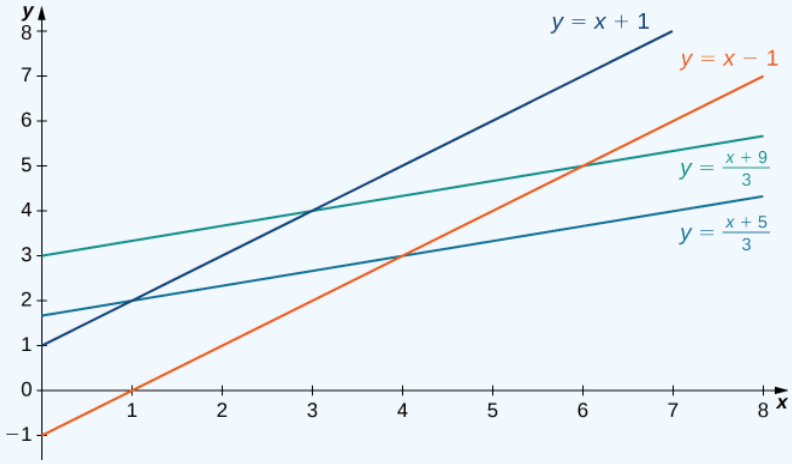


Figure 15.7.8: A parallelogram in the xy-plane that we want to transform by a change in variables.

Thus, we can choose the transformation

$$T(u,v)=\left(\frac{3u-v}{2},\ \frac{u-v}{2}\right)$$

and compute the Jacobian $J(u,v)$. We have

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 3/2 & -1/2 \\ 1/2 & -1/2 \end{vmatrix} = -\frac{3}{4} + \frac{1}{4} = -\frac{1}{2}$$

Therefore, $|J(u,v)| = \frac{1}{2}$. Also, the original integrand becomes

$$x - y = \frac{1}{2}[3u - v - u + v] = \frac{1}{2}[3u - u] = \frac{1}{2}[2u] = u.$$

Therefore, by the use of the transformation T, the integral changes to

$$\iint_R (x-y)dy\,dx = \int_{-9}^{-5}\int_{-1}^{1} J(u,v)u\,du\,dv = \int_{-9}^{-5}\int_{-1}^{1}\left(\frac{1}{2}\right)u\,du\,dv,$$

which is much simpler to compute.

Exercise 15.7.4

Make appropriate changes of variables in the integral

$$\iint_R \frac{4}{(x-y)^2}dy\,dx,$$

where R is the trapezoid bounded by the lines $x - y = 2$, $x - y = 4$, $x = 0$, and $y = 0$. Write the resulting integral.

Hint

Follow the steps in the previous example.

Answer

$x = \frac{1}{2}(v+u)$ and $y = \frac{1}{2}(v-u)$

and

$$\int_2^4 \int_{-u}^{u}\left(\frac{1}{2}\right)\cdot\frac{4}{u^2}\,dv\,du.$$

We are ready to give a problem-solving strategy for change of variables.

Problem-Solving Strategy: Change of Variables

1. Sketch the region given by the problem in the xy-plane and then write the equations of the curves that form the boundary.
2. Depending on the region or the integrand, choose the transformations $x = g(u,v)$ and $y = h(u,v)$.
3. Determine the new limits of integration in the uv-plane.
4. Find the Jacobian $J(u,v)$.
5. In the integrand, replace the variables to obtain the new integrand.
6. Replace $dy\,dx$ or $dx\,dy$, whichever occurs, by $J(u,v)du\,dv$.

In the next example, we find a substitution that makes the integrand much simpler to compute.

Example 15.7.5: Evaluating an Integral

Using the change of variables $u = x - y$ and $v = x + y$, evaluate the integral

$$\iint_R (x-y)e^{x^2-y^2}dA,$$

where R is the region bounded by the lines $x + y = 1$ and $x + y = 3$ and the curves $x^2 - y^2 = -1$ and $x^2 - y^2 = 1$ (see the first region in Figure 15.7.9).

Solution

As before, first find the region R and picture the transformation so it becomes easier to obtain the limits of integration after the transformations are made (Figure 15.7.9).

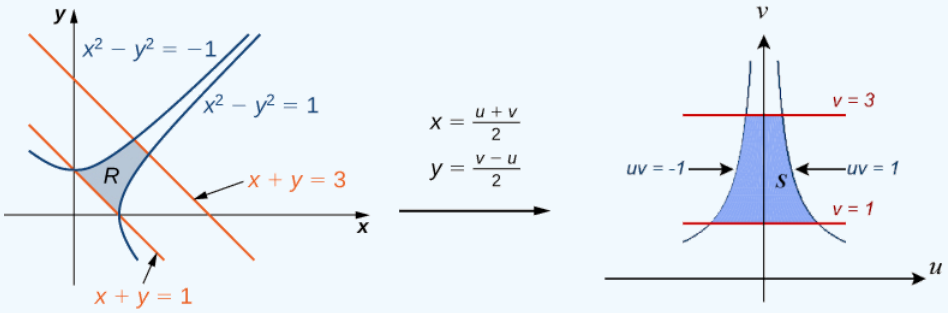


Figure 15.7.9: Transforming the region R into the region S to simplify the computation of an integral.

Given $u = x - y$ and $v = x + y$, we have $x = \frac{u+v}{2}$ and $y = \frac{v-u}{2}$ and hence the transformation to use is $T(u, v) = (\frac{u+v}{2}, \frac{v-u}{2})$. The lines $x + y = 1$ and $x + y = 3$ become $v = 1$ and $v = 3$, respectively. The curves $x^2 - y^2 = 1$ and $x^2 - y^2 = -1$ become $uv = 1$ and $uv = -1$, respectively.

Thus we can describe the region S (see the second region Figure 15.7.9) as

$$S = \left\{(u, v)|1 \leq v \leq 3,\ \frac{-1}{v} \leq u \leq \frac{1}{v}\right\}.$$

The Jacobian for this transformation is

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{vmatrix} = \frac{1}{2}.$$

Therefore, by using the transformation T, the integral changes to

$$\iint_R (x - y)e^{x^2-y^2} dA = \frac{1}{2}\int_1^3 \int_{-1/v}^{1/v} ue^{uv} du\, dv.$$

Doing the evaluation, we have

$$\frac{1}{2}\int_1^3 \int_{-1/v}^{1/v} ue^{uv} du\, dv = \frac{2}{3e} \approx 0.245.$$

? Exercise 15.7.5

Using the substitutions $x = v$ and $y = \sqrt{u + v}$, evaluate the integral $\iint_R y\ \sin(y^2 - x)\ dA$, where R is the region bounded by the lines $y = \sqrt{x}$, $x = 2$ and $y = 0$.

Hint

Sketch a picture and find the limits of integration.

Answer

$\frac{1}{2}(\sin 2 - 2)$

Change of Variables for Triple Integrals

Changing variables in triple integrals works in exactly the same way. Cylindrical and spherical coordinate substitutions are special cases of this method, which we demonstrate here.

Suppose that G is a region in uvw-space and is mapped to D in xyz-space (Figure 15.7.10) by a one-to-one C^1 transformation $T(u,v,w) = (x,y,z)$ where $x = g(u,v,w)$, $y = h(u,v,w)$, and $z = k(u,v,w)$.

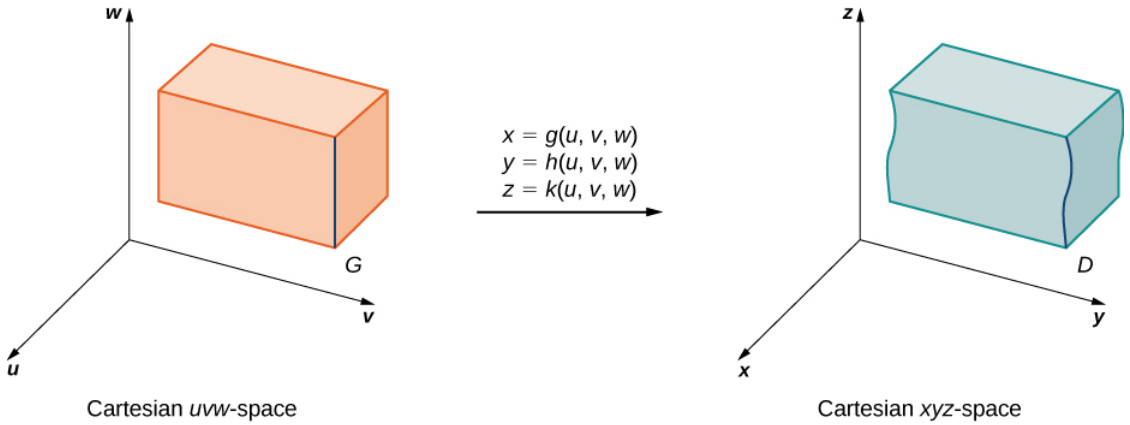


Figure 15.7.10: A region G in uvw-space mapped to a region D in xyz-space.

Then any function $F(x,y,z)$ defined on D can be thought of as another function $H(u,v,w)$ that is defined on G:

$$F(x,y,z) = F(g(u,v,w),\, h(u,v,w),\, k(u,v,w)) = H(u,v,w).$$

Now we need to define the Jacobian for three variables.

Definition: Jacobian determinant

The *Jacobian determinant* $J(u,v,w)$ in three variables is defined as follows:

$$J(u,v,w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

This is also the same as

$$J(u,v,w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

The Jacobian can also be simply denoted as $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.

With the transformations and the Jacobian for three variables, we are ready to establish the theorem that describes change of variables for triple integrals.

Change of Variables for Triple Integrals

Let $T(u,v,w) = (x,y,z)$ where $x = g(u,v,w)$, $y = h(u,v,w)$, and $z = k(u,v,w)$, be a one-to-one C^1 transformation, with a nonzero Jacobian, that maps the region G in the uvw-space into the region D in the xyz-space. As in the two-dimensional case, if F is continuous on D, then

$$\iiint_D F(x,y,z)dV = \iiint_G f(g(u,v,w)\, h(u,v,w),\, k(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\, dv\, dw \tag{15.7.1}$$

$$= \iiint_G H(u,v,w)|J(u,v,w)|du\, dv\, dw. \tag{15.7.2}$$

Let us now see how changes in triple integrals for cylindrical and spherical coordinates are affected by this theorem. We expect to obtain the same formulas as in Triple Integrals in Cylindrical and Spherical Coordinates.

Example 15.7.6A: Obtaining Formulas in Triple Integrals for Cylindrical and Spherical Coordinates

Derive the formula in triple integrals for

a. cylindrical and
b. spherical coordinates.

Solution

A.

For cylindrical coordinates, the transformation is $T(r,\theta,z) = (x,y,z)$ from the Cartesian $r\theta z$-space to the Cartesian xyz-space (Figure 15.7.11). Here $x = r\cos\theta$, $y = r\sin\theta$ and $z = z$. The Jacobian for the transformation is

$$J(r,\theta,z) = \frac{\partial(x,y,z)}{\partial(r,\theta,z)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix}$$

$$\begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

We know that $r \geq 0$, so $|J(r,\theta,z)| = r$. Then the triple integral is

$$\iiint_D f(x,y,z)dV = \iiint_G f(r\cos\theta,\, r\sin\theta,\, z)r\,dr\,d\theta\,dz.$$

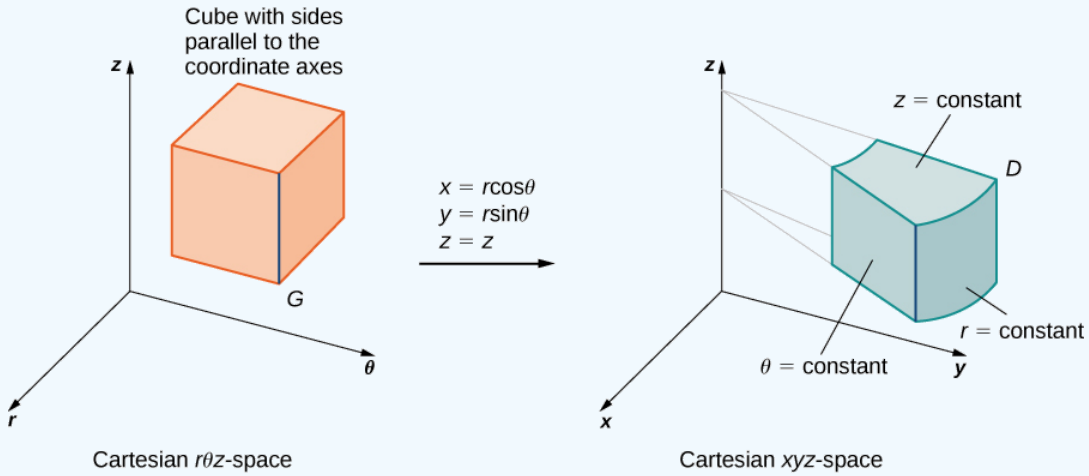


Figure 15.7.11: The transformation from rectangular coordinates to cylindrical coordinates can be treated as a change of variables from region G in $r\theta z$-space to region D in xyz-space.

B.

For spherical coordinates, the transformation is $T(\rho,\theta,\varphi)$ from the Cartesian $\rho\theta\varphi$-space to the Cartesian xyz-space (Figure 15.7.12). Here $x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, and $z = \rho\cos\varphi$. The Jacobian for the transformation is

$$J(\rho,\theta,\varphi) = \frac{\partial(x,y,z)}{\partial(\rho,\theta,\varphi)} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial \rho} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{vmatrix} = \begin{vmatrix} \sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta & \rho\cos\varphi\cos\theta \\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta & \rho\cos\varphi\sin\theta \\ \cos\varphi & 0 & -\rho\sin\varphi \end{vmatrix}.$$

Expanding the determinant with respect to the third row:

$$= \cos\varphi \begin{vmatrix} -\rho\sin\varphi\sin\theta & \rho\cos\varphi\cos\theta \\ \rho\sin\varphi\cos\theta & \rho\cos\varphi\sin\theta \end{vmatrix} - \rho\sin\varphi \begin{vmatrix} \sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta \end{vmatrix}$$

$$= \cos\varphi(-\rho^2\sin\varphi\,\cos\varphi\,\sin^2\theta - \rho^2\,\sin\varphi\,\cos\varphi\,\cos^2\theta)$$
$$-\rho\sin\varphi(\rho\sin^2\varphi\cos^2\theta + \rho\sin^2\varphi\sin^2\theta)$$

$$= -\rho^2\sin\varphi\cos^2\varphi(\sin^2\theta + \cos^2\theta) - \rho^2\sin\varphi\sin^2\varphi(\sin^2\theta + \cos^2\theta)$$

$$= -\rho^2\sin\varphi\cos^2\varphi - \rho^2\sin\varphi\sin^2\varphi$$

$$= -\rho\sin\varphi(\cos^2\varphi + \sin^2\varphi) = -\rho^2\sin\varphi.$$

Since $0 \le \varphi \le \pi$, we must have $\sin\varphi \ge 0$. Thus $|J(\rho,\theta,\varphi)| = |-\rho^2\sin\varphi| = \rho^2\sin\varphi$.

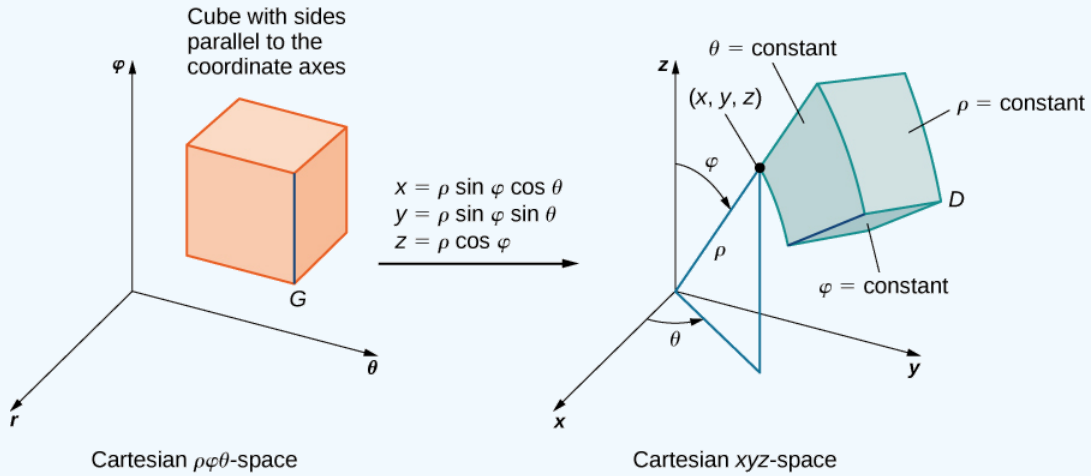


Figure 15.7.12: The transformation from rectangular coordinates to spherical coordinates can be treated as a change of variables from region G in $\rho\theta\varphi$-space to region D in xyz-space.

Then the triple integral becomes

$$\iiint_D f(x,y,z)dV = \iiint_G f(\rho\,\sin\varphi\,\cos\theta,\,\rho\,\sin\varphi\,\sin\theta,\rho\,\cos\varphi)\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta.$$

Let's try another example with a different substitution.

Example 15.7.6B: Evaluating a Triple Integral with a Change of Variables

Evaluate the triple integral

$$\int_0^3\int_0^4\int_{y/2}^{(y/2)+1}\left(x+\frac{z}{3}\right)dx\,dy\,dz$$

In xyz-space by using the transformation

$u = (2x - y)/2$, $v = y/2$, and $w = z/3$.

Then integrate over an appropriate region in uvw-space.

Solution

As before, some kind of sketch of the region G in xyz-space over which we have to perform the integration can help identify the region D in uvw-space (Figure 15.7.13). Clearly G in xyz-space is bounded by the planes $x = y/2$, $x = (y/2) + 1$, $y = 0$, $y = 4$, $z = 0$, and $z = 4$. We also know that we have to use $u = (2x - y)/2$, $v = y/2$, and $w = z/3$ for the transformations. We need to solve for x, y and z. Here we find that $x = u + v$, $y = 2v$, and $z = 3w$.

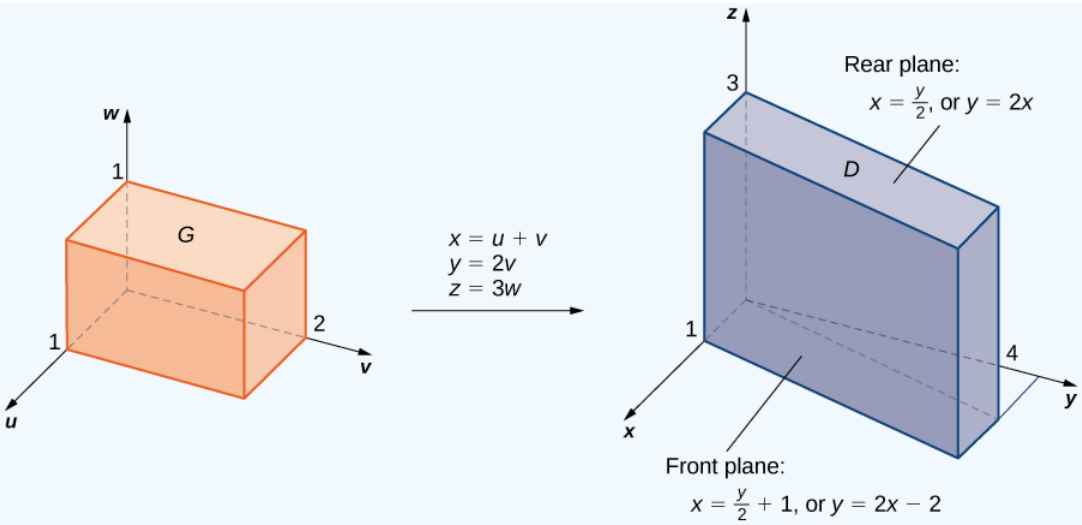


Figure 15.7.13: The region G in uvw-space is transformed to region D in xyz-space.

Using elementary algebra, we can find the corresponding surfaces for the region G and the limits of integration in uvw-space. It is convenient to list these equations in a table.

Equations in xyz for the region D	Corresponding equations in uvw for the region G	Limits for the integration in uvw
$x = y/2$	$u + v = 2v/2 = v$	$u = 0$
$x = y/2$	$u + v = (2v/2) + 1 = v + 1$	$u = 1$
$y = 0$	$2v = 0$	$v = 0$
$y = 4$	$2v = 4$	$v = 2$
$z = 0$	$3w = 0$	$w = 0$
$z = 3$	$3w = 3$	$w = 1$

Now we can calculate the Jacobian for the transformation:

$$J(u,v,w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 6.$$

The function to be integrated becomes

$$f(x,y,z) = x + \frac{z}{3} = u + v + \frac{3w}{3} = u + v + w.$$

We are now ready to put everything together and complete the problem.

$$\int_0^3 \int_0^4 \int_{y/2}^{(y/2)+1} \left(x + \frac{z}{3}\right) dx\, dy\, dz = \int_0^1 \int_0^2 \int_0^1 (u+v+w)|J(u,v,w)|\, du\, dv\, dw$$
$$= \int_0^1 \int_0^2 \int_0^1 (u+v+w)|6|\, du\, dv\, dw$$
$$= 6\int_0^1 \int_0^2 \int_0^1 (u+v+w)\, du\, dv\, dw$$
$$= 6\int_0^1 \int_0^2 \left[\frac{u^2}{2} + vu + wu\right]_0^1 dv\, dw$$
$$= 6\int_0^1 \int_0^2 \left(\frac{1}{2} + v + u\right) dv\, dw$$
$$= 6\int_0^1 \left[\frac{1}{2}v + \frac{v^2}{2} + wv\right]_0^2 dw$$
$$= 6\int_0^1 (3+2w)\, dw = 6\left[3w + w^2\right]_0^1 = 24.$$

? Exercise 15.7.6

Let D be the region in xyz-space defined by $1 \le x \le 2,\ 0 \le xy \le 2$, and $0 \le z \le 1$.

Evaluate $\iiint_D (x^2y + 3xyz)\, dx\, dy\, dz$ by using the transformation $u = x$, $v = xy$, and $w = 3z$.

Hint

Make a table for each surface of the regions and decide on the limits, as shown in the example.

Answer

$$\int_0^3 \int_0^2 \int_1^2 \left(\frac{v}{3} + \frac{vw}{3u}\right) du\, dv\, dw = 2 + \ln 8$$

Key Concepts

- A transformation T is a function that transforms a region G in one plane (space) into a region R. in another plane (space) by a change of variables.
- A transformation $T : G \to R$ defined as $T(u,v) = (x,y)$ (or $T(u,v,w) = (x,y,z)$)is said to be a one-to-one transformation if no two points map to the same image point.
- If f is continuous on R, then

$$\iint_R f(x,y)dA = \iint_S f(g(u,v),\ h(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv.$$

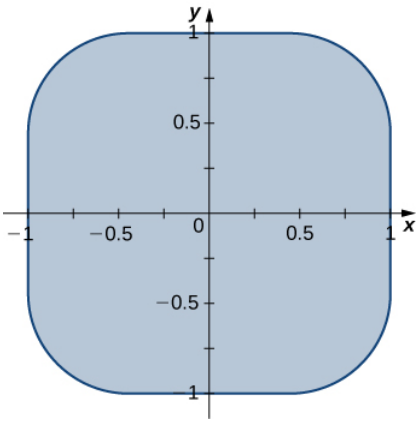
- If F is continuous on R, then

$$\iiint_R F(x,y,z)\, dV = \iiint_G F(g(u,v,w),\ h(u,v,w),\ k(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\, dv\, dw$$
$$= \iiint_G H(u,v,w)|J(u,v,w)|\, du\, dv\, dw.$$

[T] Lamé ovals (or superellipses) are plane curves of equations $(\frac{x}{a})^n + (\frac{y}{b})^n = 1$, where *a*, *b*, and *n* are positive real numbers.

a. Use a CAS to graph the regions R bounded by Lamé ovals for $a = 1,\ b = 2,\ n = 4$ and $n = 6$ respectively.

b. Find the transformations that map the region R bounded by the Lamé oval $x^4 + y^4 = 1$ also called a squircle and graphed in the following figure, into the unit disk.



c. Use a CAS to find an approximation of the area $A(R)$ of the region R bounded by $x^4 + y^4 = 1$. Round your answer to two decimal places.

[T] Lamé ovals have been consistently used by designers and architects. For instance, Gerald Robinson, a Canadian architect, has designed a parking garage in a shopping center in Peterborough, Ontario, in the shape of a superellipse of the equation $(\frac{x}{a})^n + (\frac{y}{b})^n = 1$ with $\frac{a}{b} = \frac{9}{7}$ and $n = e$. Use a CAS to find an approximation of the area of the parking garage in the case $a = 900$ yards, $b = 700$ yards, and $n = 2.72$ yards.

[Hide Solution]

$A(R) \simeq 83,999.2$

Chapter Review Exercises

True or False? Justify your answer with a proof or a counterexample.

$$\int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dy\,dx$$

Fubini's theorem can be extended to three dimensions, as long as f is continuous in all variables.

[Hide solution]

True.

The integral

$$\int_0^{2\pi} \int_0^1 \int_0^1 dz\,dr\,d\theta$$

represents the volume of a right cone.

The Jacobian of the transformation for $x = u^2 - 2v,\ y = 3v - 2uv$ is given by $-4u^2 + 6u + 4v$.

[Hide Solution]

False.

Evaluate the following integrals.

$$\iint_R (5x^3y^2 - y^2)\,dA,\ R = \{(x,y) | 0 \le x \le 2,\ 1 \le y \le 4\}$$

$$\iint_D \frac{y}{3x^2+1} dA,\ D = \{(x,y) | 0 \le x \le 1,\ -x \le y \le x\}$$

[Hide Solution]

0

$$\iint_D \sin(x^2+y^2)dA$$

where D is a disk of radius 2 centered at the origin

$$\int_0^1 \int_0^1 xye^{x^2}dx\,dy$$

[Hide Solution]

$\frac{1}{4}$

$$\int_{-1}^1 \int_0^z \int_0^{x-z} 6dy\,dx\,dz$$

$$\iiint_R 3y\,dV,$$

where $R = \{(x,y,z)|0 \le x \le 1,\ 0 \le y \le x,\ 0 \le z \le \sqrt{9-y^2}\}$

[Hide Solution]

1.475

$$\int_0^2 \int_0^{2\pi} \int_r^1 r\,dz\,d\theta\,dr$$

$$\int_0^{2\pi} \int_0^{\pi/2} \int_1^3 \rho^2\,\sin(\varphi)d\rho\,d\varphi,\ d\theta$$

[Hide Solution]

$\frac{52}{3}\pi$

$$\int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz\,dy\,sx$$

For the following problems, find the specified area or volume.

The area of region enclosed by one petal of $r = \cos(4\theta)$.

[Hide Solution]

$\frac{\pi}{16}$

The volume of the solid that lies between the paraboloid $z = 2x^2 + 2y^2$ and the plane $z = 8$.

The volume of the solid bounded by the cylinder $x^2 + y^2 = 16$ and from $z = 1$ to $z + x = 2$.

[Hide Solution]

93.291

The volume of the intersection between two spheres of radius 1, the top whose center is $(0, 0, 0.25)$ and the bottom, which is centered at $(0, 0, 0)$.

For the following problems, find the center of mass of the region.

$\rho(x, y) = xy$ on the circle with radius 1 in the first quadrant only.

[Hide Solution]

$(\frac{8}{15}, \frac{8}{15})$

$\rho(x, y) = (y + 1)\sqrt{x}$ in the region bounded by $y = e^x$, $y = 0$, and $x = 1$.

$\rho(x, y, z) = z$ on the inverted cone with radius 2 and height 2.

$(0, 0, \frac{8}{5})$

The volume an ice cream cone that is given by the solid above $z = \sqrt{(x^2 + y^2)}$ and below $z^2 + x^2 + y^2 = z$.

The following problems examine Mount Holly in the state of Michigan. Mount Holly is a landfill that was converted into a ski resort. The shape of Mount Holly can be approximated by a right circular cone of height 1100 ft and radius 6000 ft.

If the compacted trash used to build Mount Holly on average has a density $400\ lb/ft^3$, find the amount of work required to build the mountain.

[Hide Solution]

$1.452\pi \times 10^{15}$ ft-lb

In reality, it is very likely that the trash at the bottom of Mount Holly has become more compacted with all the weight of the above trash. Consider a density function with respect to height: the density at the top of the mountain is still density $400\ lb/ft^3$ and the density increases. Every 100 feet deeper, the density doubles. What is the total weight of Mount Holly?

The following problems consider the temperature and density of Earth's layers.

[T] The temperature of Earth's layers is exhibited in the table below. Use your calculator to fit a polynomial of degree 3 to the temperature along the radius of the Earth. Then find the average temperature of Earth. (*Hint*: begin at 0 in the inner core and increase outward toward the surface)

Layer	Depth from center (km)	Temperature ^oC
Rocky Crust	0 to 40	0
Upper Mantle	40 to 150	870
Mantle	400 to 650	870
Inner Mantel	650 to 2700	870
Molten Outer Core	2890 to 5150	4300
Inner Core	5150 to 6378	7200

Source: http://www.enchantedlearning.com/sub...h/Inside.shtml

[Hide Solution]

$y = -1.238 \times 10^{-7}x^3 + 0.001196x^2 - 3.666x + 7208$; average temperature approximately 2800^oC

[T] The density of Earth's layers is displayed in the table below. Using your calculator or a computer program, find the best-fit quadratic equation to the density. Using this equation, find the total mass of Earth.

Layer	Depth from center (km)	Density (g/cm^3)
Inner Core	0	12.95
Outer Core	1228	11.05
Mantle	3488	5.00
Upper Mantle	6338	3.90
Crust	6378	2.55

Source: http://hyperphysics.phy-astr.gsu.edu...rthstruct.html

The following problems concern the Theorem of Pappus (see Moments and Centers of Mass for a refresher), a method for calculating volume using centroids. Assuming a region R, when you revolve around the x-axis the volume is given by V_x

$= 2\pi A\bar{y}$, and when you revolve around the y-axis the volume is given by $V_y = 2\pi A\bar{x}$, where A is the area of R. Consider the region bounded by $x^2 + y^2 = 1$ and above $y = x + 1$.

Find the volume when you revolve the region around the x-axis.

[Hide Solution]

$\frac{\pi}{3}$

Find the volume when you revolve the region around the y-axis.

Glossary

Jacobian

the Jacobian $J(u, v)$ in two variables is a 2×2 determinant:

$$J(u, v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix};$$

the Jacobian $J(u, v, w)$ in three variables is a 3×3 determinant:

$$J(u, v, w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix}$$

one-to-one transformation

a transformation $T : G \to R$ defined as $T(u, v) = (x, y)$ is said to be one-to-one if no two points map to the same image point

planar transformation

a function T that transforms a region G in one plane into a region R in another plane by a change of variables

transformation

a function that transforms a region GG in one plane into a region RR in another plane by a change of variables

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15.7E: EXERCISES FOR SECTION 15.7

In exercises 1 - 6, the function $T: S \to R,\ T(u,v) = (x,y)$ on the region $S = \{(u,v) \mid 0 \le u \le 1,\ 0 \le v \le 1\}$ bounded by the unit square is given, where $R \in R^2$ is the image of S under T.

a. Justify that the function T is a C^1 transformation.

b. Find the images of the vertices of the unit square S through the function T.

c. Determine the image R of the unit square S and graph it.

1. $x = 2u,\ y = 3v$

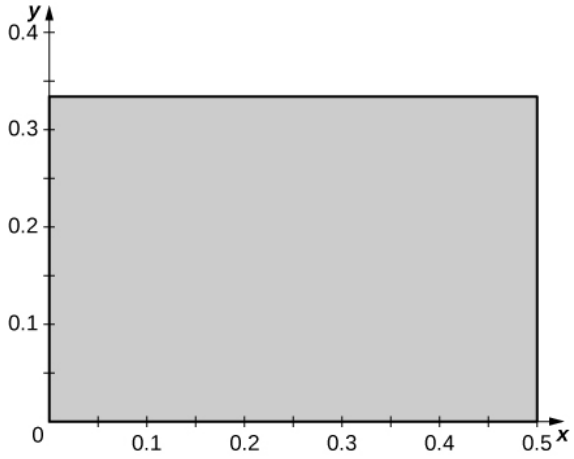
2. $x = \frac{u}{2},\ y = \frac{v}{3}$

Answer

a. $T(u,v) = (g(u,v),\ h(u,v),\ x = g(u,v) = \frac{u}{2}$ and $y = h(u,v) = \frac{v}{3}$. The functions g and h are continuous and differentiable, and the partial derivatives $g_u(u,v) = \frac{1}{2},\ g_v(u,v) = 0,\ h_u(u,v) = 0$ and $h_v(u,v) = \frac{1}{3}$ are continuous on S;

b. $T(0,0) = (0,0),\ T(1,0) = \left(\frac{1}{2}, 0\right),\ T(0,1) = \left(0, \frac{1}{3}\right)$, and $T(1,1) = \left(\frac{1}{2}, \frac{1}{3}\right)$;

c. R is the rectangle of vertices $(0,0),\ \left(0, \frac{1}{3}\right),\ \left(\frac{1}{2}, \frac{1}{3}\right)$, and $\left(0, \frac{1}{3}\right)$ in the xy-plane; the following figure.



3. $x = u - v,\ y = u + v$

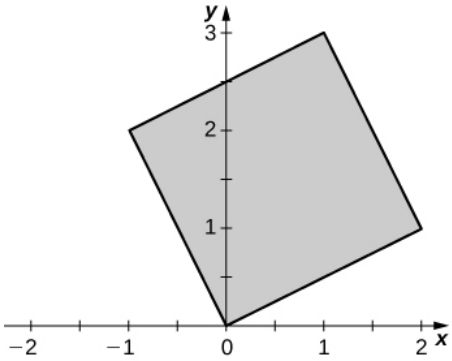
4. $x = 2u - v,\ y = u + 2v$

Answer

a. $T(u,v) = (g(u,v),\ h(u,v),\ x = g(u,v) = 2u - v$ and $y = h(u,v) = u + 2v$. The functions g and h are continuous and differentiable, and the partial derivatives $g_u(u,v) = 2,\ g_v(u,v) = -1,\ h_u(u,v) = 1$ and $h_v(u,v) = 2$ are continuous on S;

b. $T(0,0) = (0,0),\ T(1,0) = (2,1),\ T(0,1) = (-1,2)$, and $T(1,1) = (1,3)$;

c. R is the parallelogram of vertices $(0,0),\ (2,1)\ (1,3)$, and $(-1,2)$ in the xy-plane; the following figure.



5. $x = u^2,\ y = v^2$

https://math.libretexts.org/@go/page/66957

6. $x = u^3,\ y = v^3$

Answer

a. $T(u,v) = (g(u,v),\ h(u,v),\ x = g(u,v) = u^3$ and $y = h(u,v) = v^3$. The functions g and h are continuous and differentiable, and the partial derivatives $g_u(u,v) = 3u^2,\ g_v(u,v) = 0,\ h_u(u,v) = 0$ and $h_v(u,v) = 3v^2$ are continuous on S;

b. $T(0,0) = (0,0),\ T(1,0) = (1,0),\ T(0,1) = (0,1)$, and $T(1,1) = (1,1)$;

c. R is the unit square in the xy-plane see the figure in the answer to the previous exercise.

In exercises 7 - 12, determine whether the transformations $T : S \to R$ are one-to-one or not.

7. $x = u^2,\ y = v^2$, where S is the rectangle of vertices $(-1,0),\ (1,0),\ (1,1)$, and $(-1,1)$.

8. $x = u^4,\ y = u^2 + v$, where S is the triangle of vertices $(-2,0),\ (2,0)$, and $(0,2)$.

Answer

T is not one-to-one: two points of S have the same image. Indeed, $T(-2,0) = T(2,0) = (16,4)$.

9. $x = 2u,\ y = 3v$, where S is the square of vertices $(-1,1),\ (-1,-1),\ (1,-1)$, and $(1,1)$.

10. $T(u,v) = (2u - v, u)$, where S is the triangle with vertices $(-1,1),\ (-1,-1)$, and $(1,-1)$.

Answer

T is one-to-one: We argue by contradiction. $T(u_1,v_1) = T(u_2,v_2)$ implies $2u_1 - v_1 = 2u_2 - v_2$ and $u_1 = u_2$. Thus, $u_1 = u + 2$ and $v_1 = v_2$.

11. $x = u + v + w,\ y = u + v,\ z = w$, where $S = R = R^3$.

12. $x = u^2 + v + w,\ y = u^2 + v,\ z = w$, where $S = R = R^3$.

Answer

T is not one-to-one: $T(1,v,w) = (-1,v,w)$

In exercises 13 - 18, the transformations $T : R \to S$ are one-to-one. Find their related inverse transformations $T^{-1} : R \to S$.

13. $x = 4u,\ y = 5v$, where $S = R = R^2$.

14. $x = u + 2v,\ y = -u + v$, where $S = R = R^2$.

Answer

$u = \frac{x-2y}{3},\ v = \frac{x+y}{3}$

15. $x = e^{2u+v},\ y = e^{u-v}$, where $S = R^2$ and $R = \{(x,y) \mid x > 0,\ y > 0\}$

16. $x = \ln u,\ y = \ln(uv)$, where $S = \{(u,v) \mid u > 0,\ v > 0\}$ and $R = R^2$.

Answer

$u = e^x,\ v = e^{-x+y}$

17. $x = u + v + w,\ y = 3v,\ z = 2w$, where $S = R = R^3$.

18. $x = u + v,\ y = v + w,\ z = u + w$, where $S = R = R^3$.

Answer

$u = \frac{x-y+z}{2},\ v = \frac{x+y-z}{2},\ w = \frac{-x+y+z}{2}$

In exercises 19 - 22, the transformation $T : S \to R,\ T(u,v) = (x,y)$ and the region $R \subset R^2$ are given. Find the region $S \subset R^2$.

19. $x = au,\ y = bv,\ R = \{(x,y) \mid x^2 + y^2 \le a^2b^2\}$ where $a,b > 0$

20. $x = au,\ y = bc,\ R = \{(x,y) \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\}$, where $a,b > 0$

Answer

$S = \{(u,v) \mid u^2 + v^2 \le 1\}$

21. $x = \frac{u}{a},\ y = \frac{v}{b},\ z = \frac{w}{c},\ R = \{(x,y) \mid x^2 + y^2 + z^2 \le 1\}$, where $a,b,c > 0$

22. $x = au,\ y = bv,\ z = cw,\ R = \left\{(x,y)\,|\,\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} \le 1,\ z > 0\right\}$, where $a, b, c > 0$

Answer

$R = \left\{(u,v,w)\,|\,u^2 - v^2 - w^2 \le 1,\ w > 0\right\}$

In exercises 23 - 32, find the Jacobian J of the transformation.

23. $x = u + 2v,\ y = -u + v$

24. $x = \frac{u^3}{2},\ y = \frac{v}{u^2}$

Answer

$\frac{3}{2}$

25. $x = e^{2u-v},\ y = e^{u+v}$

26. $x = ue^v,\ y = e^{-v}$

Answer

-1

27. $x = u\ \cos(e^v),\ y = u\ \sin(e^v)$

28. $x = v\ \sin(u^2),\ y = v\ \cos(u^2)$

Answer

$2uv$

29. $x = u\ \cosh v,\ y = u\ \sinh v,\ z = w$

30. $x = v\ \cosh\left(\frac{1}{u}\right),\ y = v\ \sinh\left(\frac{1}{u}\right),\ z = u + w^2$

Answer

$\frac{v}{u^2}$

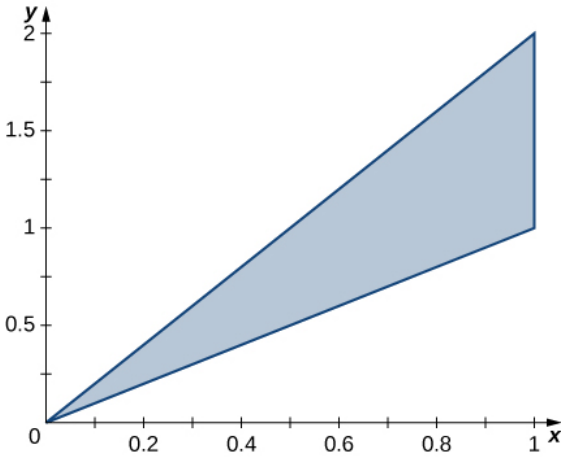
31. $x = u + v,\ y = v + w,\ z = u$

32. $x = u - v,\ y = u + v,\ z = u + v + w$

Answer

2

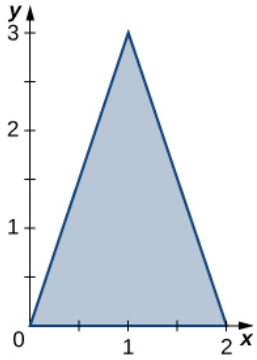
33. The triangular region R with the vertices $(0,0),\ (1,1)$, and $(1,2)$ is shown in the following figure.



a. Find a transformation $T : S \to R,\ T(u,v) = (x,y) = (au + bv + dv)$, where a, b, c, and d are real numbers with $ad - bc \neq 0$ such that $T^{-1}(0,0) = (0,0),\ T^{-1}(1,1) = (1,0)$, and $T^{-1}(1,2) = (0,1)$.

b. Use the transformation T to find the area $A(R)$ of the region R.

34. The triangular region R with the vertices $(0,0),\ (2,0)$, and $(1,3)$ is shown in the following figure.



a. Find a transformation $T: S \to R,\ T(u,v) = (x,y) = (au + bv + dv)$, where a, b, c, and d are real numbers with $ad - bc \neq 0$ such that $T^{-1}(0,0) = (0,0),\ T^{-1}(2,0) = (1,0)$, and $T^{-1}(1,3) = (0,1)$.

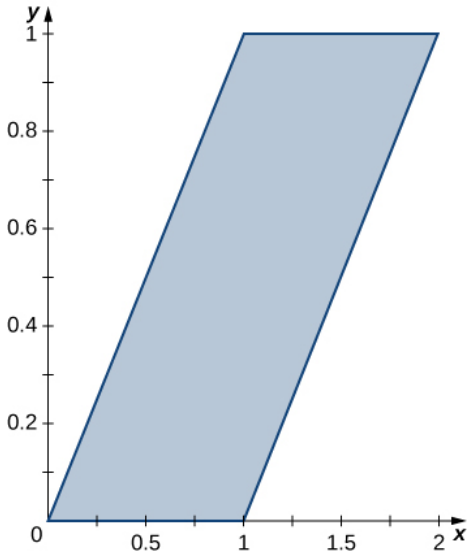
b. Use the transformation T to find the area $A(R)$ of the region R.

Answer

a. $T(u,v) = (2u + v,\ 3v)$

b. The area of R is $A(R) = \int_0^3 \int_{y/3}^{(6-y)/3} dx\, dy = \int_0^1 \int_0^{1-u} \left| \frac{\partial(x,y)}{\partial(u,v)} \right| dv\, du = \int_0^1 \int_0^{1-u} 6\, dv\, du = 3.$

In exercises 35 - 36, use the transformation $u = y - x,\ v = y$, to evaluate the integrals on the parallelogram R of vertices $(0,0)$, $(1,0)$, $(2,1)$, and $(1,1)$ shown in the following figure.



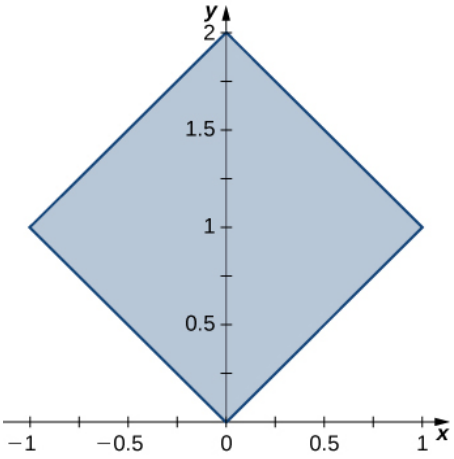
35. $\iint_R (y - x)\, dA$

36. $\iint_R (y^2 - xy)\, dA$

Answer

$-\frac{1}{4}$

In exercises 37 - 38, use the transformation $y = x = u,\ x + y = v$ to evaluate the integrals on the square R determined by the lines $y = x,\ y = -x + 2,\ y = x + 2$, and $y = -x$ shown in the following figure.



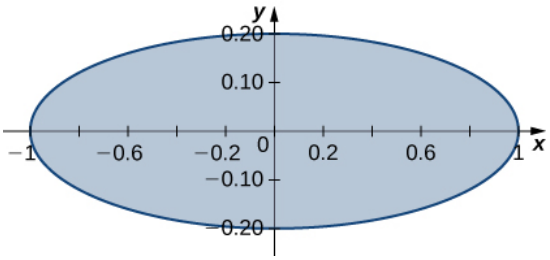
37. $\iint_R e^{x+y}\,dA$

38. $\iint_R \sin(x-y)\,dA$

Answer

$-1+\cos 2$

In exercises 39 - 40, use the transformation $x=u$, $5y=v$ to evaluate the integrals on the region R bounded by the ellipse $x^2+25y^2=1$ shown in the following figure.



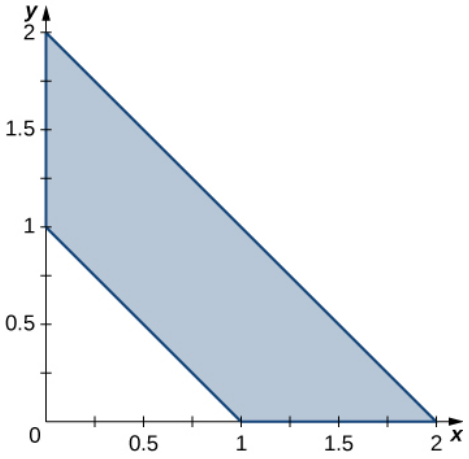
39. $\iint_R \sqrt{x^2+25y^2}\,dA$

40. $\iint_R (x^2+25y^2)^2\,dA$

Answer

$\frac{\pi}{15}$

In exercises 41 - 42, use the transformation $u=x+y$, $v=x-y$ to evaluate the integrals on the trapezoidal region R determined by the points $(1,0)$, $(2,0)$, $(0,2)$, and $(0,1)$ shown in the following figure.



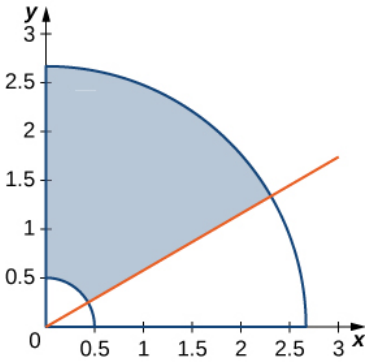
41. $\iint_R (x^2 - 2xy + y^2)\, e^{x+y}\, dA$

42. $\iint_R (x^3 + 3x^2y + 3xy^2 + y^3)\, dA$

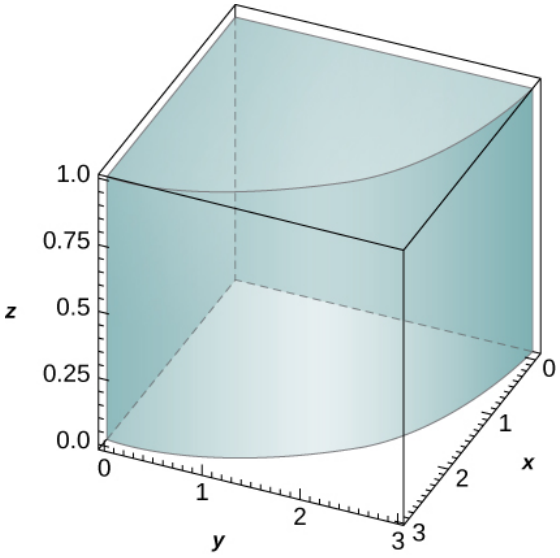
Answer

$\frac{31}{5}$

43. The circular annulus sector R bounded by the circles $4x^2 + 4y^2 = 1$ and $9x^2 + 9y^2 = 64$, the line $x = y\sqrt{3}$, and the y-axis is shown in the following figure. Find a transformation T from a rectangular region S in the $r\theta$-plane to the region R in the xy-plane. Graph S.



44. The solid R bounded by the circular cylinder $x^2 + y^2 = 9$ and the planes $z = 0,\ z = 1,\ x = 0$, and $y = 0$ is shown in the following figure. Find a transformation T from a cylindrical box S in $r\theta z$-space to the solid R in xyz-space.



https://math.libretexts.org/@go/page/66957

Answer

$T(r,\theta,z) = (r\ \cos\theta,\ r\ \sin\theta,\ z);\ S = [0,3] \times [0,\frac{\pi}{2}] \times [0,1]$ in the $r\theta z$-space

45. Show that

$$\iint_R f\left(\sqrt{\frac{x^2}{3} + \frac{y^2}{3}}\right) dA = 2\pi\sqrt{15}\int_0^1 f(\rho)\rho\ d\rho,$$

where f is a continuous function on $[0,1]$ and R is the region bounded by the ellipse $5x^2 + 3y^2 = 15$.

46. Show that

$$\iiint_R f\left(\sqrt{16x^2 + 4y^2 + z^2}\right) dV = \frac{\pi}{2}\int_0^1 f(\rho)\rho^2 d\rho,$$

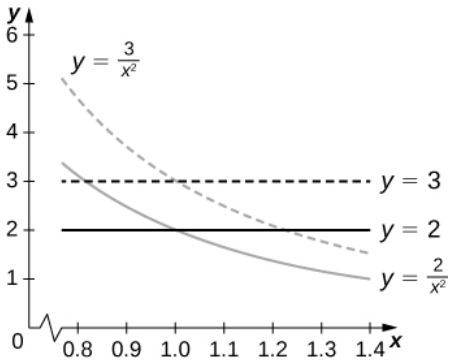
where f is a continuous function on $[0,1]$ and R is the region bounded by the ellipsoid $16x^2 + 4y^2 + z^2 = 1$.

47. [T] Find the area of the region bounded by the curves $xy = 1,\ xy = 3,\ y = 2x$, and $y = 3x$ by using the transformation $u = xy$ and $v = \frac{y}{x}$. Use a computer algebra system (CAS) to graph the boundary curves of the region R.

48. [T] Find the area of the region bounded by the curves $x^2y = 2,\ x^2y = 3,\ y = x$, and $y = 2x$ by using the transformation $u = x^2y$ and $v = \frac{y}{x}$. Use a CAS to graph the boundary curves of the region R.

Answer

The area of R is $10 - 4\sqrt{6}$; the boundary curves of R are graphed in the following figure.



49. Evaluate the triple integral

$$\int_0^1 \int_1^2 \int_z^{z+1} (y+1)\ dx\ dy\ dz$$

by using the transformation $u = x - z,\ v = 3y$, and $w = \frac{z}{2}$.

50. Evaluate the triple integral

$$\int_0^2 \int_4^6 \int_{3z}^{3z+2} (5 - 4y)\ dx\ dy\ dz$$

by using the transformation $u = x - 3z,\ v = 4y$, and $w = z$.

Answer

8

51. A transformation $T : R^2 \to R^2,\ T(u,v) = (x,y)$ of the form $x = au + bv,\ y = cu + dv$, where a, b, c, and d are real numbers, is called linear. Show that a linear transformation for which $ad - bc \neq 0$ maps parallelograms to parallelograms.

52. A transformation $T_\theta : R^2 \to R^2,\ T_\theta(u,v) = (x,y)$ of the form $x = u\ \cos\theta - v\ \sin\theta,\ y = u\ \sin\theta + v\ \cos\theta$, is called a rotation angle θ. Show that the inverse transformation of T_θ satisfies $T_\theta^{-1} = T_{-\theta}$ where $T_{-\theta}$ is the rotation of angle $-\theta$.

53. [T] Find the region S in the uv-plane whose image through a rotation of angle $\frac{\pi}{4}$ is the region R enclosed by the ellipse $x^2 + 4y^2 = 1$. Use a CAS to answer the following questions.

a. Graph the region S.

b. Evaluate the integral $\iint_S e^{-2uv}\,du\,dv$. Round your answer to two decimal places.

54. [T] The transformations $T_i : \mathbb{R}^2 \to \mathbb{R}^2$, $i = 1, ..., 4$, defined by $T_1(u,v) = (u,-v)$, $T_2(u,v) = (-u,v)$, $T_3(u,v) = (-u,-v)$, and $T_4(u,v) = (v,u)$ are called reflections about the x-axis, y-axis origin, and the line $y = x$, respectively.

a. Find the image of the region $S = \{(u,v) \mid u^2 + v^2 - 2u - 4v + 1 \le 0\}$ in the xy-plane through the transformation $T_1 \circ T_2 \circ T_3 \circ T_4$.

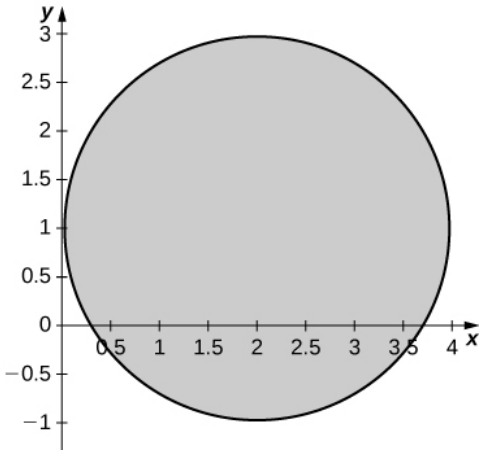
b. Use a CAS to graph R.

c. Evaluate the integral $\iint_S \sin(u^2)\,du\,dv$ by using a CAS. Round your answer to two decimal places.

Answer

a. $R = \{(x,y) \mid y^2 + x^2 - 2y - 4x + 1 \le 0\}$;

b. R is graphed in the following figure;



c. **3.16**

55. [T] The transformations $T_{k,1,1} : \mathbb{R}^3 \to \mathbb{R}^3$, $T_{k,1,1}(u,v,w) = (x,y,z)$ of the form $x = ku$, $y = v$, $z = w$, where $k \neq 1$ is a positive real number, is called a stretch if $k > 1$ and a compression if $0 < k < 1$ in the x-direction. Use a CAS to evaluate the integral $\iiint_S e^{-(4x^2+9y^2+25z^2)}\,dx\,dy\,dz$ on the solid $S = \{(x,y,z) \mid 4x^2 + 9y^2 + 25z^2 \le 1\}$ by considering the compression $T_{2,3,5}(u,v,w) = (x,y,z)$ defined by $x = \frac{u}{2}$, $y = \frac{v}{3}$, and $z = \frac{w}{5}$. Round your answer to four decimal places.

56. [T] The transformation $T_{a,0} : \mathbb{R}^2 \to \mathbb{R}^2$, $T_{a,0}(u,v) = (u + av, v)$, where $a \neq 0$ is a real number, is called a shear in the x-direction. The transformation, $T_{b,0} : R^2 \to R^2$, $T_{o,b}(u,v) = (u, bu + v)$, where $b \neq 0$ is a real number, is called a shear in the y-direction.

a. Find transformations $T_{0,2} \circ T_{3,0}$.

b. Find the image R of the trapezoidal region S bounded by $u = 0$, $v = 0$, $v = 1$, and $v = 2 - u$ through the transformation $T_{0,2} \circ T_{3,0}$.

c. Use a CAS to graph the image R in the xy-plane.

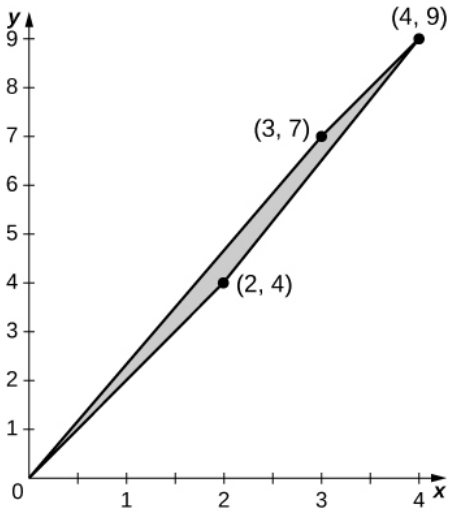
d. Find the area of the region R by using the area of region S.

Answer

a. $T_{0,2} \circ T_{3,0}(u,v) = (u + 3v, 2u + 7v)$;

b. The image S is the quadrilateral of vertices $(0,0)$, $(3,7)$, $(2,4)$, and $(4,9)$;

c. S is graphed in the following figure;



d. $\frac{3}{2}$

57. Use the transformation, $x = au,\ y = av,\ z = cw$ and spherical coordinates to show that the volume of a region bounded by the spheroid $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ is $\frac{4\pi a^2 c}{3}$.

58. Find the volume of a football whose shape is a spheroid $\frac{x^2+y^2}{a^2} + \frac{z^2}{c^2} = 1$ whose length from tip to tip is 11 inches and circumference at the center is 22 inches. Round your answer to two decimal places.

Answer

$\frac{2662}{3\pi} \approx 282.45\ in^3$

CONTRIBUTORS

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

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15.R: CHAPTER 15 REVIEW EXERCISES

True or False? **Justify your answer with a proof or a counterexample.**

1. $\int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dy\,dx$

2. Fubini's theorem can be extended to three dimensions, as long as f is continuous in all variables.

Answer

True

3. The integral $\int_0^{2\pi} \int_0^1 \int_r^1 dz\,dr\,d\theta$ represents the volume of a right cone.

4. The Jacobian of the transformation for $x = u^2 - 2v$, $y = 3v - 2uv$ is given by $-4u^2 + 6u + 4v$.

Answer

False

Evaluate the following integrals.

5. $\iint_R (5x^3y^2 - y^2)\,dA$, where $R = \{(x,y) \mid 0 \le x \le 2,\ 1 \le y \le 4\}$

6. $\iint_D \dfrac{y}{3x^2+1}\,dA$, where $D = \{(x,y) \mid 0 \le x \le 1,\ -x \le y \le x\}$

Answer

0

7. $\iint_D \sin(x^2+y^2)\,dA$ where D is a disk of radius 2 centered at the origin.

8. $\int_0^1 \int_y^1 xye^{x^2}\,dx\,dy$

Answer

$\frac{1}{4}$

9. $\int_{-1}^1 \int_0^z \int_0^{x-z} 6\,dy\,dx\,dz$

10. $\iiint_R 3y\,dV$, where $R = \{(x,y,z) \mid 0 \le x \le 1,\ 0 \le y \le x,\ 0 \le z \le 9 - y^2\}$

Answer

1.475

11. $\int_0^2 \int_0^{2\pi} \int_r^1 r\,dz\,d\theta\,dr$

12. $\int_0^{2\pi} \int_0^{\pi/2} \int_1^3 \rho^2 \sin(\varphi)\,d\rho\,d\varphi\,d\theta$

Answer

$\frac{52\pi}{3}$

13. $\int_0^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz\,dy\,dx$

For the following problems, find the specified area or volume.

14. The area of region enclosed by one petal of $r = \cos(4\theta)$.

Answer

$\frac{\pi}{16}$ units^3

15. The volume of the solid that lies between the paraboloid $z = 2x^2 + 2y^2$ and the plane $z = 8$.

16. The volume of the solid bounded by the cylinder $x^2 + y^2 = 16$ and from $z = 1$ to $z + x = 2$.

Answer

93.291 units^3

17. The volume of the intersection between two spheres of radius 1, the top whose center is $(0, 0, 0.25)$ and the bottom, which is centered at $(0, 0, 0)$.

For the following problems, find the center of mass of the region.

18. $\rho(x, y) = xy$ on the circle with radius 1 in the first quadrant only.

Answer

$\left(\frac{8}{15}, \frac{8}{15}\right)$

19. $\rho(x, y) = (y + 1)\sqrt{x}$ in the region bounded by $y = e^x$, $y = 0$, and $x = 1$.

20. $\rho(x, y, z) = z$ on the inverted cone with radius 2 and height 2.

Answer

$\left(0, 0, \frac{8}{5}\right)$

21. The volume an ice cream cone that is given by the solid above $z = \sqrt{x^2 + y^2}$ and below $z^2 + x^2 + y^2 = z$.

The following problems examine Mount Holly in the state of Michigan. Mount Holly is a landfill that was converted into a ski resort. The shape of Mount Holly can be approximated by a right circular cone of height 1100 ft and radius 6000 ft.

22. If the compacted trash used to build Mount Holly on average has a density 400 lb/ft^3, find the amount of work required to build the mountain.

Answer

$1.452\pi \times 10^{15}$ ft-lb

23. In reality, it is very likely that the trash at the bottom of Mount Holly has become more compacted with all the weight of the above trash. Consider a density function with respect to height: the density at the top of the mountain is still density 400 lb/ft^3, and the density increases. Every 100 feet deeper, the density doubles. What is the total weight of Mount Holly?

The following problems consider the temperature and density of Earth's layers.

24. **[T]** The temperature of Earth's layers is exhibited in the table below. Use your calculator to fit a polynomial of degree 3 to the temperature along the radius of the Earth. Then find the average temperature of Earth. (*Hint*: begin at 0 in the inner core and increase outward toward the surface)

Source: http://www.enchantedlearning.com/sub...h/Inside.shtml

Layer	Depth from center (km)	Temperature °C
Rocky Crust	0 to 40	0
Upper Mantle	40 to 150	870
Mantle	400 to 650	870
Inner Mantel	650 to 2700	870
Molten Outer Core	2890 to 5150	4300
Inner Core	5150 to 6378	7200

Answer

$y = -1.238 \times 10^{-7}x^3 + 0.001196x^2 - 3.666x + 7208$;
The average temperature is approximately 2800 °C.

25. **[T]** The density of Earth's layers is displayed in the table below. Using your calculator or a computer program, find the best-fit quadratic equation to the density. Using this equation, find the total mass of Earth.

Source: http://hyperphysics.phy-astr.gsu.edu...rthstruct.html

Layer	Depth from center (km)	Density ($\mathbf{g/cm^3}$)
Inner Core	0	12.95
Outer Core	1228	11.05
Mantle	3488	5.00
Upper Mantle	6338	3.90
Crust	6378	2.55

The following problems concern the Theorem of Pappus (see Moments and Centers of Mass for a refresher), a method for calculating volume using centroids. Assuming a region R, when you revolve around the x-axis the volume is given by $V_x = 2\pi A\bar{y}$, and when you revolve around the y-axis the volume is given by $V_y = 2\pi A\bar{x}$, where A is the area of R. Consider the region bounded by $x^2 + y^2 = 1$ and above $y = x + 1$.

26. Find the volume when you revolve the region around the x-axis.

Answer

$\frac{\pi}{3}$ $\mathbf{units^3}$

27. Find the volume when you revolve the region around the y-axis.

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CHAPTER OVERVIEW

16: Vector Calculus

In this chapter, we learn to model new kinds of integrals over fields such as magnetic fields, gravitational fields, or velocity fields. We also learn how to calculate the work done on a charged particle traveling through a magnetic field, the work done on a particle with mass traveling through a gravitational field, and the volume per unit time of water flowing through a net dropped in a river. All these applications are based on the concept of a vector field.

16.0: Prelude to Vector Calculus
16.1: Vector Fields
16.1E: Exercises for Section 16.1
16.2: Line Integrals
16.2E: Exercises for Section 16.2
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16.3E: Exercises for Section 16.3
16.4: Green's Theorem
16.4E: Exercises for Section 16.4
16.5: Divergence and Curl
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16.8: The Divergence Theorem
16.8E: Exercises for Section 16.8
16.R: Chapter 16 Review Exercises

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16.0: Prelude to Vector Calculus

Hurricanes are huge storms that can produce tremendous amounts of damage to life and property, especially when they reach land. Predicting where and when they will strike and how strong the winds will be is of great importance for preparing for protection or evacuation. Scientists rely on studies of rotational vector fields for their forecasts.

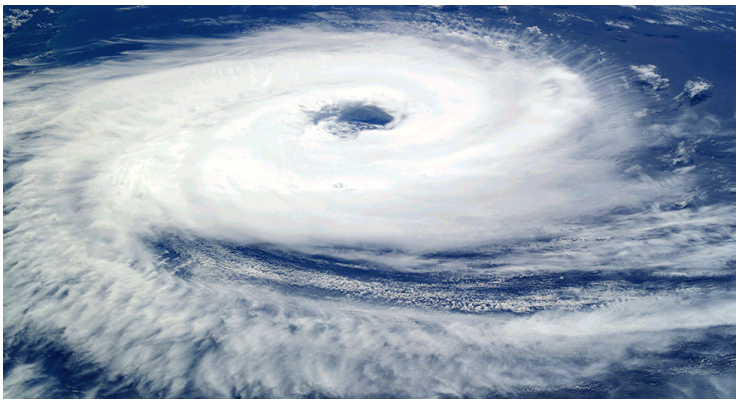


Figure 16.0.1: Hurricanes form from rotating winds driven by warm temperatures over the ocean. Meteorologists forecast the motion of hurricanes by studying the rotating vector fields of their wind velocity. Shown is Cyclone Catarina in the South Atlantic Ocean in 2004, as seen from the International Space Station. (credit: modification of work by NASA)

In this chapter, we learn to model new kinds of integrals over fields such as magnetic fields, gravitational fields, or velocity fields. We also learn how to calculate the work done on a charged particle traveling through a magnetic field, the work done on a particle with mass traveling through a gravitational field, and the volume per unit time of water flowing through a net dropped in a river.

All these applications are based on the concept of a vector field, which we explore in this chapter. Vector fields have many applications because they can be used to model real fields such as electromagnetic or gravitational fields. A deep understanding of physics or engineering is impossible without an understanding of vector fields. Furthermore, vector fields have mathematical properties that are worthy of study in their own right. In particular, vector fields can be used to develop several higher-dimensional versions of the Fundamental Theorem of Calculus.

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16.1: Vector Fields

Learning Objectives

- Recognize a vector field in a plane or in space.
- Sketch a vector field from a given equation.
- Identify a conservative field and its associated potential function.

Vector fields are an important tool for describing many physical concepts, such as gravitation and electromagnetism, which affect the behavior of objects over a large region of a plane or of space. They are also useful for dealing with large-scale behavior such as atmospheric storms or deep-sea ocean currents. In this section, we examine the basic definitions and graphs of vector fields so we can study them in more detail in the rest of this chapter.

Examples of Vector Fields

How can we model the gravitational force exerted by multiple astronomical objects? How can we model the velocity of water particles on the surface of a river? Figure 16.1.1 gives visual representations of such phenomena.

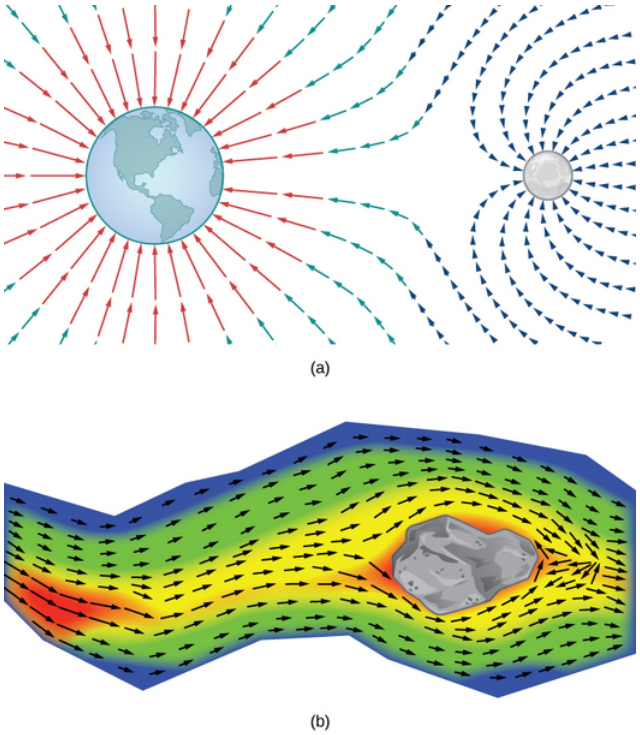


Figure 16.1.1 (a) The gravitational field exerted by two astronomical bodies on a small object. (b) The vector velocity field of water on the surface of a river shows the varied speeds of water. Red indicates that the magnitude of the vector is greater, so the water flows more quickly; blue indicates a lesser magnitude and a slower speed of water flow.

Figure 16.1.1a shows a gravitational field exerted by two astronomical objects, such as a star and a planet or a planet and a moon. At any point in the figure, the vector associated with a point gives the net gravitational force exerted by the two objects on an object of unit mass. The vectors of largest magnitude in the figure are the vectors closest to the larger object. The larger object has greater mass, so it exerts a gravitational force of greater magnitude than the smaller object.

Figure 16.1.1b shows the velocity of a river at points on its surface. The vector associated with a given point on the river's surface gives the velocity of the water at that point. Since the vectors to the left of the figure are small in magnitude, the water is flowing slowly on that part of the surface. As the water moves from left to right, it encounters some rapids around a rock. The speed of the water increases, and a whirlpool occurs in part of the rapids.

Each figure illustrates an example of a vector field. Intuitively, a vector field is a map of vectors. In this section, we study vector fields in $\mathbb{R}^2$ and $\mathbb{R}^3$.

DEFINITION: Vector Field

- A vector field $\vec{\mathbf{F}}$ in $\mathbb{R}^2$ is an assignment of a two-dimensional vector $\vec{\mathbf{F}}(x,y)$ to each point (x,y) of a subset D of $\mathbb{R}^2$. The subset D is the domain of the vector field.
- A vector field $\vec{\mathbf{F}}$ in $\mathbb{R}^3$ is an assignment of a three-dimensional vector $\vec{\mathbf{F}}(x,y,z)$ to each point (x,y,z) of a subset D of $\mathbb{R}^3$. The subset D is the domain of the vector field.

Vector Fields in $\mathbb{R}^2$

A vector field in $\mathbb{R}^2$ can be represented in either of two equivalent ways. The first way is to use a vector with components that are two-variable functions:

$$\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y) \rangle$$

The second way is to use the standard unit vectors:

$$\vec{\mathbf{F}}(x,y) = P(x,y)\,\hat{\mathbf{i}} + Q(x,y)\,\hat{\mathbf{j}}.$$

A vector field is said to be *continuous* if its component functions are continuous.

Example 16.1.1: Finding a Vector Associated with a Given Point

Let $\vec{\mathbf{F}}(x,y) = (2y^2 + x - 4)\,\hat{\mathbf{i}} + \cos(x)\,\hat{\mathbf{j}}$ be a vector field in $\mathbb{R}^2$. Note that this is an example of a continuous vector field since both component functions are continuous. What vector is associated with point $(0,-1)$?

Solution

Substitute the point values for x and y:

$$\begin{aligned}\vec{\mathbf{F}}(0,-1) &= (2(-1)^2 + 0 - 4)\,\hat{\mathbf{i}} + \cos(0)\,\hat{\mathbf{j}} \\ &= -2\,\hat{\mathbf{i}} + \hat{\mathbf{j}}.\end{aligned}$$

Exercise 16.1.1

Let $\vec{\mathbf{G}}(x,y) = x^2 y\,\hat{\mathbf{i}} - (x+y)\,\hat{\mathbf{j}}$ be a vector field in $\mathbb{R}^2$. What vector is associated with the point $(-2,3)$?

Hint

Substitute the point values into the vector function.

Answer

$\vec{\mathbf{G}}(-2,3) = 12\hat{\mathbf{i}} - \hat{\mathbf{j}}$

Drawing a Vector Field

We can now represent a vector field in terms of its components of functions or unit vectors, but representing it visually by sketching it is more complex because the domain of a vector field is in $\mathbb{R}^2$, as is the range. Therefore the "graph" of a vector field in $\mathbb{R}^2$ lives in four-dimensional space. Since we cannot represent four-dimensional space visually, we instead draw vector fields in $\mathbb{R}^2$ in a plane itself. To do this, draw the vector associated with a given point at the point in a plane. For example, suppose the vector associated with point $(4,-1)$ is $\langle 3,1 \rangle$. Then, we would draw vector $\langle 3,1 \rangle$ at point $(4,-1)$.

We should plot enough vectors to see the general shape, but not so many that the sketch becomes a jumbled mess. If we were to plot the image vector at each point in the region, it would fill the region completely and is useless. Instead, we can choose points at the intersections of grid lines and plot a sample of several vectors from each quadrant of a rectangular coordinate system in $\mathbb{R}^2$.

There are two types of vector fields in $\mathbb{R}^2$ on which this chapter focuses: radial fields and rotational fields. Radial fields model certain gravitational fields and energy source fields, and rotational fields model the movement of a fluid in a vortex. In a radial field, all vectors either point directly toward or directly away from the origin. Furthermore, the magnitude of any vector depends only on its distance from the origin. In a radial field, the vector located at point (x, y) is perpendicular to the circle centered at the origin that contains point (x, y), and all other vectors on this circle have the same magnitude.

Example 16.1.2: Drawing a Radial Vector Field

Sketch the vector field $\vec{\mathbf{F}}(x, y) = \frac{x}{2}\hat{\mathbf{i}} + \frac{y}{2}\hat{\mathbf{j}}$.

Solution

To sketch this vector field, choose a sample of points from each quadrant and compute the corresponding vector. The following table gives a representative sample of points in a plane and the corresponding vectors.

Table 16.1.1

(x, y)	$\vec{\mathbf{F}}(x, y)$	(x, y)	$\vec{\mathbf{F}}(x, y)$	(x, y)	$\vec{\mathbf{F}}(x, y)$
$(1, 0)$	$\langle \frac{1}{2}, 0 \rangle$	$(2, 0)$	$\langle 1, 0 \rangle$	$(1, 1)$	$\langle \frac{1}{2}, \frac{1}{2} \rangle$
$(0, 1)$	$\langle 0, \frac{1}{2} \rangle$	$(0, 2)$	$\langle 0, 1 \rangle$	$(-1, 1)$	$\langle -\frac{1}{2}, \frac{1}{2} \rangle$
$(-1, 0)$	$\langle -\frac{1}{2}, 0 \rangle$	$(-2, 0)$	$\langle -1, 0 \rangle$	$(-1, -1)$	$\langle -\frac{1}{2}, -\frac{1}{2} \rangle$
$(0, -1)$	$\langle 0, -\frac{1}{2} \rangle$	$(0, -2)$	$\langle 0, -1 \rangle$	$(1, -1)$	$\langle \frac{1}{2}, -\frac{1}{2} \rangle$

Figure 16.1.2a shows the vector field. To see that each vector is perpendicular to the corresponding circle, Figure 16.1.2b shows circles overlain on the vector field.

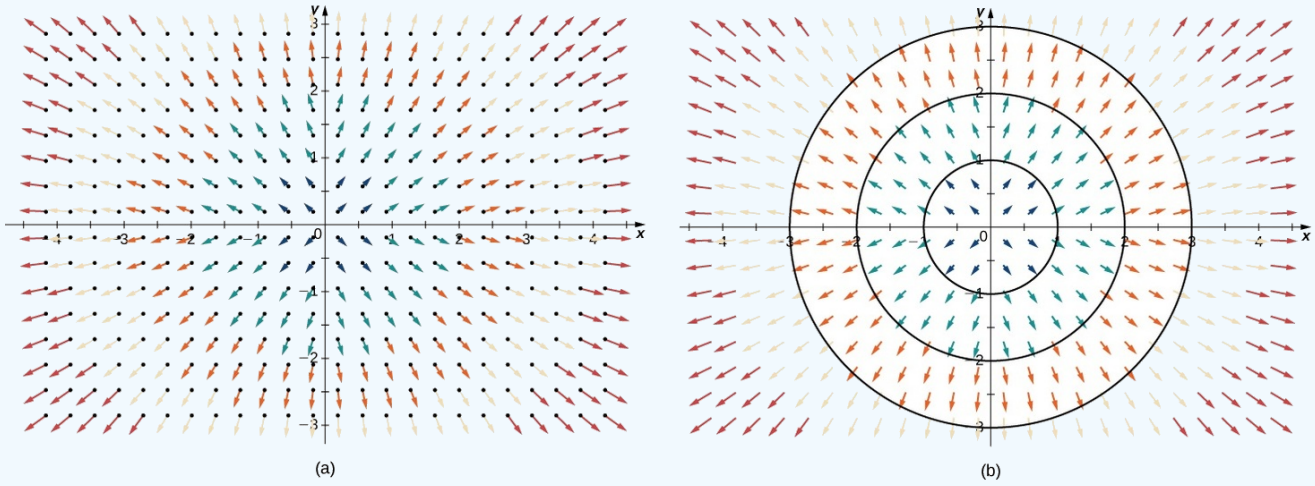


Figure 16.1.2: (a) A visual representation of the radial vector field $\vec{\mathbf{F}}(x, y) = \frac{x}{2}\hat{\mathbf{i}} + \frac{y}{2}\hat{\mathbf{j}}$. (b) The radial vector field $\vec{F}(x, y) = \frac{x}{2}\hat{\mathbf{i}} + \frac{y}{2}\hat{\mathbf{j}}$ with overlaid circles. Notice that each vector is perpendicular to the circle on which it is located.

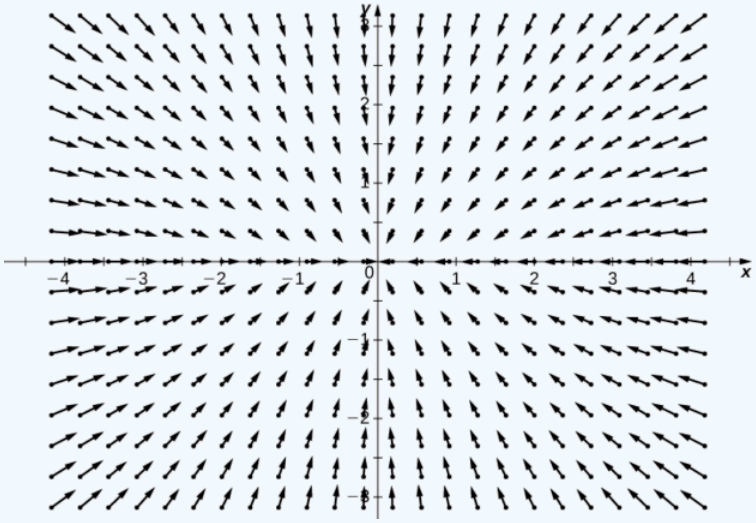
Exercise 16.1.2

Draw the radial field $\vec{\mathbf{F}}(x, y) = -\frac{x}{3}\hat{\mathbf{i}} - \frac{y}{3}\hat{\mathbf{j}}$.

Hint

Sketch enough vectors to get an idea of the shape.

Answer



In contrast to radial fields, in a *rotational field*, the vector at point (x, y) is tangent (not perpendicular) to a circle with radius $r = \sqrt{x^2 + y^2}$. In a standard rotational field, all vectors point either in a clockwise direction or in a counterclockwise direction, and the magnitude of a vector depends only on its distance from the origin. Both of the following examples are clockwise rotational fields, and we see from their visual representations that the vectors appear to rotate around the origin.

Example 16.1.3: Drawing a Rotational Vector Field

Sketch the vector field $\vec{\mathbf{F}}(x, y) = \langle y, -x \rangle$.

Solution

Create a table (see the one that follows) using a representative sample of points in a plane and their corresponding vectors. Figure 16.1.3 shows the resulting vector field.

Table 16.1.2

(x, y)	$\vec{\mathbf{F}}(x, y)$	(x, y)	$\vec{\mathbf{F}}(x, y)$	(x, y)	$\vec{\mathbf{F}}(x, y)$
$(1, 0)$	$\langle 0, -1 \rangle$	$(2, 0)$	$\langle 0, -2 \rangle$	$(1, 1)$	$\langle 1, -1 \rangle$
$(0, 1)$	$\langle 1, 0 \rangle$	$(0, 2)$	$\langle 2, 0 \rangle$	$(-1, 1)$	$\langle 1, 1 \rangle$
$(-1, 0)$	$\langle 0, 1 \rangle$	$(-2, 0)$	$\langle 0, 2 \rangle$	$(-1, -1)$	$\langle -1, 1 \rangle$
$(0, -1)$	$\langle -1, 0 \rangle$	$(0, -2)$	$\langle -2, 0 \rangle$	$(1, -1)$	$\langle -1, -1 \rangle$

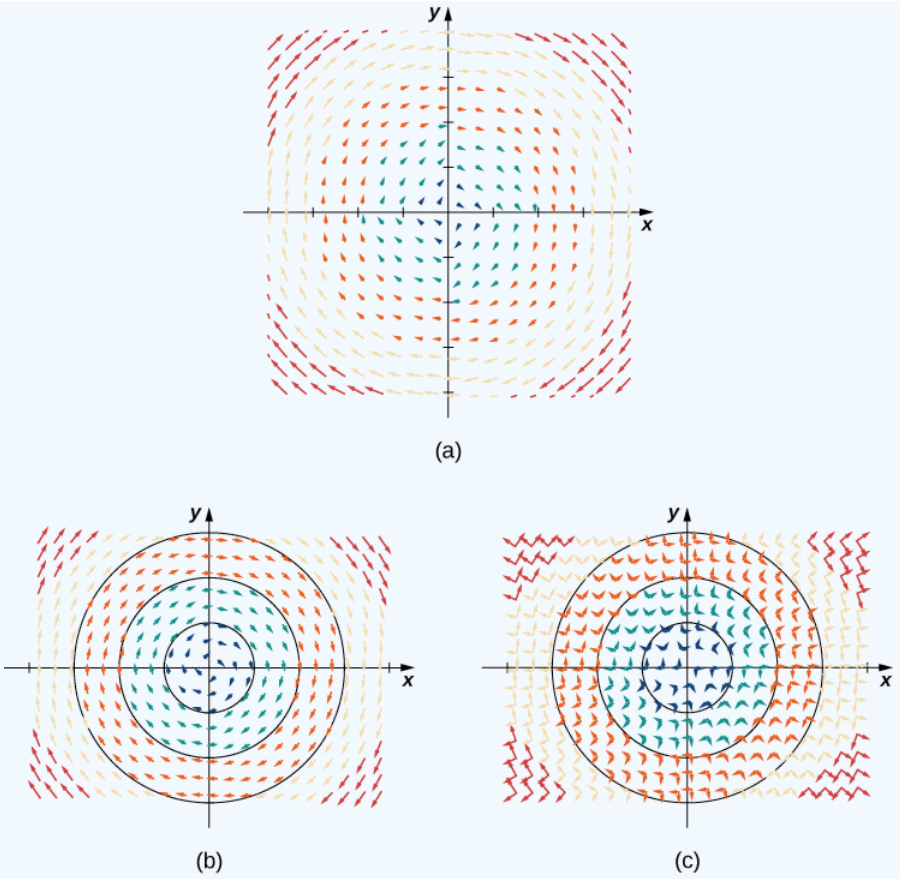


Figure 16.1.3: (a) A visual representation of vector field $\vec{\mathbf{F}}(x,y) = \langle y, -x \rangle$. (b) Vector field $\vec{\mathbf{F}}(x,y) = \langle y, -x \rangle$ with circles centered at the origin. (c) Vector $\vec{\mathbf{F}}(a,b)$ is perpendicular to radial vector $\langle a, b \rangle$ at point (a,b).

Analysis

Note that vector $\vec{\mathbf{F}}(a,b) = \langle b, -a \rangle$ points clockwise and is perpendicular to radial vector $\langle a, b \rangle$. (We can verify this assertion by computing the dot product of the two vectors: $\langle a, b \rangle \cdot \langle -b, a \rangle = -ab + ab = 0$.) Furthermore, vector $\langle b, -a \rangle$ has length $r = \sqrt{a^2 + b^2}$. Thus, we have a complete description of this rotational vector field: the vector associated with point (a,b) is the vector with length r tangent to the circle with radius r, and it points in the clockwise direction.

Sketches such as that in Figure 16.1.3 are often used to analyze major storm systems, including hurricanes and cyclones. In the northern hemisphere, storms rotate counterclockwise; in the southern hemisphere, storms rotate clockwise. (This is an effect caused by Earth's rotation about its axis and is called the Coriolis Effect.)



Figure 16.1.4: (credit: modification of work by NASA)

Example 16.1.4: Sketching a Vector Field

Sketch vector field $\vec{\mathbf{F}}(x,y) = \dfrac{y}{x^2+y^2}\hat{\mathbf{i}}, -\dfrac{x}{x^2+y^2}\hat{\mathbf{j}}$.

Solution

To visualize this vector field, first note that the dot product $\vec{\mathbf{F}}(a,b) \cdot (a\,\hat{\mathbf{i}} + b\,\hat{\mathbf{j}})$ is zero for any point (a,b). Therefore, each vector is tangent to the circle on which it is located. Also, as $(a,b) \to (0,0)$, the magnitude of $\vec{\mathbf{F}}(a,b)$ goes to infinity. To see this, note that

$$||\vec{\mathbf{F}}(a,b)|| = \sqrt{\frac{a^2+b^2}{(a^2+b^2)^2}} = \sqrt{\frac{1}{a^2+b^2}}.$$

Since $\dfrac{1}{a^2+b^2} \to \infty$ as $(a,b) \to (0,0)$, then $||\vec{\mathbf{F}}(a,b)|| \to \infty$ as $(a,b) \to (0,0)$. This vector field looks similar to the vector field in Example 16.1.3, but in this case the magnitudes of the vectors close to the origin are large. Table 16.1.3 shows a sample of points and the corresponding vectors, and Figure 16.1.5 shows the vector field. Note that this vector field models the whirlpool motion of the river in Figure 16.1.5(b). The domain of this vector field is all of $\mathbb{R}^2$ except for point $(0,0)$.

Table 16.1.3

(x,y)	$\vec{\mathbf{F}}(x,y)$	(x,y)	$\vec{\mathbf{F}}(x,y)$	(x,y)	$\vec{\mathbf{F}}(x,y)$
$(1,0)$	$\langle 0,-1\rangle$	$(2,0)$	$\langle 0,-\frac{1}{2}\rangle$	$(1,1)$	$\langle \frac{1}{2},-\frac{1}{2}\rangle$
$(0,1)$	$\langle 1,0\rangle$	$(0,2)$	$\langle \frac{1}{2},0\rangle$	$(-1,1)$	$\langle \frac{1}{2},\frac{1}{2}\rangle$
$(-1,0)$	$\langle 0,1\rangle$	$(-2,0)$	$\langle 0,\frac{1}{2}\rangle$	$(-1,-1)$	$\langle -\frac{1}{2},\frac{1}{2}\rangle$
$(0,-1)$	$\langle -1,0\rangle$	$(0,-2)$	$\langle -\frac{1}{2},0\rangle$	$(1,-1)$	$\langle -\frac{1}{2},-\frac{1}{2}\rangle$

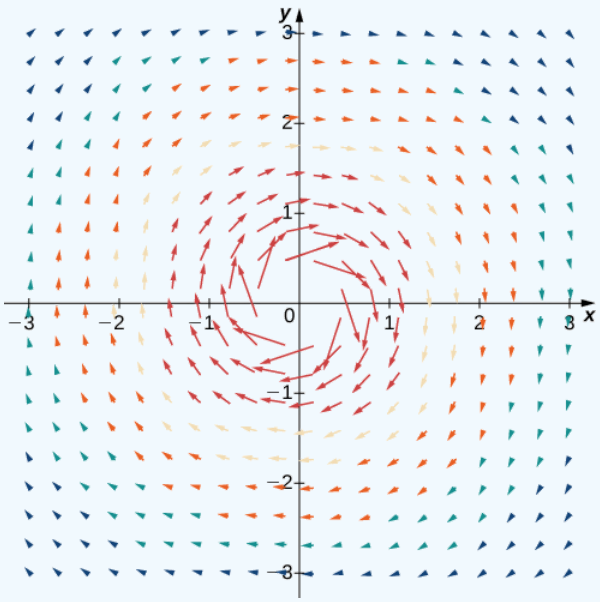


Figure 16.1.5: A visual representation of vector field $\vec{\mathbf{F}}(x,y) = \dfrac{y}{x^2+y^2}\hat{\mathbf{i}} - \dfrac{x}{x^2+y^2}\hat{\mathbf{j}}$. *This vector field could be used to model whirlpool motion of a fluid.*

? Exercise 16.1.4

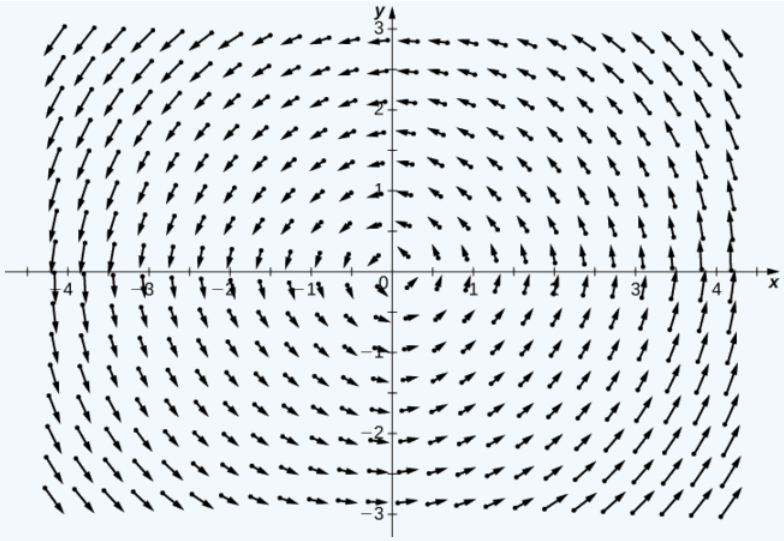
Sketch vector field $\vec{\mathbf{F}}(x,y) = \langle -2y,\, 2x\rangle$. Is the vector field radial, rotational, or neither?

Hint

Substitute enough points into $\vec{\mathbf{F}}$ to get an idea of the shape.

Answer

Rotational



Example 16.1.5: Velocity Field of a Fluid

Suppose that $\vec{\mathbf{v}}(x,y) = -\dfrac{2y}{x^2+y^2}\hat{\mathbf{i}} + \dfrac{2x}{x^2+y^2}\hat{\mathbf{j}}$ is the velocity field of a fluid. How fast is the fluid moving at point $(1,-1)$? (Assume the units of speed are meters per second.)

Solution

To find the velocity of the fluid at point $(1,-1)$, substitute the point into $\vec{\mathbf{v}}$:

$$\vec{\mathbf{v}}(1,-1) = \frac{-2(-1)}{1+1}\hat{\mathbf{i}} + \frac{2(1)}{1+1}\hat{\mathbf{j}} = \hat{\mathbf{i}} + \hat{\mathbf{j}}.$$

The speed of the fluid at $(1,-1)$ is the magnitude of this vector. Therefore, the speed is $||\hat{\mathbf{i}} + \hat{\mathbf{j}}|| = \sqrt{2}$ m/sec.

Exercise 16.1.5

Vector field $\vec{\mathbf{v}}(x,y) = \langle 4|x|,\ 1\rangle$ models the velocity of water on the surface of a river. What is the speed of the water at point $(2,3)$? Use meters per second as the units.

Hint

Remember, speed is the magnitude of velocity.

Answer

$\sqrt{65}$ m/sec

We have examined vector fields that contain vectors of various magnitudes, but just as we have unit vectors, we can also have a unit vector field. A vector field $\vec{\mathbf{F}}$ is a **unit vector field** if the magnitude of each vector in the field is 1. In a unit vector field, the only relevant information is the direction of each vector.

Example 16.1.6: A Unit Vector Field

Show that vector field $\vec{\mathbf{F}}(x,y) = \left\langle \dfrac{y}{\sqrt{x^2+y^2}}, -\dfrac{x}{\sqrt{x^2+y^2}} \right\rangle$ is a unit vector field.

Solution

To show that $\vec{\mathbf{F}}$ is a unit field, we must show that the magnitude of each vector is 1. Note that

$$\sqrt{\left(\frac{y}{\sqrt{x^2+y^2}}\right)^2+\left(-\frac{x}{\sqrt{x^2+y^2}}\right)^2}=\sqrt{\frac{y^2}{x^2+y^2}+\frac{x^2}{x^2+y^2}}$$
$$=\sqrt{\frac{x^2+y^2}{x^2+y^2}}$$
$$=1$$

Therefore, $\vec{\mathbf{F}}$ is a unit vector field.

? Exercise 16.1.6

Is vector field $\vec{\mathbf{F}}(x,y)=\langle -y,\, x\rangle$ a unit vector field?

Hint

Calculate the magnitude of $\vec{\mathbf{F}}$ at an arbitrary point (x,y).

Answer

No.

Why are unit vector fields important? Suppose we are studying the flow of a fluid, and we care only about the direction in which the fluid is flowing at a given point. In this case, the speed of the fluid (which is the magnitude of the corresponding velocity vector) is irrelevant, because all we care about is the direction of each vector. Therefore, the unit vector field associated with velocity is the field we would study.

If $\vec{\mathbf{F}}=\langle P,Q,R\rangle$ is a vector field, then the corresponding unit vector field is $\langle \frac{P}{||\vec{\mathbf{F}}||},\frac{Q}{||\vec{\mathbf{F}}||},\frac{R}{||\vec{\mathbf{F}}||}\rangle$. Notice that if $\vec{\mathbf{F}}(x,y)=\langle y,\,-x\rangle$ is the vector field from Example 16.1.6, then the magnitude of $\vec{\mathbf{F}}$ is $\sqrt{x^2+y^2}$, and therefore the corresponding unit vector field is the field $\vec{\mathbf{G}}$ from the previous example.

If $\vec{\mathbf{F}}$ is a vector field, then the process of dividing $\vec{\mathbf{F}}$ by its magnitude to form unit vector field $\vec{\mathbf{F}}/||\vec{\mathbf{F}}||$ is called normalizing the field $\vec{\mathbf{F}}$.

Vector Fields in $\mathbb{R}^3$

We have seen several examples of vector fields in $\mathbb{R}^2$; let's now turn our attention to vector fields in $\mathbb{R}^3$. These vector fields can be used to model gravitational or electromagnetic fields, and they can also be used to model fluid flow or heat flow in three dimensions. A two-dimensional vector field can really only model the movement of water on a two-dimensional slice of a river (such as the river's surface). Since a river flows through three spatial dimensions, to model the flow of the entire depth of the river, we need a vector field in three dimensions.

The extra dimension of a three-dimensional field can make vector fields in $\mathbb{R}^3$ more difficult to visualize, but the idea is the same. To visualize a vector field in $\mathbb{R}^3$, plot enough vectors to show the overall shape. We can use a similar method to visualizing a vector field in $\mathbb{R}^2$ by choosing points in each octant.

Just as with vector fields in $\mathbb{R}^2$, we can represent vector fields in $\mathbb{R}^3$ with component functions. We simply need an extra component function for the extra dimension. We write either

$$\vec{\mathbf{F}}(x,y,z)=\langle P(x,y,z),Q(x,y,z),R(x,y,z)\rangle$$

or

$$\vec{\mathbf{F}}(x,y,z)=P(x,y,z)\hat{\mathbf{i}}+Q(x,y,z)\hat{\mathbf{j}}+R(x,y,z)\hat{\mathbf{k}}.$$

Example 16.1.7: Sketching a Vector Field in Three Dimensions

Describe vector field $\vec{\mathbf{F}}(x, y, z) = \langle 1, 1, z \rangle$.

Solution

For this vector field, the x- and y-components are constant, so every point in $\mathbb{R}^3$ has an associated vector with x- and y-components equal to one. To visualize $\vec{\mathbf{F}}$, we first consider what the field looks like in the xy-plane. In the xy-plane, $z = 0$. Hence, each point of the form $(a, b, 0)$ has vector $\langle 1, 1, 0 \rangle$ associated with it. For points not in the xy-plane but slightly above it, the associated vector has a small but positive z-component, and therefore the associated vector points slightly upward. For points that are far above the xy-plane, the z-component is large, so the vector is almost vertical. Figure 16.1.6 shows this vector field.

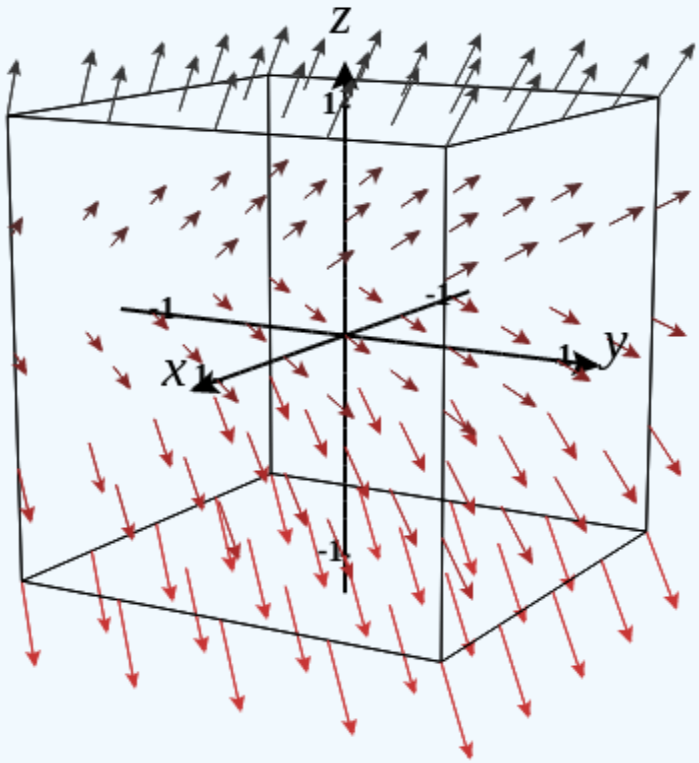


Figure 16.1.6: A visual representation of vector field $\vec{\mathbf{F}}(x, y, z) = \langle 1, 1, z \rangle$.

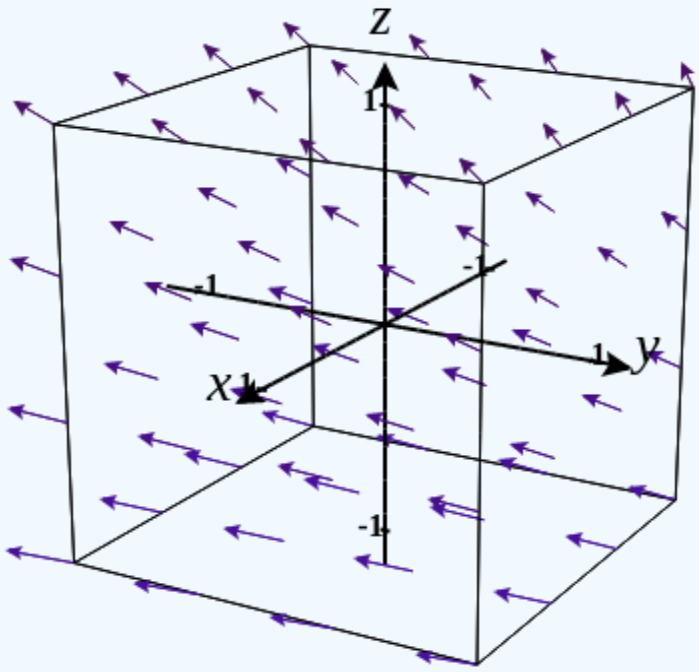
Exercise 16.1.7

Sketch vector field $\vec{\mathbf{G}}(x, y, z) = \langle 2, \frac{z}{2}, 1 \rangle$.

Hint

Substitute enough points into the vector field to get an idea of the general shape.

Answer



In the next example, we explore one of the classic cases of a three-dimensional vector field: a gravitational field.

Example 16.1.8: Describing a Gravitational Vector Field

Newton's law of gravitation states that $\vec{\mathbf{F}} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$, where G is the universal gravitational constant. It describes the gravitational field exerted by an object (object 1) of mass m_1 located at the origin on another object (object 2) of mass m_2 located at point (x, y, z). Field $\vec{\mathbf{F}}$ denotes the gravitational force that object 1 exerts on object 2, r is the distance between the two objects, and $\hat{\mathbf{r}}$ indicates the unit vector from the first object to the second. The minus sign shows that the gravitational force attracts toward the origin; that is, the force of object 1 is attractive. Sketch the vector field associated with this equation.

Solution

Since object 1 is located at the origin, the distance between the objects is given by $r = \sqrt{x^2 + y^2 + z^2}$. The unit vector from object 1 to object 2 is $\hat{\mathbf{r}} = \frac{\langle x, y, z\rangle}{||\langle x, y, z\rangle||}$, and hence $\hat{\mathbf{r}} = \langle \frac{x}{r}, \frac{y}{r}, \frac{z}{r}\rangle$. Therefore, gravitational vector field $\vec{\mathbf{F}}$ exerted by object 1 on object 2 is

$$\vec{\mathbf{F}} = -Gm_1 m_2 \langle \frac{x}{r^3}, \frac{y}{r^3}, \frac{z}{r^3}\rangle.$$

This is an example of a radial vector field in $\mathbb{R}^3$.

Figure 16.1.7 shows what this gravitational field looks like for a large mass at the origin. Note that the magnitudes of the vectors increase as the vectors get closer to the origin.

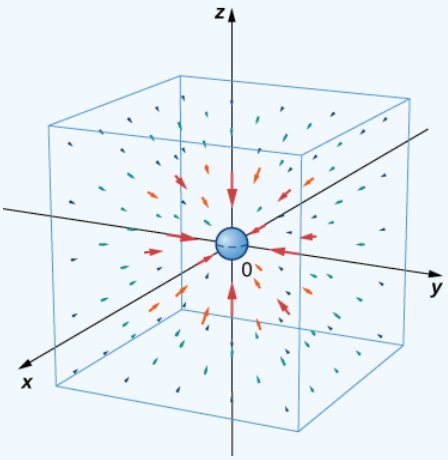


Figure 16.1.7: A visual representation of gravitational vector field $\vec{\mathbf{F}} = -Gm_1m_2\langle \frac{x}{r^3}, \frac{y}{r^3}, \frac{z}{r^3} \rangle$ for a large mass at the origin.

Exercise 16.1.8

The mass of asteroid 1 is 750,000 kg and the mass of asteroid 2 is 130,000 kg. Assume asteroid 1 is located at the origin, and asteroid 2 is located at $(15, -5, 10)$, measured in units of 10 to the eighth power kilometers. Given that the universal gravitational constant is $G = 6.67384 \times 10^{-11} m^3 kg^{-1} s^{-2}$, find the gravitational force vector that asteroid 1 exerts on asteroid 2.

Hint

Follow Example 16.1.8 and first compute the distance between the asteroids.

Answer

1.49063×10^{-18}, 4.96876×10^{-19}, 9.93752×10^{-19} N

Gradient Fields (Conservative Fields)

In this section, we study a special kind of vector field called a gradient field or a **conservative field**. These vector fields are extremely important in physics because they can be used to model physical systems in which energy is conserved. Gravitational fields and electric fields associated with a static charge are examples of gradient fields.

Recall that if f is a (scalar) function of x and y, then the gradient of f is

$$\text{grad } f = \vec{\nabla} f(x, y) = f_x(x, y)\hat{\mathbf{i}} + f_y(x, y)\hat{\mathbf{j}}.$$

We can see from the form in which the gradient is written that $\vec{\nabla} f$ is a vector field in $\mathbb{R}^2$. Similarly, if f is a function of x, y, and z, then the gradient of f is

$$\text{grad } f = \vec{\nabla} f(x, y, z) = f_x(x, y, z)\hat{\mathbf{i}} + f_y(x, y, z)\hat{\mathbf{j}} + f_z(x, y, z)\hat{\mathbf{k}}.$$

The gradient of a three-variable function is a vector field in $\mathbb{R}^3$. A gradient field is a vector field that can be written as the gradient of a function, and we have the following definition.

DEFINITION: Gradient Field

A vector field $\vec{\mathbf{F}}$ in $\mathbb{R}^2$ or in $\mathbb{R}^3$ is a gradient field if there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$.

Example 16.1.9: Sketching a Gradient Vector Field

Use technology to plot the gradient vector field of $f(x, y) = x^2 y^2$.

Solution

The gradient of f is $\vec{\nabla} f(x,y) = \langle 2xy^2,\ 2x^2y \rangle$. To sketch the vector field, use a computer algebra system such as Mathematica. Figure 16.1.8 shows $\vec{\nabla} f$.

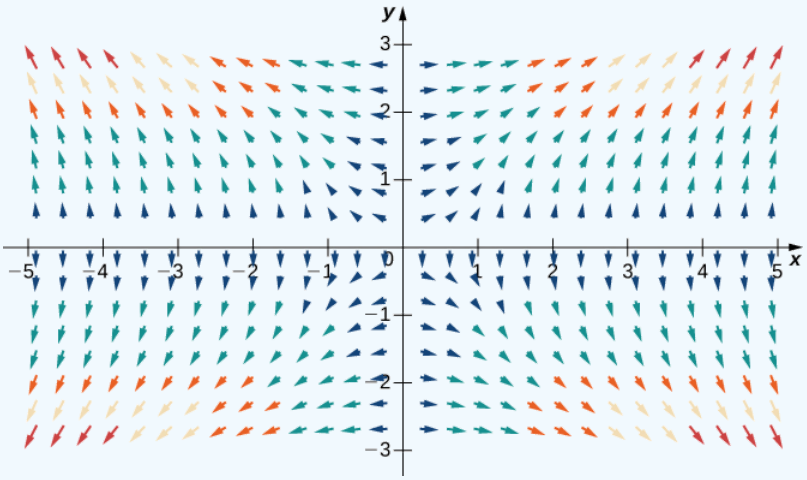


Figure 16.1.8: The gradient vector field is $\vec{\nabla} f$, where $f(x,y) = x^2y^2$.

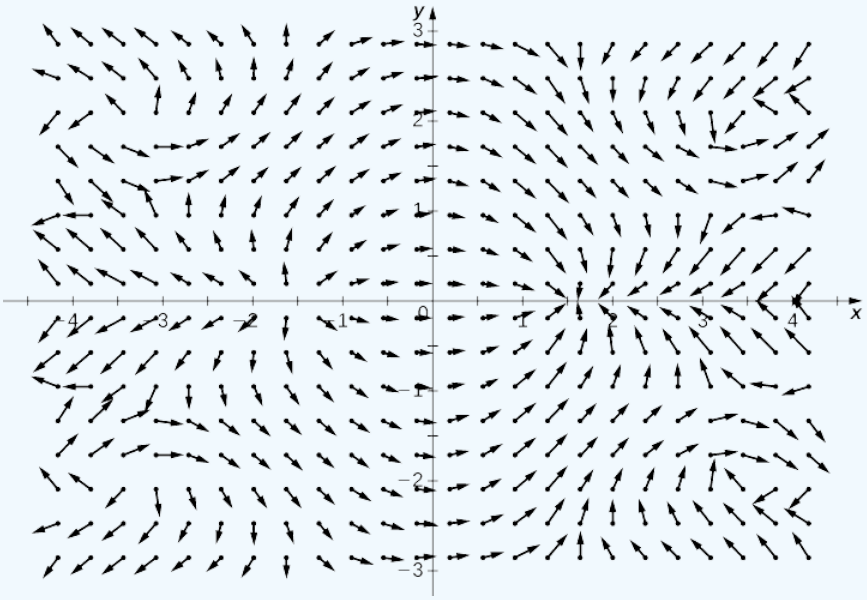
? Exercise 16.1.9

Use technology to plot the gradient vector field of $f(x,y) = \sin x \cos y$.

Hint

Find the gradient of f.

Answer



Consider the function $f(x,y) = x^2y^2$ from Example 16.1.9. Figure 16.1.9 shows the level curves of this function overlaid on the function's gradient vector field. The gradient vectors are perpendicular to the level curves, and the magnitudes of the vectors get larger as the level curves get closer together, because closely grouped level curves indicate the graph is steep, and the magnitude of the gradient vector is the largest value of the directional derivative. Therefore, you can see the local steepness of a graph by investigating the corresponding function's gradient field.

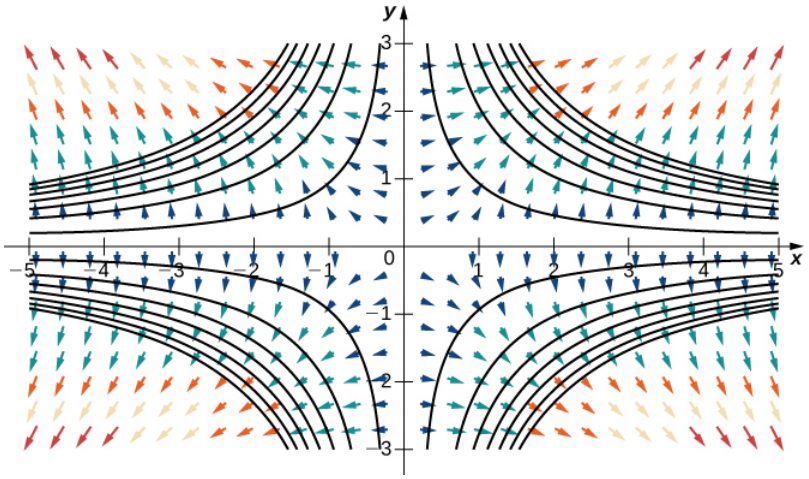


Figure 16.1.9: The gradient field of $f(x,y) = x^2y^2$ and several level curves of f. Notice that as the level curves get closer together, the magnitude of the gradient vectors increases.

As we learned earlier, a vector field $\vec{\mathbf{F}}$ is a conservative vector field, or a gradient field if there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$. In this situation, f is called a **potential function** for $\vec{\mathbf{F}}$. Conservative vector fields arise in many applications, particularly in physics. The reason such fields are called *conservative* is that they model forces of physical systems in which energy is conserved. We study conservative vector fields in more detail later in this chapter.

You might notice that, in some applications, a potential function f for $\vec{\mathbf{F}}$ is defined instead as a function such that $-\vec{\nabla} f = \vec{\mathbf{F}}$. This is the case for certain contexts in physics, for example.

✔ Example 16.1.10: Verifying a Potential Function

Is $f(x,y,z) = x^2yz - \sin(xy)$ a potential function for vector field

$$\vec{\mathbf{F}}(x,y,z) = \langle 2xyz - y\cos(xy), x^2z - x\cos(xy), x^2y\rangle?$$

Solution

We need to confirm whether $\vec{\nabla} f = \vec{\mathbf{F}}$. We have

$$\begin{aligned} f_x(x,y) &= 2xyz - y\cos(xy) \\ f_y(x,y) &= x^2z - x\cos(xy). \\ f_z(x,y) &= x^2y \end{aligned}$$

Therefore, $\vec{\nabla} f = \vec{\mathbf{F}}$ and f is a potential function for $\vec{\mathbf{F}}$.

? Exercise 16.1.10

Is $f(x,y,z) = x^2\cos(yz) + y^2z^2$ a potential function for $\vec{\mathbf{F}}(x,y,z) = \langle 2x\cos(yz), -x^2z\sin(yz) + 2yz^2, y^2\rangle$?

Hint

Compute the gradient of f.

Answer

No

✔ Example 16.1.11: Verifying a Potential Function

The velocity of a fluid is modeled by field $\vec{\mathbf{v}}(x,y) = \langle xy, \frac{x^2}{2} - y\rangle$. Verify that $f(x,y) = \dfrac{x^2y}{2} - \dfrac{y^2}{2}$ is a potential function for $\vec{\mathbf{v}}$.

Solution

To show that f is a potential function, we must show that $\vec{\nabla} f = \vec{\mathbf{v}}$. Note that $f_x(x, y) = xy$ and $f_y(x, y) = \frac{x^2}{2} - y$. Therefore, $\vec{\nabla} f(x, y) = \langle xy, \frac{x^2}{2} - y \rangle$ and f is a potential function for $\vec{\mathbf{v}}$ (Figure 16.1.10).

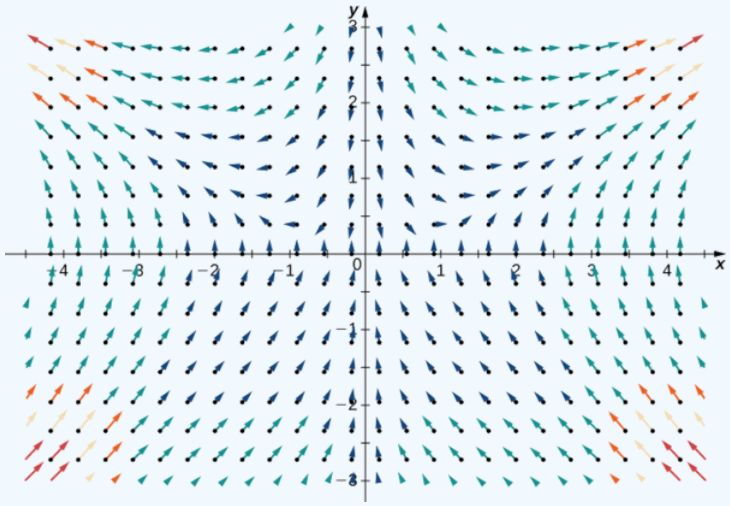


Figure 16.1.10: Velocity field $\vec{\mathbf{v}}(x, y)$ has a potential function and is a conservative field.

? Exercise 16.1.11

Verify that $f(x, y) = x^3y^2 + x$ is a potential function for velocity field $\vec{\mathbf{v}}(x, y) = \langle 3x^2y^2 + 1, 2x^3y \rangle$.

Hint

Calculate the gradient.

Answer

$\vec{\nabla} f(x, y) = \vec{\mathbf{v}}(x, y)$

If $\vec{\mathbf{F}}$ is a conservative vector field, then there is at least one potential function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$. But, could there be more than one potential function? If so, is there any relationship between two potential functions for the same vector field? Before answering these questions, let's recall some facts from single-variable calculus to guide our intuition. Recall that if $k(x)$ is an integrable function, then k has infinitely many antiderivatives. Furthermore, if $\vec{\mathbf{F}}$ and $\vec{\mathbf{G}}$ are both antiderivatives of k, then $\vec{\mathbf{F}}$ and $\vec{\mathbf{G}}$ **differ only by a constant**. That is, there is some number C such that $\vec{\mathbf{F}}(x) = \vec{\mathbf{G}}(x) + C$.

Now let $\vec{\mathbf{F}}$ be a conservative vector field and let f and g be potential functions for $\vec{\mathbf{F}}$. Since the gradient is like a derivative, $\vec{\mathbf{F}}$ being conservative means that $\vec{\mathbf{F}}$ is "integrable" with "antiderivatives" f and g. Therefore, if the analogy with single-variable calculus is valid, we expect there is some constant C such that $f(x) = g(x) + C$. The next theorem says that this is indeed the case.

To state the next theorem with precision, we need to assume the domain of the vector field is connected and open. To be connected means if P_1 and P_2 are any two points in the domain, then you can walk from P_1 to P_2 along a path that stays entirely inside the domain.

UNIQUENESS OF POTENTIAL FUNCTIONS

Let $\vec{\mathbf{F}}$ be a conservative vector field on an open and connected domain and let f and g be functions such that $\vec{\nabla} f = \vec{\mathbf{F}}$ and $\vec{\nabla} g = \vec{\mathbf{F}}$. Then, there is a constant C such that $f = g + C$.

Proof

Since f and g are both potential functions for $\vec{\mathbf{F}}$, then $\vec{\nabla}(f-g) = \vec{\nabla}f - \vec{\nabla}g = \vec{\mathbf{F}} - \vec{\mathbf{F}} = \vec{\mathbf{0}}$. Let $h = f - g$, then we have $\vec{\nabla}h = \vec{\mathbf{0}}$.We would like to show that h is a constant function.

Assume h is a function of x and y (the logic of this proof extends to any number of independent variables). Since $\vec{\nabla}h = \vec{\mathbf{0}}$, we have $h_x(x,y) = 0$ and $h_y(x,y) = 0$. The expression $h_x(x,y) = 0$ implies that h is a constant function with respect to x—that is, $h(x,y) = k_1(y)$ for some function k_1. Similarly, $h_y(x,y) = 0$ implies $h(x,y) = k_2(x)$ for some function k_2. Therefore, function h depends only on y and also depends only on x. Thus, $h(x,y) = C$ for some constant C on the connected domain of $\vec{\mathbf{F}}$. Note that we really do need connectedness at this point; if the domain of $\vec{\mathbf{F}}$ came in two separate pieces, then k could be a constant C_1 on one piece but could be a different constant C_2 on the other piece. Since $f - g = h = C$, we have that $f = g + C$, as desired.

□

Conservative vector fields also have a special property called the ***cross-partial property***. This property helps test whether a given vector field is conservative.

THE CROSS-PARTIAL PROPERTY OF CONSERVATIVE VECTOR FIELDS

Let $\vec{\mathbf{F}}$ be a vector field in two or three dimensions such that the component functions of $\vec{\mathbf{F}}$ have continuous second-order mixed-partial derivatives on the domain of $\vec{\mathbf{F}}$.

If $\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y)\rangle$ is a conservative vector field in $\mathbb{R}^2$, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

If $\vec{\mathbf{F}}(x,y,z) = \langle P(x,y,z), Q(x,y,z), R(x,y,z)\rangle$ is a conservative vector field in $\mathbb{R}^3$, then

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$
$$\frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}$$
$$\frac{\partial R}{\partial x} = \frac{\partial P}{\partial z}.$$

Proof

Since $\vec{\mathbf{F}}$ is conservative, there is a function $f(x,y)$ such that $\vec{\nabla}f = \vec{\mathbf{F}}$. Therefore, by the definition of the gradient, $f_x = P$ and $f_y = Q$. By Clairaut's theorem, $f_{xy} = f_{yx}$, But, $f_{xy} = P_y$ and $f_{yx} = Q_x$, and thus $P_y = Q_x$.

□

Clairaut's theorem gives a fast proof of the cross-partial property of conservative vector fields in $\mathbb{R}^3$, just as it did for vector fields in $\mathbb{R}^2$.

The Cross-Partial Property of Conservative Vector Fields shows that most vector fields are not conservative. The cross-partial property is difficult to satisfy in general, so most vector fields won't have equal cross-partials.

Show that rotational vector field $\vec{\mathbf{F}}(x,y) = \langle y, -x\rangle$ is not conservative.

Solution

Let $P(x,y) = y$ and $Q(x,y) = -x$. If $\vec{\mathbf{F}}$ is conservative, then the cross-partials would be equal—that is, P_y would equal Q_x.Therefore, to show that $\vec{\mathbf{F}}$ is not conservative, check that $P_y \neq Q_x$. Since $P_y = 1$ and $Q_x = -1$, the vector field is not conservative.

Exercise 16.1.12

Show that vector field $\vec{\mathbf{F}}(x,y) = xy\,\hat{\mathbf{i}} - x^2y\,\hat{\mathbf{j}}$ is not conservative.

Hint

Check the cross-partials.

Answer

$P_y(x,y) = x$ and $Q_x(x,y) = -2xy$. Since $P_y(x,y) \neq Q_x(x,y)$, $\vec{\mathbf{F}}$ is not conservative.

Example 16.1.13: Showing a Vector Field Is Not Conservative

Is vector field $\vec{\mathbf{F}}(x,y,z) = \langle 7, -2, x^3\rangle$ conservative?

Solution

Let $P(x,y,z) = 7$, $Q(x,y,z) = -2$, and $R(x,y,z) = x^3$. If $\vec{\mathbf{F}}$ is conservative, then all three cross-partial equations will be satisfied —that is, if $\vec{\mathbf{F}}$ is conservative, then P_y would equal Q_x, Q_z would equal R_y, and R_x would equal P_z. Note that

$$P_y = Q_x = R_y = Q_z = 0$$

so the first two necessary equalities hold. However, $R_x(x,y,z) = x^3$ and $P_z(x,y,z) = 0$ so $R_x \neq P_z$. Therefore, $\vec{\mathbf{F}}$ is not conservative.

Exercise 16.1.13

Is vector field $\vec{\mathbf{G}}(x,y,z) = \langle y,\ x,\ xyz\rangle$ conservative?

Hint

Check the cross-partials.

Answer

No

We conclude this section with a word of warning: The Cross-Partial Property of Conservative Vector Fields says that if $\vec{\mathbf{F}}$ is conservative, then $\vec{\mathbf{F}}$ has the cross-partial property. The theorem does *not* say that, if $\vec{\mathbf{F}}$ has the cross-partial property, then $\vec{\mathbf{F}}$ is conservative (the converse of an implication is not logically equivalent to the original implication). In other words, The Cross-Partial Property of Conservative Vector Fields can only help determine that a field is not conservative; it does not let you conclude that a vector field is conservative.

For example, consider vector field $\vec{\mathbf{F}}(x,y) = \langle x^2y, \frac{x^3}{3}\rangle$. This field has the cross-partial property, so it is natural to try to use The Cross-Partial Property of Conservative Vector Fields to conclude this vector field is conservative. However, this is a misapplication of the theorem. We learn later how to conclude that $\vec{\mathbf{F}}$ is conservative.

Key Concepts

- A vector field assigns a vector $\vec{\mathbf{F}}(x,y)$ to each point (x,y) in a subset D of $\mathbb{R}^2$ or $\mathbb{R}^3$. $\vec{\mathbf{F}}(x,y,z)$ to each point (x,y,z) in a subset D of $\mathbb{R}^3$.
- Vector fields can describe the distribution of vector quantities such as forces or velocities over a region of the plane or of space. They are in common use in such areas as physics, engineering, meteorology, oceanography.
- We can sketch a vector field by examining its defining equation to determine relative magnitudes in various locations and then drawing enough vectors to determine a pattern.
- A vector field $\vec{\mathbf{F}}$ is called conservative if there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$.

Key Equations

- **Vector field in $\mathbb{R}^2$**
  $\vec{\mathbf{F}}(x, y) = \langle P(x, y),\ Q(x, y)\rangle$
  or
  $\vec{\mathbf{F}}(x, y) = P(x, y)\,\hat{\mathbf{i}} + Q(x, y)\,\hat{\mathbf{j}}$
- **Vector field in $\mathbb{R}^3$**
  $\vec{\mathbf{F}}(x, y, z) = \langle P(x, y, z),\ Q(x, y, z),\ R(x, y, z)\rangle$
  or
  $\vec{\mathbf{F}}(x, y, z) = P(x, y, z)\,\hat{\mathbf{i}} + Q(x, y, z)\,\hat{\mathbf{j}} + R(x, y, z)\,\hat{\mathbf{k}}$

Glossary

conservative field

a vector field for which there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$

gradient field

a vector field $\vec{\mathbf{F}}$ for which there exists a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$; in other words, a vector field that is the gradient of a function; such vector fields are also called *conservative*

potential function

a scalar function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$

radial field

a vector field in which all vectors either point directly toward or directly away from the origin; the magnitude of any vector depends only on its distance from the origin

rotational field

a vector field in which the vector at point (x, y) is tangent to a circle with radius $r = \sqrt{x^2 + y^2}$; in a rotational field, all vectors flow either clockwise or counterclockwise, and the magnitude of a vector depends only on its distance from the origin

unit vector field

a vector field in which the magnitude of every vector is 1

vector field

measured in $\mathbb{R}^2$, an assignment of a vector $\vec{\mathbf{F}}(x, y)$ to each point (x, y) of a subset D of $\mathbb{R}^2$; in $\mathbb{R}^3$, an assignment of a vector $\vec{\mathbf{F}}(x, y, z)$ to each point (x, y, z) of a subset D of $\mathbb{R}^3$

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16.1E: Exercises for Section 16.1

1. The domain of vector field $\vec{\mathbf{F}} = \vec{\mathbf{F}}(x, y)$ is a set of points (x, y) in a plane, and the range of $\vec{\mathbf{F}}$ is a set of *what* in the plane?

For exercises 2 - 4, determine whether the statement is *true or false*.

2. Vector field $\vec{\mathbf{F}} = \langle 3x^2, 1\rangle$ is a gradient field for both $\phi_1(x, y) = x^3 + y$ and $\phi_2(x, y) = y + x^3 + 100$.

3. Vector field $\vec{\mathbf{F}} = \dfrac{\langle y, x\rangle}{\sqrt{x^2 + y^2}}$ is constant in direction and magnitude on a unit circle.

4. Vector field $\vec{\mathbf{F}} = \dfrac{\langle y, x\rangle}{\sqrt{x^2 + y^2}}$ is neither a radial field nor a rotation field.

For exercises 5 - 13, describe each vector field by drawing some of its vectors.

5. [T] $\vec{\mathbf{F}}(x, y) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$

6. [T] $\vec{\mathbf{F}}(x, y) = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$

7. [T] $\vec{\mathbf{F}}(x, y) = x\,\hat{\mathbf{i}} - y\,\hat{\mathbf{j}}$

8. [T] $\vec{\mathbf{F}}(x, y) = \hat{\mathbf{i}} + \hat{\mathbf{j}}$

9. [T] $\vec{\mathbf{F}}(x, y) = 2x\,\hat{\mathbf{i}} + 3y\,\hat{\mathbf{j}}$

10. [T] $\vec{\mathbf{F}}(x, y) = 3\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$

11. [T] $\vec{\mathbf{F}}(x, y) = y\,\hat{\mathbf{i}} + \sin x\,\hat{\mathbf{j}}$

12. [T] $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$

13. [T] $\vec{\mathbf{F}}(x, y, z) = 2x\,\hat{\mathbf{i}} - 2y\,\hat{\mathbf{j}} - 2z\,\hat{\mathbf{k}}$

14. [T] $\vec{\mathbf{F}}(x, y, z) = yz\,\hat{\mathbf{i}} - xz\,\hat{\mathbf{j}}$

For exercises 15 - 20, find the gradient vector field of each function f.

15. $f(x, y) = x \sin y + \cos y$

16. $f(x, y, z) = ze^{-xy}$

17. $f(x, y, z) = x^2y + xy + y^2z$

18. $f(x, y) = x^2 \sin(5y)$

19. $f(x, y) = \ln(1 + x^2 + 2y^2)$

20. $f(x, y, z) = x \cos\left(\frac{y}{z}\right)$

21. What is vector field $\vec{\mathbf{F}}(x, y)$ with a value at (x, y) that is of unit length and points toward $(1, 0)$?

For exercises 22 - 24, write formulas for the vector fields with the given properties.

22. All vectors are parallel to the x-axis and all vectors on a vertical line have the same magnitude.

23. All vectors point toward the origin and have constant length.

24. All vectors are of unit length and are perpendicular to the position vector at that point.

25. Give a formula $\vec{\mathbf{F}}(x, y) = M(x, y)\,\hat{\mathbf{i}} + N(x, y)\,\hat{\mathbf{j}}$ for the vector field in a plane that has the properties that $\vec{\mathbf{F}} = \vec{\mathbf{0}}$ at $(0, 0)$ and that at any other point (a, b), $\vec{\mathbf{F}}$ is tangent to circle $x^2 + y^2 = a^2 + b^2$ and points in the clockwise direction with magnitude $\|\vec{\mathbf{F}}\| = \sqrt{a^2 + b^2}$.

26. Is vector field $\vec{\mathbf{F}}(x, y) = (P(x, y), Q(x, y)) = (\sin x + y)\,\hat{\mathbf{i}} + (\cos y + x)\,\hat{\mathbf{j}}$ a gradient field?

27. Find a formula for vector field $\vec{\mathbf{F}}(x, y) = M(x, y)\,\hat{\mathbf{i}} + N(x, y)\,\hat{\mathbf{j}}$ given the fact that for all points (x, y), $\vec{\mathbf{F}}$ points toward the origin and $\|\vec{\mathbf{F}}\| = \dfrac{10}{x^2 + y^2}$.

For exercises 28 - 29, assume that an electric field in the xy-plane caused by an infinite line of charge along the x-axis is a gradient field with potential function $V(x,y) = c\ln\left(\frac{r_0}{\sqrt{x^2+y^2}}\right)$, where $c > 0$ is a constant and r_0 is a reference distance at which the potential is assumed to be zero.

28. Find the components of the electric field in the x- and y-directions, where $\vec{\mathbf{E}}(x,y) = -\vec{\nabla}V(x,y)$.

29. Show that the electric field at a point in the xy-plane is directed outward from the origin and has magnitude $\|\vec{\mathbf{E}}\| = \frac{c}{r}$, where $r = \sqrt{x^2+y^2}$.

A ***flow line*** (or *streamline*) of a vector field $\vec{\mathbf{F}}$ is a curve $\vec{\mathbf{r}}(t)$ such that $d\vec{\mathbf{r}}/dt = \vec{\mathbf{F}}(\vec{\mathbf{r}}(t))$. If $\vec{\mathbf{F}}$ represents the velocity field of a moving particle, then the flow lines are paths taken by the particle. Therefore, flow lines are tangent to the vector field.

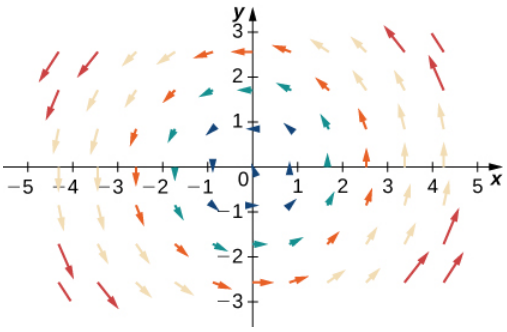
For exercises 30 and 31, show that the given curve $\vec{\mathbf{c}}(t)$ is a flow line of the given velocity vector field $\vec{\mathbf{F}}(x,y,z)$.

30. $\vec{\mathbf{c}}(t) = \langle e^{2t}, \ln|t|, \frac{1}{t}\rangle,\ t \neq 0; \quad \vec{\mathbf{F}}(x,y,z) = \langle 2x, z, -z^2\rangle$

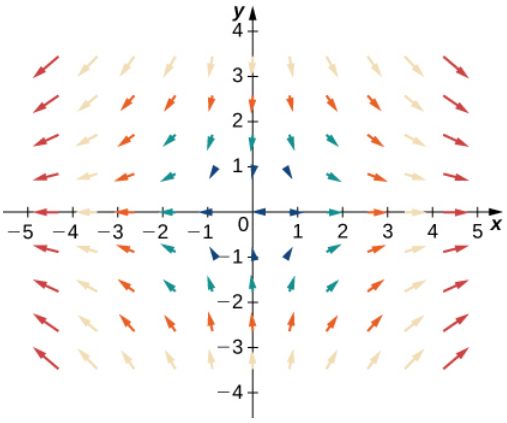
31. $\vec{\mathbf{c}}(t) = \langle \sin t, \cos t, e^t\rangle; \quad \vec{\mathbf{F}}(x,y,z) = \langle y, -x, z\rangle$

For exercises 32 - 34, let $\vec{\mathbf{F}} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$, $\vec{\mathbf{G}} = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$, and $\vec{\mathbf{H}} = x\,\hat{\mathbf{i}} - y\,\hat{\mathbf{j}}$. Match $\vec{\mathbf{F}}$, $\vec{\mathbf{G}}$, and $\vec{\mathbf{H}}$ with their graphs.

32.

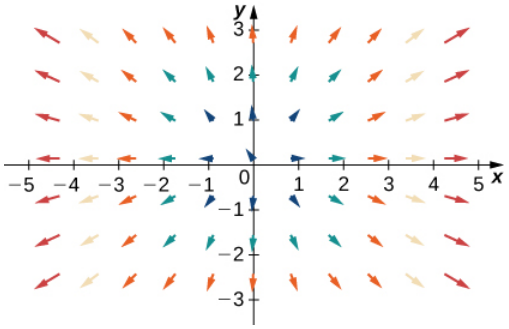


33.



34.

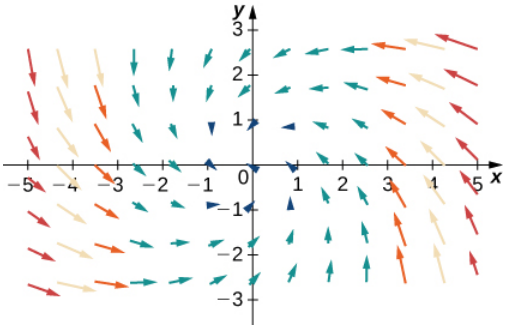
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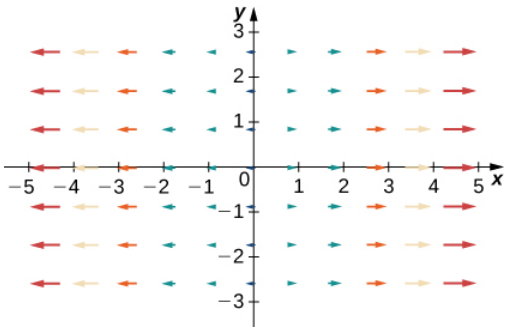
For exercises 35 - 38, let $\vec{\mathbf{F}} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$, $\vec{\mathbf{G}} = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$, and $\vec{\mathbf{H}} = -x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$. Match the vector fields with their graphs in (I)−(IV).

a. $\vec{\mathbf{F}} + \vec{\mathbf{G}}$
b. $\vec{\mathbf{F}} + \vec{\mathbf{H}}$
c. $\vec{\mathbf{G}} + \vec{\mathbf{H}}$
d. $-\vec{\mathbf{F}} + \vec{\mathbf{G}}$

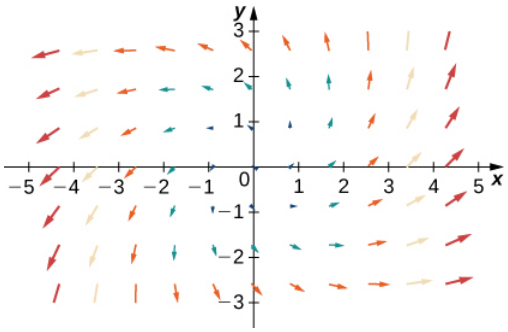
35.



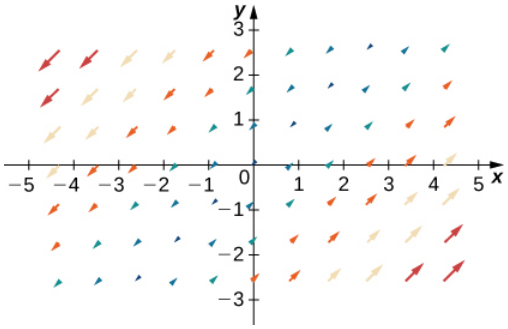
36.



37.



38.



Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

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16.2: Line Integrals

Learning Objectives

- Calculate a scalar line integral along a curve.
- Calculate a vector line integral along an oriented curve in space.
- Use a line integral to compute the work done in moving an object along a curve in a vector field.
- Describe the flux and circulation of a vector field.

We are familiar with single-variable integrals of the form $\int_a^b f(x)\,dx$, where the domain of integration is an interval $[a,b]$. Such an interval can be thought of as a curve in the xy-plane, since the interval defines a line segment with endpoints $(a,0)$ and $(b,0)$—in other words, a line segment located on the x-axis. Suppose we want to integrate over *any* curve in the plane, not just over a line segment on the x-axis. Such a task requires a new kind of integral, called a *line integral.*

Line integrals have many applications to engineering and physics. They also allow us to make several useful generalizations of the Fundamental Theorem of Calculus. And, they are closely connected to the properties of vector fields, as we shall see.

Scalar Line Integrals

A line integral gives us the ability to integrate multivariable functions and vector fields over arbitrary curves in a plane or in space. There are two types of line integrals: scalar line integrals and vector line integrals. Scalar line integrals are integrals of a scalar function over a curve in a plane or in space. Vector line integrals are integrals of a vector field over a curve in a plane or in space. Let's look at scalar line integrals first.

A scalar line integral is defined just as a single-variable integral is defined, except that for a scalar line integral, the integrand is a function of more than one variable and the domain of integration is a curve in a plane or in space, as opposed to a curve on the x-axis.

For a scalar line integral, we let C be a smooth curve in a plane or in space and let f be a function with a domain that includes C. We chop the curve into small pieces. For each piece, we choose point P in that piece and evaluate f at P. (We can do this because all the points in the curve are in the domain of f.) We multiply $f(P)$ by the arc length of the piece Δs, add the product $f(P)\Delta s$ over all the pieces, and then let the arc length of the pieces shrink to zero by taking a limit. The result is the scalar line integral of the function over the curve.

For a formal description of a scalar line integral, let C be a smooth curve in space given by the parameterization $\vec{\mathbf{r}}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$. Let $f(x,y,z)$ be a function with a domain that includes curve C. To define the line integral of the function f over C, we begin as most definitions of an integral begin: we chop the curve into small pieces. Partition the parameter interval $[a,b]$ into n subintervals $[t_{i-l}, t_i]$ of equal width for $1 \le i \le n$, where $t_0 = a$ and $t_n = b$ (Figure 16.2.1). Let t_i^* be a value in the ith interval $[t_{i-l}, t_i]$. Denote the endpoints of $\vec{\mathbf{r}}(t_0)$, $\vec{\mathbf{r}}(t_1)$,…, $\vec{\mathbf{r}}(t_n)$ by P_0,…, P_n. Points P_i divide curve C into n pieces C_1, C_2,…, C_n,with lengths Δs_1, Δs_2,…, Δs_n, respectively. Let P_i^* denote the endpoint of $\vec{\mathbf{r}}(t_i^*)$ for $1 \le i \le n$. Now, we evaluate the function f at point P_i^* for $1 \le i \le n$. Note that P_i^* is in piece C_1, and therefore P_i^* is in the domain of f. Multiply $f(P_i^*)$ by the length Δs_1 of C_1, which gives the area of the "sheet" with base C_1, and height $f(P_i^*)$. This is analogous to using rectangles to approximate area in a single-variable integral. Now, we form the sum $\sum_{i=1}^{n} f(P_i^*)\,\Delta s_i$.

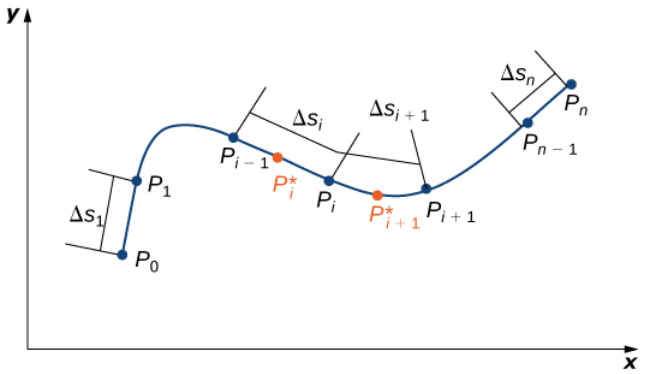


Figure 16.2.1: Curve C has been divided into n pieces, and a point inside each piece has been chosen.

Note the similarity of this sum versus a Riemann sum; in fact, this definition is a generalization of a Riemann sum to arbitrary curves in space. Just as with Riemann sums and integrals of form $\int_a^b g(x)\,dx$, we define an integral by letting the width of the pieces of the curve shrink to zero by taking a limit. The result is the scalar line integral of f along C.

You may have noticed a difference between this definition of a scalar line integral and a single-variable integral. In this definition, the arc lengths Δs_1, Δs_2,…, Δs_n aren't necessarily the same; in the definition of a single-variable integral, the curve in the x-axis is partitioned into pieces of equal length. This difference does not have any effect in the limit. As we shrink the arc lengths to zero, their values become close enough that any small difference becomes irrelevant.

DEFINITION: Scalar Line Integral

Let f be a function with a domain that includes the smooth curve C that is parameterized by $\vec{\mathbf{r}}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$. The *scalar line integral* of f along C is

$$\int_C f(x,y,z)\,ds = \lim_{n\to\infty} \sum_{i=1}^{n} f(P_i^*)\,\Delta s_i \tag{16.2.1}$$

if this limit exists (t_i^* and Δs_i are defined as in the previous paragraphs). If C is a planar curve, then C can be represented by the parametric equations $x = x(t)$, $y = y(t)$, and $a \le t \le b$. If C is smooth and $f(x,y)$ is a function of two variables, then the scalar line integral of f along C is defined similarly as

$$\int_C f(x,y)\,ds = \lim_{n\to\infty} \sum_{i=1}^{n} f(P_i^*)\,\Delta s_i, \tag{16.2.2}$$

if this limit exists.

If f is a continuous function on a smooth curve C, then $\int_C f\,ds$ always exists. Since $\int_C f\,ds$ is defined as a limit of Riemann sums, the continuity of f is enough to guarantee the existence of the limit, just as the integral $\int_a^b g(x)\,dx$ exists if g is continuous over $[a,b]$.

Before looking at how to compute a line integral, we need to examine the geometry captured by these integrals. Suppose that $f(x,y) \ge 0$ for all points (x,y) on a smooth planar curve C. Imagine taking curve C and projecting it "up" to the surface defined by $f(x,y)$, thereby creating a new curve C' that lies in the graph of $f(x,y)$ (Figure 16.2.2). Now we drop a "sheet" from C' down to the xy-plane. The area of this sheet is $\int_C f(x,y)ds$. If $f(x,y) \le 0$ for some points in C, then the value of $\int_C f(x,y)\,ds$ is the area above the xy-plane less the area below the xy-plane. (Note the similarity with integrals of the form $\int_a^b g(x)\,dx$.)

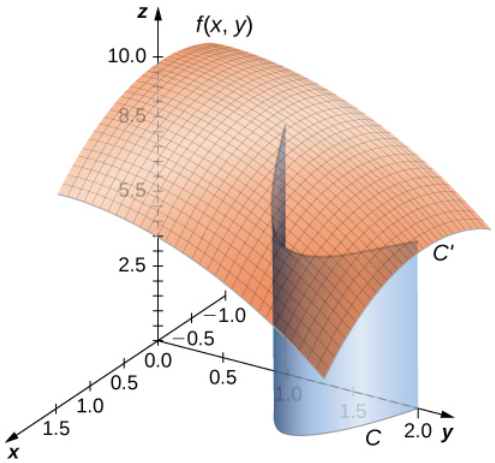


Figure 16.2.2: The area of the blue sheet is $\int_C f(x,y)\,ds$.

From this geometry, we can see that line integral $\int_C f(x,y)\,ds$ does not depend on the parameterization $\vec{\mathbf{r}}(t)$ of C. As long as the curve is traversed exactly once by the parameterization, the area of the sheet formed by the function and the curve is the same. This same kind of geometric argument can be extended to show that the line integral of a three-variable function over a curve in space does not depend on the parameterization of the curve.

Example 16.2.1: Finding the Value of a Line Integral

Find the value of integral $\int_C 2\,ds$, where C is the upper half of the unit circle.

Solution

The integrand is $f(x,y) = 2$. Figure 16.2.3 shows the graph of $f(x,y) = 2$, curve C, and the sheet formed by them. Notice that this sheet has the same area as a rectangle with width π and length 2. Therefore, $\int_C 2\,ds = 2\pi$ units^2.

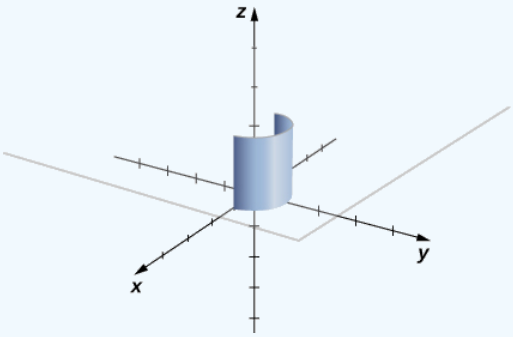


Figure 16.2.3: The sheet that is formed by the upper half of the unit circle in a plane and the graph of $f(x,y) = 2$.

To see that $\int_C 2\,ds = 2\pi$ using the definition of line integral, we let $\vec{\mathbf{r}}(t)$ be a parameterization of C. Then, $f(\vec{\mathbf{r}}(t_i)) = 2$ for any number t_i in the domain of $\vec{\mathbf{r}}$. Therefore,

$$\begin{aligned}\int_C f\,ds &= \lim_{n\to\infty}\sum_{i=1}^{n} f(\vec{\mathbf{r}}(t_i^*))\,\Delta s_i \\ &= \lim_{n\to\infty}\sum_{i=1}^{n} 2\,\Delta s_i \\ &= 2\lim_{n\to\infty}\sum_{i=1}^{n} \Delta s_i \\ &= 2(\text{length of } C) \\ &= 2\pi \text{ units}^2.\end{aligned}$$

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Exercise 16.2.1

Find the value of $\int_C (x+y)\,ds$, where C is the curve parameterized by $x=t$, $y=t$, $0\le t\le 1$.

Hint

Find the shape formed by C and the graph of function $f(x,y)=x+y$.

Answer

$\sqrt{2}$

Note that in a scalar line integral, the integration is done with respect to arc length s, which can make a scalar line integral difficult to calculate. To make the calculations easier, we can translate $\int_C f\,ds$ to an integral with a variable of integration that is t.

Let $\vec{\mathbf{r}}(t)=\langle x(t),y(t),z(t)\rangle$ for $a\le t\le b$ be a parameterization of C. Since we are assuming that C is smooth, $\vec{\mathbf{r}}\,'(t)=\langle x'(t),y'(t),z'(t)\rangle$ is continuous for all t in $[a,b]$. In particular, $x'(t)$, $y'(t)$, and $z'(t)$ exist for all t in $[a,b]$. According to the arc length formula, we have

$$\text{length}(C_i)=\Delta s_i=\int_{t_{i-1}}^{t_i}\|\vec{\mathbf{r}}\,'(t)\|\,dt.$$

If width $\Delta t_i=t_i-t_{i-1}$ is small, then function $\int_{t_{i-1}}^{t_i}\|\vec{\mathbf{r}}\,'(t)\|\,dt\approx\|\vec{\mathbf{r}}\,'(t_i^*)\|\,\Delta t_i$, $\|\vec{\mathbf{r}}\,'(t)\|$ is almost constant over the interval $[t_{i-1},t_i]$.Therefore,

$$\int_{t_{i-1}}^{t_i}\|\vec{\mathbf{r}}\,'(t)\|\,dt\approx\|\vec{\mathbf{r}}\,'(t_i^*)\|\,\Delta t_i, \tag{16.2.3}$$

and we have

$$\sum_{i=1}^{n}f(\vec{\mathbf{r}}(t_i^*))\,\Delta s_i\approx\sum_{i=1}^{n}f(\vec{\mathbf{r}}(t_i^*))\|\vec{\mathbf{r}}\,'(t_i^*)\|\,\Delta t_i.$$

See Figure 16.2.4.

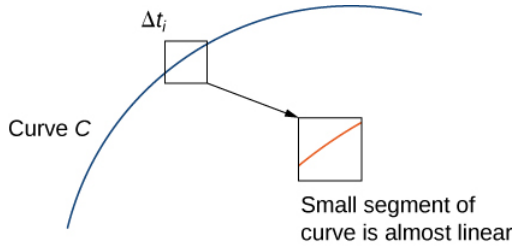


Figure 16.2.4: If we zoom in on the curve enough by making Δt_i very small, then the corresponding piece of the curve is approximately linear.

Note that

$$\lim_{n\to\infty}\sum_{i=1}^{n}f(\vec{\mathbf{r}}(t_i^*))\|\vec{\mathbf{r}}\,'(t_i^*)\|\,\Delta t_i=\int_a^b f(\vec{\mathbf{r}}(t))\|\vec{\mathbf{r}}\,'(t)\|\,dt.$$

In other words, as the widths of intervals $[t_{i-1},t_i]$ shrink to zero, the sum $\sum_{i=1}^{n}f(\vec{\mathbf{r}}(t_i^*))\|\vec{\mathbf{r}}\,'(t_i^*)\|\,\Delta t_i$ converges to the integral $\int_a^b f(\vec{\mathbf{r}}(t))\|\vec{\mathbf{r}}\,'(t)\|\,dt$. Therefore, we have the following theorem.

Theorem: Evaluating a Scalar Line Integral

Let f be a continuous function with a domain that includes the smooth curve C with parameterization $\vec{\mathbf{r}}(t)$, $a \le t \le b$. Then

$$\int_C f\,ds = \int_a^b f(\vec{\mathbf{r}}(t))\|\vec{\mathbf{r}}'(t)\|\,dt. \tag{16.2.4}$$

Although we have labeled Equation 16.2.3 as an equation, it is more accurately considered an approximation because we can show that the left-hand side of Equation 16.2.3 approaches the right-hand side as $n \to \infty$. In other words, letting the widths of the pieces shrink to zero makes the right-hand sum arbitrarily close to the left-hand sum. Since

$$\|\vec{\mathbf{r}}'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2},$$

we obtain the following theorem, which we use to compute scalar line integrals.

Theorem: Scalar Line Integral Calculation

Let f be a continuous function with a domain that includes the smooth curve C with parameterization $\vec{\mathbf{r}}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$. Then

$$\int_C f(x,y,z)\,ds = \int_a^b f(\vec{\mathbf{r}}(t))\sqrt{(x\prime(t))^2 + (y'(t))^2 + (z'(t))^2}\,dt.$$

Similarly,

$$\int_C f(x,y)\,ds = \int_a^b f(\vec{\mathbf{r}}(t))\sqrt{(x'(t))^2 + (y'(t))^2}\,dt$$

if C is a planar curve and f is a function of two variables.

Note that a consequence of this theorem is the equation $ds = \|\vec{\mathbf{r}}'(t)\|\,dt$. In other words, the change in arc length can be viewed as a change in the t-domain, scaled by the magnitude of vector $\vec{\mathbf{r}}'(t)$.

Example 16.2.2: Evaluating a Line Integral

Find the value of integral $\int_C (x^2 + y^2 + z)\,ds$, where C is part of the helix parameterized by $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t, t\rangle$, $0 \le t \le 2\pi$.

Solution

To compute a scalar line integral, we start by converting the variable of integration from arc length s to t. Then, we can use Equation 16.2.1 to compute the integral with respect to t. Note that

$$f(\vec{\mathbf{r}}(t)) = \cos^2 t + \sin^2 t + t = 1 + t$$

and

$$\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} = \sqrt{(-\sin(t))^2 + \cos^2(t) + 1} = \sqrt{2}.$$

Therefore,

$$\int_C (x^2 + y^2 + z)\,ds = \int_0^{2\pi} (1+t)\sqrt{2}\,dt.$$

Notice that Equation 16.2.1 translated the original difficult line integral into a manageable single-variable integral. Since

$$\begin{aligned}\int_0^{2\pi} (1+t)\sqrt{2}\,dt &= \left[\sqrt{2}t + \frac{\sqrt{2}t^2}{2}\right]_0^{2\pi} \\ &= 2\sqrt{2}\pi + 2\sqrt{2}\pi^2,\end{aligned}$$

we have

$$\int_C (x^2+y^2+z)\,ds = 2\sqrt{2}\pi + 2\sqrt{2}\pi^2.$$

Exercise 16.2.2

Evaluate $\int_C (x^2+y^2+z)ds$, where C is the curve with parameterization $\vec{\mathbf{r}}(t) = \langle \sin(3t), \cos(3t)\rangle$, $0 \le t \le \frac{\pi}{4}$.

Hint

Use the two-variable version of scalar line integral definition (Equation 16.2.2).

Answer

$$\frac{1}{3} + \frac{\sqrt{2}}{6} + \frac{3\pi}{4}$$

Example 16.2.3: Independence of Parameterization

Find the value of integral $\int_C (x^2+y^2+z)\,ds$, where C is part of the helix parameterized by $\vec{\mathbf{r}}(t) = \langle \cos(2t), \sin(2t), 2t\rangle$, $0 \le t \le \pi$. Notice that this function and curve are the same as in the previous example; the only difference is that the curve has been reparameterized so that time runs twice as fast.

Solution

As with the previous example, we use Equation 16.2.1 to compute the integral with respect to t. Note that $f(\vec{\mathbf{r}}(t)) = \cos^2(2t) + \sin^2(2t) + 2t = 2t + 1$ and

$$\begin{aligned}\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} &= \sqrt{(-\sin t + \cos t + 4)} \\ &= 22\end{aligned}$$

so we have

$$\begin{aligned}\int_C (x^2+y^2+z)ds &= 2\sqrt{2}\int_0^{\pi}(1+2t)dt \\ &= 2\sqrt{2}\Big[t+t^2\Big]_0^{\pi} \\ &= 2\sqrt{2}(\pi+\pi^2).\end{aligned}$$

Notice that this agrees with the answer in the previous example. Changing the parameterization did not change the value of the line integral. Scalar line integrals are independent of parameterization, as long as the curve is traversed exactly once by the parameterization.

Exercise 16.2.3

Evaluate line integral $\int_C (x^2+yz)\,ds$, where C is the line with parameterization $\vec{\mathbf{r}}(t) = \langle 2t, 5t, -t\rangle$, $0 \le t \le 10$. Reparameterize C with parameterization $s(t) = \langle 4t, 10t, -2t\rangle$, $0 \le t \le 5$, recalculate line integral $\int_C (x^2+yz)\,ds$, and notice that the change of parameterization had no effect on the value of the integral.

Hint

Use Equation 16.2.1.

Answer

Both line integrals equal $-\frac{1000\sqrt{30}}{3}$.

Now that we can evaluate line integrals, we can use them to calculate arc length. If $f(x,y,z)=1$, then

$$\begin{aligned}\int_C f(x,y,z)\,ds &= \lim_{n\to\infty}\sum_{i=1}^{n} f(t_i^*)\,\Delta s_i \\ &= \lim_{n\to\infty}\sum_{i=1}^{n}\Delta s_i \\ &= \lim_{n\to\infty}\text{length}(C) \\ &= \text{length}(C).\end{aligned}$$

Therefore, $\int_C 1\,ds$ is the arc length of C.

Example 16.2.4: Calculating Arc Length

A wire has a shape that can be modeled with the parameterization $\vec{\mathbf{r}}(t)=\langle\cos t,\sin t,\frac{2}{3}t^{3/2}\rangle$, $0\le t\le 4\pi$. Find the length of the wire.

Solution

The length of the wire is given by $\int_C 1\,ds$, where C is the curve with parameterization $\vec{\mathbf{r}}$. Therefore,

$$\begin{aligned}\text{The length of the wire} &= \int_C 1\,ds \\ &= \int_0^{4\pi}||\vec{\mathbf{r}}'(t)||\,dt \\ &= \int_0^{4\pi}\sqrt{(-\sin t)^2+\cos^2 t+t}dt \\ &= \int_0^{4\pi}\sqrt{1+t}dt \\ &= \frac{2(1+t)^{\frac{3}{2}}}{3}\Bigg|_0^{4\pi} \\ &= \frac{2}{3}\left((1+4\pi)^{3/2}-1\right).\end{aligned}$$

Exercise 16.2.4

Find the length of a wire with parameterization $\vec{\mathbf{r}}(t)=\langle 3t+1,4-2t,5+2t\rangle$, $0\le t\le 4$.

Hint

Find the line integral of one over the corresponding curve.

Answer

$4\sqrt{17}$

Vector Line Integrals

The second type of line integrals are vector line integrals, in which we integrate along a curve through a vector field. For example, let

$$\vec{\mathbf{F}}(x,y,z)=P(x,y,z)\,\hat{\mathbf{i}}+Q(x,y,z)\,\hat{\mathbf{j}}+R(x,y,z)\,\hat{\mathbf{k}}$$

be a continuous vector field in $\mathbb{R}^3$ that represents a force on a particle, and let C be a smooth curve in $\mathbb{R}^3$ contained in the domain of $\vec{\mathbf{F}}$. How would we compute the work done by $\vec{\mathbf{F}}$ in moving a particle along C?

To answer this question, first note that a particle could travel in two directions along a curve: a forward direction and a backward direction. The work done by the vector field depends on the direction in which the particle is moving. Therefore, we must specify a direction along curve C; such a specified direction is called an orientation of a curve. The specified direction is the positive direction along C; the opposite direction is the negative direction along C. When C has been given an orientation, C is called an oriented curve (Figure 16.2.5). The work done on the particle depends on the direction along the curve in which the particle is moving.

A closed curve is one for which there exists a parameterization $\vec{\mathbf{r}}(t)$, $a \le t \le b$, such that $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$, and the curve is traversed exactly once. In other words, the parameterization is one-to-one on the domain (a, b).

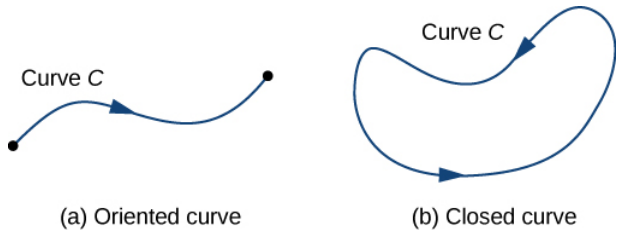


Figure 16.2.5: (a) An oriented curve between two points. (b) A closed oriented curve.

Let $\vec{\mathbf{r}}(t)$ be a parameterization of C for $a \le t \le b$ such that the curve is traversed exactly once by the particle and the particle moves in the positive direction along C. Divide the parameter interval $[a, b]$ into n subintervals $[t_{i-1}, t_i]$, $0 \le i \le n$, of equal width. Denote the endpoints of $r(t_0)$, $r(t_1)$,..., $r(t_n)$ by P_0,...,P_n. Points P_i divide C into n pieces. Denote the length of the piece from P_{i-1} to P_i by Δs_i. For each i, choose a value t_i^* in the subinterval $[t_{i-1}, t_i]$. Then, the endpoint of $\vec{\mathbf{r}}(t_i^*)$ is a point in the piece of C between P_{i-1} and P_i (Figure 16.2.6). If Δs_i is small, then as the particle moves from P_{i-1} to P_i along C, it moves approximately in the direction of $\vec{\mathbf{T}}(P_i)$, the unit tangent vector at the endpoint of $\vec{\mathbf{r}}(t_i^*)$. Let P_i^* denote the endpoint of $\vec{\mathbf{r}}(t_i^*)$. Then, the work done by the force vector field in moving the particle from P_{i-1} to P_i is $\vec{\mathbf{F}}(P_i^*) \cdot (\Delta s_i \vec{\mathbf{T}}(P_i^*))$, so the total work done along C is

$$\sum_{i=1}^{n} \vec{\mathbf{F}}(P_i^*) \cdot (\Delta s_i \vec{\mathbf{T}}(P_i^*)) = \sum_{i=1}^{n} \vec{\mathbf{F}}(P_i^*) \cdot \vec{\mathbf{T}}(P_i^*)\, \Delta s_i.$$

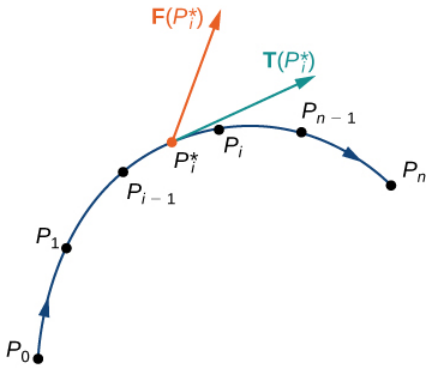


Figure 16.2.6: Curve C is divided into n pieces, and a point inside each piece is chosen. The dot product of any tangent vector in the ith piece with the corresponding vector $\vec{\mathbf{F}}$ is approximated by $\vec{\mathbf{F}}(P_i^*) \cdot \vec{\mathbf{T}}(P_i^*)$.

Letting the arc length of the pieces of C get arbitrarily small by taking a limit as $n \to \infty$ gives us the work done by the field in moving the particle along C. Therefore, the work done by $\vec{\mathbf{F}}$ in moving the particle in the positive direction along C is defined as

$$W = \int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds,$$

which gives us the concept of a vector line integral.

DEFINITION: Line Integral of a Vector Field

The vector line integral of vector field $\vec{\mathbf{F}}$ along oriented smooth curve C is

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = \lim_{n \to \infty} \sum_{i=1}^{n} \vec{\mathbf{F}}(P_i^*) \cdot \vec{\mathbf{T}}(P_i^*) \Delta s_i$$

if that limit exists.

With scalar line integrals, neither the orientation nor the parameterization of the curve matters. As long as the curve is traversed exactly once by the parameterization, the value of the line integral is unchanged. With vector line integrals, the orientation of the curve does matter. If we think of the line integral as computing work, then this makes sense: if you hike up a mountain, then the gravitational force of Earth does negative work on you. If you walk down the mountain by the exact same path, then Earth's gravitational force does positive work on you. In other words, reversing the path changes the work value from negative to positive in this case. Note that if C is an oriented curve, then we let $-C$ represent the same curve but with opposite orientation.

As with scalar line integrals, it is easier to compute a vector line integral if we express it in terms of the parameterization function $\vec{\mathbf{r}}$ and the variable t. To translate the integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}} ds$ in terms of t, note that unit tangent vector $\vec{\mathbf{T}}$ along C is given by $\vec{\mathbf{T}} = \dfrac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|}$ (assuming $\|\vec{\mathbf{r}}'(t)\| \neq 0$). Since $ds = \|\vec{\mathbf{r}}'(t)\|\, dt$, as we saw when discussing scalar line integrals, we have

$$\vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \frac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|} \|\vec{\mathbf{r}}'(t)\| dt = \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t)\, dt.$$

Thus, we have the following formula for computing vector line integrals:

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t)\, dt. \tag{16.2.5}$$

Because of Equation 16.2.5, we often use the notation $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ for the line integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$.

If $\vec{\mathbf{r}}(t) = \langle x(t), y(t), z(t) \rangle$, then $\dfrac{d\vec{\mathbf{r}}}{dt}$ denotes vector $\langle x'(t), y'(t), z'(t) \rangle$, and $d\vec{\mathbf{r}} = \vec{\mathbf{r}}'(t)\, dt$.

Example 16.2.5: Evaluating a Vector Line Integral

Find the value of integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is the semicircle parameterized by $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t \rangle$, $0 \leq t \leq \pi$ and $\vec{\mathbf{F}} = \langle -y, x \rangle$.

Solution

We can use Equation 16.2.5 to convert the variable of integration from s to t. We then have

$$\vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) = \langle -\sin t, \cos t \rangle \text{ and } \vec{\mathbf{r}}'(t) = \langle -\sin t, \cos t \rangle.$$

Therefore,

$$\begin{aligned} \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^\pi \langle -\sin t, \cos t \rangle \cdot \langle -\sin t, \cos t \rangle\, dt \\ &= \int_0^\pi \sin^2 t + \cos^2 t\, dt \\ &= \int_0^\pi 1\, dt = \pi. \end{aligned}$$

See Figure 16.2.7.

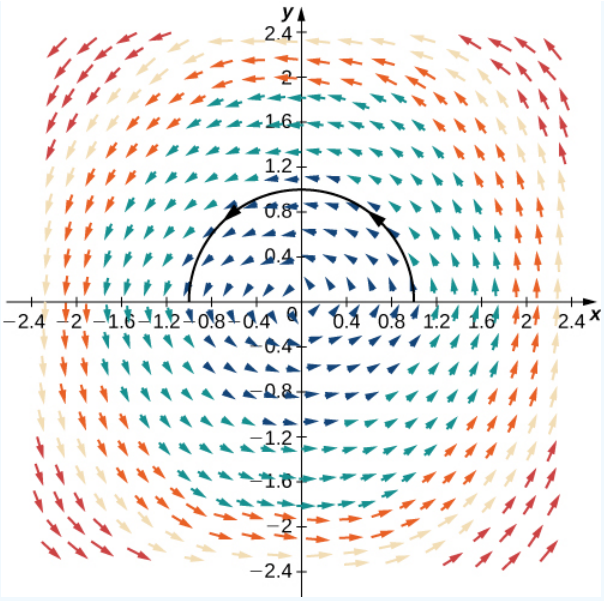


Figure 16.2.7: This figure shows curve $\vec{\mathbf{r}}(t) = \langle \cos t,\ \sin t \rangle$, $0 \le t \le \pi$ in vector field $\vec{\mathbf{F}} = \langle -y,\ x \rangle$.

Example 16.2.6: Reversing Orientation

Find the value of integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is the semicircle parameterized by $\vec{\mathbf{r}}(t) = \langle \cos(t + \pi), \sin t \rangle$, $0 \le t \le \pi$ and $\vec{\mathbf{F}} = \langle -y, x \rangle$.

Solution

Notice that this is the same problem as Example 16.2.5, except the orientation of the curve has been traversed. In this example, the parameterization starts at $\vec{\mathbf{r}}(0) = \langle -1, 0 \rangle$ and ends at $\vec{\mathbf{r}}(\pi) = \langle 1, 0 \rangle$. By Equation 16.2.5,

$$\begin{aligned} \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^\pi \langle -\sin t, \cos(t+\pi) \rangle \cdot \langle -\sin(t+\pi), \cos t \rangle dt \\ &= \int_0^\pi \langle -\sin t, -\cos t \rangle \cdot \langle \sin t, \cos t \rangle dt \\ &= \int_0^\pi (-\sin^2 t - \cos^2 t) dt \\ &= \int_0^\pi -1 dt \\ &= -\pi. \end{aligned}$$

Notice that this is the negative of the answer in Example 16.2.5. It makes sense that this answer is negative because the orientation of the curve goes against the "flow" of the vector field.

Let C be an oriented curve and let $-C$ denote the same curve but with the orientation reversed. Then, the previous two examples illustrate the following fact:

$$\int_{-C} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = -\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

That is, reversing the orientation of a curve changes the sign of a line integral.

Exercise 16.2.6

Let $\vec{\mathbf{F}} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$ be a vector field and let C be the curve with parameterization $\langle t, t^2 \rangle$ for $0 \le t \le 2$. Which is greater: $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$ or $\int_{-C} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$?

Hint

Imagine moving along the path and computing the dot product $\vec{\mathbf{F}} \cdot \vec{\mathbf{T}}$ as you go.

Answer

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$$

Another standard notation for integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ is $\int_C P\, dx + Q\, dy + R\, dz$. In this notation, P, Q, and R are functions, and we think of $d\vec{\mathbf{r}}$ as vector $\langle dx, dy, dz \rangle$. To justify this convention, recall that $d\vec{\mathbf{r}} = \vec{\mathbf{T}}\, ds = \vec{\mathbf{r}}'(t)\, dt = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt$. Therefore,

$$\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \langle P, Q, R \rangle \cdot \langle dx, dy, dz \rangle = P\, dx + Q\, dy + R\, dz.$$

If $d\vec{\mathbf{r}} = \langle dx, dy, dz \rangle$, then $\frac{dr}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle$, which implies that $d\vec{\mathbf{r}} = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt$. Therefore

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_C P\, dx + Q\, dy + R\, dz \tag{16.2.6}$$

$$= \int_a^b \left(P\big(\vec{\mathbf{r}}(t)\big) \frac{dx}{dt} + Q\big(\vec{\mathbf{r}}(t)\big) \frac{dy}{dt} + R\big(\vec{\mathbf{r}}(t)\big) \frac{dz}{dt} \right) dt. \tag{16.2.7}$$

Example 16.2.7: Finding the Value of an Integral of the Form $\int_C P\, dx + Q\, dy + R\, dz$

Find the value of integral $\int_C z\, dx + x\, dy + y\, dz$, where C is the curve parameterized by $\vec{\mathbf{r}}(t) = \langle t^2, \sqrt{t}, t \rangle$, $1 \le t \le 4$.

Solution

As with our previous examples, to compute this line integral we should perform a change of variables to write everything in terms of t. In this case, Equation 16.2.7 allows us to make this change:

$$\begin{aligned} \int_C z\, dx + x\, dy + y\, dz &= \int_1^4 \left(t(2t) + t^2 \left(\frac{1}{2\sqrt{t}} \right) + \sqrt{t} \right) dt \\ &= \int_1^4 \left(2t^2 + \frac{t^{3/2}}{2} + \sqrt{t} \right) dt \\ &= \left[\frac{2t^3}{3} + \frac{t^{5/2}}{5} + \frac{2t^{3/2}}{3} \right]_{t=1}^{t=4} \\ &= \frac{793}{15}. \end{aligned}$$

Exercise 16.2.7

Find the value of $\int_C 4x\, dx + z\, dy + 4y^2\, dz$, where C is the curve parameterized by $\vec{\mathbf{r}}(t) = \langle 4\cos(2t), 2\sin(2t), 3 \rangle$, $0 \le t \le \frac{\pi}{4}$.

Hint

Write the integral in terms of t using Equation 16.2.7.

Answer

-26

We have learned how to integrate smooth oriented curves. Now, suppose that C is an oriented curve that is not smooth, but can be written as the union of finitely many smooth curves. In this case, we say that C is a piecewise smooth curve. To be precise, curve C is piecewise smooth if C can be written as a union of n smooth curves C_1, C_2,…, C_n such that the endpoint of C_i is the starting point of C_{i+1} (Figure 16.2.8). When curves C_i satisfy the condition that the endpoint of C_i is the starting point of C_{i+1}, we write their union as $C_1 + C_2 + \cdots + C_n$.

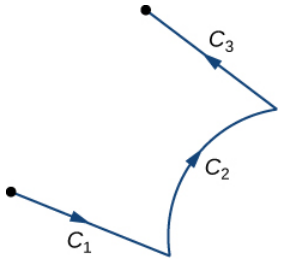


Figure 16.2.8: The union of C_1, C_2, C_3 is a piecewise smooth curve.

The next theorem summarizes several key properties of vector line integrals.

Theorem: Properties of Vector Line Integrals

Let $\vec{\mathbf{F}}$ and $\vec{\mathbf{G}}$ be continuous vector fields with domains that include the oriented smooth curve C. Then

1. $\displaystyle\int_C (\vec{\mathbf{F}} + \vec{\mathbf{G}}) \cdot d\vec{\mathbf{r}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_C \vec{\mathbf{G}} \cdot d\vec{\mathbf{r}}$
2. $\displaystyle\int_C k\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = k \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where k is a constant
3. $\displaystyle\int_{-C} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = -\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$
4. Suppose instead that C is a piecewise smooth curve in the domains of $\vec{\mathbf{F}}$ and $\vec{\mathbf{G}}$, where $C = C_1 + C_2 + \cdots + C_n$ and $C_1, C_2, \ldots, C_n$ are smooth curves such that the endpoint of C_i is the starting point of C_{i+1}. Then

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \cdots + \int_{C_n} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

Notice the similarities between these items and the properties of single-variable integrals. Properties i. and ii. say that line integrals are linear, which is true of single-variable integrals as well. Property iii. says that reversing the orientation of a curve changes the sign of the integral. If we think of the integral as computing the work done on a particle traveling along C, then this makes sense. If the particle moves backward rather than forward, then the value of the work done has the opposite sign. This is analogous to the equation $\displaystyle\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$. Finally, if $[a_1, a_2]$, $[a_2, a_3]$,…, $[a_{n-1}, a_n]$ are intervals, then

$$\int_{a_1}^{a_n} f(x)\,dx = \int_{a_1}^{a_2} f(x)\,dx + \int_{a_1}^{a_3} f(x)\,dx + \cdots + \int_{a_{n-1}}^{a_n} f(x)\,dx,$$

which is analogous to property iv.

Example 16.2.8: Using Properties to Compute a Vector Line Integral

Find the value of integral $\displaystyle\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\,ds$, where C is the rectangle (oriented counterclockwise) in a plane with vertices $(0,0)$, $(2,0)$, $(2,1)$, and $(0,1)$, and where $\vec{\mathbf{F}} = \langle x - 2y, y - x \rangle$ (Figure 16.2.9).

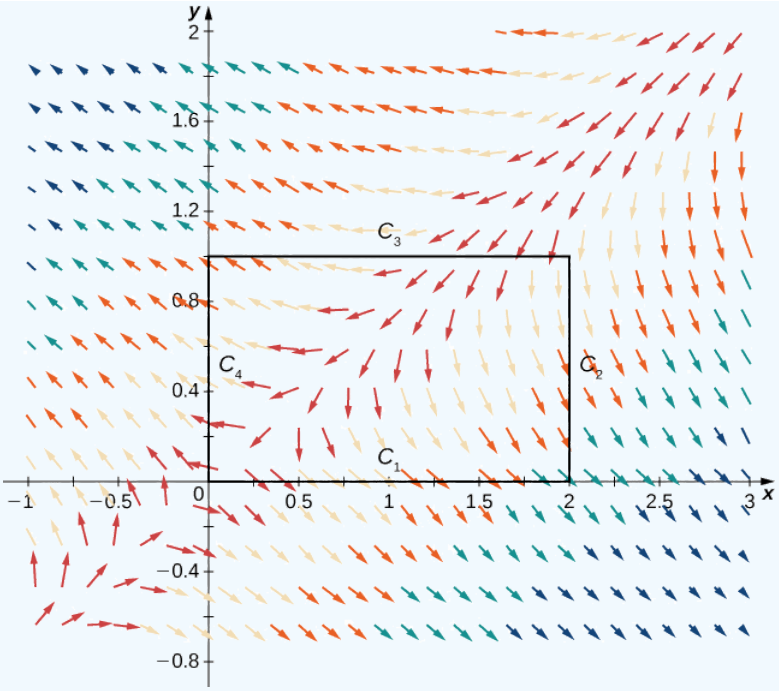


Figure 16.2.9: Rectangle and vector field for Example 16.2.8.

Solution

Note that curve C is the union of its four sides, and each side is smooth. Therefore C is piecewise smooth. Let C_1 represent the side from $(0,0)$ to $(2,0)$, let C_2 represent the side from $(2,0)$ to $(2,1)$, let C_3 represent the side from $(2,1)$ to $(0,1)$, and let C_4 represent the side from $(0,1)$ to $(0,0)$ (Figure 16.2.9). Then,

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr = \int_{C_1} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr + \int_{C_2} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr + \int_{C_3} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr + \int_{C_4} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr.$$

We want to compute each of the four integrals on the right-hand side using Equation 16.2.1. Before doing this, we need a parameterization of each side of the rectangle. Here are four parameterizations (note that they traverse C counterclockwise):

$$\begin{aligned} C_1 &: \langle t, 0\rangle, 0 \le t \le 2 \\ C_2 &: \langle 2, t\rangle, 0 \le t \le 1 \\ C_3 &: \langle 2-t, 1\rangle, 0 \le t \le 2 \\ C_4 &: \langle 0, 1-t\rangle, 0 \le t \le 1. \end{aligned}$$

Therefore,

$$\begin{aligned} \int_{C_1} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, dr &= \int_0^2 \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}\,'(t)\, dt \\ &= \int_0^2 \langle t - 2(0), 0 - t\rangle \cdot \langle 1, 0\rangle\, dt = \int_0^2 t\, dt \\ &= \left[\tfrac{t^2}{2}\right]_0^2 = 2. \end{aligned}$$

Notice that the value of this integral is positive, which should not be surprising. As we move along curve C_1 from left to right, our movement flows in the general direction of the vector field itself. At any point along C_1, the tangent vector to the curve and the corresponding vector in the field form an angle that is less than 90°. Therefore, the tangent vector and the force vector have a positive dot product all along C_1, and the line integral will have positive value.

The calculations for the three other line integrals are done similarly:

$$\int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_0^1 \langle 2-2t, t-2\rangle \cdot \langle 0,1\rangle \, dt$$
$$= \int_0^1 (t-2)\, dt$$
$$= \left[\tfrac{t^2}{2} - 2t\right]_0^1 = -\frac{3}{2},$$

$$\int_{C_3} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = \int_0^2 \langle (2-t)-2, 1-(2-t)\rangle \cdot \langle -1,0\rangle \, dt$$
$$= \int_0^2 t\, dt = 2,$$

and

$$\int_{C_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_0^1 \langle -2(1-t), 1-t\rangle \cdot \langle 0,-1\rangle \, dt$$
$$= \int_0^1 (t-1)\, dt$$
$$= \left[\tfrac{t^2}{2} - t\right]_0^1 = -\frac{1}{2}.$$

Thus, we have $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 2$.

? Exercise 16.2.8

Calculate line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}$ is vector field $\langle y^2, 2xy+1\rangle$ and C is a triangle with vertices $(0,0)$, $(4,0)$, and $(0,5)$, oriented counterclockwise.

Hint

Write the triangle as a union of its three sides, then calculate three separate line integrals.

Answer

0

Applications of Line Integrals

Scalar line integrals have many applications. They can be used to calculate the length or mass of a wire, the surface area of a sheet of a given height, or the electric potential of a charged wire given a linear charge density. Vector line integrals are extremely useful in physics. They can be used to calculate the work done on a particle as it moves through a force field, or the flow rate of a fluid across a curve. Here, we calculate the mass of a wire using a scalar line integral and the work done by a force using a vector line integral.

Suppose that a piece of wire is modeled by curve C in space. The mass per unit length (the linear density) of the wire is a continuous function $\rho(x,y,z)$. We can calculate the total mass of the wire using the scalar line integral $\int_C \rho(x,y,z)\, ds$. The reason is that mass is density multiplied by length, and therefore the density of a small piece of the wire can be approximated by $\rho(x^*,y^*,z^*)\ \Delta s$ for some point (x^*,y^*,z^*) in the piece. Letting the length of the pieces shrink to zero with a limit yields the line integral $\int_C \rho(x,y,z)\, ds$.

✓ Example 16.2.9: Calculating the Mass of a Wire

Calculate the mass of a spring in the shape of a curve parameterized by $\langle t, 2\cos t, 2\sin t\rangle$, $0 \le t \le \frac{\pi}{2}$, with a density function given by $\rho(x,y,z) = e^x + yz$ kg/m (Figure 16.2.10).

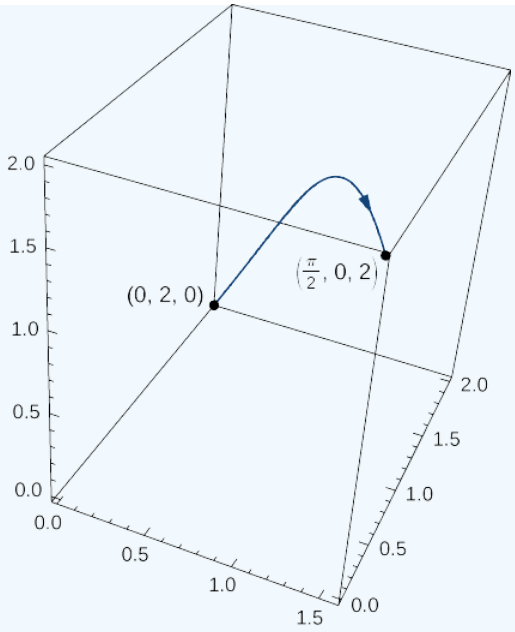


Figure 16.2.10: The wire from Example 16.2.9.

Solution

To calculate the mass of the spring, we must find the value of the scalar line integral $\int_C (e^x + yz)\, ds$, where C is the given helix. To calculate this integral, we write it in terms of t using Equation 16.2.1:

$$\begin{aligned}\int_C (e^x + yz)\, ds &= \int_0^{\frac{\pi}{2}} \left((e^t + 4\cos t \sin t)\sqrt{1 + (-2\cos t)^2 + (2\sin t)^2}\right) dt \\ &= \int_0^{\frac{\pi}{2}} \left((e^t + 4\cos t \sin t)\sqrt{5}\right) dt \\ &= \sqrt{5}\left[e^t + 2\sin^2 t\right]_{t=0}^{t=\pi/2} \\ &= \sqrt{5}(e^{\pi/2} + 1).\end{aligned}$$

Therefore, the mass is $\sqrt{5}(e^{\pi/2} + 1)$ kg.

Exercise 16.2.9

Calculate the mass of a spring in the shape of a helix parameterized by $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t, t\rangle$, $0 \le t \le 6\pi$, with a density function given by $\rho(x, y, z) = x + y + z$ kg/m.

Hint

Calculate the line integral of ρ over the curve with parameterization $\vec{\mathbf{r}}$.

Answer

$18\sqrt{2}\pi^2$ kg

When we first defined vector line integrals, we used the concept of work to motivate the definition. Therefore, it is not surprising that calculating the **work done by a vector field** representing a force is a standard use of vector line integrals. Recall that if an object moves along curve C in force field $\vec{\mathbf{F}}$, then the work required to move the object is given by $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$.

Example 16.2.10: Calculating Work

How much work is required to move an object in vector force field $\vec{\mathbf{F}} = \langle yz, xy, xz \rangle$ along path $\vec{\mathbf{r}}(t) = \langle t^2, t, t^4 \rangle$, $0 \le t \le 1$? See Figure 16.2.11.

Solution

Let C denote the given path. We need to find the value of $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. To do this, we use Equation 16.2.5 :

$$\begin{aligned} \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^1 (\langle t^5, t^3, t^6 \rangle \cdot \langle 2t, 1, 4t^3 \rangle)\, dt \\ &= \int_0^1 (2t^6 + t^3 + 4t^9)\, dt \\ &= \Big[\frac{2t^7}{7} + \frac{t^4}{4} + \frac{2t^{10}}{5}\Big]_{t=0}^{t=1} = \frac{131}{140} \text{ units of work.} \end{aligned}$$

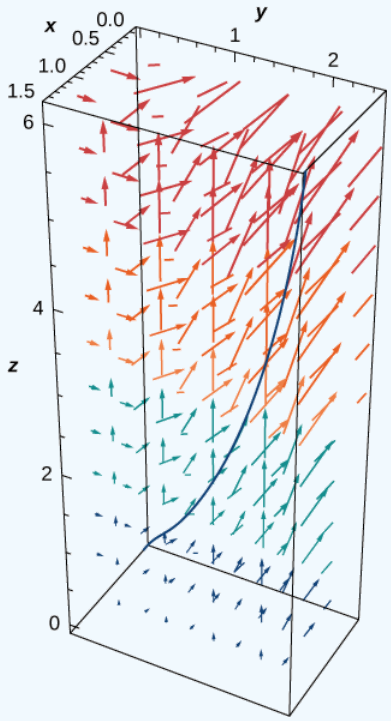


Figure 16.2.11: The curve and vector field for Example 16.2.10.

Flux

We close this section by discussing two key concepts related to line integrals: flux across a plane curve and circulation along a plane curve. Flux is used in applications to calculate fluid flow across a curve, and the concept of circulation is important for characterizing conservative gradient fields in terms of line integrals. Both these concepts are used heavily throughout the rest of this chapter. The idea of flux is especially important for Green's theorem, and in higher dimensions for Stokes' theorem and the divergence theorem.

Let C be a plane curve and let $\vec{\mathbf{F}}$ be a vector field in the plane. Imagine C is a membrane across which fluid flows, but C does not impede the flow of the fluid. In other words, C is an idealized membrane invisible to the fluid. Suppose $\vec{\mathbf{F}}$ represents the velocity field of the fluid. How could we quantify the rate at which the fluid is crossing C?

Recall that the line integral of $\vec{\mathbf{F}}$ along C is $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$—in other words, the line integral is the dot product of the vector field with the unit tangential vector with respect to arc length. If we replace the unit tangential vector with unit normal vector $\vec{\mathbf{N}}(t)$ and instead

compute integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$, we determine the flux across C. To be precise, the definition of integral $\displaystyle\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$ is the same as integral $\displaystyle\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$, except the $\vec{\mathbf{T}}$ in the Riemann sum is replaced with $\vec{\mathbf{N}}$. Therefore, the flux across C is defined as

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \lim_{n\to\infty} \sum_{i=1}^{n} \vec{\mathbf{F}}(P_i^*) \cdot \vec{\mathbf{N}}(P_i^*)\, \Delta s_i,$$

where P_i^* and Δs_i are defined as they were for integral $\displaystyle\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$. Therefore, a flux integral is an integral that is *perpendicular* to a vector line integral, because $\vec{\mathbf{N}}$ and $\vec{\mathbf{T}}$ are perpendicular vectors.

If $\vec{\mathbf{F}}$ is a velocity field of a fluid and C is a curve that represents a membrane, then the flux of $\vec{\mathbf{F}}$ across C is the quantity of fluid flowing across C per unit time, or the rate of flow.

More formally, let C be a plane curve parameterized by $\vec{\mathbf{r}}(t) = \langle x(t),\, y(t)\rangle$, $a \le t \le b$. Let $\vec{\mathbf{n}}(t) = \langle y'(t),\, -x'(t)\rangle$ be the vector that is normal to C at the endpoint of $\vec{\mathbf{r}}(t)$ and points to the right as we traverse C in the positive direction (Figure 16.2.12). Then, $\vec{\mathbf{N}}(t) = \dfrac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}$ is the unit normal vector to C at the endpoint of $\vec{\mathbf{r}}(t)$ that points to the right as we traverse C.

DEFINITION: Flux

The *flux* of $\vec{\mathbf{F}}$ across C is line integral

$$\int_C \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}\, ds.$$

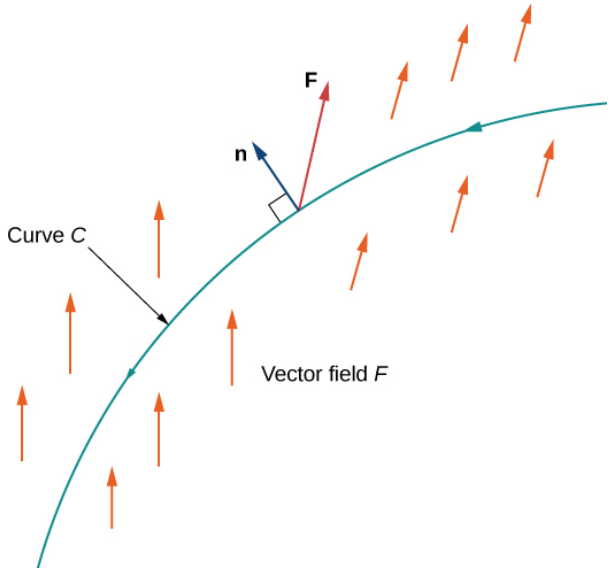


Figure 16.2.12: The flux of vector field $\vec{\mathbf{F}}$ *across curve* C *is computed by an integral similar to a vector line integral.*

We now give a formula for calculating the flux across a curve. This formula is analogous to the formula used to calculate a vector line integral (see Equation 16.2.5).

Theorem: Calculating Flux Across a Curve

Let $\vec{\mathbf{F}}$ be a vector field and let C be a smooth curve with parameterization $r(t) = \langle x(t), y(t)\rangle$, $a \le t \le b$.Let $\vec{\mathbf{n}}(t) = \langle y'(t), -x'(t)\rangle$. The flux of $\vec{\mathbf{F}}$ across C is

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{n}}(t)\, dt. \tag{16.2.8}$$

Proof

Before deriving the formula, note that

$$\|\vec{\mathbf{n}}(t)\| = \|\langle y'(t), -x'(t)\rangle\| = \sqrt{(y'(t))^2 + (x'(t))^2} = \|\vec{\mathbf{r}}'(t)\|.$$

Therefore,

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds &= \int_C \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}\, ds \\ &= \int_a^b \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|} \|\vec{\mathbf{r}}'(t)\|\, dt \\ &= \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{n}}(t)\, dt.\end{aligned}$$

□

Example 16.2.11: Flux across a Curve

Calculate the flux of $\vec{\mathbf{F}} = \langle 2x, 2y\rangle$ across a unit circle oriented counterclockwise (Figure 16.2.13).

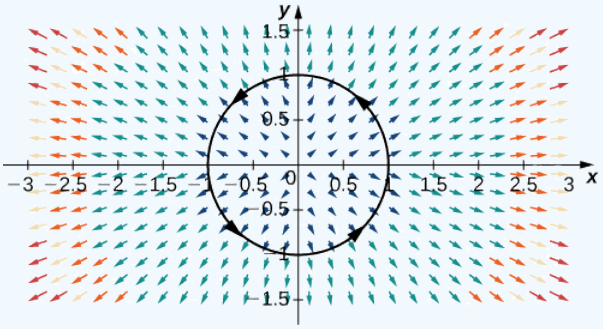


Figure 16.2.13: A unit circle in vector field $\vec{\mathbf{F}} = \langle 2x, 2y\rangle$.

Solution

To compute the flux, we first need a parameterization of the unit circle. We can use the standard parameterization $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le 2\pi$. The normal vector to a unit circle is $\langle \cos t, \sin t\rangle$. Therefore, the flux is

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds &= \int_0^{2\pi} \langle 2\cos t, 2\sin t\rangle \cdot \langle \cos t, \sin t\rangle\, dt \\ &= \int_0^{2\pi} (2\cos^2 t + 2\sin^2 t)\, dt \\ &= 2\int_0^{2\pi} (\cos^2 t + \sin^2 t)\, dt \\ &= 2\int_0^{2\pi} dt = 4\pi.\end{aligned}$$

Exercise 16.2.11

Calculate the flux of $\vec{\mathbf{F}} = \langle x + y, 2y\rangle$ across the line segment from $(0, 0)$ to $(2, 3)$, where the curve is oriented from left to right.

Hint

Use Equation 16.2.8.

Answer

$3/2$

Let $\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y)\rangle$ be a two-dimensional vector field. Recall that integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$ is sometimes written as $\int_C P\, dx + Q\, dy$. Analogously, flux $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$ is sometimes written in the notation $\int_C -Q\, dx + P\, dy$, because the unit normal vector $\vec{\mathbf{N}}$ is perpendicular to the unit tangent $\vec{\mathbf{T}}$. Rotating the vector $d\vec{\mathbf{r}} = \langle dx, dy\rangle$ by 90° results in vector $\langle dy, -dx\rangle$. Therefore, the line integral in Example 16.2.8 can be written as $\int_C -2y\, dx + 2x\, dy$.

Circulation

Now that we have defined flux, we can turn our attention to circulation. The line integral of vector field $\vec{\mathbf{F}}$ along an oriented closed curve is called the circulation of $\vec{\mathbf{F}}$ along C. Circulation line integrals have their own notation: $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$. The circle on the integral symbol denotes that C is "circular" in that it has no endpoints. Example 16.2.5 shows a calculation of circulation.

To see where the term *circulation* comes from and what it measures, let $\vec{\mathbf{v}}$ represent the velocity field of a fluid and let C be an oriented closed curve. At a particular point P, the closer the direction of $\vec{\mathbf{v}}(P)$ is to the direction of $\vec{\mathbf{T}}(P)$, the larger the value of the dot product $\vec{\mathbf{v}}(P) \cdot \vec{\mathbf{T}}(P)$. The maximum value of $\vec{\mathbf{v}}(P) \cdot \vec{\mathbf{T}}(P)$ occurs when the two vectors are pointing in the exact same direction; the minimum value of $\vec{\mathbf{v}}(P) \cdot \vec{\mathbf{T}}(P)$ occurs when the two vectors are pointing in opposite directions. Thus, the value of the circulation $\oint_C \vec{\mathbf{v}} \cdot \vec{\mathbf{T}}\, ds$ measures the tendency of the fluid to move in the direction of C.

Example 16.2.12: Calculating Circulation

Let $\vec{\mathbf{F}} = \langle -y, x\rangle$ be the vector field from Example 16.2.3 and let C represent the unit circle oriented counterclockwise. Calculate the circulation of $\vec{\mathbf{F}}$ along C.

Solution

We use the standard parameterization of the unit circle: $\vec{\mathbf{r}}(t) = \langle \cos t, \sin t\rangle$, $0 \le t \le 2\pi$. Then, $\vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) = \langle -\sin t, \cos t\rangle$ and $\vec{\mathbf{r}}\,'(t) = \langle -\sin t, \cos t\rangle$. Therefore, the circulation of $\vec{\mathbf{F}}$ along C is

$$\begin{aligned}\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds &= \int_0^{2\pi} \langle -\sin t, \cos t\rangle \cdot \langle -\sin t, \cos t\rangle\, dt \\ &= \int_0^{2\pi} (\sin^2 t + \cos^2 t)\, dt \\ &= \int_0^{2\pi} dt = 2\pi \text{ units of work.}\end{aligned}$$

Notice that the circulation is positive. The reason for this is that the orientation of C "flows" with the direction of $\vec{\mathbf{F}}$. At any point along the circle, the tangent vector and the vector from $\vec{\mathbf{F}}$ form an angle of less than 90°, and therefore the corresponding dot product is positive.

In Example 16.2.12, what if we had oriented the unit circle clockwise? We denote the unit circle oriented clockwise by $-C$. Then

$$\oint_{-C} \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = -\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = -2\pi \text{ units of work.}$$

Notice that the circulation is negative in this case. The reason for this is that the orientation of the curve flows against the direction of $\vec{\mathbf{F}}$.

Exercise 16.2.12

Calculate the circulation of $\vec{\mathbf{F}}(x,y) = \langle -\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}\rangle$ along a unit circle oriented counterclockwise.

Hint

Use Equation 16.2.8.

Answer

2π units of work

Example 16.2.13: Calculating Work

Calculate the work done on a particle that traverses circle C of radius 2 centered at the origin, oriented counterclockwise, by field $\vec{\mathbf{F}}(x, y) = \langle -2,\ y\rangle$. Assume the particle starts its movement at $(1,\ 0)$.

Solution

The work done by $\vec{\mathbf{F}}$ on the particle is the circulation of $\vec{\mathbf{F}}$ along C: $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$. We use the parameterization $\vec{\mathbf{r}}(t) = \langle 2\cos t,\ 2\sin t\rangle$, $0 \le t \le 2\pi$ for C. Then, $\vec{\mathbf{r}}'(t) = \langle -2\sin t,\ 2\cos t\rangle$ and $\vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) = \langle -2,\ 2\sin t\rangle$. Therefore, the circulation of $\vec{\mathbf{F}}$ along C is

$$\begin{aligned}\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds &= \int_0^{2\pi} \langle -2, 2\sin t\rangle \cdot \langle -2\sin t, 2\cos t\rangle\, dt \\ &= \int_0^{2\pi} (4\sin t + 4\sin t\cos t)\, dt \\ &= \Big[-4\cos t + 4\sin^2 t\Big]_0^{2\pi} \\ &= \big(-4\cos(2\pi) + 2\sin^2(2\pi)\big) - \big(-4\cos(0) + 4\sin^2(0)\big) \\ &= -4 + 4 = 0 \text{ units of work.}\end{aligned}$$

The force field does zero work on the particle.

Notice that the circulation of $\vec{\mathbf{F}}$ along C is zero. Furthermore, notice that since $\vec{\mathbf{F}}$ is the gradient of $f(x, y) = -2x + \dfrac{y^2}{2}$, $\vec{\mathbf{F}}$ is conservative. We prove in a later section that under certain broad conditions, the circulation of a conservative vector field along a closed curve is zero.

Exercise 16.2.14

Calculate the work done by field $\vec{\mathbf{F}}(x, y) = \langle 2x,\ 3y\rangle$ on a particle that traverses the unit circle. Assume the particle begins its movement at $(-1,\ 0)$.

Hint

Use Equation 16.2.8.

Answer

0 units of work

Key Concepts

- Line integrals generalize the notion of a single-variable integral to higher dimensions. The domain of integration in a single-variable integral is a line segment along the x-axis, but the domain of integration in a line integral is a curve in a plane or in space.
- If C is a curve, then the length of C is $\displaystyle\int_C ds$.
- There are two kinds of line integral: scalar line integrals and vector line integrals. Scalar line integrals can be used to calculate the mass of a wire; vector line integrals can be used to calculate the work done on a particle traveling through a field.
- Scalar line integrals can be calculated using Equation 16.2.1; vector line integrals can be calculated using Equation 16.2.5.
- Two key concepts expressed in terms of line integrals are flux and circulation. Flux measures the rate that a field crosses a given line; circulation measures the tendency of a field to move in the same direction as a given closed curve.

Key Equations

- **Calculating a scalar line integral**

  $$\int_C f(x, y, z)\, ds = \int_a^b f(\vec{\mathbf{r}}(t))\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\, dt$$

- **Calculating a vector line integral**
  $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds = \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t)\, dt$
  or
  $\int_C P\, dx + Q\, dy + R\, dz = \int_a^b \left(P\big(\vec{\mathbf{r}}(t)\big) \frac{dx}{dt} + Q\big(\vec{\mathbf{r}}(t)\big) \frac{dy}{dt} + R\big(\vec{\mathbf{r}}(t)\big) \frac{dz}{dt} \right) dt$
- **Calculating flux**
  $\int_C \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}\, ds = \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{n}}(t)\, dt$

Glossary

circulation

the tendency of a fluid to move in the direction of curve C. If C is a closed curve, then the circulation of $\vec{\mathbf{F}}$ along C is line integral $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$, which we also denote $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$.

closed curve

a curve for which there exists a parameterization $\vec{\mathbf{r}}(t), a \leq t \leq b$, such that $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$, and the curve is traversed exactly once

flux

the rate of a fluid flowing across a curve in a vector field; the flux of vector field $\vec{\mathbf{F}}$ across plane curve C is line integral $\int_C \vec{\mathbf{F}} \cdot \frac{\vec{\mathbf{n}}(t)}{\|\vec{\mathbf{n}}(t)\|}\, ds$

line integral

the integral of a function along a curve in a plane or in space

orientation of a curve

the orientation of a curve C is a specified direction of C

piecewise smooth curve

an oriented curve that is not smooth, but can be written as the union of finitely many smooth curves

scalar line integral

the scalar line integral of a function f along a curve C with respect to arc length is the integral $\int_C f\, ds$, it is the integral of a scalar function f along a curve in a plane or in space; such an integral is defined in terms of a Riemann sum, as is a single-variable integral

vector line integral

the vector line integral of vector field $\vec{\mathbf{F}}$ along curve C is the integral of the dot product of $\vec{\mathbf{F}}$ with unit tangent vector $\vec{\mathbf{T}}$ of C with respect to arc length, $\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\, ds$; such an integral is defined in terms of a Riemann sum, similar to a single-variable integral

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16.2E: Exercises for Section 16.2

1. *True or False?* Line integral $\int_C f(x,y)\,ds$ is equal to a definite integral if C is a smooth curve defined on $[a,b]$ and if function f is continuous on some region that contains curve C.

2. *True or False?* Vector functions $\vec{\mathbf{r}}_1 = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}}, \quad 0 \le t \le 1$, and $\vec{\mathbf{r}}_2 = (1-t)\,\hat{\mathbf{i}} + (1-t)^2\,\hat{\mathbf{j}}, \quad 0 \le t \le 1$, define the same oriented curve.

3. *True or False?* $\int_{-C}(P\,dx + Q\,dy) = \int_C (P\,dx - Q\,dy)$

4. *True or False?* A piecewise smooth curve C consists of a finite number of smooth curves that are joined together end to end.

5. *True or False?* If C is given by $x(t) = t, \quad y(t) = t, \quad 0 \le t \le 1$, then $\int_C xy\,ds = \int_0^1 t^2\,dt$.

For the following exercises, use a computer algebra system (CAS) to evaluate the line integrals over the indicated path.

6. [T] $\int_C (x+y)\,ds$

$C: x = t, y = (1-t), z = 0$ from $(0,1,0)$ to $(1,0,0)$

7. [T] $\int_C (x-y)ds$

$C: \vec{\mathbf{r}}(t) = 4t\,\hat{\mathbf{i}} + 3t\,\hat{\mathbf{j}}$ when $0 \le t \le 2$

8. [T] $\int_C (x^2+y^2+z^2)\,ds$

$C: \vec{\mathbf{r}}(t) = \sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}} + 8t\,\hat{\mathbf{k}}$ when $0 \le t \le \dfrac{\pi}{2}$

9. [T] Evaluate $\int_C xy^4\,ds$, where C is the right half of circle $x^2+y^2=16$ and is traversed in the clockwise direction.

10. [T] Evaluate $\int_C 4x^3 ds$, where C is the line segment from $(-2,-1)$ to $(1,2)$.

For the following exercises, find the work done.

11. Find the work done by vector field $\vec{\mathbf{F}}(x,y,z) = x\,\hat{\mathbf{i}} + 3xy\,\hat{\mathbf{j}} - (x+z)\,\hat{\mathbf{k}}$ on a particle moving along a line segment that goes from $(1,4,2)$ to $(0,5,1)$.

12. Find the work done by a person weighing 150 lb walking exactly one revolution up a circular, spiral staircase of radius 3 ft if the person rises 10 ft.

13. Find the work done by force field $\vec{\mathbf{F}}(x,y,z) = -\dfrac{1}{2}x\,\hat{\mathbf{i}} - \dfrac{1}{2}y\,\hat{\mathbf{j}} + \dfrac{1}{4}\,\hat{\mathbf{k}}$ on a particle as it moves along the helix $\vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$ from point $(1,0,0)$ to point $(-1,0,3\pi)$.

14. Find the work done by vector field $\vec{\mathbf{F}}(x,y) = y\,\hat{\mathbf{i}} + 2x\,\hat{\mathbf{j}}$ in moving an object along path C, which joins points $(1,0)$ and $(0,1)$.

15. Find the work done by force $\vec{\mathbf{F}}(x,y) = 2y\,\hat{\mathbf{i}} + 3x\,\hat{\mathbf{j}} + (x+y)\,\hat{\mathbf{k}}$ in moving an object along curve $\vec{\mathbf{r}}(t) = \cos(t)\,\hat{\mathbf{i}} + \sin(t)\,\hat{\mathbf{j}} + 16\,\hat{\mathbf{k}}$, where $0 \le t \le 2\pi$.

16. Find the mass of a wire in the shape of a circle of radius 2 centered at (3, 4) with linear mass density $\rho(x,y) = y^2$.

For the following exercises, evaluate the line integrals.

17. Evaluate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x,y) = -1\,\hat{\mathbf{j}}$, and C is the part of the graph of $y = \frac{1}{2}x^3 - x$ from $(2,2)$ to $(-2,-2)$.

18. Evaluate $\int_\gamma (x^2+y^2+z^2)^{-1} ds$, where γ is the helix $x = \cos t, y = \sin t, z = t$, with $0 \le t \le T$.

19. Evaluate $\int_C yz\,dx + xz\,dy + xy\,dz$ over the line segment from $(1,1,1)$ to $(3,2,0)$.

20. Let C be the line segment from point (0, 1, 1) to point (2, 2, 3). Evaluate line integral $\int_C y\,ds$.

21. [T] Use a computer algebra system to evaluate the line integral $\int_C y^2\,dx + x\,dy$, where C is the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$.

22. [T] Use a computer algebra system to evaluate the line integral $\int_C (x + 3y^2)\,dy$ over the path C given by $x = 2t, y = 10t$, where $0 \le t \le 1$.

23. [T] Use a CAS to evaluate line integral $\int_C xy\,dx + y\,dy$ over path C given by $x = 2t, y = 10t$, where $0 \le t \le 1$.

24. Evaluate line integral $\int_C (2x - y)\,dx + (x + 3y)\,dy$, where C lies along the x-axis from $x = 0$ to $x = 5$.

26. [T] Use a CAS to evaluate $\int_C \dfrac{y}{2x^2 - y^2}\,ds$, where C is defined by the parametric equations $x = t, y = t$, for $1 \le t \le 5$.

27. [T] Use a CAS to evaluate $\int_C xy\,ds$, where C is defined by the parametric equations $x = t^2, y = 4t$, for $0 \le t \le 1$.

In the following exercises, find the work done by force field $\vec{\mathbf{F}}$ on an object moving along the indicated path.

28. $\vec{\mathbf{F}}(x, y) = -x\,\hat{\mathbf{i}} - 2y\,\hat{\mathbf{j}}$

$C : y = x^3$ from $(0, 0)$ to $(2, 8)$

29. $\vec{\mathbf{F}}(x, y) = 2x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$

$<C$: counterclockwise around the triangle with vertices $(0, 0), (1, 0)$, and $(1, 1)$

30. $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} - 5z\,\hat{\mathbf{k}}$

$C : \vec{\mathbf{r}}(t) = 2\cos t\,\hat{\mathbf{i}} + 2\sin t\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}},\ 0 \le t \le 2\pi$

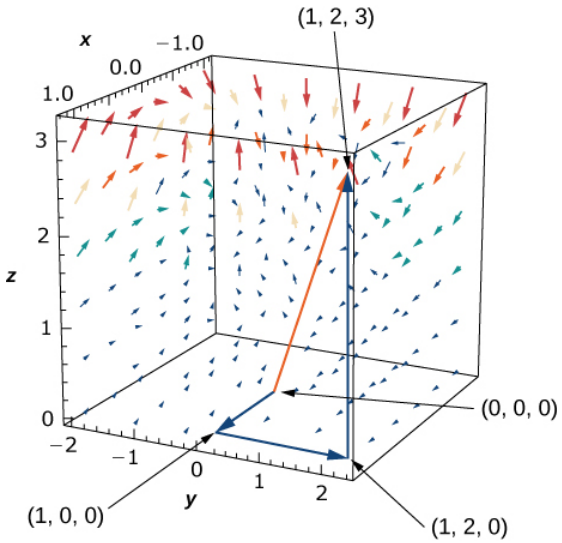
31. Let $\vec{\mathbf{F}}$ be vector field $\vec{\mathbf{F}}(x, y) = (y^2 + 2xe^y + 1)\,\hat{\mathbf{i}} + (2xy + x^2e^y + 2y)\,\hat{\mathbf{j}}$. Compute the work of integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is the path $\vec{\mathbf{r}}(t) = \sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}}, \quad 0 \le t \le \dfrac{\pi}{2}$.

32. Compute the work done by force $\vec{\mathbf{F}}(x, y, z) = 2x\,\hat{\mathbf{i}} + 3y\,\hat{\mathbf{j}} - z\,\hat{\mathbf{k}}$ along path $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + t^3\,\hat{\mathbf{k}}$, where $0 \le t \le 1$.

33. Evaluate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = \dfrac{1}{x + y}\,\hat{\mathbf{i}} + \dfrac{1}{x + y}\,\hat{\mathbf{j}}$ and C is the segment of the unit circle going counterclockwise from $(1, 0)$ to $(0, 1)$.

34. Force $\vec{\mathbf{F}}(x, y, z) = zy\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}} + z^2x\,\hat{\mathbf{k}}$ acts on a particle that travels from the origin to point $(1, 2, 3)$. Calculate the work done if the particle travels:

a. along the path $(0, 0, 0) \to (1, 0, 0) \to (1, 2, 0) \to (1, 2, 3)$ along straight-line segments joining each pair of endpoints;
b. along the straight line joining the initial and final points.
c. Is the work the same along the two paths?



35. Find the work done by vector field $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + 3xy\,\hat{\mathbf{j}} - (x + z)\,\hat{\mathbf{k}}$ on a particle moving along a line segment that goes from $(1, 4, 2)$ to $(0, 5, 1)$.

36. How much work is required to move an object in vector field $\vec{\mathbf{F}}(x, y) = y\,\hat{\mathbf{i}} + 3x\,\hat{\mathbf{j}}$ along the upper part of ellipse $\frac{x^2}{4} + y^2 = 1$ from $(2, 0)$ to $(-2, 0)$?

37. A vector field is given by $\vec{\mathbf{F}}(x, y) = (2x + 3y)\,\hat{\mathbf{i}} + (3x + 2y)\,\hat{\mathbf{j}}$. Evaluate the line integral of the field around a circle of unit radius traversed in a clockwise fashion.

38. Evaluate the line integral of scalar function xy along parabolic path $y = x^2$ connecting the origin to point $(1, 1)$.

39. Find $\int_C y^2\,dx + (xy - x^2)\,dy$ along $C : y = 3x$ from $(0, 0)$ to $(1, 3)$.

40. Find $\int_C y^2\,dx + (xy - x^2)\,dy$ along $C : y^2 = 9x$ from $(0, 0)$ to $(1, 3)$.

For the following exercises, use a CAS to evaluate the given line integrals.

41. [T] Evaluate $\vec{\mathbf{F}}(x, y, z) = x^2z\,\hat{\mathbf{i}} + 6y\,\hat{\mathbf{j}} + yz^2\,\hat{\mathbf{k}}$, where C is represented by $\vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + \ln t\,\hat{\mathbf{k}}, 1 \le t \le 3$.

42. [T] Evaluate line integral $\int_\gamma xe^y\,ds$ where, γ is the arc of curve $x = e^y$ from $(1, 0)$ to $(e, 1)$.

43. [T] Evaluate the integral $\int_\gamma xy^2\,ds$, where γ is a triangle with vertices $(0, 1, 2), (1, 0, 3)$, and $(0, -1, 0)$.

44. [T] Evaluate line integral $\int_\gamma (y^2 - xy)\,dx$, where γ is curve $y = \ln x$ from $(1, 0)$ toward $(e, 1)$.

45. [T] Evaluate line integral $\int_\gamma xy^4\,ds$, where γ is the right half of circle $x^2 + y^2 = 16$.

46. [T] Evaluate $\int C\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}, \int C\vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y, z) = x^2y\,\hat{\mathbf{i}} + (x - z)\,\hat{\mathbf{j}} + xyz\,\hat{\mathbf{k}}$ and

$C : \vec{\mathbf{r}}(t) = t\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + 2\,\hat{\mathbf{k}}, 0 \le t \le 1.$

47. Evaluate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = 2x \sin y\,\hat{\mathbf{i}} + (x^2 \cos y - 3y^2)\,\hat{\mathbf{j}}$ and

C is any path from $(-1, 0)$ to $(5, 1)$.

48. Find the line integral of $\vec{\mathbf{F}}(x, y, z) = 12x^2\,\hat{\mathbf{i}} - 5xy\,\hat{\mathbf{j}} + xz\,\hat{\mathbf{k}}$ over path C defined by $y = x^2, z = x^3$ from point $(0, 0, 0)$ to point $(2, 4, 8)$.

49. Find the line integral of $\int_C (1 + x^2y)\,ds$, where C is ellipse $\vec{\mathbf{r}}(t) = 2\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}}$ from $0 \le t \le \pi$.

For the following exercises, find the flux.

50. Compute the flux of $\vec{\mathbf{F}} = x^2\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$ across a line segment from $(0,0)$ to $(1,2)$.

51. Let $\vec{\mathbf{F}} = 5\,\hat{\mathbf{i}}$ and let C be curve $y = 0$, with $0 \le x \le 4$. Find the flux across C.

52. Let $\vec{\mathbf{F}} = 5\,\hat{\mathbf{j}}$ and let C be curve $y = 0$, with $0 \le x \le 4$. Find the flux across C.

53. Let $\vec{\mathbf{F}} = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$ and let $C : \vec{\mathbf{r}}(t) = \cos t\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{j}}$ for $0 \le t \le 2\pi$. Calculate the flux across C.

54. Let $\vec{\mathbf{F}} = (x^2 + y^3)\,\hat{\mathbf{i}} + (2xy)\,\hat{\mathbf{j}}$. Calculate flux $\vec{\mathbf{F}}$ orientated counterclockwise across the curve $C : x^2 + y^2 = 9$.

Complete the rest of the exercises as stated.

55. Find the line integral of $\int_C z^2\,dx + y\,dy + 2y\,dz$, where C consists of two parts: C_1 and C_2. C_1 is the intersection of cylinder $x^2 + y^2 = 16$ and plane $z = 3$ from $(0,4,3)$ to $(-4,0,3)$. C_2 is a line segment from $(-4,0,3)$ to $(0,1,5)$.

56. A spring is made of a thin wire twisted into the shape of a circular helix $x = 2\cos t,\ y = 2\sin t,\ z = t$. Find the mass of two turns of the spring if the wire has a constant mass density of ρ grams per cm.

57. A thin wire is bent into the shape of a semicircle of radius a. If the linear mass density at point P is directly proportional to its distance from the line through the endpoints, find the mass of the wire.

58. An object moves in force field $\vec{\mathbf{F}}(x,y,z) = y^2\,\hat{\mathbf{i}} + 2(x+1)y\,\hat{\mathbf{j}}$ counterclockwise from point $(2,0)$ along elliptical path $x^2 + 4y^2 = 4$ to $(-2,0)$, and back to point $(2,0)$ along the x-axis. How much work is done by the force field on the object?

59. Find the work done when an object moves in force field $\vec{\mathbf{F}}(x,y,z) = 2x\,\hat{\mathbf{i}} - (x+z)\,\hat{\mathbf{j}} + (y-x)\,\hat{\mathbf{k}}$ along the path given by $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + (t^2 - t)\,\hat{\mathbf{j}} + 3\,\hat{\mathbf{k}},\ 0 \le t \le 1$.

60. If an inverse force field $\vec{\mathbf{F}}$ is given by $\vec{\mathbf{F}}(x,y,z) = \dfrac{k}{\|r\|^3} r$, where k is a constant, find the work done by $\vec{\mathbf{F}}$ as its point of application moves along the x-axis from $A(1,0,0)$ to $B(2,0,0)$.

61. David and Sandra plan to evaluate line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ along a path in the xy-plane from $(0,0)$ to $(1,1)$. The force field is $\vec{\mathbf{F}}(x,y) = (x+2y)\,\hat{\mathbf{i}} + (-x+y^2)\,\hat{\mathbf{j}}$. David chooses the path that runs along the x-axis from $(0,0)$ to $(1,0)$ and then runs along the vertical line $x = 1$ from $(1,0)$ to the final point $(1,1)$. Sandra chooses the direct path along the diagonal line $y = x$ from $(0,0)$ to $(1,1)$. Whose line integral is larger and by how much?

Contributors

Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.

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16.3: Conservative Vector Fields

Learning Objectives

- Describe simple and closed curves; define connected and simply connected regions.
- Explain how to find a potential function for a conservative vector field.
- Use the Fundamental Theorem for Line Integrals to evaluate a line integral in a vector field.
- Explain how to test a vector field to determine whether it is conservative.

In this section, we continue the study of conservative vector fields. We examine the Fundamental Theorem for Line Integrals, which is a useful generalization of the Fundamental Theorem of Calculus to line integrals of conservative vector fields. We also discover show how to test whether a given vector field is conservative, and determine how to build a potential function for a vector field known to be conservative.

Curves and Regions

Before continuing our study of conservative vector fields, we need some geometric definitions. The theorems in the subsequent sections all rely on integrating over certain kinds of curves and regions, so we develop the definitions of those curves and regions here. We first define two special kinds of curves: closed curves and simple curves. As we have learned, a closed curve is one that begins and ends at the same point. A simple curve is one that does not cross itself. A curve that is both closed and simple is a simple closed curve (Figure 16.3.1).

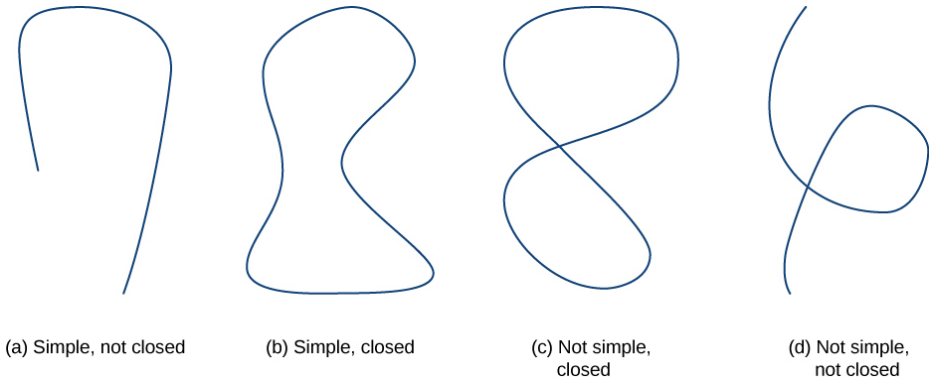


Figure 16.3.1. Types of curves that are simple or not simple and closed or not closed.

DEFINITION: Closed Curves

Curve C is a *closed curve* if there is a parameterization $\vec{\mathbf{r}}(t)$, $a \leq t \leq b$ of C such that the parameterization traverses the curve exactly once and $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$. Curve C is a simple curve if C does not cross itself. That is, C is simple if there exists a parameterization $\vec{\mathbf{r}}(t)$, $a \leq t \leq b$ of C such that $\vec{\mathbf{r}}$ is one-to-one over (a, b). It is possible for $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$, meaning that the simple curve is also closed.

Example 16.3.1: Determining Whether a Curve Is Simple and Closed

Is the curve with parameterization $\vec{\mathbf{r}}(t) = \left\langle \cos t, \frac{\sin(2t)}{2} \right\rangle$, $0 \leq t \leq 2\pi$ a simple closed curve?

Solution

Note that $\vec{\mathbf{r}}(0) = \langle 1, 0 \rangle = \vec{\mathbf{r}}(2\pi)$; therefore, the curve is closed. The curve is not simple, however. To see this, note that $\vec{\mathbf{r}}\left(\frac{\pi}{2}\right) = \langle 0, 0 \rangle = \vec{\mathbf{r}}\left(\frac{3\pi}{2}\right)$, and therefore the curve crosses itself at the origin (Figure 16.3.2).

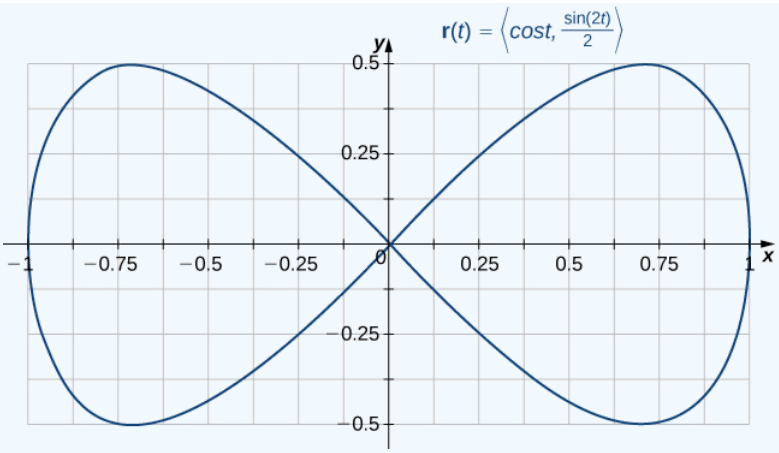


Figure 16.3.2. *A curve that is closed but not simple.*

? Exercise 16.3.1

Is the curve given by parameterization $\vec{\mathbf{r}}(t) = \langle 2\cos t, 3\sin t\rangle$, $0 \le t \le 6\pi$, a simple closed curve?

Hint

Sketch the curve.

Answer

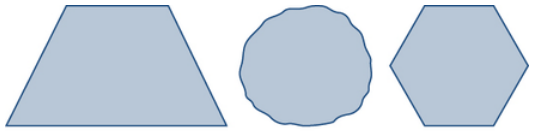
Yes

Many of the theorems in this chapter relate an integral over a region to an integral over the boundary of the region, where the region's boundary is a simple closed curve or a union of simple closed curves. To develop these theorems, we need two geometric definitions for regions: that of a connected region and that of a simply connected region. A connected region is one in which there is a path in the region that connects any two points that lie within that region. A simply connected region is a connected region that does not have any holes in it. These two notions, along with the notion of a simple closed curve, allow us to state several generalizations of the Fundamental Theorem of Calculus later in the chapter. These two definitions are valid for regions in any number of dimensions, but we are only concerned with regions in two or three dimensions.

DEFINITION: connected regions

A region D is a *connected region* if, for any two points P_1 and P_2, there is a path from P_1 to P_2 with a trace contained entirely inside D. A region D is a simply connected region if D is connected for any simple closed curve C that lies inside D, and curve C can be shrunk continuously to a point while staying entirely inside D. In two dimensions, a region is simply connected if it is connected and has no holes.

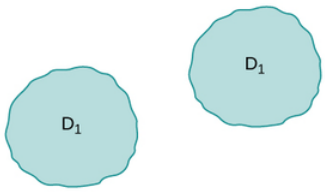
All simply connected regions are connected, but not all connected regions are simply connected (Figure 16.3.3).



(a) Simply connected regions



(b) Connected regions that are not simply connected

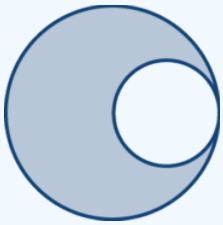


(c) A region that is not connected

Figure 16.3.3: Not all connected regions are simply connected. (a) Simply connected regions have no holes. (b) Connected regions that are not simply connected may have holes but you can still find a path in the region between any two points. (c) A region that is not connected has some points that cannot be connected by a path in the region.

? Exercise 16.3.2

Is the region in the below image connected? Is the region simply connected?



Hint

Consider the definitions.

Answer

The region in the figure is connected. The region in the figure is not simply connected.

Fundamental Theorem for Line Integrals

Now that we understand some basic curves and regions, let's generalize the Fundamental Theorem of Calculus to line integrals. Recall that the Fundamental Theorem of Calculus says that if a function f has an antiderivative F, then the integral of f from a to b depends only on the values of F at a and at b—that is,

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

If we think of the gradient as a derivative, then the same theorem holds for vector line integrals. We show how this works using a motivational example.

Example 16.3.2: Evaluating a Line Integral and the Antiderivatives of the Endpoints

Let $\vec{\mathbf{F}}(x,y) = \langle 2x, 4y\rangle$. Calculate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is the line segment from $(0,0)$ to $(2,2)$ (Figure 16.3.4).

Solution

We use the method from the previous section to calculate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. Curve C can be parameterized by $\vec{\mathbf{r}}(t) = \langle 2t, 2t\rangle$, $0 \le t \le 1$. Then, $\vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) = \langle 4t, 8t\rangle$ and $\vec{\mathbf{r}}'(t) = \langle 2, 2\rangle$, which implies that

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^1 \langle 4t, 8t\rangle \cdot \langle 2, 2\rangle dt \\ &= \int_0^1 (8t + 16t) dt = \int_0^1 24t dt \\ &= \left[12t^2\right]_0^1 = 12.\end{aligned}$$

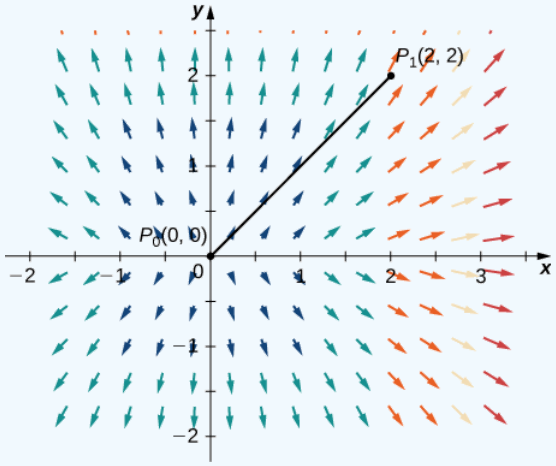


Figure 16.3.4: The value of line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ *depends only on the value of the potential function of* $\vec{\mathbf{F}}$ *at the endpoints of the curve.*

Notice that $\vec{\mathbf{F}} = \vec{\nabla} f$, where $f(x,y) = x^2 + 2y^2$. If we think of the gradient as a derivative, then f is an "antiderivative" of $\vec{\mathbf{F}}$. In the case of single-variable integrals, the integral of derivative $g'(x)$ is $g(b) - g(a)$, where a is the start point of the interval of integration and b is the endpoint. If vector line integrals work like single-variable integrals, then we would expect integral $\vec{\mathbf{F}}$ to be $f(P_1) - f(P_0)$, where P_1 is the endpoint of the curve of integration and P_0 is the start point. Notice that this is the case for this example:

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = 12$$

and

$$f(2,2) - f(0,0) = 4 + 8 - 0 = 12.$$

In other words, the integral of a "derivative" can be calculated by evaluating an "antiderivative" at the endpoints of the curve and subtracting, just as for single-variable integrals.

The following theorem says that, under certain conditions, what happened in the previous example holds for any gradient field. The same theorem holds for vector line integrals, which we call the **Fundamental Theorem for Line Integrals**.

Theorem: THE FUNDAMENTAL THEOREM FOR LINE INTEGRALS

Let C be a piecewise smooth curve with parameterization $\vec{\mathbf{r}}(t)$, $a \le t \le b$. Let f be a function of two or three variables with first-order partial derivatives that exist and are continuous on C. Then,

$$\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a)). \tag{16.3.1}$$

Proof

First,

$$\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = \int_a^b \vec{\nabla} f(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t)\, dt.$$

By the chain rule,

$$\frac{d}{dt}(f(\vec{\mathbf{r}}(t)) = \vec{\nabla} f(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t)$$

Therefore, by the Fundamental Theorem of Calculus,

$$\begin{aligned} \int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} &= \int_a^b \vec{\nabla} f(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}'(t) dt \\ &= \int_a^b \frac{d}{dt}(f(\vec{\mathbf{r}}(t)) dt \\ &= \left[f(\vec{\mathbf{r}}(t)) \right]_{t=a}^{t=b} \\ &= f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a)). \end{aligned}$$

□

We know that if $\vec{\mathbf{F}}$ is a conservative vector field, there is a potential function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$. Therefore

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a)).$$

In other words, just as with the Fundamental Theorem of Calculus, computing the line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}$ is conservative, is a two-step process:

1. Find a potential function ("antiderivative") f for $\vec{\mathbf{F}}$ and
2. Compute the value of f at the endpoints of C and calculate their difference $f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a))$.

Keep in mind, however, there is one major difference between the Fundamental Theorem of Calculus and the Fundamental Theorem for Line Integrals:
A function of one variable that is continuous must have an antiderivative. However, a vector field, even if it is continuous, does not need to have a potential function.

Example 16.3.3: Applying the Fundamental Theorem

Calculate integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y, z) = \langle 2x \ln y, \frac{x^2}{y} + z^2, 2yz \rangle$ and C is a curve with parameterization $\vec{\mathbf{r}}(t) = \langle t^2, t, t \rangle$, $1 \leq t \leq e$

a. without using the Fundamental Theorem of Line Integrals and
b. using the Fundamental Theorem of Line Integrals.

Solution

1. First, let's calculate the integral without the Fundamental Theorem for Line Integrals and instead use the method we learned in the previous section:

$$\begin{aligned}\int_C \vec{\mathbf{F}}\cdot dr &= \int_1^e \vec{\mathbf{F}}(\vec{\mathbf{r}}(t))\cdot \vec{\mathbf{r}}\,'(t)\,dt \\ &= \int_1^e \langle 2t^2\ln t, \frac{t^4}{t}+t^2, 2t^2\rangle \cdot \langle 2t,1,1\rangle\,dt \\ &= \int_1^e (4t^3\ln t + t^3 + 3t^2)\,dt \\ &= \int_1^e 4t^3\ln t\,dt + \int_1^e (t^3+3t^2)\,dt \\ &= \int_1^e 4t^3\ln t\,dt + \left[\frac{t^4}{4}+t^3\right]_1^e \\ &= \int_1^e 4t^3\ln t\,dt + \frac{e^4}{4}+e^3-\frac{1}{4}-1 \\ &= \int_1^e 4t^3\ln t\,dt + \frac{e^4}{4}+e^3-\frac{5}{4}\end{aligned}$$

Integral $\int_1^e t^3\ln t\,dt$ requires integration by parts. Let $u=\ln t$ and $dv=t^3$. Then $u=\ln t$, $dv=t^3$ and

$$du=\frac{1}{t}\,dt,\quad v=\frac{t^4}{4}.$$

Therefore,

$$\begin{aligned}\int_1^e t^3\ln t\,dt &= \left[\frac{t^4}{4}\ln t\right]_1^e - \frac{1}{4}\int_1^e t^3\,dt \\ &= \frac{e^4}{4}-\frac{1}{4}\left(\frac{e^4}{4}-\frac{1}{4}\right).\end{aligned}$$

Thus,

$$\begin{aligned}\int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} &= 4\int_1^e t^3\ln t\,dt \quad + \quad \frac{e^4}{4}+e^3-\frac{5}{4} \\ &= 4\left(\frac{e^4}{4}-\frac{1}{4}\left(\frac{e^4}{4}-\frac{1}{4}\right)\right)+\frac{e^4}{4}+e^3-\frac{5}{4} \\ &= e^4-\frac{e^4}{4}+\frac{1}{4}+\frac{e^4}{4}+e^3-\frac{5}{4} \\ &= e^4+e^3-1.\end{aligned}$$

2. Given that $f(x,y,z)=x^2\ln y+yz^2$ is a potential function for $\vec{\mathbf{F}}$, let's use the Fundamental Theorem for Line Integrals to calculate the integral. Note that

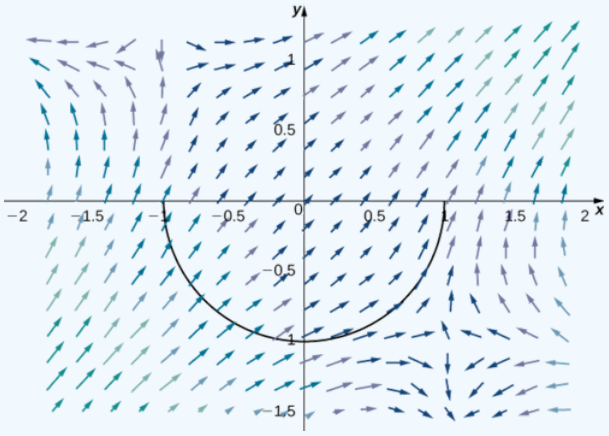
$$\begin{aligned}\int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} &= \int_C \vec{\nabla} f\cdot d\vec{\mathbf{r}} \\ &= f(\vec{\mathbf{r}}(e))-f(\vec{\mathbf{r}}(1)) \\ &= f(e^2,e,e)-f(1,1,1) \\ &= e^4+e^3-1.\end{aligned}$$

This calculation is much more straightforward than the calculation we did in (a). As long as we have a potential function, calculating a line integral using the Fundamental Theorem for Line Integrals is much easier than calculating without the theorem.

Example 16.3.3 illustrates a nice feature of the Fundamental Theorem of Line Integrals: it allows us to calculate more easily many vector line integrals. As long as we have a potential function, calculating the line integral is only a matter of evaluating the potential function at the endpoints and subtracting.

Exercise 16.3.3

Given that $f(x,y) = (x-1)^2 y + (y+1)^2 x$ is a potential function for $\vec{\mathbf{F}}(x,y) = \langle 2xy - 2y + (y+1)^2, (x-1)^2 + 2yx + 2x \rangle$, calculate integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is the lower half of the unit circle oriented counterclockwise.



Hint

The Fundamental Theorem for Line Intervals says this integral depends only on the value of f at the endpoints of C.

Answer

2

The Fundamental Theorem for Line Integrals has two important consequences. The first consequence is that if $\vec{\mathbf{F}}$ is conservative and C is a closed curve, then the circulation of $\vec{\mathbf{F}}$ along C is zero—that is, $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$. To see why this is true, let f be a potential function for $\vec{\mathbf{F}}$. Since C is a closed curve, the terminal point $\vec{\mathbf{r}}(b)$ of C is the same as the initial $\vec{\mathbf{r}}(a)$ of C—that is, $\vec{\mathbf{r}}(a) = \vec{\mathbf{r}}(b)$. Therefore, by the Fundamental Theorem for Line Integrals,

$$\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \oint_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} \tag{16.3.2}$$

$$= f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a)) \tag{16.3.3}$$

$$= f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(b)) \tag{16.3.4}$$

$$= 0. \tag{16.3.5}$$

Recall that the reason a conservative vector field $\vec{\mathbf{F}}$ is called "conservative" is because such vector fields model forces in which energy is conserved. We have shown gravity to be an example of such a force. If we think of vector field $\vec{\mathbf{F}}$ in integral $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ as a gravitational field, then the equation $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$ follows. If a particle travels along a path that starts and ends at the same place, then the work done by gravity on the particle is zero.

The second important consequence of the Fundamental Theorem for Line Integrals (Equation 16.3.1) is that line integrals of conservative vector fields are independent of path—meaning, they depend only on the endpoints of the given curve, and do not depend on the path between the endpoints.

DEFINITION: Path Independence

Let $\vec{\mathbf{F}}$ be a vector field with domain D; it is independent of path (or path independent) if

$$\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

for any paths C_1 and C_2 in D with the same initial and terminal points.

The second consequence is stated formally in the following theorem.

Theorem: CONSERVATIVE FIELDS

If $\vec{\mathbf{F}}$ is a *conservative vector field*, then $\vec{\mathbf{F}}$ is independent of path.

Proof

Let D denote the domain of $\vec{\mathbf{F}}$ and let C_1 and C_2 be two paths in D with the same initial and terminal points (Figure 16.3.5). Call the initial point P_1 and the terminal point P_2. Since $\vec{\mathbf{F}}$ is conservative, there is a potential function f for $\vec{\mathbf{F}}$. By the Fundamental Theorem for Line Integrals,

$$\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = f(P_2) - f(P_1) = \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

Therefore, $\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ and $\vec{\mathbf{F}}$ is independent of path.

□

To visualize what independence of path means, imagine three hikers climbing from base camp to the top of a mountain. Hiker 1 takes a steep route directly from camp to the top. Hiker 2 takes a winding route that is not steep from camp to the top. Hiker 3 starts by taking the steep route but halfway to the top decides it is too difficult for him. Therefore he returns to camp and takes the non-steep path to the top. All three hikers are traveling along paths in a gravitational field. Since gravity is a force in which energy is conserved, the gravitational field is conservative. By independence of path, the total amount of work done by gravity on each of the hikers is the same because they all started in the same place and ended in the same place. The work done by the hikers includes other factors such as friction and muscle movement, so the total amount of energy each one expended is not the same, but the net energy expended against gravity is the same for all three hikers.

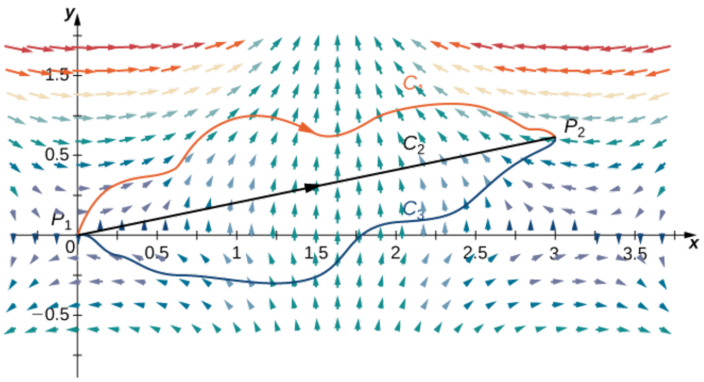


Figure 16.3.4: The vector field is conservative, and therefore independent of path.

We have shown that if $\vec{\mathbf{F}}$ is conservative, then $\vec{\mathbf{F}}$ is independent of path. It turns out that if the domain of $\vec{\mathbf{F}}$ is open and connected, then the converse is also true. That is, if $\vec{\mathbf{F}}$ is independent of path and the domain of $\vec{\mathbf{F}}$ is open and connected, then $\vec{\mathbf{F}}$ is conservative. Therefore, the set of conservative vector fields on open and connected domains is precisely the set of vector fields independent of path.

Theorem: THE PATH INDEPENDENCE TEST FOR CONSERVATIVE FIELDS

If $\vec{\mathbf{F}}$ is a continuous vector field that is independent of path and the domain D of $\vec{\mathbf{F}}$ is open and connected, then $\vec{\mathbf{F}}$ is conservative.

Proof

We prove the theorem for vector fields in $\mathbb{R}^2$. The proof for vector fields in $\mathbb{R}^3$ is similar. To show that $\vec{\mathbf{F}} = \langle P, Q \rangle$ is conservative, we must find a potential function f for $\vec{\mathbf{F}}$. To that end, let X be a fixed point in D. For any point (x, y) in D, let C be a path from X to (x, y). Define $f(x, y)$ by $f(x, y) = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. (Note that this definition of f makes sense only because $\vec{\mathbf{F}}$ is

independent of path. If $\vec{\mathbf{F}}$ was not independent of path, then it might be possible to find another path C' from X to (x, y) such that $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \neq \int_{C'} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, and in such a case $f(x, y)$ would not be a function.) We want to show that f has the property $\vec{\nabla} f = \vec{\mathbf{F}}$.

Since domain D is open, it is possible to find a disk centered at (x, y) such that the disk is contained entirely inside D. Let (a, y) with $a < x$ be a point in that disk. Let C be a path from X to (x, y) that consists of two pieces: C_1 and C_2. The first piece, C_1, is any path from C to (a, y) that stays inside D; C_2 is the horizontal line segment from (a, y) to (x, y) (Figure 16.3.6). Then

$$f(x, y) = \int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

The first integral does not depend on x, so

$$f_x(x, y) = \frac{\partial}{\partial x} \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

If we parameterize C_2 by $\vec{\mathbf{r}}(t) = \langle t, y \rangle$, $a \leq t \leq x$, then

$$\begin{aligned} f_x(x, y) &= \frac{\partial}{\partial x} \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\ &= \frac{\partial}{\partial x} \int_a^x \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \vec{\mathbf{r}}\,'(t)\, dt \\ &= \frac{\partial}{\partial x} \int_a^x \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \frac{d}{dt}(\langle t, y \rangle)\, dt \\ &= \frac{\partial}{\partial x} \int_a^x \vec{\mathbf{F}}(\vec{\mathbf{r}}(t)) \cdot \langle 1, 0 \rangle\, dt \\ &= \frac{\partial}{\partial x} \int_a^x P(t, y)\, dt. \end{aligned}$$

By the Fundamental Theorem of Calculus (part 1),

$$f_x(x, y) = \frac{\partial}{\partial x} \int_a^x P(t, y)\, dt = P(x, y).$$

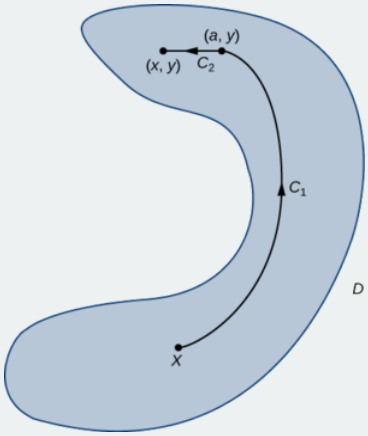


Figure 16.3.6. Here, C_1 is any path from C to (a, y) that stays inside D, and C_2 is the horizontal line segment from (a, y) to (x, y).

A similar argument using a vertical line segment rather than a horizontal line segment shows that $f_y(x, y) = Q(x, y)$.

Therefore $\vec{\nabla} f = \vec{\mathbf{F}}$ and $\vec{\mathbf{F}}$ is conservative.

□

We have spent a lot of time discussing and proving the theorems above, but we can summarize them simply: a vector field $\vec{\mathbf{F}}$ on an open and connected domain is conservative if and only if it is independent of path. This is important to know because conservative vector fields are extremely important in applications, and these theorems give us a different way of viewing what it means to be conservative using path independence.

Example 16.3.4: Showing That a Vector Field Is Not Conservative

Use path independence to show that vector field $\vec{\mathbf{F}}(x,y) = \langle x^2 y, y+5\rangle$ is not conservative.

Solution

We can indicate that $\vec{\mathbf{F}}$ is not conservative by showing that $\vec{\mathbf{F}}$ is not path independent. We do so by giving two different paths, C_1 and C_2, that both start at $(0,0)$ and end at $(1,1)$, and yet $\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \neq \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$.

Let C_1 be the curve with parameterization $\vec{\mathbf{r}}_1(t) = \langle t,\ t\rangle$, $0 \le t \le 1$ and let C_2 be the curve with parameterization $\vec{\mathbf{r}}_2(t) = \langle t,\ t^2\rangle$, $0 \le t \le 1$ (Figure 16.3.7.). Then

$$\begin{aligned}\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^1 \vec{\mathbf{F}}(\vec{\mathbf{r}}_1(t)) \cdot \vec{\mathbf{r}}_1{}'(t)\, dt \\ &= \int_0^1 \langle t^3, t+5\rangle \cdot \langle 1,1\rangle\, dt = \int_0^1 (t^3 + t + 5)\, dt \\ &= \left[\frac{t^4}{4} + \frac{t^2}{2} + 5t\right]_0^1 = \frac{23}{4}\end{aligned}$$

and

$$\begin{aligned}\int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^1 \vec{\mathbf{F}}(\vec{\mathbf{r}}_2(t)) \cdot \vec{\mathbf{r}}_2{}'(t)\, dt \\ &= \int_0^1 \langle t^4, t^2+5\rangle \cdot \langle 1, 2t\rangle\, dt = \int_0^1 (t^4 + 2t^3 + 10t)\, dt \\ &= \left[\frac{t^5}{5} + \frac{t^4}{2} + 5t^2\right]_0^1 = \frac{57}{10}.\end{aligned}$$

Since $\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \neq \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, the value of a line integral of $\vec{\mathbf{F}}$ depends on the path between two given points. Therefore, $\vec{\mathbf{F}}$ is not independent of path, and $\vec{\mathbf{F}}$ is not conservative.

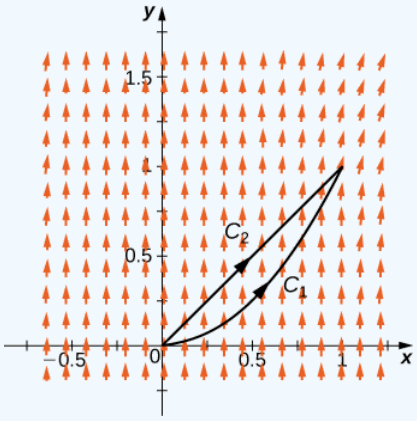


Figure 16.3.7: Curves C_1 and C_2 are both oriented from left to right.

Exercise 16.3.4

Show that $\vec{\mathbf{F}}(x,y) = \langle xy,\ x^2y^2\rangle$ is not path independent by considering the line segment from $(0,0)$ to $(0,2)$ and the piece of the graph of $y = \dfrac{x^2}{2}$ that goes from $(0,0)$ to $(0,2)$.

Hint

Calculate the corresponding line integrals.

Answer

If C_1 and C_2 represent the two curves, then

$$\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \neq \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

Conservative Vector Fields and Potential Functions

As we have learned, the Fundamental Theorem for Line Integrals says that if $\vec{\mathbf{F}}$ is conservative, then calculating $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ has two steps: first, find a potential function f for $\vec{\mathbf{F}}$ and, second, calculate $f(P_1) - f(P_0)$, where P_1 is the endpoint of C and P_0 is the starting point. To use this theorem for a conservative field $\vec{\mathbf{F}}$, we must be able to find a potential function f for $\vec{\mathbf{F}}$. Therefore, we must answer the following question: Given a conservative vector field $\vec{\mathbf{F}}$, how do we find a function f such that $\vec{\nabla} f = \vec{\mathbf{F}}$? Before giving a general method for finding a potential function, let's motivate the method with an example.

Example 16.3.5: Finding a Potential Function

Find a potential function for $\vec{\mathbf{F}}(x, y) = \langle 2xy^3, 3x^2y^2 + \cos(y) \rangle$, thereby showing that $\vec{\mathbf{F}}$ is conservative.

Solution

Suppose that $f(x, y)$ is a potential function for $\vec{\mathbf{F}}$. Then, $\vec{\nabla} f = \vec{\mathbf{F}}$, and therefore

$$f_x(x, y) = 2xy^3 \quad \text{and} \quad f_y(x, y) = 3x^2y^2 + \cos y.$$

Integrating the equation $f_x(x, y) = 2xy^3$ with respect to x yields the equation

$$f(x, y) = x^2y^3 + h(y).$$

Notice that since we are integrating a two-variable function with respect to x, we must add a constant of integration that is a constant with respect to x, but may still be a function of y. The equation $f(x, y) = x^2y^3 + h(y)$ can be confirmed by taking the partial derivative with respect to x:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2y^3) + \frac{\partial}{\partial x}(h(y)) = 2xy^3 + 0 = 2xy^3.$$

Since f is a potential function for $\vec{\mathbf{F}}$,

$$f_y(x, y) = 3x^2y^2 + \cos(y),$$

and therefore

$$3x^2y^2 + g'(y) = 3x^2y^2 + \cos(y).$$

This implies that $h'(y) = \cos y$, so $h(y) = \sin y + C$. Therefore, *any* function of the form $f(x, y) = x^2y^3 + \sin(y) + C$ is a potential function. Taking, in particular, $C = 0$ gives the potential function $f(x, y) = x^2y^3 + \sin(y)$.

To verify that f is a potential function, note that $\vec{\nabla} f(x, y) = \langle 2xy^3, 3x^2y^2 + \cos y \rangle = \vec{\mathbf{F}}$.

Exercise 16.3.5

Find a potential function for $\vec{\mathbf{F}}(x, y) = \langle e^x y^3 + y, 3e^x y^2 + x \rangle$.

Hint

Follow the steps in Example 16.3.5.

Answer

$f(x, y) = e^x y^3 + xy$

The logic of the previous example extends to finding the potential function for any conservative vector field in $\mathbb{R}^2$. Thus, we have the following problem-solving strategy for finding potential functions:

PROBLEM-SOLVING STRATEGY: FINDING A POTENTIAL FUNCTION FOR A CONSERVATIVE VECTOR FIELD $\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y)\rangle$

1. Integrate P with respect to x. This results in a function of the form $g(x,y) + h(y)$, where $h(y)$ is unknown.
2. Take the partial derivative of $g(x,y) + h(y)$ with respect to y, which results in the function $gy(x,y) + h'(y)$.
3. Use the equation $gy(x,y) + h'(y) = Q(x,y)$ to find $h'(y)$.
4. Integrate $h'(y)$ to find $h(y)$.
5. Any function of the form $f(x,y) = g(x,y) + h(y) + C$, where C is a constant, is a potential function for $\vec{\mathbf{F}}$.

We can adapt this strategy to find potential functions for vector fields in $\mathbb{R}^3$, as shown in the next example.

Example 16.3.6: Finding a Potential Function in $\mathbb{R}^3$

Find a potential function for $F(x,y,z) = \langle 2xy, x^2 + 2yz^3, 3y^2z^2 + 2z\rangle$, thereby showing that $\vec{\mathbf{F}}$ is conservative.

Solution

Suppose that f is a potential function. Then, $\vec{\nabla} f = \vec{\mathbf{F}}$ and therefore $f_x(x,y,z) = 2xy$. Integrating this equation with respect to x yields the equation $f(x,y,z) = x^2y + g(y,z)$ for some function g. Notice that, in this case, the constant of integration with respect to x is a function of y and z.

Since f is a potential function,

$$x^2 + 2yz^3 = f_y(x,y,z) = x^2 + g_y(y,z).$$

Therefore,

$$g_y(y,z) = 2yz^3.$$

Integrating this function with respect to y yields

$$g(y,z) = y^2z^3 + h(z)$$

for some function $h(z)$ of z alone. (Notice that, because we know that g is a function of only y and z, we do not need to write $g(y,z) = y^2z^3 + h(x,z)$.) Therefore,

$$f(x,y,z) = x^2y + g(y,z) = x^2y + y^2z^3 + h(z).$$

To find f, we now must only find h. Since f is a potential function,

$$3y^2z^2 + 2z = g_z(y,z) = 3y^2z^2 + h'(z).$$

This implies that $h'(z) = 2z$, so $h(z) = z^2 + C$. Letting $C = 0$ gives the potential function

$$f(x,y,z) = x^2y + y^2z^3 + z^2.$$

To verify that f is a potential function, note that $\vec{\nabla} f(x,y,z) = \langle 2xy, x^2 + 2yz^3, 3y^2z^2 + 2z\rangle = \vec{\mathbf{F}}(x,y,z)$.

Exercise 16.3.6

Find a potential function for $\vec{\mathbf{F}}(x,y,z) = \langle 12x^2, \cos y \cos z, 1 - \sin y \sin z\rangle$.

Hint

Following Example 16.3.6, begin by integrating with respect to x.

Answer

$f(x,y,z) = 4x^3 + \sin y \cos z + z$

We can apply the process of finding a potential function to a gravitational force. Recall that, if an object has unit mass and is located at the origin, then the gravitational force in $\mathbb{R}^2$ that the object exerts on another object of unit mass at the point (x,y) is

given by vector field

$$\vec{\mathbf{F}}(x,y) = -G\left\langle \frac{x}{(x^2+y^2)^{3/2}}, \frac{y}{(x^2+y^2)^{3/2}} \right\rangle,$$

where G is the universal gravitational constant. In the next example, we build a potential function for $\vec{\mathbf{F}}$, thus confirming what we already know: that gravity is conservative.

Example 16.3.7: Finding a Potential Function

Find a potential function f for $\vec{\mathbf{F}}(x,y) = -G\left\langle \frac{x}{(x^2+y^2)^{3/2}}, \frac{y}{(x^2+y^2)^{3/2}} \right\rangle$.

Solution

Suppose that f is a potential function. Then, $\vec{\nabla} f = \vec{\mathbf{F}}$ and therefore

$$f_x(x,y) = \frac{-Gx}{(x^2+y^2)^{3/2}}.$$

To integrate this function with respect to x, we can use u-substitution. If $u = x^2+y^2$, then $\frac{du}{2} = x\,dx$, so

$$\begin{aligned}\int \frac{-Gx}{(x^2+y^2)^{3/2}}\,dx &= \int \frac{-G}{2u^{3/2}}\,du \\ &= \frac{G}{\sqrt{u}} + h(y) \\ &= \frac{G}{\sqrt{x^2+y^2}} + h(y)\end{aligned}$$

for some function $h(y)$. Therefore,

$$f(x,y) = \frac{G}{\sqrt{x^2+y^2}} + h(y).$$

Since f is a potential function for $\vec{\mathbf{F}}$,

$$f_y(x,y) = \frac{-Gy}{(x^2+y^2)^{3/2}}$$

.

Since $f(x,y) = \frac{G}{\sqrt{x^2+y^2}} + h(y)$, $f_y(x,y)$ also equals $\frac{-Gy}{(x^2+y^2)^{3/2}} + h'(y)$.

Therefore,

$$\frac{-Gy}{(x^2+y^2)^{3/2}} + h'(y) = \frac{-Gy}{(x^2+y^2)^{3/2}},$$

which implies that $h'(y) = 0$. Thus, we can take $h(y)$ to be any constant; in particular, we can let $h(y) = 0$. The function

$$f(x,y) = \frac{G}{\sqrt{x^2+y^2}}$$

is a potential function for the gravitational field $\vec{\mathbf{F}}$. To confirm that f is a potential function, note that

$$\begin{aligned}\vec{\nabla} f(x,y) &= \langle -\frac{1}{2}\frac{G}{(x^2+y^2)^{3/2}}(2x), -\frac{1}{2}\frac{G}{(x^2+y^2)^{3/2}}(2y)\rangle \\ &= \langle \frac{-Gx}{(x^2+y^2)^{3/2}}, \frac{-Gy}{(x^2+y^2)^{3/2}}\rangle \\ &= \vec{\mathbf{F}}(x,y).\end{aligned}$$

Exercise 16.3.7

Find a potential function f for the three-dimensional gravitational force $\vec{\mathbf{F}}(x,y,z) = \left\langle \frac{-Gx}{(x^2+y^2+z^2)^{3/2}}, \frac{-Gy}{(x^2+y^2+z^2)^{3/2}}, \frac{-Gz}{(x^2+y^2+z^2)^{3/2}} \right\rangle$.

Hint

Follow the Problem-Solving Strategy.

Answer

$$f(x,y,z) = \frac{G}{\sqrt{x^2+y^2+z^2}}$$

Testing a Vector Field

Until now, we have worked with vector fields that we know are conservative, but if we are not told that a vector field is conservative, we need to be able to test whether it is conservative. Recall that, if $\vec{\mathbf{F}}$ is conservative, then $\vec{\mathbf{F}}$ has the cross-partial property (see The Cross-Partial Property of Conservative Vector Fields). That is, if $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ is conservative, then $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$. So, if $\vec{\mathbf{F}}$ has the cross-partial property, then is $\vec{\mathbf{F}}$ conservative? If the domain of $\vec{\mathbf{F}}$ is open and simply connected, then the answer is yes.

Theorem: THE CROSS-PARTIAL TEST FOR CONSERVATIVE FIELDS

If $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ is a vector field on an open, simply connected region D and $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$ throughout D, then $\vec{\mathbf{F}}$ is conservative.

Although a proof of this theorem is beyond the scope of the text, we can discover its power with some examples. Later, we see why it is necessary for the region to be simply connected.

Combining this theorem with the cross-partial property, we can determine whether a given vector field is conservative:

Theorem: CROSS-PARTIAL PROPERTY OF CONSERVATIVE FIELDS

Let $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ be a vector field on an open, simply connected region D. Then $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$ throughout D if and only if $\vec{\mathbf{F}}$ is conservative.

The version of this theorem in $\mathbb{R}^2$ is also true. If $\vec{\mathbf{F}}(x,y) = \langle P, Q \rangle$ is a vector field on an open, simply connected domain in $\mathbb{R}^2$, then $\vec{\mathbf{F}}$ is conservative if and only if $P_y = Q_x$.

Example 16.3.8: Determining Whether a Vector Field Is Conservative

Determine whether vector field $\vec{\mathbf{F}}(x,y,z) = \langle xy^2z, x^2yz, z^2 \rangle$ is conservative.

Solution

Note that the domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^2$ and $\mathbb{R}^3$ is simply connected. Therefore, we can use **The Cross-Partial Property of Conservative Vector Fields** to determine whether $\vec{\mathbf{F}}$ is conservative. Let

$$P(x,y,z) = xy^2z$$

$$Q(x,y,z) = x^2yz$$

and

$$R(x,y,z) = z^2.$$

Since $Q_z(x,y,z) = x^2y$ and $R_y(x,y,z) = 0$, the vector field is not conservative.

Example 16.3.9: Determining Whether a Vector Field Is Conservative

Determine vector field $\vec{\mathbf{F}}(x,y) = \langle x\ln(y),\ \frac{x^2}{2y}\rangle$ is conservative.

Solution

Note that the domain of $\vec{\mathbf{F}}$ is the part of $\mathbb{R}^2$ in which $y > 0$. Thus, the domain of $\vec{\mathbf{F}}$ is part of a plane above the x-axis, and this domain is simply connected (there are no holes in this region and this region is connected). Therefore, we can use **The Cross-Partial Property of Conservative Vector Fields** to determine whether $\vec{\mathbf{F}}$ is conservative. Let

$$P(x,y) = x\ln(y) \quad \text{and} \quad Q(x,y) = \frac{x^2}{2y}.$$

Then $P_y(x,y) = \frac{x}{y} = Q_x(x,y)$ and thus $\vec{\mathbf{F}}$ is conservative.

Exercise 16.3.8

Determine whether $\vec{\mathbf{F}}(x,y) = \langle \sin x\cos y,\ \cos x\sin y\rangle$ is conservative.

Hint

Use **The Cross-Partial Property of Conservative Vector Fields** from the previous section.

Answer

It is conservative.

When using **The Cross-Partial Property of Conservative Vector Fields**, it is important to remember that a theorem is a tool, and like any tool, it can be applied only under the right conditions. In the case of **The Cross-Partial Property of Conservative Vector Fields**, the theorem can be applied only if the domain of the vector field is simply connected.

To see what can go wrong when misapplying the theorem, consider the vector field from Example 16.3.4:

$$\vec{\mathbf{F}}(x,y) = \frac{y}{x^2+y^2}\,\hat{\mathbf{i}} + \frac{-x}{x^2+y^2}\,\hat{\mathbf{j}}.$$

This vector field satisfies the cross-partial property, since

$$\frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right) = \frac{(x^2+y^2)-y(2y)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

and

$$\frac{\partial}{\partial x}\left(\frac{-x}{x^2+y^2}\right) = \frac{-(x^2+y^2)+x(2x)}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

Since $\vec{\mathbf{F}}$ satisfies the cross-partial property, we might be tempted to conclude that $\vec{\mathbf{F}}$ is conservative. However, $\vec{\mathbf{F}}$ is not conservative. To see this, let

$$\vec{\mathbf{r}}(t) = \langle \cos t, \sin t\rangle, \quad 0 \le t \le \pi$$

be a parameterization of the upper half of a unit circle oriented counterclockwise (denote this C_1) and let

$$\vec{\mathbf{s}}(t) = \langle \cos t, -\sin t\rangle, \quad 0 \le t \le \pi$$

be a parameterization of the lower half of a unit circle oriented clockwise (denote this C_2). Notice that C_1 and C_2 have the same starting point and endpoint. Since $\sin^2 t + \cos^2 t = 1$,

$$\vec{\mathbf{F}}(\vec{\mathbf{r}}(t))\cdot\vec{\mathbf{r}}\,'(t) = \langle \sin(t), -\cos(t)\rangle \cdot \langle -\sin(t), \cos(t)\rangle = -1$$

and

$$\vec{\mathbf{F}}(\vec{\mathbf{s}}(t)) \cdot \vec{\mathbf{s}}\,'(t) = \langle -\sin t, -\cos t \rangle \cdot \langle -\sin t, -\cos t \rangle = \sin^2 t + \cos^2 t = 1.$$

Therefore,

$$\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_0^{\pi} -1 \, dt = -\pi$$

and

$$\int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_0^{\pi} 1 \, dt = \pi.$$

Thus, C_1 and C_2 have the same starting point and endpoint, but $\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \neq \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. Therefore, $\vec{\mathbf{F}}$ is not independent of path and $\vec{\mathbf{F}}$ is not conservative.

To summarize: $\vec{\mathbf{F}}$ satisfies the cross-partial property and yet $\vec{\mathbf{F}}$ is not conservative. What went wrong? Does this contradict **The Cross-Partial Property of Conservative Vector Fields**? The issue is that the domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^2$ except for the origin. In other words, the domain of $\vec{\mathbf{F}}$ has a hole at the origin, and therefore the domain is not simply connected. Since the domain is not simply connected, **The Cross-Partial Property of Conservative Vector Fields** does not apply to $\vec{\mathbf{F}}$.

We close this section by looking at an example of the usefulness of the Fundamental Theorem for Line Integrals. Now that we can test whether a vector field is conservative, we can always decide whether the Fundamental Theorem for Line Integrals can be used to calculate a vector line integral. If we are asked to calculate an integral of the form $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, then our first question should be: Is $\vec{\mathbf{F}}$ conservative? If the answer is yes, then we should find a potential function and use the Fundamental Theorem for Line Integrals to calculate the integral. If the answer is no, then the Fundamental Theorem for Line Integrals cannot help us and we have to use other methods, such as using the method from the previous section (using $\vec{\mathbf{F}}(\vec{\mathbf{r}}(t))$ and $\vec{\mathbf{r}}\,'(t)$).

Example 16.3.10: Using the Fundamental Theorem for Line Integrals

Calculate line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y, z) = \langle 2xe^y z + e^x z,\ x^2 e^y z,\ x^2 e^y + e^x \rangle$ and C is any smooth curve that goes from the origin to $(1, 1, 1)$.

Solution

Before trying to compute the integral, we need to determine whether $\vec{\mathbf{F}}$ is conservative and whether the domain of $\vec{\mathbf{F}}$ is simply connected. The domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^3$, which is connected and has no holes. Therefore, the domain of $\vec{\mathbf{F}}$ is simply connected. Let

$$P(x, y, z) = 2xe^y z + e^x z, \quad Q(x, y, z) = x^2 e^y z, \quad \text{and} \quad R(x, y, z) = x^2 e^y + e^x$$

so that $\vec{\mathbf{F}}(x, y, z) = \langle P, Q, R \rangle$. Since the domain of $\vec{\mathbf{F}}$ is simply connected, we can check the cross partials to determine whether $\vec{\mathbf{F}}$ is conservative. Note that

$$\begin{aligned} P_y(x, y, z) &= 2xe^y z = Q_x(x, y, z) \\ P_z(x, y, z) &= 2xe^y + e^x = R_x(x, y, z) \\ Q_z(x, y, z) &= x^2 e^y = R_y(x, y, z). \end{aligned}$$

Therefore, $\vec{\mathbf{F}}$ is conservative.

To evaluate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ using the Fundamental Theorem for Line Integrals, we need to find a potential function f for $\vec{\mathbf{F}}$. Let f be a potential function for $\vec{\mathbf{F}}$. Then, $\vec{\nabla} f = \vec{\mathbf{F}}$, and therefore $f_x(x, y, z) = 2xe^y z + e^x z$. Integrating this equation with respect to x gives $f(x, y, z) = x^2 e^y z + e^x z + h(y, z)$ for some function h. Differentiating this equation with respect to y gives $x^2 e^y z + h_y(y, z) = Q(x, y, z) = x^2 e^y z$, which implies that $h_y(y, z) = 0$. Therefore, h is a function of z only, and $f(x, y, z) = x^2 e^y z + e^x z + h(z)$. To find h, note that $f_z = x^2 e^y + e^x + h'(z) = R = x^2 e^y + e^x$. Therefore, $h'(z) = 0$ and we can take $h(z) = 0$. A potential function for $\vec{\mathbf{F}}$ is $f(x, y, z) = x^2 e^y z + e^x z$.

Now that we have a potential function, we can use the Fundamental Theorem for Line Integrals to evaluate the integral. By the theorem,

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} \\ &= f(1,1,1) - f(0,0,0) \\ &= 2e.\end{aligned}$$

Analysis

Notice that if we hadn't recognized that $\vec{\mathbf{F}}$ is conservative, we would have had to parameterize C and use the method from the previous section. Since curve C is unknown, using the Fundamental Theorem for Line Integrals is much simpler.

Exercise 16.3.9

Calculate integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x,y) = \langle \sin x \sin y, 5 - \cos x \cos y \rangle$ and C is a semicircle with starting point $(0,\pi)$ and endpoint $(0,-\pi)$.

Hint

Use the Fundamental Theorem for Line Integrals.

Answer

-10π

Example 16.3.11: Work Done on a Particle

Let $\vec{\mathbf{F}}(x,y) = \langle 2xy^2, 2x^2y \rangle$ be a force field. Suppose that a particle begins its motion at the origin and ends its movement at any point in a plane that is not on the x-axis or the y-axis. Furthermore, the particle's motion can be modeled with a smooth parameterization. Show that $\vec{\mathbf{F}}$ does positive work on the particle.

Solution

We show that $\vec{\mathbf{F}}$ does positive work on the particle by showing that $\vec{\mathbf{F}}$ is conservative and then by using the Fundamental Theorem for Line Integrals.

To show that $\vec{\mathbf{F}}$ is conservative, suppose $f(x,y)$ were a potential function for $\vec{\mathbf{F}}$. Then, $\vec{\nabla} f(x,y) = \vec{\mathbf{F}}(x,y) = \langle 2xy^2, 2x^2y \rangle$ and therefore $f_x(x,y) = 2xy^2$ and $f_y(x,y) = 2x^2y$. The equation $fx(x,y) = 2xy^2$ implies that $f(x,y) = x^2y^2 + h(y)$. Deriving both sides with respect to y yields $f_y(x,y) = 2x^2y + h'(y)$. Therefore, $h'(y) = 0$ and we can take $h(y) = 0$.

If $f(x,y) = x^2y^2$, then note that $\vec{\nabla} f(x,y) = \langle 2xy^2, 2x^2y \rangle = \vec{\mathbf{F}}$, and therefore f is a potential function for $\vec{\mathbf{F}}$.

Let (a,b) be the point at which the particle stops is motion, and let C denote the curve that models the particle's motion. The work done by $\vec{\mathbf{F}}$ on the particle is $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. By the Fundamental Theorem for Line Integrals,

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_C \nabla f \cdot d\vec{\mathbf{r}} \\ &= f(a,b) - f(0,0) \\ &= a^2b^2.\end{aligned}$$

Since $a \neq 0$ and $b \neq 0$, by assumption, $a^2b^2 > 0$. Therefore, $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} > 0$, and $\vec{\mathbf{F}}$ does positive work on the particle.

Analysis

Notice that this problem would be much more difficult without using the Fundamental Theorem for Line Integrals. To apply the tools we have learned, we would need to give a curve parameterization and use the method from the previous section. Since the path of motion C can be as exotic as we wish (as long as it is smooth), it can be very difficult to parameterize the motion of the particle.

? Exercise 16.3.10

Let $\vec{\mathbf{F}}(x,y) = \langle 4x^3y^4, 4x^4y^3 \rangle$, and suppose that a particle moves from point $(4,4)$ to $(1,1)$ along any smooth curve. Is the work done by $\vec{\mathbf{F}}$ on the particle positive, negative, or zero?

Hint

Use the Fundamental Theorem for Line Integrals.

Answer

Negative

Key Concepts

- The theorems in this section require curves that are closed, simple, or both, and regions that are connected or simply connected.
- The line integral of a conservative vector field can be calculated using the Fundamental Theorem for Line Integrals. This theorem is a generalization of the Fundamental Theorem of Calculus in higher dimensions. Using this theorem usually makes the calculation of the line integral easier.
- Conservative fields are independent of path. The line integral of a conservative field depends only on the value of the potential function at the endpoints of the domain curve.
- Given vector field $\vec{\mathbf{F}}$, we can test whether $\vec{\mathbf{F}}$ is conservative by using the cross-partial property. If $\vec{\mathbf{F}}$ has the cross-partial property and the domain is simply connected, then $\vec{\mathbf{F}}$ is conservative (and thus has a potential function). If $\vec{\mathbf{F}}$ is conservative, we can find a potential function by using the Problem-Solving Strategy.
- The circulation of a conservative vector field on a simply connected domain over a closed curve is zero.

Key Equations

- **Fundamental Theorem for Line Integrals**
  $\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a))$
- **Circulation of a conservative field over curve *C* that encloses a simply connected region**
  $\oint_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = 0$

Glossary

closed curve

a curve that begins and ends at the same point

connected region

a region in which any two points can be connected by a path with a trace contained entirely inside the region

Fundamental Theorem for Line Integrals

the value of line integral $\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$ depends only on the value of f at the endpoints of C: $\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a))$

independence of path

a vector field $\vec{\mathbf{F}}$ has path independence if $\int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ for any curves C_1 and C_2 in the domain of $\vec{\mathbf{F}}$ with the same initial points and terminal points

simple curve

a curve that does not cross itself

simply connected region

a region that is connected and has the property that any closed curve that lies entirely inside the region encompasses points that are entirely inside the region

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16.3E: Exercises for Section 16.3

1. *True* or *False?* If vector field $\vec{\mathbf{F}}$ is conservative on the open and connected region D, then line integrals of $\vec{\mathbf{F}}$ are path independent on D, regardless of the shape of D.

2. *True* or *False?* Function $\vec{\mathbf{r}}(t) = \vec{\mathbf{a}} + t(\vec{\mathbf{b}} - \vec{\mathbf{a}})$, where $0 \le t \le 1$, parameterizes the straight-line segment from $\vec{\mathbf{a}}$ to $\vec{\mathbf{b}}$.

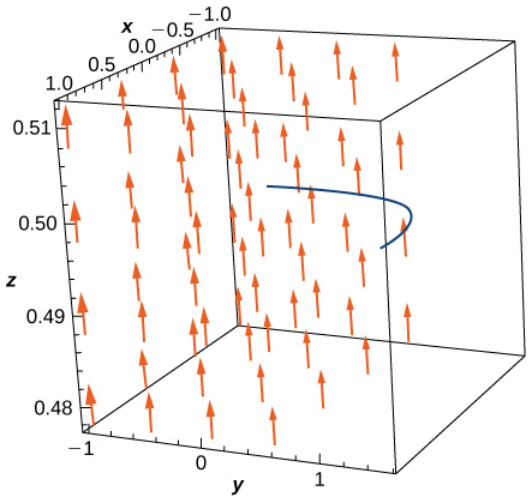
3. *True* or *False?* Vector field $\vec{\mathbf{F}}(x, y, z) = (y \sin z)\,\hat{\mathbf{i}} + (x \sin z)\,\hat{\mathbf{j}} + (xy \cos z)\,\hat{\mathbf{k}}$ is conservative.

4. *True* or *False?* Vector field $\vec{\mathbf{F}}(x, y, z) = y\,\hat{\mathbf{i}} + (x + z)\,\hat{\mathbf{j}} - y\,\hat{\mathbf{k}}$ is conservative.

5. Justify the Fundamental Theorem of Line Integrals for $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ in the case when $\vec{\mathbf{F}}(x, y) = (2x + 2y)\,\hat{\mathbf{i}} + (2x + 2y)\,\hat{\mathbf{j}}$ and C is a portion of the positively oriented circle $x^2 + y^2 = 25$ from $(5, 0)$ to $(3, 4)$.

6. [T] Find $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = (ye^{xy} + \cos x)\,\hat{\mathbf{i}} + \left(xe^{xy} + \frac{1}{y^2+1}\right)\,\hat{\mathbf{j}}$ and C is a portion of curve $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$.

7. [T] Evaluate line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = (e^x \sin y - y)\,\hat{\mathbf{i}} + (e^x \cos y - x - 2)\,\hat{\mathbf{j}}$, and C is the path given by $\vec{\mathbf{r}}(t) = (t^3 \sin \frac{\pi t}{2})\,\hat{\mathbf{i}} - (\frac{\pi}{2}\cos(\frac{\pi t}{2} + \frac{\pi}{2}))\,\hat{\mathbf{j}}$ for $0 \le t \le 1$.



For the following exercises, determine whether the vector field is conservative and, if it is, find the potential function.

8. $\vec{\mathbf{F}}(x, y) = 2xy^3\,\hat{\mathbf{i}} + 3y^2x^2\,\hat{\mathbf{j}}$

9. $\vec{\mathbf{F}}(x, y) = (-y + e^x \sin y)\,\hat{\mathbf{i}} + ((x + 2)e^x \cos y)\,\hat{\mathbf{j}}$

10. $\vec{\mathbf{F}}(x, y) = (e^{2x} \sin y)\,\hat{\mathbf{i}} + (e^{2x} \cos y)\,\hat{\mathbf{j}}$

11. $\vec{\mathbf{F}}(x, y) = (6x + 5y)\,\hat{\mathbf{i}} + (5x + 4y)\,\hat{\mathbf{j}}$

12. $\vec{\mathbf{F}}(x, y) = (2x \cos(y) - y \cos(x))\,\hat{\mathbf{i}} + (-x^2 \sin(y) - \sin(x))\,\hat{\mathbf{j}}$

13. $\vec{\mathbf{F}}(x, y) = (ye^x + \sin(y))\,\hat{\mathbf{i}} + (e^x + x \cos(y))\,\hat{\mathbf{j}}$

For the following exercises, evaluate the line integrals using the Fundamental Theorem of Line Integrals.

14. $\oint_C (y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}) \cdot d\vec{\mathbf{r}}$, where C is any path from $(0, 0)$ to $(2, 4)$

15. $\oint_C (2y\,dx + 2x\,dy)$, where C is the line segment from $(0, 0)$ to $(4, 4)$

16. [T] $\oint_C \left[\arctan\frac{y}{x} - \frac{xy}{x^2 + y^2}\right]\,dx + \left[\frac{x^2}{x^2 + y^2} + e^{-y}(1 - y)\right]\,dy$, where C is any smooth curve from $(1, 1)$ to $(-1, 2)$.

17. Find the conservative vector field for the potential function $f(x, y) = 5x^2 + 3xy + 10y^2$.

For the following exercises, determine whether the vector field is conservative and, if so, find a potential function.

18. $\vec{\mathbf{F}}(x,y) = (12xy)\,\hat{\mathbf{i}} + 6(x^2+y^2)\,\hat{\mathbf{j}}$

19. $\vec{\mathbf{F}}(x,y) = (e^x \cos y)\,\hat{\mathbf{i}} + 6(e^x \sin y)\,\hat{\mathbf{j}}$

20. $\vec{\mathbf{F}}(x,y) = (2xye^{x^2y})\,\hat{\mathbf{i}} + 6(x^2e^{x^2y})\,\hat{\mathbf{j}}$

21. $\vec{\mathbf{F}}(x,y,z) = (ye^z)\,\hat{\mathbf{i}} + (xe^z)\,\hat{\mathbf{j}} + (xye^z)\,\hat{\mathbf{k}}$

22. $\vec{\mathbf{F}}(x,y,z) = (\sin y)\,\hat{\mathbf{i}} - (x\cos y)\,\hat{\mathbf{j}} + \hat{\mathbf{k}}$

23. $\vec{\mathbf{F}}(x,y,z) = \dfrac{1}{y}\,\hat{\mathbf{i}} - \dfrac{x}{y^2}\,\hat{\mathbf{j}} + (2z-1)\,\hat{\mathbf{k}}$

24. $\vec{\mathbf{F}}(x,y,z) = 3z^2\,\hat{\mathbf{i}} - \cos y\,\hat{\mathbf{j}} + 2xz\,\hat{\mathbf{k}}$

25. $\vec{\mathbf{F}}(x,y,z) = (2xy)\,\hat{\mathbf{i}} + (x^2+2yz)\,\hat{\mathbf{j}} + y^2\,\hat{\mathbf{k}}$

For the following exercises, determine whether the given vector field is conservative and find a potential function.

26. $\vec{\mathbf{F}}(x,y) = (e^x \cos y)\,\hat{\mathbf{i}} + 6(e^x \sin y)\,\hat{\mathbf{j}}$

27. $\vec{\mathbf{F}}(x,y) = (2xye^{x^2y})\,\hat{\mathbf{i}} + (x^2e^{x^2y})\,\hat{\mathbf{j}}$

For the following exercises, evaluate the integral using the Fundamental Theorem of Line Integrals.

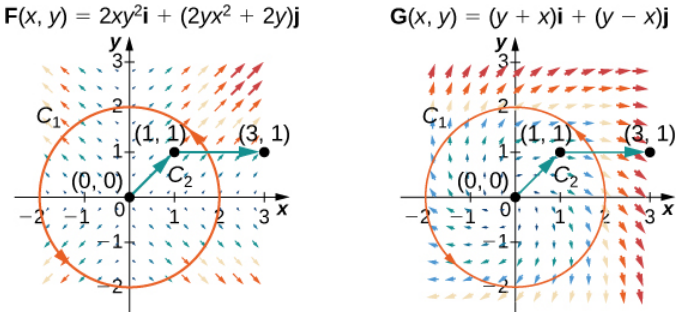
28. Evaluate $\displaystyle\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$, where $f(x,y,z) = \cos(\pi x) + \sin(\pi y) - xyz$ and C is any path that starts at $(1,12,2)$ and ends at $(2,1,-1)$.

29. [T] Evaluate $\displaystyle\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$, where $f(x,y) = xy + e^x$ and C is a straight line from $(0,0)$ to $(2,1)$.

30. [T] Evaluate $\displaystyle\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$, where $f(x,y) = x^2y - x$ and C is any path in a plane from (1, 2) to (3, 2).

31. Evaluate $\displaystyle\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}}$, where $f(x,y,z) = xyz^2 - yz$ and C has initial point $(1,2,3)$ and terminal point $(3,5,2)$.

For the following exercises, let $\vec{\mathbf{F}}(x,y) = 2xy^2\,\hat{\mathbf{i}} + (2yx^2+2y)\,\hat{\mathbf{j}}$ and $G(x,y) = (y+x)\,\hat{\mathbf{i}} + (y-x)\,\hat{\mathbf{j}}$, and let C_1 be the curve consisting of the circle of radius 2, centered at the origin and oriented counterclockwise, and C_2 be the curve consisting of a line segment from $(0,0)$ to $(1,1)$ followed by a line segment from $(1,1)$ to $(3,1)$.



32. Calculate the line integral of $\vec{\mathbf{F}}$ over C_1.

33. Calculate the line integral of $\vec{\mathbf{G}}$ over C_1.

34. Calculate the line integral of $\vec{\mathbf{F}}$ over C_2.

35. Calculate the line integral of $\vec{\mathbf{G}}$ over C_2.

36. [T] Let $\vec{\mathbf{F}}(x,y,z) = x^2\,\hat{\mathbf{i}} + z\sin(yz)\,\hat{\mathbf{j}} + y\sin(yz)\,\hat{\mathbf{k}}$. Calculate $\displaystyle\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is a path from $A = (0,0,1)$ to $B = (3,1,2)$.

37. [T] Find line integral $\displaystyle\oint_C \vec{\mathbf{F}} \cdot dr$ of vector field $\vec{\mathbf{F}}(x,y,z) = 3x^2z\,\hat{\mathbf{i}} + z^2\,\hat{\mathbf{j}} + (x^3+2yz)\,\hat{\mathbf{k}}$ along curve C parameterized by $\vec{\mathbf{r}}(t) = \left(\frac{\ln t}{\ln 2}\right)\hat{\mathbf{i}} + t^{3/2}\,\hat{\mathbf{j}} + t\cos(\pi t), 1 \le t \le 4$.

For exercises 38 - 40, show that the following vector fields are conservative. Then calculate $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ for the given curve.

38. $\vec{\mathbf{F}}(x,y) = (xy^2 + 3x^2y)\,\hat{\mathbf{i}} + (x+y)x^2\,\hat{\mathbf{j}}$; C is the curve consisting of line segments from $(1,1)$ to $(0,2)$ to $(3,0)$.

39. $\vec{\mathbf{F}}(x,y) = \dfrac{2x}{y^2+1}\,\hat{\mathbf{i}} - \dfrac{2y(x^2+1)}{(y^2+1)^2}\,\hat{\mathbf{j}}$; C is parameterized by $x = t^3 - 1,\ y = t^6 - t$, for $0 \le t \le 1$.

40. [T] $\vec{\mathbf{F}}(x,y) = [\cos(xy^2) - xy^2\sin(xy^2)]\,\hat{\mathbf{i}} - 2x^2y\sin(xy^2)\,\hat{\mathbf{j}}$; C is the curve $\langle e^t, e^{t+1}\rangle$, for $-1 \le t \le 0$.

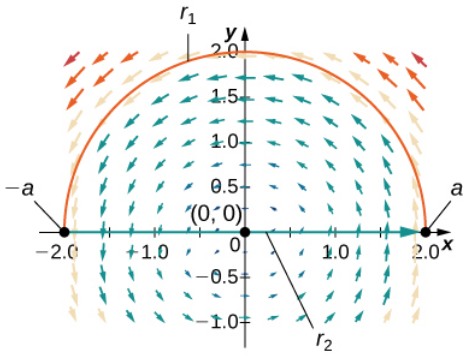
41. The mass of Earth is approximately $6 \times 10^{27} g$ and that of the Sun is 330,000 times as much. The gravitational constant is $6.7 \times 10^{-8} cm^3/s^2 \cdot g$. The distance of Earth from the Sun is about $1.5 \times 10^{12} cm$. Compute, approximately, the work necessary to increase the distance of Earth from the Sun by $1\ cm$.

42. [T] Let $\vec{\mathbf{F}}(x,y,z) = (e^x \sin y)\,\hat{\mathbf{i}} + (e^x \cos y)\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$. Evaluate the integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{r}}(t) = \langle \sqrt{t}, t^3, e^{\sqrt{t}}\rangle$, for $0 \le t \le 1$.

43. [T] Let $C : [1,2] \to \mathbb{R}^2$ be given by $x = e^{t-1}, y = \sin\left(\frac{\pi}{t}\right)$. Use a computer to compute the integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_C 2x\cos y\,dx - x^2 \sin y\,dy$, where $\vec{\mathbf{F}}(x,y) = (2x\cos y)\,\hat{\mathbf{i}} - (x^2\sin y)\,\hat{\mathbf{j}}$.

44. [T] Use a computer algebra system to find the mass of a wire that lies along the curve $\vec{\mathbf{r}}(t) = (t^2-1)\,\hat{\mathbf{j}} + 2t\,\hat{\mathbf{k}}$, where $0 \le t \le 1$, if the density is given by $d(t) = \dfrac{3}{2}t$.

45. Find the circulation and flux of field $\vec{\mathbf{F}}(x,y) = -y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}$ around and across the closed semicircular path that consists of semicircular arch $\vec{\mathbf{r}}_1(t) = (a\cos t)\,\hat{\mathbf{i}} + (a\sin t)\,\hat{\mathbf{j}}, \quad 0 \le t \le \pi$, followed by line segment $\vec{\mathbf{r}}_2(t) = t\,\hat{\mathbf{i}}, \quad -a \le t \le a$.



46. Compute $\int_C \cos x \cos y\,dx - \sin x \sin y\,dy$, where $\vec{\mathbf{r}}(t) = \langle t, t^2\rangle, \quad 0 \le t \le 1$.

47. Complete the proof of the theorem titled THE PATH INDEPENDENCE TEST FOR CONSERVATIVE FIELDS by showing that $f_y = Q(x,y)$.

Contributors

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16.4: Green's Theorem

Learning Objectives

- Apply the circulation form of Green's theorem.
- Apply the flux form of Green's theorem.
- Calculate circulation and flux on more general regions.

In this section, we examine Green's theorem, which is an extension of the Fundamental Theorem of Calculus to two dimensions. Green's theorem has two forms: a circulation form and a flux form, both of which require region D in the double integral to be simply connected. However, we will extend Green's theorem to regions that are not simply connected.

Put simply, Green's theorem relates a line integral around a simply closed plane curve C and a double integral over the region enclosed by C. The theorem is useful because it allows us to translate difficult line integrals into more simple double integrals, or difficult double integrals into more simple line integrals.

Extending the Fundamental Theorem of Calculus

Recall that the Fundamental Theorem of Calculus says that

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

As a geometric statement, this equation says that the integral over the region below the graph of $F'(x)$ and above the line segment $[a, b]$ depends only on the value of F at the endpoints a and b of that segment. Since the numbers a and b are the boundary of the line segment $[a, b]$, the theorem says we can calculate integral $\int_a^b F'(x)\,dx$ based on information about the boundary of line segment $[a, b]$ (Figure 16.4.1). The same idea is true of the Fundamental Theorem for Line Integrals:

$$\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(\vec{\mathbf{r}}(b)) - f(\vec{\mathbf{r}}(a)).$$

When we have a potential function (an "antiderivative"), we can calculate the line integral based solely on information about the boundary of curve C.

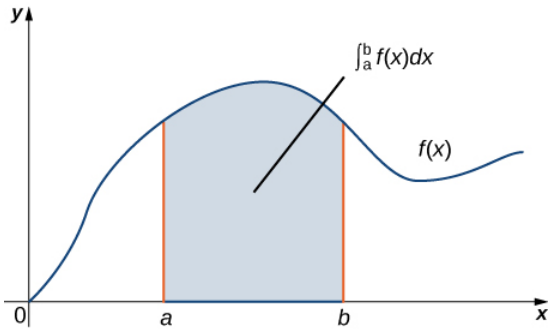


Figure 16.4.1: The Fundamental Theorem of Calculus says that the integral over line segment $[a, b]$ depends only on the values of the antiderivative at the endpoints of $[a, b]$.

Green's theorem takes this idea and extends it to calculating double integrals. Green's theorem says that we can calculate a double integral over region D based solely on information about the boundary of D. Green's theorem also says we can calculate a line integral over a simple closed curve C based solely on information about the region that C encloses. In particular, Green's theorem connects a double integral over region D to a line integral around the boundary of D.

Circulation Form of Green's Theorem

The first form of Green's theorem that we examine is the circulation form. This form of the theorem relates the vector line integral over a simple, closed plane curve C to a double integral over the region enclosed by C. Therefore, the circulation of a vector field along a simple closed curve can be transformed into a double integral and vice versa.

GREEN'S THEOREM (CIRCULATION FORM)

Let D be an open, simply connected region with a boundary curve C that is a piecewise smooth, simple closed curve oriented counterclockwise (Figure 16.4.2). Let $\vec{\mathbf{F}} = \langle P, Q \rangle$ be a vector field with component functions that have continuous partial derivatives on D. Then,

$$\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \oint_C P\,dx + Q\,dy \tag{16.4.1}$$

$$= \iint_D (Q_x - P_y)\,dA. \tag{16.4.2}$$

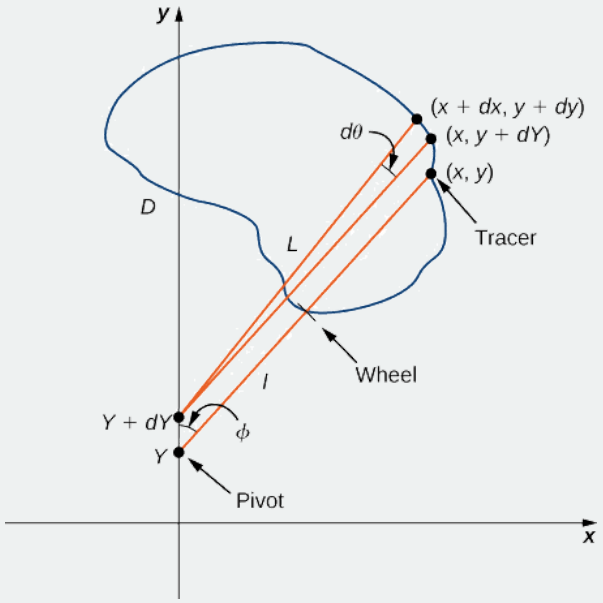


Figure 16.4.2: The circulation form of Green's theorem relates a line integral over curve C to a double integral over region D.

Notice that Green's theorem can be used only for a two-dimensional vector field $\vec{\mathbf{F}}$. If $\vec{\mathbf{F}}$ is a three-dimensional field, then Green's theorem does not apply. Since

$$\int_C P\,dx + Q\,dy = \int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}}\,ds$$

this version of Green's theorem is sometimes referred to as the ***tangential form* of Green's theorem.**

The proof of Green's theorem is rather technical, and beyond the scope of this text. Here we examine a proof of the theorem in the special case that D is a rectangle. For now, notice that we can quickly confirm that the theorem is true for the special case in which $\vec{\mathbf{F}} = \langle P, Q \rangle$ is conservative. In this case,

$$\oint_C P\,dx + Q\,dy = 0$$

because the circulation is zero in conservative vector fields. $\vec{\mathbf{F}}$ satisfies the cross-partial condition, so $P_y = Q_x$. Therefore,

$$\iint_D (Q_x - P_y)\,dA = \int_D 0\,dA = 0 = \oint_C P\,dx + Q\,dy$$

which confirms Green's theorem in the case of **conservative vector fields**.

Proof

Let's now prove that the circulation form of Green's theorem is true when the region D is a rectangle. Let D be the rectangle $[a, b] \times [c, d]$ oriented counterclockwise. Then, the boundary C of D consists of four piecewise smooth pieces C_1, C_2, C_3, and C_4 (Figure 16.4.3). We parameterize each side of D as follows:

$$C_1 : \vec{\mathbf{r}}_1(t) = \langle t, c \rangle,\ a \le t \le b$$

$$C_2 : \vec{\mathbf{r}}_2(t) = \langle b, t \rangle,\ c \le t \le d$$

$$-C_3 : \vec{\mathbf{r}}_3(t) = \langle t, d \rangle,\ a \le t \le b$$

$$-C_4 : \vec{\mathbf{r}}_4(t) = \langle a, t \rangle,\ c \le t \le d.$$

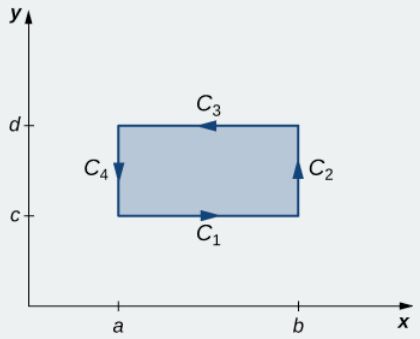


Figure 16.4.3: Rectangle D is oriented counterclockwise.

Then,

$$\begin{aligned}
\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_3} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\
&= \int_{C_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{C_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} - \int_{-C_3} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} - \int_{-C_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\
&= \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}_1(t)) \cdot \vec{\mathbf{r}}_1'(t)\, dt + \int_c^d \vec{\mathbf{F}}(\vec{\mathbf{r}}_2(t)) \cdot \vec{\mathbf{r}}_2'(t)\, dt \\
&\quad - \int_a^b \vec{\mathbf{F}}(\vec{\mathbf{r}}_3(t)) \cdot \vec{\mathbf{r}}_3'(t)\, dt - \int_c^d \vec{\mathbf{F}}(\vec{\mathbf{r}}_4(t)) \cdot \vec{\mathbf{r}}_4'(t)\, dt \\
&= \int_a^b P(t,c)\, dt + \int_c^d Q(b,t)\, dt - \int_a^b P(t,d)\, dt - \int_c^d Q(a,t)\, dt \\
&= \int_a^b (P(t,c) - P(t,d))\, dt + \int_c^d (Q(b,t) - Q(a,t))\, dt \\
&= -\int_a^b (P(t,d) - P(t,c))\, dt + \int_c^d (Q(b,t) - Q(a,t))\, dt.
\end{aligned}$$

By the Fundamental Theorem of Calculus,

$$P(t,d) - P(t,c) = \int_c^d \frac{\partial}{\partial y} P(t,y) dy$$

and

$$Q(b,t) - Q(a,t) = \int_a^b \frac{\partial}{\partial x} Q(x,t)\, dx.$$

Therefore,

$$\begin{aligned}
&-\int_a^b (P(t,d) - P(t,c))\, dt + \int_c^d (Q(b,t) - Q(a,t))\, dt \\
&= -\int_a^b \int_c^d \frac{\partial}{\partial y} P(t,y)\, dy\, dt + \int_c^d \int_a^b \frac{\partial}{\partial x} Q(x,t)\, dx\, dt.
\end{aligned}$$

But,

$$
\begin{aligned}
-\int_a^b \int_c^d \frac{\partial}{\partial y} P(t,y)\,dy\,dt + \int_c^d \int_a^b \frac{\partial}{\partial x} Q(x,t)\,dx\,dt &= -\int_a^b \int_c^d \frac{\partial}{\partial y} P(x,y)\,dy\,dx \\
&\quad + \int_c^d \int_a^b \frac{\partial}{\partial x} Q(x,y)\,dx\,dy \\
&= \int_a^b \int_c^d (Q_x - P_y)\,dy\,dx \\
&= \iint_D (Q_x - P_y)\,dA.
\end{aligned}
$$

Therefore, $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_D (Q_x - P_y)\,dA$ and we have proved Green's theorem in the case of a rectangle.

□

To prove Green's theorem over a general region D, we can decompose D into many tiny rectangles and use the proof that the theorem works over rectangles. The details are technical, however, and beyond the scope of this text.

Example 16.4.1: Applying Green's Theorem over a Rectangle

Calculate the line integral

$$\oint_C x^2 y\,dx + (y-3)\,dy,$$

where C is a rectangle with vertices $(1,1)$, $(4,1)$, $(4,5)$, and $(1,5)$ oriented counterclockwise.

Solution

Let $\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y)\rangle = \langle x^2 y, y-3\rangle$. Then, $Q_x(x,y) = 0$ and $P_y(x,y) = x^2$. Therefore, $Q_x - P_y = -x^2$.

Let D be the rectangular region enclosed by C (Figure 16.4.4). By Green's theorem,

$$
\begin{aligned}
\oint_C x^2 y\,dx + (y-3)\,dy &= \iint_D (Q_x - P_y)\,dA \\
&= \iint_D -x^2\,dA = \int_1^5 \int_1^4 -x^2\,dx\,dy \\
&= \int_1^5 -\frac{x^3}{3}\Big|_1^4\,dy \\
&= \int_1^5 \left(-\frac{64}{3} + \frac{1}{3}\right)\,dy \\
&= \int_1^5 -21\,dy = -84.
\end{aligned}
$$

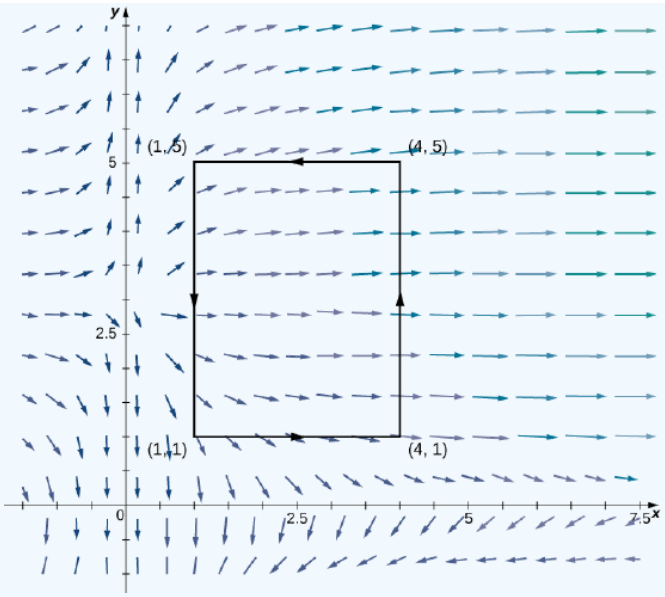


Figure 16.4.4: The line integral over the boundary of the rectangle can be transformed into a double integral over the rectangle.

Analysis

If we were to evaluate this line integral without using Green's theorem, we would need to parameterize each side of the rectangle, break the line integral into four separate line integrals, and use the methods from the section titled **Line Integrals** to evaluate each integral. Furthermore, since the vector field here is not conservative, we cannot apply the Fundamental Theorem for Line Integrals. Green's theorem makes the calculation much simpler.

Example 16.4.2: Applying Green's Theorem to Calculate Work

Calculate the work done on a particle by force field

$$\vec{\mathbf{F}}(x, y) = \langle y + \sin x, e^y - x \rangle$$

as the particle traverses circle $x^2 + y^2 = 4$ exactly once in the counterclockwise direction, starting and ending at point $(2, 0)$.

Solution

Let C denote the circle and let D be the disk enclosed by C. The work done on the particle is

$$W = \oint_C (y + \sin x)\, dx + (e^y - x)\, dy.$$

As with Example 16.4.1, this integral can be calculated using tools we have learned, but it is easier to use the double integral given by Green's theorem (Figure 16.4.5).

Let $\vec{\mathbf{F}}(x, y) = \langle P(x, y), Q(x, y) \rangle = \langle y + \sin x, e^y - x \rangle$. Then, $Q_x = -1$ and $P_y = 1$. Therefore, $Q_x - P_y = -2$.

By Green's theorem,

$$\begin{aligned} W &= \oint_C (y + \sin(x))dx + (e^y - x)\, dy \\ &= \iint_D (Q_x - P_y)\, dA \\ &= \iint_D -2\, dA \\ &= -2(area(D)) = -2\pi(2^2) = -8\pi. \end{aligned}$$

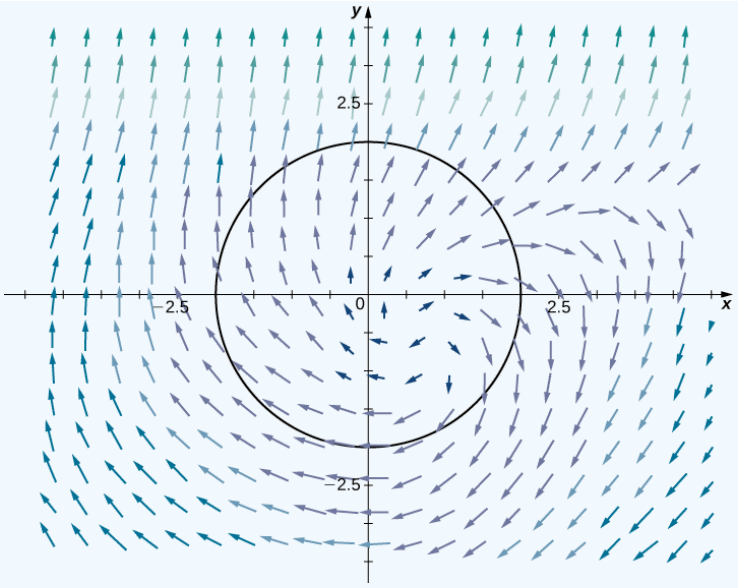


Figure 16.4.5: The line integral over the boundary circle can be transformed into a double integral over the disk enclosed by the circle.

Exercise 16.4.2

Use Green's theorem to calculate line integral

$$\oint_C \sin(x^2)\,dx + (3x - y)\,dy.$$

where C is a right triangle with vertices $(-1, 2)$, $(4, 2)$, and $(4, 5)$ oriented counterclockwise.

Hint

Transform the line integral into a double integral.

Answer

$\frac{45}{2}$

In the preceding two examples, the double integral in Green's theorem was easier to calculate than the line integral, so we used the theorem to calculate the line integral. In the next example, the double integral is more difficult to calculate than the line integral, so we use Green's theorem to translate a double integral into a line integral.

Example 16.4.3: Applying Green's Theorem over an Ellipse

Calculate the area enclosed by ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (Figure 16.4.6).

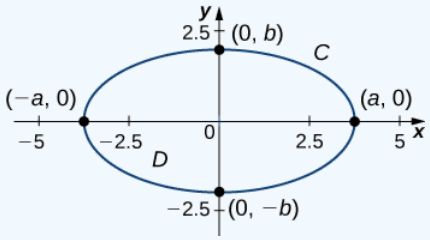


Figure 16.4.6: Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is denoted by C.

Solution

Let C denote the ellipse and let D be the region enclosed by C. Recall that ellipse C can be parameterized by

- $x = a\cos t$,
- $y = b\sin t$,
- $0 \le t \le 2\pi$.

Calculating the area of D is equivalent to computing double integral $\iint_D \, dA$. To calculate this integral without Green's theorem, we would need to divide D into two regions: the region above the x-axis and the region below. The area of the ellipse is

$$\int_{-a}^{a}\int_{0}^{\sqrt{b^2-(bx/a)^2}} \, dy\,dx + \int_{-a}^{a}\int_{-\sqrt{b^2-(bx/a)^2}}^{0} \, dy\,dx.$$

These two integrals are not straightforward to calculate (although when we know the value of the first integral, we know the value of the second by symmetry). Instead of trying to calculate them, we use Green's theorem to transform $\iint_D \, dA$ into a line integral around the boundary C.

Consider vector field

$$\vec{\mathbf{F}}(x,y) = \langle P, Q\rangle = \langle -\frac{y}{2}, \frac{x}{2}\rangle.$$

Then, $Q_x = \frac{1}{2}$ and $P_y = -\frac{1}{2}$, and therefore $Q_x - P_y = 1$. Notice that $\vec{\mathbf{F}}$ was chosen to have the property that $Q_x - P_y = 1$. Since this is the case, Green's theorem transforms the line integral of $\vec{\mathbf{F}}$ over C into the double integral of 1 over D.

By Green's theorem,

$$\begin{aligned}\iint_D dA &= \iint_D (Q_x - P_y)\, dA \\ &= \int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} = \frac{1}{2}\int_C -y\,dx + x\,dy \\ &= \frac{1}{2}\int_0^{2\pi} -b\sin t(-a\sin t) + a(\cos t)b\cos t\,dt \\ &= \frac{1}{2}\int_0^{2\pi} ab\cos^2 t + ab\sin^2 t\,dt \\ &= \frac{1}{2}\int_0^{2\pi} ab\,dt = \pi ab.\end{aligned}$$

Therefore, the area of the ellipse is πab units^2.

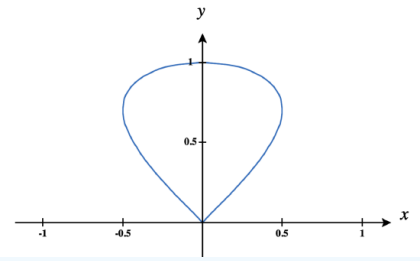
In Example 16.4.3, we used vector field $\vec{\mathbf{F}}(x,y) = \langle P, Q\rangle = \langle -\frac{y}{2}, \frac{x}{2}\rangle$ to find the area of any ellipse. The logic of the previous example can be extended to derive a formula for the area of any region D. Let D be any region with a boundary that is a simple closed curve C oriented counterclockwise. If $\vec{\mathbf{F}}(x,y) = \langle P, Q\rangle = \langle -\frac{y}{2}, \frac{x}{2}\rangle$, then $Q_x - P_y = 1$. Therefore, by the same logic as in Example 16.4.3,

$$\text{area of } D = \iint_D \, dA = \frac{1}{2}\oint_C -y\,dx + x\,dy. \tag{16.4.3}$$

It's worth noting that if $\vec{\mathbf{F}} = \langle P, Q\rangle$ is any vector field with $Q_x - P_y = 1$, then the logic of the previous paragraph works. So. Equation 16.4.3 is not the only equation that uses a vector field's mixed partials to get the area of a region.

? Exercise 16.4.3

Find the area of the region enclosed by the curve with parameterization $\vec{\mathbf{r}}(t) = \langle \sin t\cos t, \sin t\rangle$, $0 \le t \le \pi$.



Hint

Use Equation 16.4.3.

Answer

$\frac{2}{3}$ units^2.

Solution

For convenience, let's use the same vector field we used in Example 16.4.3, where $Q_x - P_y = 1$.

$$\vec{\mathbf{F}}(x,y) = \langle P, Q\rangle = \langle -\frac{y}{2}, \frac{x}{2}\rangle.$$

For the curve $r(t) = \langle \sin t \cos t, \sin t\rangle$ with $0 \le t \le \pi$, we have

$$\vec{\mathbf{r}}'(t) = \langle \cos^2 t - \sin^2 t, \cos t\rangle \text{ and}$$
$$\vec{\mathbf{F}}(t) = \left\langle -\frac{\sin t}{2}, \frac{\sin t \cos t}{2}\right\rangle$$

By Green's theorem,

$$
\begin{aligned}
\text{Area of the region } D = \iint_D dA &= \iint_D (Q_x - P_y)\, dA \\
&= \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \frac{1}{2}\int_C -y\, dx + x\, dy \\
&= \frac{1}{2}\int_0^{\pi} \Big(-\sin t(\cos^2 t - \sin^2 t) + \sin t \cos^2 t \Big)\, dt \\
&= \frac{1}{2}\int_0^{\pi} \Big(-\sin t \cos^2 t + \sin t \sin^2 t + \sin t \cos^2 t \Big)\, dt \\
&= \frac{1}{2}\int_0^{\pi} \Big(\cancel{-\sin t \cos^2 t} + \sin t(1 - \cos^2 t) + \cancel{\sin t \cos^2 t} \Big)\, dt \\
&= \frac{1}{2}\int_0^{\pi} \Big(\sin t - \cos^2 t \sin t \Big)\, dt \\
&= \frac{1}{2}\left[-\cos t + \frac{\cos^3 t}{3} \right] \Bigg|_0^{\pi} \\
&= \frac{1}{2}\left[-\cos \pi + \frac{\cos^3 \pi}{3} + \cos 0 - \frac{\cos^3 0}{3} \right] \\
&= \frac{1}{2}\left[1 - \frac{1}{3} + 1 - \frac{1}{3} \right] \\
&= \frac{1}{2}\left[2 - \frac{2}{3} \right] \\
&= \frac{1}{2}\left[\frac{6}{3} - \frac{2}{3} \right] \\
&= \frac{1}{2}\left[\frac{4}{3} \right] \\
&= \frac{2}{3} \text{ units}^2
\end{aligned}
$$

Therefore the area of this region is $\frac{2}{3}$ units^2.

Flux Form of Green's Theorem

The circulation form of Green's theorem relates a double integral over region D to line integral $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{T}} ds$, where C is the boundary of D. The flux form of Green's theorem relates a double integral over region D to the flux across boundary C. The flux of a fluid across a curve can be difficult to calculate using the flux line integral. This form of Green's theorem allows us to translate a difficult flux integral into a double integral that is often easier to calculate.

GREEN'S THEOREM (FLUX FORM)

Let D be an open, simply connected region with a boundary curve C that is a piecewise smooth, simple closed curve that is oriented counterclockwise (Figure 16.4.7). Let $\vec{\mathbf{F}} = \langle P, Q \rangle$ be a vector field with component functions that have continuous partial derivatives on an open region containing D. Then,

$$\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \iint_D P_x + Q_y\, dA. \tag{16.4.4}$$

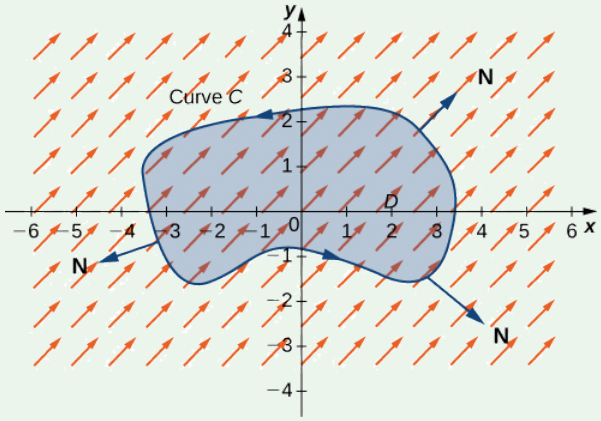


Figure 16.4.7: The flux form of Green's theorem relates a double integral over region D to the flux across curve C.

Because this form of Green's theorem contains unit normal vector $\vec{\mathbf{N}}$, it is sometimes referred to as the *normal form* of Green's theorem.

Proof

Recall that $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \oint_C -Q\, dx + P\, dy$. Let $M = -Q$ and $N = P$. By the circulation form of Green's theorem,

$$\begin{aligned} \oint_C -Q\, dx + P\, dy &= \oint_C M\, dx + N\, dy \\ &= \iint_D N_x - M_y\, dA \\ &= \iint_D P_x - (-Q)_y\, dA \\ &= \iint_D P_x + Q_y\, dA. \end{aligned}$$

□

Example 16.4.4A: Applying Green's Theorem for Flux across a Circle

Let C be a circle of radius r centered at the origin (Figure 16.4.8) and let $\vec{\mathbf{F}}(x, y) = \langle x, y \rangle$. Calculate the flux across C.

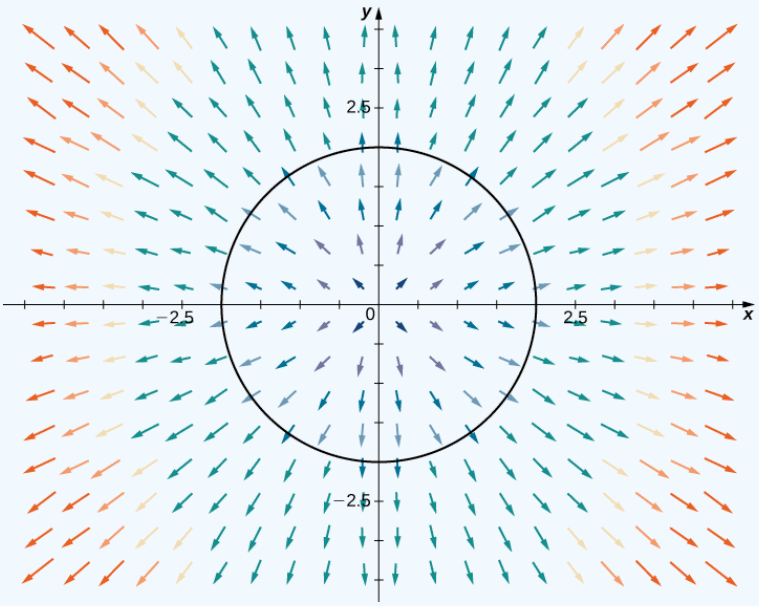


Figure 16.4.8: Curve C is a circle of radius r centered at the origin.

Solution

Let D be the disk enclosed by C. The flux across C is $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$. We could evaluate this integral using tools we have learned, but Green's theorem makes the calculation much more simple. Let $P(x,y) = x$ and $Q(x,y) = y$ so that $\vec{\mathbf{F}} = \langle P, Q\rangle$. Note that $P_x = 1 = Q_y$, and therefore $P_x + Q_y = 2$. By Green's theorem,

$$\int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \iint_D 2\, dA = 2 \iint_D dA.$$

Since $\iint_D dA$ is the area of the circle, $\iint_D dA = \pi r^2$. Therefore, the flux across C is $2\pi r^2$.

Example 16.4.4B: Applying Green's Theorem for Flux across a Triangle

Let S be the triangle with vertices $(0,0)$, $(1,0)$, and $(0,3)$ oriented clockwise (Figure 16.4.9). Calculate the flux of $\vec{\mathbf{F}}(x,y) = \langle P(x,y), Q(x,y)\rangle = \langle x^2 + e^y, x + y\rangle$ across S.

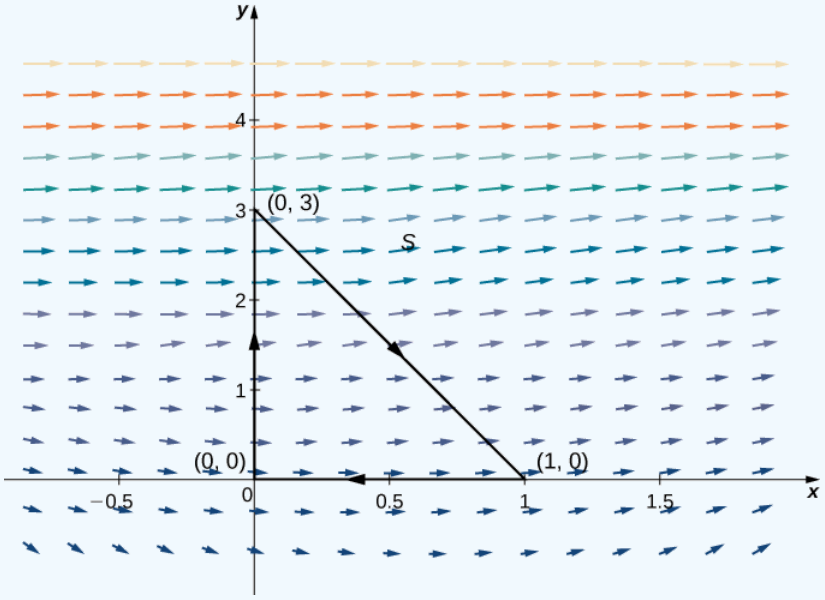


Figure 16.4.9: Curve S is a triangle with vertices $(0,0)$, $(1,0)$, and $(0,3)$ oriented clockwise.

Solution

To calculate the flux without Green's theorem, we would need to break the flux integral into three line integrals, one integral for each side of the triangle. Using Green's theorem to translate the flux line integral into a single double integral is much more simple.

Let D be the region enclosed by S. Note that $P_x = 2x$ and $Q_y = 1$; therefore, $P_x + Q_y = 2x + 1$. Green's theorem applies only to simple closed curves oriented counterclockwise, but we can still apply the theorem because $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = -\oint_{-S} \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$ and $-S$ is oriented counterclockwise. By Green's theorem, the flux is

$$\begin{aligned}\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds &= \oint_{-S} \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds \\ &= -\iint_D (P_x + Q_y)\, dA \\ &= -\iint_D (2x+1)\, dA.\end{aligned}$$

Notice that the top edge of the triangle is the line $y = -3x + 3$. Therefore, in the iterated double integral, the y-values run from $y = 0$ to $y = -3x + 3$, and we have

$$
\begin{aligned}
-\iint_D (2x+1)\,dA &= -\int_0^1 \int_0^{-3x+3} (2x+1)\,dy\,dx \\
&= -\int_0^1 (2x+1)y\Big|_0^{-3x+3}\,dx \\
&= -\int_0^1 (2x+1)(-3x+3)\,dx \\
&= -\int_0^1 (-6x^2+3x+3)\,dx \\
&= -\left[-2x^3+\frac{3x^2}{2}+3x\right]\Big|_0^1 \\
&= -\frac{5}{2}.
\end{aligned}
$$

Exercise 16.4.4

Calculate the flux of $\vec{\mathbf{F}}(x,y) = \langle x^3, y^3\rangle$ across a unit circle oriented counterclockwise.

Hint

Apply Green's theorem and use polar coordinates.

Answer

$\frac{3\pi}{2}$

Example 16.4.5: Applying Green's Theorem for Water Flow across a Rectangle

Water flows from a spring located at the origin. The velocity of the water is modeled by vector field $\vec{\mathbf{v}}(x,y) = \langle 5x+y, x+3y\rangle$ m/sec. Find the amount of water per second that flows across the rectangle with vertices $(-1,-2)$, $(1,-2)$, $(1,3)$,and $(-1,3)$, oriented counterclockwise (Figure 16.4.10).

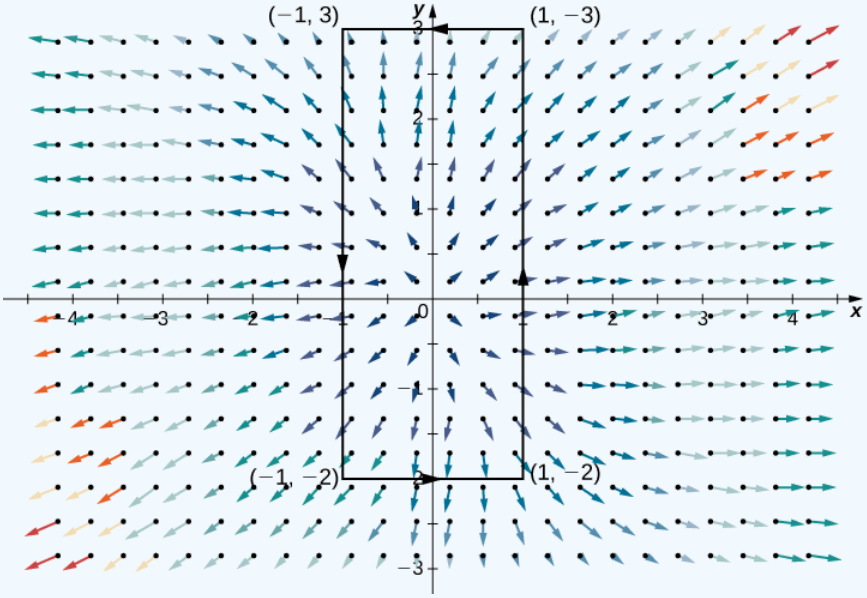


Figure 16.4.10: Water flows across the rectangle with vertices $(-1,-2)$, $(1,-2)$, $(1,3)$,and $(-1,3)$, oriented counterclockwise.

Solution

Let C represent the given rectangle and let D be the rectangular region enclosed by C. To find the amount of water flowing across C, we calculate flux $\int_C \vec{\mathbf{v}} \cdot d\vec{\mathbf{r}}$. Let $P(x,y) = 5x+y$ and $Q(x,y) = x+3y$ so that $\vec{\mathbf{v}} = \langle P, Q\rangle$. Then, $P_x = 5$ and $Q_y = 3$. By

Green's theorem,

$$\begin{aligned}\int_C \vec{\mathbf{v}} \cdot d\vec{\mathbf{r}} &= \iint_D (P_x + Q_y)\, dA \\ &= \iint_D 8\, dA \\ &= 8(area\ of\ D) = 80.\end{aligned}$$

Therefore, the water flux is 80 m^2/sec.

Recall that if vector field $\vec{\mathbf{F}}$ is conservative, then $\vec{\mathbf{F}}$ does no work around closed curves—that is, the circulation of $\vec{\mathbf{F}}$ around a closed curve is zero. In fact, if the domain of $\vec{\mathbf{F}}$ is simply connected, then $\vec{\mathbf{F}}$ is conservative if and only if the circulation of $\vec{\mathbf{F}}$ around any closed curve is zero. If we replace "circulation of $\vec{\mathbf{F}}$" with "flux of $\vec{\mathbf{F}}$," then we get a definition of a source-free vector field. The following statements are all equivalent ways of defining a source-free field $\vec{\mathbf{F}} = \langle P, Q \rangle$ on a simply connected domain (note the similarities with properties of conservative vector fields):

1. The flux $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$ across any closed curve C is zero.
2. If C_1 and C_2 are curves in the domain of $\vec{\mathbf{F}}$ with the same starting points and endpoints, then $\int_{C_1} \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \int_{C_2} \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds$. In other words, flux is independent of path.
3. There is a stream function $g(x, y)$ for $\vec{\mathbf{F}}$. A stream function for $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a function g such that $P = g_y$ and $Q = -g_x$.Geometrically, $\vec{\mathbf{F}} = \langle a, b \rangle$ is tangential to the level curve of g at (a, b). Since the gradient of g is perpendicular to the level curve of g at (a, b), stream function g has the property $\vec{\mathbf{F}}(a, b) \cdot \vec{\nabla} g(a, b) = 0$ for any point (a, b) in the domain of g. (Stream functions play the same role for source-free fields that potential functions play for conservative fields.)
4. $P_x + Q_y = 0$

Example 16.4.6: Finding a Stream Function

Verify that rotation vector field $\vec{\mathbf{F}}(x, y) = \langle y, -x \rangle$ is source free, and find a stream function for $\vec{\mathbf{F}}$.

Solution

Note that the domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^2$, which is simply connected. Therefore, to show that $\vec{\mathbf{F}}$ is source free, we can show any of items 1 through 4 from the previous list to be true. In this example, we show that item 4 is true. Let $P(x, y) = y$ and $Q(x, y) = -x$. Then $P_x = 0 = Q_y$, and therefore $P_x + Q_y = 0$. Thus, $\vec{\mathbf{F}}$ is source free.

To find a stream function for $\vec{\mathbf{F}}$, proceed in the same manner as finding a potential function for a conservative field. Let g be a stream function for $\vec{\mathbf{F}}$. Then $g_y = y$, which implies that

$$g(x, y) = \frac{y^2}{2} + h(x).$$

Since $-g_x = Q = -x$, we have $h'(x) = x$. Therefore,

$$h(x) = \frac{x^2}{2} + C.$$

Letting $C = 0$ gives stream function

$$g(x, y) = \frac{x^2}{2} + \frac{y^2}{2}.$$

To confirm that g is a stream function for $\vec{\mathbf{F}}$, note that $g_y = y = P$ and $-g_x = -x = Q$.

Notice that source-free rotation vector field $\vec{\mathbf{F}}(x, y) = \langle y, -x \rangle$ is perpendicular to conservative radial vector field $\vec{\nabla} g = \langle x, y \rangle$ (Figure 16.4.11).

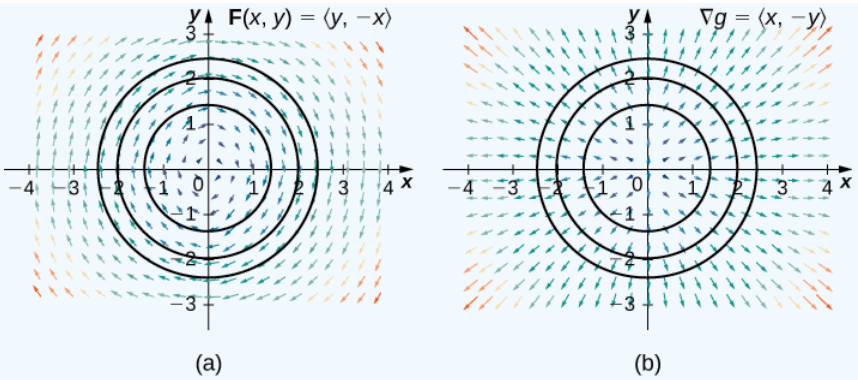


Figure 16.4.11: (a) In this image, we see the three-level curves of g and vector field $\vec{\mathbf{F}}$. Note that the $\vec{\mathbf{F}}$ vectors on a given level curve are tangent to the level curve. (b) In this image, we see the three-level curves of g and vector field $\vec{\nabla} g$. The gradient vectors are perpendicular to the corresponding level curve. Therefore, $\vec{\mathbf{F}}(a,b) \cdot \vec{\nabla} g(a,b) = 0$ for any point in the domain of g.

? Exercise 16.4.6

Find a stream function for vector field $\vec{\mathbf{F}}(x,y) = \langle x \sin y, \cos y \rangle$.

Hint

Follow the outline provided in the previous example.

Answer

$g(x,y) = -x \cos y$

Vector fields that are both conservative and source free are important vector fields. One important feature of conservative and source-free vector fields on a simply connected domain is that any potential function f of such a field satisfies Laplace's equation $f_{xx} + f_{yy} = 0$. Laplace's equation is foundational in the field of partial differential equations because it models such phenomena as gravitational and magnetic potentials in space, and the velocity potential of an ideal fluid. A function that satisfies Laplace's equation is called a *harmonic* function. Therefore any potential function of a conservative and source-free vector field is harmonic.

To see that any potential function of a conservative and source-free vector field on a simply connected domain is harmonic, let f be such a potential function of vector field $\vec{\mathbf{F}} = \langle P, Q \rangle$. Then, $f_x = P$ and $f_x = Q$ because $\vec{\nabla} f = \vec{\mathbf{F}}$. Therefore, $f_{xx} = P_x$ and $f_{yy} = Q_y$. Since $\vec{\mathbf{F}}$ is source free, $f_{xx} + f_{yy} = P_x + Q_y = 0$, and we have that f is harmonic.

✓ Example 16.4.7: Satisfying Laplace's Equation

For vector field $\vec{\mathbf{F}}(x,y) = \langle e^x \sin y, e^x \cos y \rangle$, verify that the field is both conservative and source free, find a potential function for $\vec{\mathbf{F}}$, and verify that the potential function is harmonic.

Solution

Let $P(x,y) = e^x \sin y$ and $Q(x,y) = e^x \cos y$. Notice that the domain of $\vec{\mathbf{F}}$ is all of two-space, which is simply connected. Therefore, we can check the cross-partials of $\vec{\mathbf{F}}$ to determine whether $\vec{\mathbf{F}}$ is conservative. Note that $P_y = e^x \cos y = Q_x$, so $\vec{\mathbf{F}}$ is conservative. Since $P_x = e^x \sin y$ and $Q_y = e^x \sin y$, $P_x + Q_y = 0$ and the field is source free.

To find a potential function for $\vec{\mathbf{F}}$, let f be a potential function. Then, $\vec{\nabla} f = \vec{\mathbf{F}}$, so $f_x(x,y) = e^x \sin y$. Integrating this equation with respect to x gives $f(x,y) = e^x \sin y + h(y)$. Since $f_y(x,y) = e^x \cos y$, differentiating f with respect to y gives $e^x \cos y = e^x \cos y + h'(y)$. Therefore, we can take $h(y) = 0$, and $f(x,y) = e^x \sin y$ is a potential function for f.

To verify that f is a harmonic function, note that $f_{xx}(x,y) = \dfrac{\partial}{\partial x}(e^x \sin y) = e^x \sin y$ and

$f_{yy}(x,y) = \dfrac{\partial}{\partial x}(e^x \cos y) = -e^x \sin y$. Therefore, $f_{xx} + f_{yy} = 0$, and f satisfies Laplace's equation.

Exercise 16.4.7

Is the function $f(x,y) = e^{x+5y}$ harmonic?

Hint

Determine whether the function satisfies Laplace's equation.

Answer

No

Green's Theorem on General Regions

Green's theorem, as stated, applies only to regions that are simply connected—that is, Green's theorem as stated so far cannot handle regions with holes. Here, we extend Green's theorem so that it does work on regions with finitely many holes (Figure 16.4.12).

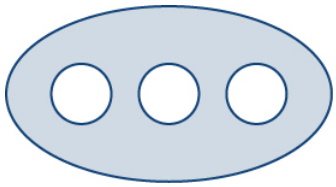


Figure 16.4.12: Green's theorem, as stated, does not apply to a nonsimply connected region with three holes like this one.

Before discussing extensions of Green's theorem, we need to go over some terminology regarding the boundary of a region. Let D be a region and let C be a component of the boundary of D. We say that C is *positively oriented* if, as we walk along C in the direction of orientation, region D is always on our left. Therefore, the counterclockwise orientation of the boundary of a disk is a positive orientation, for example. Curve C is *negatively oriented* if, as we walk along C in the direction of orientation, region D is always on our right. The clockwise orientation of the boundary of a disk is a negative orientation, for example.

Let D be a region with finitely many holes (so that D has finitely many boundary curves), and denote the boundary of D by ∂D (Figure 16.4.13). To extend Green's theorem so it can handle D, we divide region D into two regions, D_1 and D_2 (with respective boundaries ∂D_1 and ∂D_2), in such a way that $D = D_1 \cup D_2$ and neither D_1 nor D_2 has any holes (Figure 16.4.13).

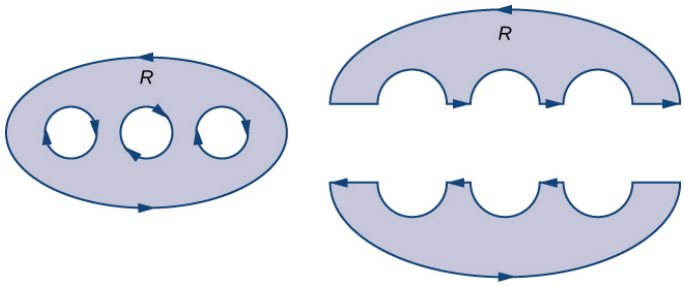


Figure 16.4.13: (a) Region D with an oriented boundary has three holes. (b) Region D split into two simply connected regions has no holes.

Assume the boundary of D is oriented as in the figure, with the inner holes given a negative orientation and the outer boundary given a positive orientation. The boundary of each simply connected region D_1 and D_2 is positively oriented. If $\vec{\mathbf{F}}$ is a vector field defined on D, then Green's theorem says that

$$\oint_{\partial D} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \oint_{\partial D_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \oint_{\partial D_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \tag{16.4.5}$$

$$= \iint_{D_1} Q_x - P_y \, dA + \iint_{D_2} Q_x - P_y \, dA \tag{16.4.6}$$

$$= \iint_{D} (Q_x - P_y) \, dA. \tag{16.4.7}$$

Therefore, Green's theorem still works on a region with holes.

To see how this works in practice, consider annulus D in Figure 16.4.14 and suppose that $F = \langle P, Q \rangle$ is a vector field defined on this annulus. Region D has a hole, so it is not simply connected. Orient the outer circle of the annulus counterclockwise and the inner circle clockwise (Figure 16.4.14) so that, when we divide the region into D_1 and D_2, we are able to keep the region on our left as we walk along a path that traverses the boundary. Let D_1 be the upper half of the annulus and D_2 be the lower half. Neither of these regions has holes, so we have divided D into two simply connected regions.

We label each piece of these new boundaries as P_i for some i, as in Figure 16.4.14. If we begin at P and travel along the oriented boundary, the first segment is P_1, then P_2, P_3, and P_4. Now we have traversed D_1 and returned to P. Next, we start at P again and traverse D_2. Since the first piece of the boundary is the same as P_4 in D_1, but oriented in the opposite direction, the first piece of D_2 is $-P_4$. Next, we have P_5, then $-P_2$, and finally P_6.

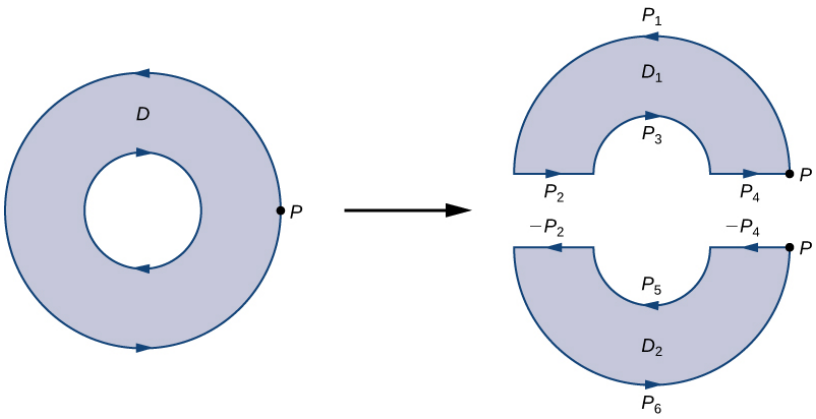


Figure 16.4.14: Breaking the annulus into two separate regions gives us two simply connected regions. The line integrals over the common boundaries cancel out.

Figure 16.4.14 shows a path that traverses the boundary of D. Notice that this path traverses the boundary of region D_1, returns to the starting point, and then traverses the boundary of region D_2. Furthermore, as we walk along the path, the region is always on our left. Notice that this traversal of the P_i paths covers the entire boundary of region D. If we had only traversed one portion of the boundary of D, then we cannot apply Green's theorem to D.

The boundary of the upper half of the annulus, therefore, is $P_1 \cup P_2 \cup P_3 \cup P_4$ and the boundary of the lower half of the annulus is $-P_4 \cup P_5 \cup -P_2 \cup P_6$. Then, Green's theorem implies

$$\begin{aligned}
\oint_{\partial D} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_{P_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_3} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{-P_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} && (16.4.8)\\
&\quad + \int_{P_5} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{-P_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_6} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\
&= \int_{P_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_3} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_4} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} && (16.4.9)\\
&\quad + \int_{P_5} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{-P_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_6} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\
&= \int_{P_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_3} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_5} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \int_{P_6} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} && (16.4.10)\\
&= \oint_{\partial D_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} + \oint_{\partial D_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} && (16.4.11)\\
&= \iint_{D_1} (Q_x - P_y)\, dA + \iint_{D_2} (Q_x - P_y)\, dA && (16.4.12)\\
&= \iint_{D} (Q_x - P_y)\, dA. && (16.4.13)
\end{aligned}$$

Therefore, we arrive at the equation found in Green's theorem—namely,

$$\oint_{\partial D} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_{D} (Q_x - P_y)\, dA.$$

The same logic implies that the flux form of Green's theorem can also be extended to a region with finitely many holes:

$$\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \iint_D (P_x + Q_y)\, dA.$$

Example 16.4.8A: Using Green's Theorem on a Region with Holes

Calculate the integral

$$\oint_{\partial D} (\sin x - \frac{y^3}{3})dx + (\frac{y^3}{3} + \sin y)dy,$$

where D is the annulus given by the polar inequalities $1 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$.

Solution

Although D is not simply connected, we can use the extended form of Green's theorem to calculate the integral. Since the integration occurs over an annulus, we convert to polar coordinates:

$$\begin{aligned}\oint_{\partial D} (\sin x - \frac{y^3}{3})\, dx + (\frac{x^3}{3} + \sin y)\, dy &= \iint_D (Q_x - P_y)\, dA \\ &= \iint_D (x^2 + y^2)\, dA \\ &= \int_0^{2\pi} \int_1^2 r^3\, dr d\theta = \int_0^{2\pi} \frac{15}{4}\, d\theta \\ &= \frac{15\pi}{2}.\end{aligned}$$

Example 16.4.8B: Using the Extended Form of Green's Theorem

Let $\vec{\mathbf{F}} = \langle P, Q\rangle = \left\langle \frac{y}{x^2+y^2}, -\frac{x}{x^2+y^2} \right\rangle$ and let C be any simple closed curve in a plane oriented counterclockwise. What are the possible values of $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$?

Solution

We use the extended form of Green's theorem to show that $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ is either 0 or -2π—that is, no matter how crazy curve C is, the line integral of $\vec{\mathbf{F}}$ along C can have only one of two possible values. We consider two cases: the case when C encompasses the origin and the case when C does not encompass the origin.

Case 1: C Does Not Encompass the Origin

In this case, the region enclosed by C is simply connected because the only hole in the domain of $\vec{\mathbf{F}}$ is at the origin. We showed in our discussion of cross-partials that $\vec{\mathbf{F}}$ satisfies the cross-partial condition. If we restrict the domain of $\vec{\mathbf{F}}$ just to C and the region it encloses, then $\vec{\mathbf{F}}$ with this restricted domain is now defined on a simply connected domain. Since $\vec{\mathbf{F}}$ satisfies the cross-partial property on its restricted domain, the field $\vec{\mathbf{F}}$ is conservative on this simply connected region and hence the circulation $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ is zero.

Case 2: C Does Encompass the Origin

In this case, the region enclosed by C is not simply connected because this region contains a hole at the origin. Let C_1 be a circle of radius a centered at the origin so that C_1 is entirely inside the region enclosed by C (Figure 16.4.15). Give C_1 a clockwise orientation.

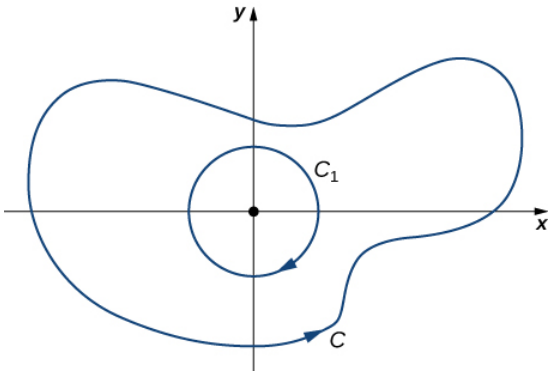


Figure 16.4.15: Choose circle C_1 centered at the origin that is contained entirely inside C.

Let D be the region between C_1 and C, and C is orientated counterclockwise. By the extended version of Green's theorem,

$$\begin{aligned}\int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} + \int_{C_1} \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} &= \iint_D Qx_P_y\, dA \\ &= \iint_D -\frac{y^2-x^2}{(x^2+y^2)^2} + \frac{y^2-x^2}{(x^2+y^2)^2} dA \\ &= 0,\end{aligned}$$

and therefore

$$\int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} = -\int_{C_1} \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}.$$

Since C_1 is a specific curve, we can evaluate $\int_{C_1} \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}$. Let

$$x = a\cos t, \quad y = a\sin t, \quad 0 \le t \le 2\pi$$

be a parameterization of C_1. Then,

$$\begin{aligned}\int_{C_1} \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} &= \int_0^{2\pi} \vec{\mathbf{F}}(\vec{\mathbf{r}}(t))\cdot \vec{\mathbf{r}}'(t)\, dt \\ &= \int_0^{2\pi} \langle -\frac{\sin(t)}{a}, -\frac{\cos(t)}{a}\rangle \cdot \langle -a\sin(t), -a\cos(t)\rangle\, dt \\ &= \int_0^{2\pi} \sin^2(t) + \cos^2(t)\, dt \\ &= \int_0^{2\pi} dt = 2\pi.\end{aligned}$$

Therefore, $\int_C \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}} = -2\pi$.

? Exercise 16.4.8

Calculate integral $\oint_{\partial D} \vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}$, where D is the annulus given by the polar inequalities $2 \le r \le 5$, $0 \le \theta \le 2\pi$, and $\vec{\mathbf{F}}(x,y) = \langle x^3, 5x + e^y \sin y\rangle$.

Hint

Use the extended version of Green's theorem.

Answer

105π

MEASURING AREA FROM A BOUNDARY: THE PLANIMETER

Imagine you are a doctor who has just received a magnetic resonance image of your patient's brain. The brain has a tumor (Figure 16.4.16). How large is the tumor? To be precise, what is the area of the red region? The red cross-section of the tumor has an irregular shape, and therefore it is unlikely that you would be able to find a set of equations or inequalities for the region and then be able to calculate its area by conventional means. You could approximate the area by chopping the region into tiny squares (a Riemann sum approach), but this method always gives an answer with some error.

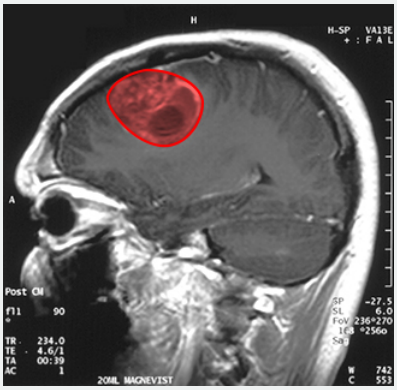


Figure 16.4.16: This magnetic resonance image of a patient's brain shows a tumor, which is highlighted in red. (credit: modification of work by Christaras A, Wikimedia Commons)

Instead of trying to measure the area of the region directly, we can use a device called a *rolling planimeter* to calculate the area of the region exactly, simply by measuring its boundary. In this project you investigate how a planimeter works, and you use Green's theorem to show the device calculates area correctly.

A rolling planimeter is a device that measures the area of a planar region by tracing out the boundary of that region (Figure 16.4.17). To measure the area of a region, we simply run the tracer of the planimeter around the boundary of the region. The planimeter measures the number of turns through which the wheel rotates as we trace the boundary; the area of the shape is proportional to this number of wheel turns. We can derive the precise proportionality equation using Green's theorem. As the tracer moves around the boundary of the region, the tracer arm rotates and the roller moves back and forth (but does not rotate).

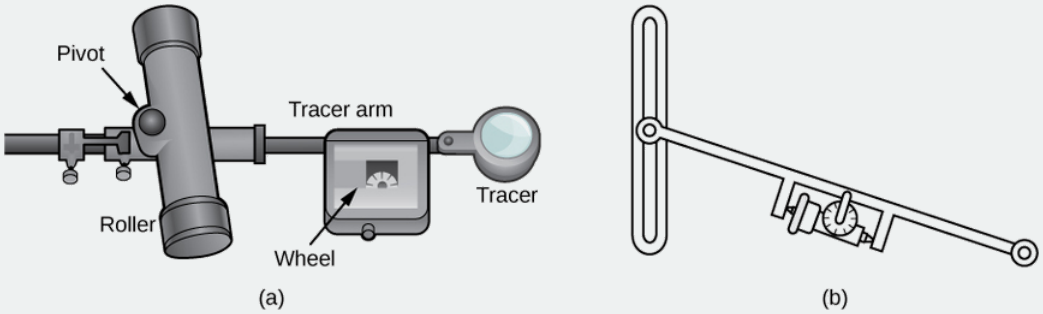


Figure 16.4.17: (a) A rolling planimeter. The pivot allows the tracer arm to rotate. The roller itself does not rotate; it only moves back and forth. (b) An interior view of a rolling planimeter. Notice that the wheel cannot turn if the planimeter is moving back and forth with the tracer arm perpendicular to the roller.

Let C denote the boundary of region D, the area to be calculated. As the tracer traverses curve C, assume the roller moves along the *y*-axis (since the roller does not rotate, one can assume it moves along a straight line). Use the coordinates (x, y) to represent points on boundary C, and coordinates $(0, Y)$ to represent the position of the pivot. As the planimeter traces C, the pivot moves along the *y*-axis while the tracer arm rotates on the pivot.

Watch a short animation of a planimeter in action.

Begin the analysis by considering the motion of the tracer as it moves from point (x, y) counterclockwise to point $(x + dx, y + dy)$ that is close to (x, y) (Figure 16.4.18). The pivot also moves, from point $(0, Y)$ to nearby point $(0, Y + dY)$. How much does the wheel turn as a result of this motion? To answer this question, break the motion into two parts. First, roll the pivot along the *y*-axis from $(0, Y)$ to $(0, Y + dY)$ without rotating the tracer arm. The tracer arm then ends up at point $(x, y + dY)$ while

maintaining a constant angle ϕ with the x-axis. Second, rotate the tracer arm by an angle $d\theta$ without moving the roller. Now the tracer is at point $(x + dx, y + dy)$. Let ll be the distance from the pivot to the wheel and let L be the distance from the pivot to the tracer (the length of the tracer arm).

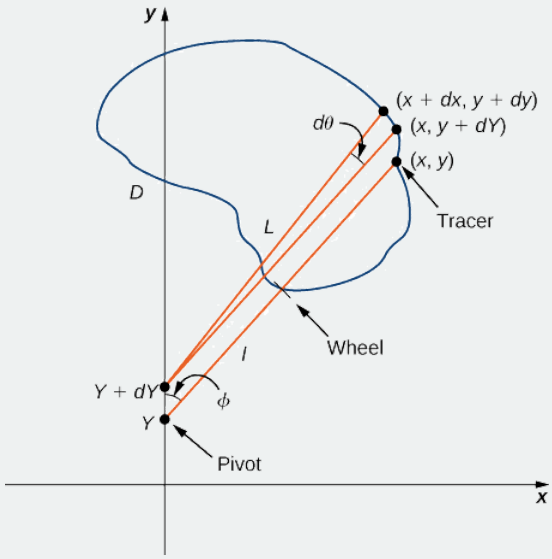


Figure 16.4.18: Mathematical analysis of the motion of the planimeter.

1. Explain why the total distance through which the wheel rolls the small motion just described is $\sin\phi dY + ld\theta = \frac{x}{L}dY + ld\theta$.
2. Show that $\oint_C d\theta = 0$.
3. Use step 2 to show that the total rolling distance of the wheel as the tracer traverses curve C is

   Total wheel roll $= \frac{1}{L}\oint_C x dY$.

   Now that you have an equation for the total rolling distance of the wheel, connect this equation to Green's theorem to calculate area D enclosed by C.
4. Show that $x^2 + (y - Y)^2 = L^2$.
5. Assume the orientation of the planimeter is as shown in Figure 16.4.18. Explain why $Y \leq y$, and use this inequality to show there is a unique value of Y for each point (x, y): $Y = y = \sqrt{L^2 - x^2}$.
6. Use step 5 to show that $dY = dy + \frac{x}{L^2 - x^2}dx$.
7. Use Green's theorem to show that $\oint_C \frac{x}{L^2 - x^2}dx = 0$.
8. Use step 7 to show that the total wheel roll is

$$\text{Total wheel roll} \quad = \quad 1L\oint_C x\,dy.$$

   It took a bit of work, but this equation says that the variable of integration Y in step 3 can be replaced with y.
9. Use Green's theorem to show that the area of D is $\oint_C x\,dy$. The logic is similar to the logic used to show that the area of D $= 12\oint_C -y\,dx + x\,dy$.
10. Conclude that the area of D equals the length of the tracer arm multiplied by the total rolling distance of the wheel.

You now know how a planimeter works and you have used Green's theorem to justify that it works. To calculate the area of a planar region D, use a planimeter to trace the boundary of the region. The area of the region is the length of the tracer arm multiplied by the distance the wheel rolled.

Key Concepts

- Green's theorem relates the integral over a connected region to an integral over the boundary of the region. Green's theorem is a version of the Fundamental Theorem of Calculus in one higher dimension.

- Green's Theorem comes in two forms: a circulation form and a flux form. In the circulation form, the integrand is $\vec{\mathbf{F}} \cdot \vec{\mathbf{T}}$. In the flux form, the integrand is $\vec{\mathbf{F}} \cdot \vec{\mathbf{N}}$.
- Green's theorem can be used to transform a difficult line integral into an easier double integral, or to transform a difficult double integral into an easier line integral.
- A vector field is source free if it has a stream function. The flux of a source-free vector field across a closed curve is zero, just as the circulation of a conservative vector field across a closed curve is zero.

Key Equations

- **Green's theorem, circulation form**
  $\oint_C P\,dx + Q\,dy = \iint_D Q_x - P_y\,dA$, where C is the boundary of D
- **Green's theorem, flux form**
  $\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\,ds = \iint_D P_x + Q_y\,dA$, where C is the boundary of D
- **Green's theorem, extended version**
  $\oint_{\partial D} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_D Q_x - P_y\,dA$

Glossary

Green's theorem

relates the integral over a connected region to an integral over the boundary of the region

stream function

if $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a source-free vector field, then stream function g is a function such that $P = g_y$ and $Q = -g_x$

Contributions

Paul Seeburger created a corrected image and the worked out solution for Exercise 16.4.3.

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16.4E: Exercises for Section 16.4

For the following exercises, evaluate the line integrals by applying Green's theorem.

1. $\int_C 2xy\,dx + (x+y)\,dy$, where C is the path from $(0,0)$ to $(1,1)$ along the graph of $y = x^3$ and from $(1,1)$ to $(0,0)$ along the graph of $y = x$ oriented in the counterclockwise direction

2. $\int_C 2xy\,dx + (x+y)\,dy$, where C is the boundary of the region lying between the graphs of $y = 0$ and $y = 4 - x^2$ oriented in the counterclockwise direction

3. $\int_C 2\arctan\left(\frac{y}{x}\right)\,dx + \ln(x^2+y^2)\,dy$, where C is defined by $x = 4 + 2\cos\theta,\ y = 4\sin\theta$ oriented in the counterclockwise direction

4. $\int_C \sin x\cos y\,dx + (xy + \cos x\sin y)\,dy$, where C is the boundary of the region lying between the graphs of $y = x$ and $y = \sqrt{x}$ oriented in the counterclockwise direction

5. $\int_C xy\,dx + (x+y)\,dy$, where C is the boundary of the region lying between the graphs of $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$ oriented in the counterclockwise direction

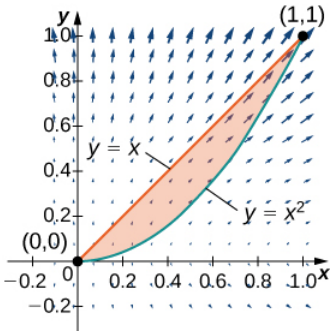
6. $\oint_C (-y\,dx + x\,dy)$, where C consists of line segment C_1 from $(-1,0)$ to $(1,0)$, followed by the semicircular arc C_2 from $(1,0)$ back to $(-1,0)$

For the following exercises, use Green's theorem.

7. Let C be the curve consisting of line segments from $(0,0)$ to $(1,1)$ to $(0,1)$ and back to $(0,0)$. Find the value of $\int_C xy\,dx + \sqrt{y^2+1}\,dy$.

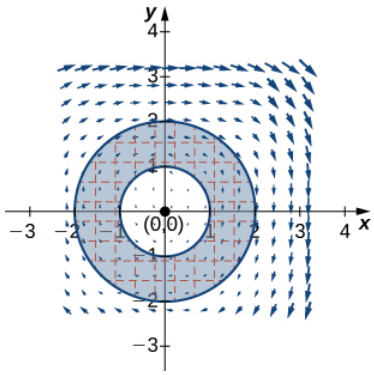
8. Evaluate line integral $\int_C xe^{-2x}\,dx + (x^4 + 2x^2y^2)\,dy$, where C is the boundary of the region between circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, and is a positively oriented curve.

9. Find the counterclockwise circulation of field $\vec{\mathbf{F}}(x,y) = xy\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}}$ around and over the boundary of the region enclosed by curves $y = x^2$ and $y = x$ in the first quadrant and oriented in the counterclockwise direction.



10. Evaluate $\oint_C y^3\,dx - x^3y^2\,dy$, where C is the positively oriented circle of radius 2 centered at the origin.

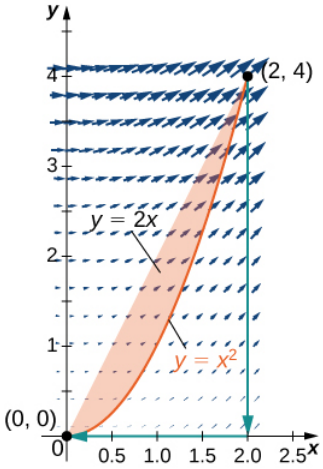
11. Evaluate $\oint_C y^3\,dx - x^3\,dy$, where C includes the two circles of radius 2 and radius 1 centered at the origin, both with positive orientation.



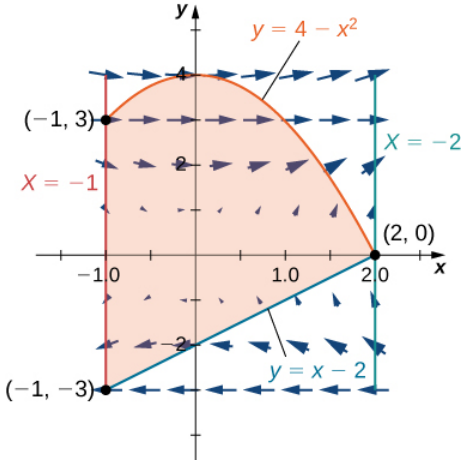
12. Calculate $\oint_C -x^2y\,dx + xy^2\,dy$, where C is a circle of radius 2 centered at the origin and oriented in the counterclockwise direction.

13. Calculate integral $\oint_C 2[y + x\sin(y)]\,dx + [x^2\cos(y) - 3y^2]\,dy$ along triangle C with vertices $(0,0)$, $(1,0)$ and $(1,1)$, oriented counterclockwise, using Green's theorem.

14. Evaluate integral $\oint_C (x^2 + y^2)\,dx + 2xy\,dy$, where C is the curve that follows parabola $y = x^2$ from $(0,0)$, $(2,4)$, then the line from $(2,4)$ to $(2,0)$, and finally the line from $(2,0)$ to $(0,0)$.



15. Evaluate line integral $\oint_C (y - \sin(y)\cos(y))\,dx + 2x\sin^2(y)\,dy$, where C is oriented in a counterclockwise path around the region bounded by $x = -1$, $x = 2$, $y = 4 - x^2$, and $y = x - 2$.



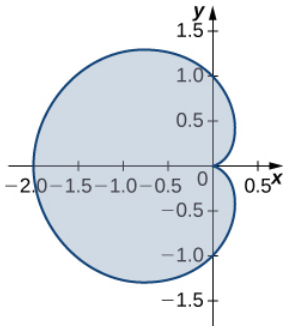
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For the following exercises, use Green's theorem to find the area.

16. Find the area between ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and circle $x^2 + y^2 = 25$.

17. Find the area of the region enclosed by parametric equation

$$\vec{\mathbf{p}}(\theta) = (\cos(\theta) - \cos^2(\theta))\,\hat{\mathbf{i}} + (\sin(\theta) - \cos(\theta)\sin(\theta))\,\hat{\mathbf{j}} \text{ for } 0 \le \theta \le 2\pi.$$



18. Find the area of the region bounded by hypocycloid $\vec{\mathbf{r}}(t) = \cos^3(t)\,\hat{\mathbf{i}} + \sin^3(t)\,\hat{\mathbf{j}}$. The curve is parameterized by $t \in [0, 2\pi]$.

19. Find the area of a pentagon with vertices $(0, 4)$, $(4, 1)$, $(3, 0)$, $(-1, -1)$, and $(-2, 2)$.

20. Use Green's theorem to evaluate $\int_{C^+} (y^2 + x^3)\,dx + x^4\,dy$, where C^+ is the perimeter of square $[0, 1] \times [0, 1]$ oriented counterclockwise.

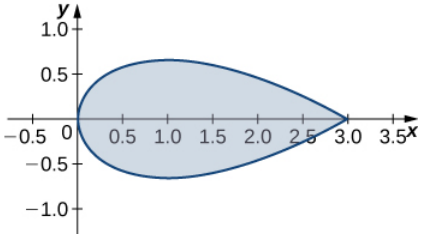
21. Use Green's theorem to prove the area of a disk with radius a is $A = \pi a^2$ units^2.

22. Use Green's theorem to find the area of one loop of a four-leaf rose $r = 3\sin 2\theta$. (*Hint*: $x\,dy - y\,dx = r^2\,d\theta$).

23. Use Green's theorem to find the area under one arch of the cycloid given by the parametric equations: $x = t - \sin t,\ y = 1 - \cos t,\ t \ge 0$.

24. Use Green's theorem to find the area of the region enclosed by curve

$$\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + \left(\tfrac{t^3}{3} - t\right)\hat{\mathbf{j}}, \text{ for } -\sqrt{3} \le t \le \sqrt{3}.$$



25. [T] Evaluate Green's theorem using a computer algebra system to evaluate the integral $\int_C xe^y\,dx + e^x\,dy$, where C is the circle given by $x^2 + y^2 = 4$ and is oriented in the counterclockwise direction.

26. Evaluate $\int_C (x^2y - 2xy + y^2)\,ds$, where C is the boundary of the unit square $0 \le x \le 1,\ 0 \le y \le 1$, traversed counterclockwise.

27. Evaluate $\int_C \dfrac{-(y+2)\,dx + (x-1)\,dy}{(x-1)^2 + (y+2)^2}$, where C is any simple closed curve with an interior that does not contain point $(1, -2)$ traversed counterclockwise.

28. Evaluate $\int_C \dfrac{x\,dx + y\,dy}{x^2 + y^2}$, where C is any piecewise, smooth simple closed curve enclosing the origin, traversed counterclockwise.

For the following exercises, use Green's theorem to calculate the work done by force $\vec{\mathbf{F}}$ on a particle that is moving counterclockwise around closed path C.